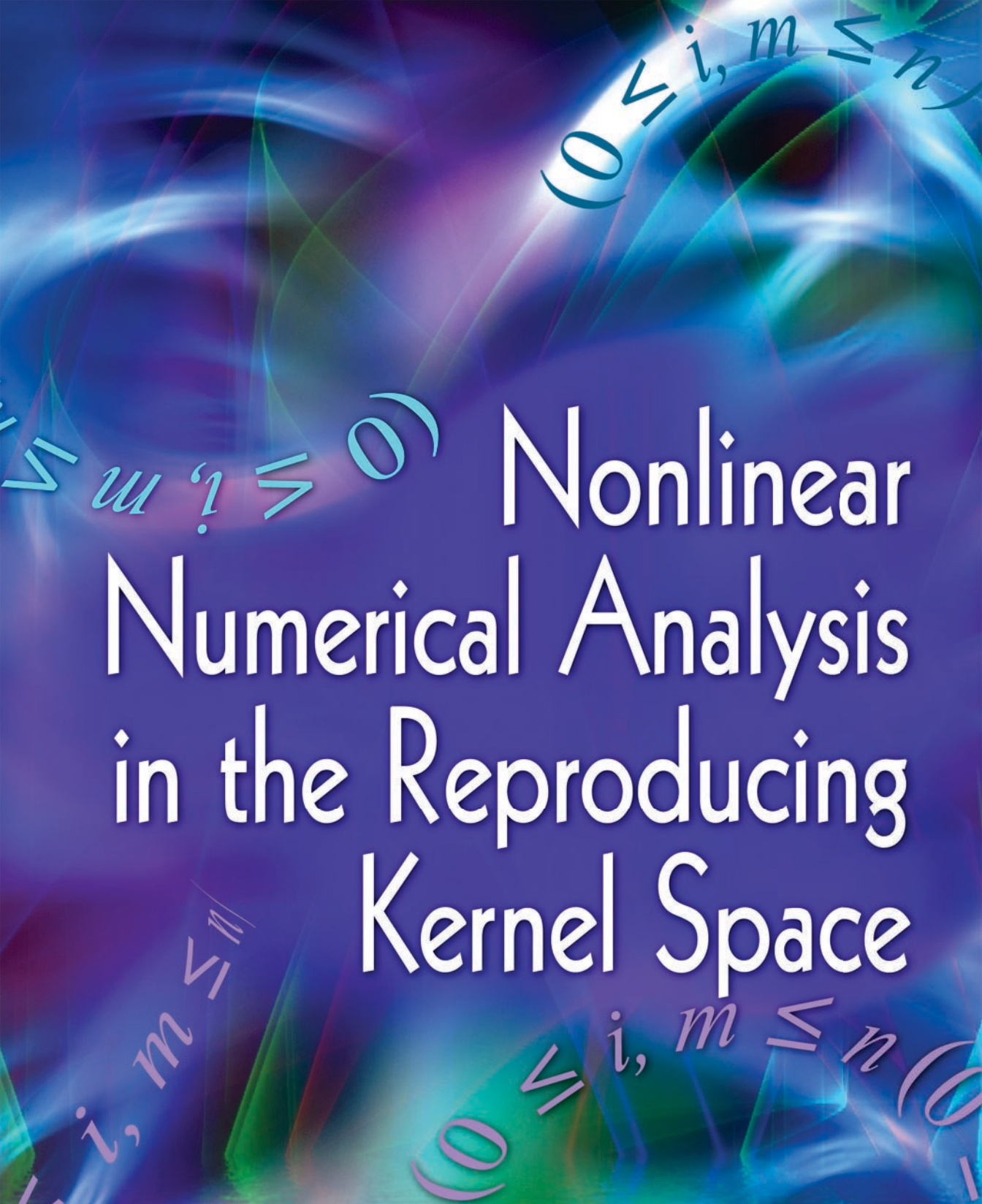


Nonlinear Numerical Analysis in the Reproducing Kernel Space

Minggen Cui · Yingzhen Lin

NOVA

Nonlinear Numerical Analysis in the Reproducing Kernel Space

By

Minggen Cui and Yingzhen Lin

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Foreword

This book is written with the invitation of Nova, the world famous publishing house. It is our pleasure that Nova has given us such a good opportunity to publish the book. We publish the book with the following two purposes: 1. Although the application of reproducing kernel has been explored in different fields in the past twenty to thirty years and the relevant researches are active in the recent five years, there is still not a book on the application of reproducing kernel. We intend to introduce a few commonplace remarks so that others may come up with valuable opinions. 2. We have been engaged in the research of reproducing kernel application for more than 20 years and worked out some feasible solutions for many problems. This book attempts to introduce to the readers engaged in mathematical application these solutions, especially the constructing theory of the reproducing kernel space that we have originally created and gradually improved. Reproducing kernel space is a special Hilbert space. We have been engaged in the constructing theory research of the reproducing kernel space since 1980's, and worked out a series of specific structural methods for reproducing kernel space and reproducing kernel functions represented by finite simple functions. In recent years, many differential equations reflecting the objective phenomena have been established, arousing the boundary value problems of the partial differential equation with various boundary value conditions, such as the periodic boundary value condition, linear boundary value condition and non-local boundary value condition. Some boundary value conditions are very complex, for instance, the integral boundary value conditions are determined by one or several given indefinite boundary value conditions, that is, the boundary is the function of time, t. We can rationally construct corresponding reproducing kernel space according to the various determining solution conditions of the differential equation. To speak more exactly, we can construct the reproducing kernel space satisfying such determining solution conditions and work out the reproducing kernel functions of the space. In this way, the determining solution problems will be half simplified, no matter how complex they are. In the second half of the task, the reproducing kernel represented by the finite simple function will almost "shoulder all the work". This is the strong application advantage of the reproducing kernel space, which the readers can figure it out in the following chapters.

In recent years, there are many papers on the solution of nonlinear problems with

reproducing kernel methods, which are published in different periodicals. We have summarized the representative concepts and methods of these papers in this book. In principle, readers can directly read any chapter after reading Chapter 1. These contents deal with the hot topics these days in the field of international mathematics.

The concept of reproducing kernel can be traced back to the paper of Zaremba [1] in 1908. It was proposed for discussing the boundary value problems of the harmonic functions. This is the first reproducing kernel corresponding to function family introduced in special case and with the reproducibility proved. Actually, in the early development stage of the reproducing kernel theory, most of the works were carried out by Bergman [2]–[8], and most of the kernels discussed in the 1930's and 1940's are Bergman kernels. Bergman put forward the corresponding kernels of the harmonic functions with one or several variables, and the corresponding kernel of the analytic function in squared metric, and applied them in the research of the boundary value problem of the elliptic partial differential equation. It can be said that this is the first stage in the development history of reproducing kernel. The second development stage of the reproducing kernel theory was initiated by Mercer [9]. Mercer discovered that the continuous kernel of the positive definite integral equation has the positive definite property:

$$\sum_{i,j=1}^{n} k(x_i, y_j)\xi_i\xi_j \geq 0.$$

He named the kernel with this property as positive definite Hermite matrix. He also discovered that the positive definite Hermite matrix corresponded to a function family, proposed a Hilbert space with inner product $\langle f, g\rangle$, and proved the reproducibility of the kernel in this space:

$$f(y) = \langle f(x), k(x, y)\rangle.$$

The third development stage of the reproducing kernel theory belonged to N. Aronszajn. In 1950, he summarized the works of the predecessors and worked out a systematic reproducing kernel theory including the Bergman kernel function ([13]). This reproducing kernel theory has laid a good foundation for the research of each special case and greatly simplified the proof. In this theory, the reproducibility of the kernel functions plays an important role. It has also proved that the reproducing kernel has positive definite Hermite matrix, thus unifying the Bergman concept in the first stage of reproducing kernel development and the Bercer concept in the second stage.

Since then, many mathematicians have used the reproducing kernel theory to solve the theoretical problems of many special fields. For example, Brylinski and Raneea [11] constructed a reproducing kernel in the quantization Hilbert space to research the Hamilton quantization of energy operator. Wilson and Raja made a research on the holomorphic discrete series in the space SU_{PA} [12]. Alpay used the

generic structure theorem to exemplify the relationship between the reproducing kernel space and the positive branch polynomial [13].

In 1970, Larkin put forward the optimal approximation principle in the Hilbert function space with reproducing kernel [14]. In 1974, Chawala proposed the optimal approximation rules for the Hilbert function space with reproducing kernel and polynomial precision [15].

There are many examples of using the reproducing kernel to solve the application problems. And there are many successful application examples in such fields as signal processing [16], stochastic process [17], estimation theory [18], [19] and wavelet transform [20]. It can be said that the most attractive function space in the various function spaces for numerical analysis is the reproducing kernel space. A simple example below will help to prove this.

Assuming that $R_x^{(j)}(y)$, x, $y \in [0,1]$ is the reproducing kernel under the inner product $\langle \cdot, \cdot \rangle_{H_j}$ in the reproducing kernel space H_j, $\{x_i\}_{i=0}^{\infty}$ is the dense point set of $[0,1]$, and $\varphi_i(x) \overset{def}{=} R_{x_i}^{(1)}(x)$, we will discuss the solution problem of the operator equation $\mathbb{L}u = f$, $u \in H_2$, $f \in H_1$, where $\mathbb{L}$ is the bounded linear operator of $H_2 \longrightarrow H_1$. We use $\mathbb{L}^*$ to represent the conjugate operator of $\mathbb{L}$, and set $\psi_i(x) = \mathbb{L}^*\varphi_i(x)$. If $\overline{\psi}_i(x)$ is used to represent a normal orthogonal system (this is a complete system of H_2) obtained using Gram–Schmidt orthogonalization process of $\psi_i(x)$,

$$\overline{\psi}_i(x) = \sum_{k=1}^{i} \beta_{ik}\varphi_k(x).$$

The solution of this operator equation can be represented as:

$$\begin{aligned} u(x) &= \sum_{i=1}^{\infty} \langle u, \overline{\psi}_i \rangle_{H_2} \overline{\psi}_i(x) \\ &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} \langle \mathbb{L}u, \varphi_k \rangle_{H_1} \overline{\psi}_i(x) \\ &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} f(x_k) \overline{\psi}_i(x). \end{aligned}$$

We use three steps to complete the solution process of this operator equation:

(1) Construct the reproducing kernel space and corresponding reproducing kernel satisfying all the conditions for determining solution. The ability of the reproducing kernel space to absorb all the conditions for determining solution reflects the feature of the reproducing kernel method.

(2) Use $\psi_i(x) = \mathbb{L}^*\varphi_i(x)$ to construct the base of the space. It is hard to express the conjugate operator $\mathbb{L}^*$ in generic space. However, in reproducing kernel space, $\mathbb{L}^*\varphi_i(x)$ is a definite expression represented by elementary function.

(3) Use the reproducibility of $\varphi(x) = R_{x_i}(x)$: $\langle f, \varphi_i \rangle_{H_1} = f(x_i)$. As shown above, we have precisely expressed the solution of the linear operator equation in the form of series through the function value at the right side of the equation.

The successful solution of the linear problem is the base for solving the nonlinear problems. In the past, it was considered impossible to precisely express the nonlinear problems. However, we have now precisely expressed the solution of the nonlinear differential equation with elementary function in the form of series. More and more papers are published on this field.

As mentioned above, one of the keys to solve the nonlinear problem with reproducing kernel method is to construct the reproducing kernel space satisfying the conditions for determining solution of the nonlinear problem. And another key is to use the property of reproducing kernel to further describe the approximate solution. We can prove through strict argumentation that the solution and its derivative of corresponding order are both uniformly convergent.

The first task for reflecting the reproducing kernel construction theory shall be collating the relevant literature [21]. Later, especially in recent years, the reproducing kernel has been greatly simplified through the effort of the second author of this book. As a result, the reproducing kernel with various boundary conditions can be expressed in the spline form of polynomial. The computing speed has been greatly increased. This is an essential improvement of the reproducing kernel construction theory. We will summarize this in Chapter 1.

This book includes two parts, the first of which is prepared by the second author, while the second by the first author. It is divided into 7 chapters. Chapter 1 introduces the reproducing kernel construction theory. Its content is different from that of the paper "Theory of Reproducing Kernel", and there is no repetition. Chapter 2 discusses the linear problems, including the singularly perturbation problem and obstacle problem which are the hot topics of recent years. Linear problem is a basic problem and the base for discussing the nonlinear problem in the subsequent chapters. Chapter 3 discusses the algebraic equation problems, including the infinite linear system of equation and infinite system of quadratic equation. In principle, a nonlinear differential equation can be converted to an infinite system of algebraic equation. The discussion of indefinite system of algebraic equation is of extensive significance. Chapter 4 discusses the solution problem of nonlinear integral equation. The theory and algorithm of the linear integral equation are very mature, and it is a typical problem. However, the discussion on the qualitative problem of nonlinear integral equation is very active in recent years. Comparatively, discussion of numerical method research is much less. This book will discuss the numerical method on several typical nonlinear integral equations. Chapter 5 discusses the nonlinear differential equation. In terms of nonlinear ordinary differential equation and partial differential equation, both the qualitative problem and the numerical problems are the most concerned and active research directions. We do not want to superficially describe the theories and methods familiar to people or introduce

the theories and methods appearing in the papers one by one in recent years. We will only introduce the reproducing kernel method on the typical hot topics and problems in the past and recent years rather than focusing on specific problems. Chapter 6 discusses the solution method for a typical quadrate nonlinear operator equation ($\mathbb{A}u\mathbb{B}u + \mathbb{C}u = f$, $\mathbb{A}$, $\mathbb{B}$, $\mathbb{C}$ is the linear operator). It is easy to apply this method to the solution of n-th power nonlinear operator equation. This chapter will discuss how to convert the solution of nonlinear operator equation into the solution of nonlinear system of operator equation. Chapter 7 discusses the coefficient inverse problem of differential equation. It is a topic with strong application background, and there are many theories and qualitative researches on the inverse problem. The two inverse problems discussed in this chapter are used to exemplify the solution of nonlinear problem with reproducing kernel method. This book does not aim to be involved with specific problems.

This book is the summary of the first author's over twenty years of work on the reproducing kernel space numerical fields after he proposed the reproducing kernel expression for a specific reproducing kernel space $W_2^1[a, b]$ in 1986. The title *Nonlinear Numerical Analysis of Reproducing Kernel Space* seemed to be too "big", but we later took it as appropriate. The reason is that although this book does not cover all the contents of numerical analysis, it discusses the most important content of numerical analysis – the numerical method of differential equation, and that we have provided complete theory system and algorithm for each problem in the book based on the special property and skills of the reproducing kernel space, and proved the effectiveness of these algorithms through numerous experiments. The initial research results of the reproducing kernel space were obtained when Minggen Cui was studying for the doctor's degree in Harbin Institute of Technology in the 1980's. The author hereby expresses his sincere gratitude to his supervisors Professor Zhang Bo and Professor Wu Congxin for their care and guidance. We also want to express thanks to our doctorate students Geng Fazhan, Du Hong, Chen Zhong, Zhou Yongfang, Yao Huanmin and Yang Lihong because they have provided many valuable materials for this book. We'd appreciate it if any comment or advice will be given on the book.

December 2007

Minggen Cui, Yingzhen Lin

Part I

1

Chapter 1

Fundamental Concepts of Reproducing Kernel Space

In this chapter, we introduce some concepts, properties and theorems of the reproducing kernel, and provide relevant proof and examples to help with the reading and understanding of the readers. Especially, we will provide detailed deduction for the solution of the reproducing kernel function $R_y(x)$ of the reproducing kernel space under different conditions for the understanding and application of the readers. After reading Chapters 2 through 7, readers will feel the advantages of the reproducing kernel space in solving many actual problems.

1.1 Definition of Reproducing Kernel Space

Definition 1.1.1. Let

$$\mathcal{H} = \Big\{ f(x) |\ f(x) \text{ is a real value function or complex function,} \\ x \in \mathrm{X},\ \mathrm{X} \text{ is a abstract set} \Big\}$$

is a Hilbert space, whth inner product

$$\langle f(x), g(x) \rangle_{\mathcal{H}}, \quad (f(x),\, g(x) \in \mathcal{H}).$$

If there exists a function $R_y(x)$, for each fixed $y \in \mathrm{X}$, then $R_y(x) \in \mathcal{H}$, and any $f(x) \in \mathcal{H}$, which satisfies

$$\langle f(x), R_y(x) \rangle_{\mathcal{H}} = f(y), \tag{1.1.1}$$

then $R_y(x)$ is called the reproducing kernel of $\mathcal{H}$ and Hilbert space $\mathcal{H}$ is called the reproducing kernel space.

Example 1.1.1. n-dimensional Euclidean $\mathbb{R}^n$ is a reproducing kernel space.

$\mathbb{R}^n = \{\mathbf{a}|\ \mathbf{a} = (a_1, a_2, \ldots, a_n),\ a_i \in \mathbb{R}\}$.

$\mathbf{e}_m(i) = \begin{cases} 0, & i \neq m \\ 1, & i = m \end{cases}$ $(0 \leq i,\ m \leq n)$ is a reproducing kernel of $\mathbb{R}^n$.

In fact, $\mathbf{e}_m = (0, \ldots, \overset{m}{1}, \ldots, 0)$, and $\langle \mathbf{a}, \mathbf{e}_m \rangle = a_m$.

Similarly,

$$\ell^2\text{-space} = \left\{ \mathbf{a}|\ \mathbf{a} = (a_1, a_2, \ldots),\ a_i \in \mathbb{R},\ \sum_{i=1}^{\infty} a_i^2 < \infty \right\}$$

is a reproducing kernel space.

Example 1.1.2. Let $\mathcal{H}$ is a n-dimensional Hilbert space, $\{e_i\}_{i=1}^n$ is the normal orthogonal basis, that is

$$\langle e_i(t), e_j(t) \rangle_{\mathcal{H}} = \begin{cases} 0, & i \neq j, \\ 1, & i = j, \end{cases}$$

then $R_s(t) = \sum_{i=1}^{n} e_i(t)\overline{e_i(s)}$ is the reproducing kernel of $\mathcal{H}$.

In fact, for every $f(t) \in \mathcal{H}$,

$$\begin{aligned} f(t) &= \sum_{i=1}^{n} a_i e_i(t) \ \ (a_i \ \text{are complex numbers}), \\ \langle f(t), R_s(t) \rangle_{\mathcal{H}} &= \left\langle \sum_{i=1}^{n} a_i e_i(t), \sum_{j=1}^{n} e_j(t)\overline{e_j(s)} \right\rangle_{\mathcal{H}} = \sum_{i=1}^{n} a_i e_i(s) = f(s). \end{aligned}$$

1.2 Fundamental Properties of Reproducing Kernel

Property 1.2.1. *If $\mathcal{H}$ is a reproducing kernel space, then reproducing kernel $R_y(x)$ in $\mathcal{H}$ is conjugate symmetric, that is $R_y(x) = \overline{R_x(y)}$.*

Proof.

$$R_y(x) = \langle R_y(\cdot), R_x(\cdot) \rangle_{\mathcal{H}} = \overline{\langle R_x(\cdot), R_y(\cdot) \rangle_{\mathcal{H}}} = \overline{R_x(y)}. \qquad \square$$

Property 1.2.2. *If $\mathcal{H}$ is a reproducing kernel space, then reproducing kernel $R_y(x)$ in $\mathcal{H}$ is unique.*

Proof. Let $Q_y(x)$ be also a reproducing kernel of $\mathcal{H}$, then

$$Q_y(x) = \langle Q_y(\cdot), R_x(\cdot) \rangle_{\mathcal{H}} = \overline{\langle R_x(\cdot), Q_y(\cdot) \rangle_{\mathcal{H}}} = \overline{R_x(y)} = R_y(x). \qquad \square$$

Property 1.2.3. *If $R_y(x)$ is the reproducing kernel in $\mathcal{H}$, then for each $x \in \mathrm{X}$ $R_x(x) \geq 0$ and $R_x(x) = 0$ if and only if $\mathcal{H} = \{0\}$.*

Proof.

$$R_x(x) = \langle R_x(\cdot\,), R_x(\cdot\,)\rangle_{\mathcal{H}} = \|R_x(\cdot\,)\|^2_{\mathcal{H}}.$$ □

Property 1.2.4. *Reproducing kernel $R_y(x)$ is a positive definite kernel, that is,*

$$\sum_{i,j=1}^{n} R_{x_i}(x_j)\overline{\xi_i}\xi_j \geq 0 \ \text{ for every } \ x_i \in \mathrm{X},$$

ξ_i are any complex numbers $i, j = 1, \ldots, n$).

See [10]

Property 1.2.5. *Hilbert function space $\mathcal{H}$ is a reproducing kernel space if and only if for any fixed $x \in \mathrm{X}$, the linear functional $\mathbf{I}(f) = f(x)$ is bounded.*

Proof. Necessary. Since $\mathcal{H}$ is a reproducing kernel space, there exists a reproducing kernel $R_y(x)$. For every linear functional $\mathbf{I}(f) = f(x)$, we have

$$|\mathbf{I}(f)| = f(x) = \left|\langle f(\cdot\,), R_x(\cdot\,)\rangle_{\mathcal{H}}\right| \leq \|f(\cdot\,)\|_{\mathcal{H}}\|R_x(\cdot\,)\|_{\mathcal{H}} \tag{1.2.1}$$

$$= \|f\|_{\mathcal{H}}\sqrt{R_x(x)} = M_x\|f\|_{\mathcal{H}} \ \ (M_x > 0). \tag{1.2.2}$$

Sufficiency. For $\forall\, f(x) \in \mathcal{H}$, due to $\mathbf{I}(f) = f(x)$ is a linear functional, by F. Riesz theorem, there exists a unique $R_x(\cdot\,) \in \mathcal{H}$, so that

$$f(x) = \mathbf{I}(f) = \langle f, R_x\rangle_{\mathcal{H}}.$$

Therefore $R_y(x)$ is the unique reproducing kernel of $\mathcal{H}$. □

1.3 Reproducing Kernel Space $W_2^m[a, b]$ and its Reproducing Kernel Function

1.3.1 Absolutely Continuous Function and Some Properties

Definition 1.3.1. Given a function $f(x)$ on interval $[a, b]$, let $\{(a_k, b_k)\}_{k=1}^n$ is a set of mutually disjoint open intervals $(a_k, b_k) \subset [a, b]$. If for $\forall\,\varepsilon$, $\exists\,\delta$ which has no relation whit n, such that

$$\sum_{i=1}^{n} \left|f(b_i) - f(a_i)\right| < \varepsilon \ \text{ for } \ \sum_{i=1}^{n}(b_i - a_i) < \delta,$$

then the function $f(x)$ is said to absolutely continuous on interval $[a, b]$.

Proposition 1.3.1. *A absolutely continuous function on interval $[a,b]$ is the continuous function.*

Proposition 1.3.2. *Given a function $f(x)$ on interval $[a,b]$, if $|f'(x)| \le M$, $x \in [a,b]$, then $f(x)$ is absolutely continuous.*

Proof. For arbitrary $\varepsilon > 0$, choose $\delta = \frac{\varepsilon}{M}$. Let $\{(a_k,b_k)\}_{k=1}^n$ be a set of mutually disjoint open intervals $(a_k,b_k) \subset [a,b]$, satisfying $\sum_{i=1}^n (b_i - a_i) < \delta$. Note that

$$\big|f(b_k) - f(a_k)\big| = |f'(\xi_k)|(b_k - a_k) < M(b_k - a_k), \;\; \xi_k \in (b_k, a_k) \subset [a,b].$$

We have

$$\sum_{k=1}^n \big|f(b_k) - f(a_k)\big| < M \sum_{k=1}^n (b_k - a_k) < M\,\frac{\varepsilon}{M} = \varepsilon. \qquad \square$$

Corollary 1.3.1. *Suppose that a function $f'(x)$ is continuous on $[a,b]$, then $f(x)$ is absolutely continuous on $[a,b]$.*

Proposition 1.3.3. *If function $f(x)$ is integrable on interval $[a,b]$, then function $\int_a^x f(t)dt$ is absolutely continuous on $[a,b]$.*

Proof. By absolute continuity of integral, for arbitrary $\varepsilon > 0$, there exists a $\delta > 0$, such that $\int_E f(t)dt < \varepsilon$ for $m(E) < \delta$. Assume $\{(a_k,b_k)\}_{k=1}^n$ is a set of mutually disjoint open intervals $(a_k,b_k) \subset [a,b]$, satisfying $\sum_{i=1}^n (b_i - a_i) < \delta$. It is easy to see that

$$\begin{aligned} \sum_{k=1}^n \left| \int_a^{b_k} f(t)dt - \int_a^{a_k} f(t)\,dt \right| &= \sum_{k=1}^n \left| \int_{a_k}^{b_k} f(t)\,dt \right| \\ \le \sum_{k=1}^n \int_{a_k}^{b_k} |f(t)|\,dt &= \int_{\bigcup_{k=1}^n (a_k,b_k)} |f(t)|\,dt < \varepsilon. \end{aligned} \tag{1.3.1}$$

The proof is complete. $\square$

1.3.2 Function Space $W_2^m[a,b]$ is a Hilbert Space

The function space $W_2^m[a,b]$ is defined as follows:

$$W_2^m[a,b] = \Big\{ f(x) |\; f^{(m-1)}(x) \text{ is absolutely continuous, } f^{(m)}(x) \in L^2[a,b],\; x \in [a,b] \Big\}. \tag{1.3.2}$$

The inner product and the norm in the function space $W_2^m[a,b]$ are defined as follows.

For any functions $f(x)$, $g(x) \in W_2^m[a,b]$,

$$\langle f, g\rangle_{W_2^m} = \sum_{i=0}^{m-1} f^{(i)}(a)g^{(i)}(a) + \int_a^b f^{(m)}(x)g^{(m)}(x)\,dx, \tag{1.3.3}$$

$$\|f\|_{W_2^m} = \sqrt{\langle f, f\rangle_{W_2^m}}. \tag{1.3.4}$$

It is easy to prove that $W_2^m[a,b]$ is an inner space.

Theorem 1.3.1. *Function space $W_2^m[a,b]$ is a Hilbert Space.*

Proof. Suppose $f_n(x)$ $(n = 1, 2, \ldots)$ is a Cauchy sequence in $W_2^m[a,b]$, i.e.,

$$\|f_{n+p} - f_n\|_{W_2^m}^2 = \sum_{i=0}^{m-1} \left[f_{n+p}^{(i)}(a) - f_n^{(i)}(a)\right]^2 + \int_a^b \left[f_{n+p}^{(m)}(x) - f_n^{(m)}(x)\right]^2 dx \underset{n\to\infty}{\longrightarrow} 0.$$

Therefore, we have

$$f_{n+p}^{(i)}(a) - f_n^{(i)}(a) \underset{n\to\infty}{\longrightarrow} 0, \quad i = 0, 1, \ldots, m-1$$

and

$$\int_a^b \left[f_{n+p}^{(m)}(x) - f_n^{(m)}(x)\right]^2 dx \underset{n\to\infty}{\longrightarrow} 0,$$

which indicates that for any i $(0 \le i \le m-1)$, the sequence $f_n^{(i)}(a)$ $(n = 1, 2, \ldots)$ is a Cauchy sequence in $\mathbb{R}$ and $f_n^{(m)}(x)$ $(n = 1, 2, \ldots)$ is a Cauchy sequence in space $L^2[a,b]$. So, there exist unique real number λ_i $(i = 0, 1, \ldots, m-1)$ and unique function $h(x) \in L^2[a,b]$, satisfying:

$$\lim_{n\to\infty} f_n^{(i)}(a) \longrightarrow \lambda_i \quad (i = 0, 1, \ldots, m-1)$$

and

$$\lim_{n\to\infty} \int_a^b [f_n^{(m)}(x) - h(x)]^2\,dx \longrightarrow 0.$$

Suppose

$$g(x) = \sum_{i=0}^{m-1} \frac{\lambda_k}{k!}(x-a)^k + \overbrace{\int_a^x \int_a^x \cdots \int_a^x}^{m} h(x)(dx)^m.$$

Since $h(x) \in L^2[a,b]$,

$$g^{(m-1)}(x) = \lambda_{m-1} + \int_a^x h(x)\,dx$$

is absolutely continuous in $[a,b]$ and

$$g^{(m)}(x) = h(x) \in L^2[a,b]$$

is true almost everywhere in $[a,b]$. We must have, $g(x) \in W_2^m[a,b]$ and $g^{(i)}(a) = \lambda_i$ $(i = 0, 1, \ldots, m-1)$. Moreover, we have

$$\begin{aligned} \|f_n(x) - g(x)\|^2_{W_2^m} &= \sum_{i=0}^{m-1} \left[f_n^{(i)}(a) - g^{(i)}(a)\right]^2 + \int_a^b [f_n^{(m)}(x) - g^{(m)}(x)]^2\,dx \\ &= \sum_{i=0}^{m-1} \left[f_n^{(i)}(a) - \lambda_i\right]^2 + \int_a^b [f_n^{(m)}(x) - h(x)]^2 dx \underset{n\to\infty}{\longrightarrow} 0. \end{aligned}$$

Hence, function space $W_2^m[a,b]$ is a Hilbert Space. □

1.3.3 Function Space $W_2^m[a,b]$ is a Reproducing Kernel Space

Theorem 1.3.2. *Function space $W_2^m[a,b]$ is a reproducing kernel space.*

Proof. Suppose $\mathbf{I}(f) = f(x)$, $x \in [a,b]$ is a linear functional of $W_m^2[a,b]$ and $f(x) \in W_2^m[a,b]$. We have

$$f^{(m-1)}(x) = f^{(m-1)}(a) + \int_a^x f^{(m)}(x)\,dx$$

and

$$|f^{(m-1)}(x)| \le |f^{(m-1)}(a)| + \int_a^x |f^{(m)}(x)|\,dx \le |f^{(m-1)}(a)| + \int_a^b |f^{(m)}(x)|\,dx.$$

Note that

$$\begin{aligned} \int_a^b |f^{(m)}(x)|\,dx &\le \left[(b-a)\int_a^b |f^{(m)}(x)|^2\,dx\right]^{1/2} = M_0\left[\int_a^b |f^{(m)}(x)|^2\,dx\right]^{1/2} \\ &\le M_0\left[\sum_{i=0}^{m-1}[f^{(i)}(a)]^2 + \int_a^b |f^{(m)}(x)|^2\,dx\right]^{1/2} = M_0\|f\|_{W_2^m}, \end{aligned}$$

and for any $i (0 \le i \le m-1)$,

$$|f^{(i)}(a)| \le \left[\sum_{i=0}^{m-1}[f^{(i)}(a)]^2 + \int_a^b |f^{(m)}(x)|^2\,dx\right]^{1/2} = \|f\|_{W_2^m}. \tag{1.3.5}$$

Therefore

$$|f^{(m-1)}(x)| \leq M_1 \|f\|_{W_2^m}. \tag{1.3.6}$$

Noting that

$$\begin{aligned} |f^{(m-2)}(x)| &\leq |f^{(m-2)}(a)| + \int_a^x |f^{(m-1)}(x)|\, dx \\ &\leq |f^{(m-2)}(a)| + \int_a^b |f^{(m-1)}(x)|\, dx, \end{aligned}$$

from (1.3.5) and (1.3.6), we have

$$|f^{(m-2)}(x)| \leq \|f\|_{W_2^m} + (b-a)M_1\|f\|_{W_2^m} = M_2\|f\|_{W_2^m}. \tag{1.3.7}$$

Similarly, we have

$$|\mathbf{I}(f)| = |f(x)| \leq M_m \|f\|_{W_2^m}. \tag{1.3.8}$$

So, I is bounded functional in $W_2^m[a,b]$. By Property 1.2.5, we know $W_2^m[a,b]$ is a reproducing kernel space. □

Now, let's find out the expression form of the reproducing kernel function $R_y(x)$ in $W_2^m[a,b]$.

Suppose $R_y(x)$ is the reproducing kernel function of $W_2^m[a,b]$, then for any fixed $y \in [a,b]$ and any $f(x) \in W_2^m[a,b]$, $R_y(x)$ must satisfy

$$\langle f(x), R_y(x)\rangle_{W_2^m} = f(y). \tag{1.3.9}$$

Applying (1.3.3), we have

$$\langle f(x), R_y(x)\rangle_{W_2^m} = \sum_{i=0}^{m-1} f^{(i)}(a)\, \frac{\partial^i R_y(a)}{\partial x^i} + \int_a^b f^{(m)}(x)\, \frac{\partial^m R_y(x)}{\partial x^m}\, dx,$$

and

$$\begin{aligned} &\int_a^b f^{(m)}(x) \frac{\partial^m R_y(x)}{\partial x^m}\, dx \\ &\qquad = \sum_{i=0}^{m-1} (-1)^i f^{(m-i-1)}(x)\ \frac{\partial^{m+i} R_y(x)}{\partial x^{m+i}}\bigg|_{x=a}^{b} + (-1)^m \int_a^b f(x)\, \frac{\partial^{2m} R_y(x)}{\partial x^{2m}}\, dx. \end{aligned}$$

By variable substitution, one obtains

$$\sum_{i=0}^{m-1} (-1)^i f^{(m-i-1)}(x) \frac{\partial^{m+i} R_y(x)}{\partial x^{m+i}} = \sum_{i=0}^{m-1} (-1)^{m-i-1} f^{(i)}(x) \frac{\partial^{2m-i-1} R_y(x)}{\partial x^{2m-i-1}}.$$

Moreover,

$$\begin{aligned}\langle f(x), R_y(x)\rangle &= \sum_{i=0}^{m-1} f^{(i)}(a)\left[\frac{\partial^i R_y(a)}{\partial x^i} - (-1)^{m-i-1}\frac{\partial^{2m-i-1}R_y(a)}{\partial x^{2m-i-1}}\right] \\ &+ \sum_{i=0}^{m-1}(-1)^{m-i-1} f^{(i)}(b)\frac{\partial^{2m-i-1}R_y(b)}{\partial x^{2m-i-1}} \\ &+ (-1)^m \int_a^b f(x)\frac{\partial^{2m}R_y(x)}{\partial x^{2m}}\,dx.\end{aligned}$$

Therefore, $R_y(x)$ is the solution of the following generalized differential equation

$$\begin{cases} (-1)^m\dfrac{\partial^{2m}R_y(x)}{\partial x^{2m}} = \delta(x-y), \\ \dfrac{\partial^i R_y(a)}{\partial x^i} - (-1)^{m-i-1}\dfrac{\partial^{2m-i-1}R_y(a)}{\partial x^{2m-i-1}} = 0, \\ \dfrac{\partial^{2m-i-1}R_y(b)}{\partial x^{2m-i-1}} = 0, \quad i = 0, 1, \ldots, m-1. \end{cases} \tag{1.3.10}$$

While $x \neq y$, it is easy to know that $R_y(x)$ is the solution of the following constant linear homogeneous differential equation with $2m$ orders, i.e.,

$$(-1)^m\frac{\partial^{2m}R_y(x)}{\partial x^{2m}} = 0, \tag{1.3.11}$$

with the boundary conditions:

$$\begin{cases} \dfrac{\partial^i R_y(a)}{\partial x^i} - (-1)^{m-i-1}\dfrac{\partial^{2m-i-1}R_y(a)}{\partial x^{2m-i-1}} = 0, \\ \dfrac{\partial^{2m-i-1}R_y(b)}{\partial x^{2m-i-1}} = 0, \quad i = 0, 1, \ldots, m-1. \end{cases} \tag{1.3.12}$$

We know that equation (1.3.11) has characteristic equation $\lambda^{2m} = 0$, and the eigenvalue $\lambda = 0$ is a root whose multiplicity is $2m$. Therefore, the general solution of Eq. (1.3.10) is

$$R_y(x) = \begin{cases} lR_y(x) = \displaystyle\sum_{i=1}^{2m} c_i(y)x^{i-1}, & x < y, \\ rR_y(x) = \displaystyle\sum_{i=1}^{2m} d_i(y)x^{i-1}, & x > y. \end{cases} \tag{1.3.13}$$

Now we are ready to calculate the coefficients $c_i(y)$ and $d_i(y)$, $i = 1, 2, \ldots, 2m$.

Since

$$(-1)^m \frac{\partial^{2m} R_y(x)}{\partial x^{2m}} = \delta(x-y),$$

we have

$$\frac{\partial^i l R_y(y)}{\partial x^i} = \frac{\partial^i r R_y(y)}{\partial x^i}, \quad i = 0, 1, \ldots, 2m-2, \tag{1.3.14}$$

and

$$(-1)^m \left(\frac{\partial^{2m-1} l R_y(y^+)}{\partial x^{2m-1}} - \frac{\partial^{2m-1} r R_y(y^-)}{\partial x^{2m-1}} \right) = 1. \tag{1.3.15}$$

The above $2m$ equations in (1.3.14) and (1.3.15) provided $2m$ conditions for solving the coefficients $c_i(y)$ and $d_i(y)$ $(i = 1, 2, \ldots, 2m)$ in Eq. (1.3.13). Noting that Eq. (1.3.12) provided $2m$ boundary conditions, we have $4m$ equations, i.e., (1.3.12), (1.3.14) and (1.3.15). It is easy to know these $4m$ equations are linear equations with the variables $c_i(y)$ and $d_i(y)$, and $c_i(y)$ and $d_i(y)$ could be calculated by many methods. As long as the coefficients $c_i(y)$ and $d_i(y)$ are known, the exact expression of the reproducing kernel function $R_y(x)$ could be calculated from Eq. (1.3.13).

Proposition 1.3.4. *Let $R_y(x)$ is the reproducing kernel of the space $W_2^m[a,b]$, then*

$$\frac{\partial^{i+j} R_y(x)}{\partial x^i \partial y^j} \in L^2[a,b], \quad i+j = 2m-1$$

with respect to x or y.

Proof. By (1.3.10), for arbitrary x, $y \in [a,b]$,

$$\frac{\partial^{2m} R_y(x)}{\partial x^{2m}} = (-1)^{2m} \delta(x-y),$$

therefore

$$\frac{\partial^{2m-1} R_y(x)}{\partial x^{2m-1}} = (-1)^{2m} (x-y)_+^0 + C_1,$$

where

$$(x-y)_+^0 = \begin{cases} 1, & x > y, \\ 0, & x \le y, \end{cases}$$

C_1 is a constant. Note that

$$\frac{\partial^{2m} R_y(x)}{\partial x^{2m-1} \partial y} = (-1)^{2m-1} \delta(x-y).$$

Therefore

$$\frac{\partial^{2m-1} R_y(x)}{\partial x^{2m-2} \partial y} = (-1)^{2m-1} (x-y)_+^0 + C_2,$$

C_2 is a constant, namely,

$$\frac{\partial^{2m-1} R_y(x)}{\partial x^{2m-2} \partial y} \in L^2[a,b].$$

Using this method, one can obtain that

$$\frac{\partial^{i+j} R_y(x)}{\partial x^i \partial y^j} \in L^2[a,b], \;\; i+j = 2m-1$$

with respect to x or y. □

Corollary 1.3.2.

$$\frac{\partial^{i+j} R_y(x)}{\partial x^i \partial y^j}, \;\; 0 \le i+j \le 2m-2$$

is a absolutely continuous function in $[a,b]$ *with respect to* x *or* y.

Base on Proposition 1.3.4 and Corollary 1.3.2, we have

Theorem 1.3.3. *Let* $R_y(x)$ *is the reproducing kernel of the space* $W_2^m[a,b]$*, then*

$$\frac{\partial^{i+j} R_y(x)}{\partial x^i \partial y^j} \in W_2^m[a,b], \;\; i+j = m-1$$

with respect to x *or* y.

Theorem 1.3.4. *Let* $W_2^m[a,b]$ *is a reproducing kernel space and* $f_n(x), f(x) \in W_2^m[a,b]$ $(n = 1,2,\ldots)$*. If* $f_n(x)$ *converges to* $f(x)$ *in the sense of* $\|\cdot\|_{W_2^m}$*, then* $f_n^{(k)}(x)$ *converges to* $f^{(k)}(x)$ *uniformly,* $0 \le k \le m-1$.

Proof. Let $R_y(x)$ is the reproducing kernel of the space $W_2^m[a,b]$. Because

$$\begin{aligned} f_n(x) - f(x) &= \langle f_n(y) - f(y), R_x(y) \rangle_{W_2^m}, \\ f_n^{(k)}(x) - f^{(k)}(x) &= (f_n(x) - f(x))^{(k)} \\ &= \frac{\partial^k}{\partial x^k} \left[\langle f_n(y) - f(y), R_x(y) \rangle_{W_2^m} \right] \\ &= \left\langle f_n(y) - f(y), \frac{\partial^k}{\partial x^k} R_x(y) \right\rangle_{W_2^m} \end{aligned}$$

and $\frac{\partial^k}{\partial x^k} R_x(y) \in W_2^m[a,b]$, by Theory 1.3.3, one obtains

$$\left| f_n^{(k)}(x) - f^{(k)}(x) \right| \le \|f_n(y) - f(y)\|_{W_2^m} \left\| \frac{\partial^k}{\partial x^k} R_x(y) \right\|_{W_2^m}.$$

Also $\|\frac{\partial^k}{\partial x^k} R_x(y)\|_{W_2^m}$ is continuous with respect to x in $[a,b]$, then

$$\left| f_n^{(k)}(x) - f^{(k)}(x) \right| \le M \|f_n(y) - f(y)\|_{W_2^m},$$

where M is a positive number. So that

$$\lim_{n\to\infty} f_n(x) \underset{\|\cdot\|_{W_2^m}}{=} f(x) \Longrightarrow \lim_{n\to\infty} f_n(x) \underset{unif.}{=} f(x).$$

□

1.3.4 Closed Subspaces of the Reproducing Kernel Space $W_2^m[a,b]$

We may construct the closed subspaces ${}^oW_2^m[a,b]$ of the reproducing kernel space $W_2^m[a,b]$ by imposing several homogeneous boundary conditions on ${}^oW_2^m[a,b]$.

Definition 1.3.2. Let a function space

$$ {}^oW_2^m[a,b] = \Big\{ f(x) | f(x) \in W_2^m[a,b] \text{ by (1.3.2)},\ f'(a) = 0,\ f(b) = 0 \Big\}, $$

Similar to the proof of Section 1.3.2, we can prove it is a Hilbert reproducing kernel space.

Let us try to find the reproducing kernel function $Q_y(x)$ of ${}^oW_2^m[a,b]$. The method is completely similar to the method of Section 1.3.2, that is, $Q_y(x)$ should satisfy

$$ \begin{aligned} \langle f(x), Q_y(x) \rangle_{{}^oW_2^m} &= \sum_{i=0}^{m-1} f^{(i)}(a) \frac{\partial^i Q_y(a)}{\partial x^i} - \sum_{i=0}^{m-1} f^{(i)}(a)(-1)^{m-i-1} \frac{\partial^{2m-i-1} Q_y(a)}{\partial x^{2m-i-1}} \\ &+ \sum_{i=0}^{m-1} (-1)^{m-i-1} f^{(i)}(b) \frac{\partial^{2m-i-1} Q_y(b)}{\partial x^{2m-i-1}} \\ &+ (-1)^m \int_a^b f(x) \frac{\partial^{2m} Q_y(x)}{\partial x^{2m}}\, dx. \end{aligned} $$

Therefore, $Q_y(x)$ is the solution of the following generalized differential equation

$$ \begin{cases} (-1)^m \dfrac{\partial^{2m} Q_y(x)}{\partial x^{2m}} = \delta(x-y), \\ \dfrac{\partial^i Q_y(a)}{\partial x^i} - (-1)^{m-i-1} \dfrac{\partial^{2m-i-1} Q_y(a)}{\partial x^{2m-i-1}} = 0, \quad i = 0, 2, 3, \ldots, m-1, \\ \dfrac{\partial^{2m-i-1} Q_y(b)}{\partial x^{2m-i-1}} = 0, \qquad\qquad i = 1, 2, \ldots, m-1, \\ Q_y(b) = 0, \\ \dfrac{\partial Q_y(a)}{\partial x} = 0. \end{cases} \tag{1.3.16} $$

Note that $Q_y(x)$ in here is different form $R_y(x)$ in Section 1.3.2, because of some limitation conditions in ${}^oW_2^m[a,b]$.

While $x \neq y$, it is easy to know that $Q_y(x)$ is the solution of the following linear homogeneous differential equation with $2m$ orders, i.e.,

$$ (-1)^m \frac{\partial^{2m} Q_y(x)}{\partial x^{2m}} = 0, \tag{1.3.17} $$

with the boundary conditions:

$$\begin{cases} \dfrac{\partial^i Q_y(a)}{\partial x^i} - (-1)^{m-i-1}\dfrac{\partial^{2m-i-1} Q_y(a)}{\partial x^{2m-i-1}} = 0, & i = 0, 2, 3, \ldots, m-1, \\ \dfrac{\partial^{2m-i-1} Q_y(b)}{\partial x^{2m-i-1}} = 0, & i = 1, 2, \ldots, m-1, \\ Q_y(b) = 0, \\ \dfrac{\partial Q_y(a)}{\partial x} = 0. \end{cases} \tag{1.3.18}$$

We know that equation (1.3.17) has characteristic equation $\lambda^{2m} = 0$, and the eigenvalue $\lambda = 0$ is a root whose multiplicity is $2m$. Therefore, the general solution of Eq. (1.3.16) is

$$Q_y(x) = \begin{cases} lQ_y(x) = \displaystyle\sum_{i=1}^{2m} c_i(y) x^{i-1}, & x < y, \\ rQ_y(x) = \displaystyle\sum_{i=1}^{2m} d_i(y) x^{i-1}, & x > y. \end{cases} \tag{1.3.19}$$

Now we are ready to calculate the coefficients $c_i(y)$ and $d_i(y)$, $i = 1, 2, \ldots, 2m$. Since

$$(-1)^m \frac{\partial^{2m} Q_y(x)}{\partial x^{2m}} = \delta(x - y),$$

we have

$$\frac{\partial^i l Q_y(y)}{\partial x^i} = \frac{\partial^i r Q_y(y)}{\partial x^i}, \quad i = 0, 1, \ldots, 2m-2, \tag{1.3.20}$$

and

$$(-1)^m \left(\frac{\partial^{2m-1} l Q_y(y^+)}{\partial x^{2m-1}} - \frac{\partial^{2m-1} r Q_y(y^-)}{\partial x^{2m-1}} \right) = 1. \tag{1.3.21}$$

The above equations in (1.3.20) and (1.3.21) provided $2m$ conditions for solving the coefficients $c_i(y)$ and $d_i(y)$ ($i = 1, 2, \ldots, 2m$) in Eq. (1.3.19). Note that Eq. (1.3.18) provided $2m$ boundary conditions, so we have $4m$ equations, i.e., (1.3.18), (1.3.20) and (1.3.21). It is easy to know these $4m$ equations are linear equations with the variables $c_i(y)$ and $d_i(y)$, and $c_i(y)$ and $d_i(y)$ could be calculated by many methods. As long as the coefficients $c_i(y)$ and $d_i(y)$ are obtained, the exact expression of the reproducing kernel function $Q_y(x)$ could be calculated from Eq. (1.3.19).

Theorem 1.3.5. *The reproducing kernel space ${}^oW_2^m[a, b]$ is a closed subspaces of $W_2^m[a, b]$.*

Proof. Let $f_n(x) \in {}^oW_2^m[a,b]$, $n = 1, 2, \ldots$, $f(x) \in W_2^m[a,b]$,

$$\lim_{n\to\infty} \|f_n(x) - f(x)\|_{W_2^m} = 0.$$

By Theory 1.3.4, $f_n(x)$, $f_n'(x)$ converge to $f(x)$, $f'(x)$ uniformly. That is

$$\begin{aligned} \lim_{n\to\infty} f_n(b) &= f(b), \\ \lim_{n\to\infty} f_n'(a) &= f'(a). \end{aligned}$$

Because $f_n(x) \in {}^oW_2^m[a,b]$, $f_n(b) = f_n'(a) = 0$, $n = 1, 2, \ldots$, we obtain $f(b) = f'(a) = 0$, namely $f(x) \in {}^oW_2^m[a,b]$. So ${}^oW_2^m[a,b]$ is a closed subspaces of $W_2^m[a,b]$. □

The reproducing kernel space ${}^oW_2^m[a,b]$ may contain various diversified homogeneous boundary conditions. Of course, we will obtain different reproducing kernel spaces and corresponding reproducing kernel functions. Sometimes several non-local homogeneous conditions can be also imposed on the space. For instance.

Theorem 1.3.6. *The space*

$${}^oW_2^m[a,b] = \left\{ f(x) | f(x) \in W_2^m[a,b], \int_a^b \rho(x) f(x)\, dx = 0 \right\},$$

is a reproducing kernel space, where $\rho(x) > 0$ is a weighting function.

The proof is left to readers.

It is a key to construct the reproducing kernel space with different boundary conditions for solving different practical problems. In this book, we shall solve various problems in the reproducing kernel space.

1.3.5 Two Notes About Reproducing Kernel Space $W_2^m[a,b]$

From the definition of reproducing kernel space $W_2^m[a,b]$, we can see that function $u^{(m-1)}(x)$ is required to be a real absolute continuous function on the closed interval $[a,b]$. Next we will discuss the following two situations:

(1) If $u(x)$ is a real continuous function on the closed interval $[a,b]$, and $u'(x) \in L^2[a,b]$, can we deduce that $u(x)$ is a real absolutely continuous function on the closed interval $[a,b]$?

(2) In the definition of the reproducing kernel space $W_2^1[a,b]$, If we replace "absolutely continuous" with "continuous" or "bounded variation", can the function space still be a reproducing kernel space?

Note(1)

Even if $u(x)$ satisfies following conditions:

(i) $u(x)$ is continuous on the closed interval $[a,b]$, differentiable almost everywhere, and $u'(x) \in L^2[a,b]$;

(ii) $u(x)$ is a bounded variation function on the closed interval $[a,b]$, differentiable almost everywhere, and $u'(x) \in L^2[a,b]$,

it still can not imply $u(x)$ is absolute continuous on the closed interval $[a,b]$.

Let us consider: Suppose P_0 is the Cantor set on the closed interval $[a,b]$, that is, P_0 is made up of such dots. First, divide $[a,b]$ into three equal parts, remove the middle part (except two endpoints), then divide the remaining two parts into three equal parts respectively, remove the middle part of each one (except two endpoints), do it like this, every time we divide all the closed intervals remained into three equal parts respectively and remove the middle part of each closed interval, at last, we get the closed set P_0, its measure is zero. Let G_0 be the complement of P_0 on the closed interval$[a,b]$. Now, define a function, $\theta(x)$, $x \in [a,b]$, on the 2^{k-1} intervals that are removed k times, $\theta(x)$ take values in the following forms from left to right orderly :

$$\frac{1}{2^k}, \frac{3}{2^k}, \frac{5}{2^k}, \ldots, \frac{2^k-1}{2^k}.$$

So $\theta(x)$ has definition on G_0, it is a constant function on each small interval of G_0, and also an increasing function on G_0. Extend $\theta(x)$ to the entire interval $[a,b]$ continuously as follows:

$$\theta(x) = \sup_{t \le x,\ t \in G_0} \theta(t), \quad \theta(a) = 0, \quad \theta(b) = 1.$$

From the above practice, we can see that $\theta(x)$ is an increasing function on the closed interval $[a,b]$, therefore $\theta(x)$ is a bounded variation function on $[a,b]$. Besides, $\theta(x)$ is a continuous function. In fact, it is not continuous at the point x_0, each numbers of the open interval $(\theta(x_0-0), \theta(x_0+0))$ will not be value of $\theta(x)$ except $\theta(x_0)$. It is contradictory with the fact that the function values of $\theta(x)$ are dense on the closed interval $[a,b]$ (When x_0 is the endpoint a or b, the open interval $(\theta(x_0-0), \theta(x_0+0))$ should be replaced by the interval $[\theta(a), \theta(a+0))$ or $(\theta(b-), \theta(b)])$. As $\theta'(x) = 0$, a.e., so $\theta(x)$ is continuous, $\theta'(x) \in L^2[a,b]$ exists almost everywhere. But

$$\int_{[a,b]} \theta'(x)\, dx = 0 < 1 = \theta(b) - \theta(a).$$

So $\theta(x)$ is not absolute continuous.

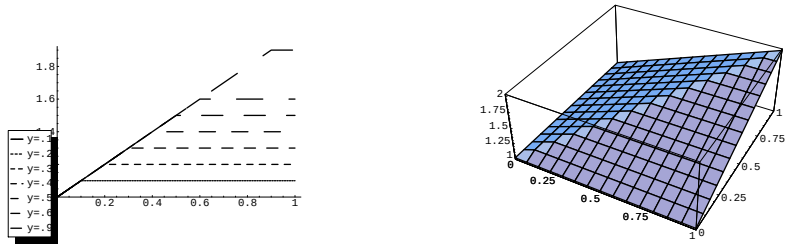


Figure 1.1: Left is curves of $R_{y_i}(x)$, Right is curves of $R_y(x)$. $R_y(x)$ is the reproducing kernel of $W_2^1[0,1]$

Note(2)

We must not replace the condition that $u(x)$ is absolute continuous in the reproducing kernel space $W_2^1[a,b]$ with $u(x)$ is continuous or bounded variation function.

In the same way, we consider the function $\theta(x)$ on $[a,b]$. By process of $\theta(x)$ being constructed, we can known that $\theta(x)$ is continuous and is a bounded variation function, $\theta'(x) = 0$ a.e., $\theta(x) \neq 0$ a.e. Suppose that $\theta(x) \in W_2^1[a,b]$, then

$$\|\theta(x)\|^2_{W_2^1} = \theta(x)^2(a) + \int_a^b [\theta(x)'(x)]^2\,dx = 0 \ \text{ iff } \ \theta(x) = 0, \ \text{ a.e.}$$

This is inconsistent.

1.4 Several Expressions of the Reproducing Kernel of $W_2^m[0,1]$ or ${}^oW_2^m[0,1]$

We are ready to present some expressions of reproducing kernel functions in $W_2^m[0,1]$ by using the approaches proposed in the above sections.

1^0 The reproducing kernel function $R_y(x)$ in $W_2^1[0,1]$,

$$R_y(x) = \begin{cases} 1+x, & x \leq y, \\ 1+y, & x > y. \end{cases} \tag{1.4.1}$$

2^0 The reproducing kernel function $R_y(x)$ in $W_2^3[0,1]$,

$$R_y(x) = \begin{cases} 1 + \dfrac{x^5}{120} + \dfrac{1}{12}x^2y^2(3+x) + xy\left(1 - \dfrac{x^4}{24}\right), & x \leq y, \\ 1 + \dfrac{y^5}{120} + \dfrac{1}{12}x^2y^2(3+y) + xy\left(1 - \dfrac{y^4}{24}\right), & x > y. \end{cases}$$

3^0 The reproducing kernel function $R_y(x)$ in $W_2^7[0,1]$,

$$\begin{cases} R_y(x) = \dfrac{1}{51011754393600}\Big(8192y^7(1716x^6-1287x^5y+715x^4y^2 \\ \quad -286x^3y^3+78x^2y^4-13xy^5+y^6)+8192\big(6227020800+x^{13} \\ \quad -13x(x^{11}-479001600)y+78x^2(19958400+x^9)y^2 \\ \quad -286x^3(x^7-604800)y^3+715x^4(15120+x^5)y^4-1287x^5(x^3-336)y^5 \\ \quad +1716x^6(7+x)y^6\big)-7028736x^6y^6(x+y+|y-x|) \\ \quad +1317888x^5y^5(x+y+|y-x|)^3-183040x^4y^4(x+y+|y-x|)^5 \\ \quad +18304x^3y^3(x+y+|y-x|)^7-1248x^2y^2(x+y+|y-x|)^9 \\ \quad +52xy(x+y+|y-x|)^{11}-(x+y+|y-x|)^{13}\Big). \end{cases}$$

4^0 The reproducing kernel function $R_y(x)$ in ${}^oW_2^2[0,1]$, where

$$\begin{aligned} {}^oW_2^3[0,1] &= \Big\{f(x)|f(x)\in W_2^2[0,1],\ f'(1)=0\Big\}, \\ R_y(x) &= \begin{cases} 1-\dfrac{x^3}{6}+\dfrac{1}{2}yx(2+x), & x\le y, \\ 1-\dfrac{y^3}{6}+\dfrac{1}{2}yx(2+y), & x>y. \end{cases} \end{aligned}$$

1.5 The Binary Reproducing Kernel Space $W_2^{(m,n)}(\mathcal{D})$

1.5.1 The Binary Completely Continuous Functions and Some Properties

Let set $\mathcal{D}=[a,b]\times[c,d]\subset\mathbb{R}^2$.

Definition 1.5.1. Given a function $f(x,y)$ on interval $\mathcal{D}$, let

$$\Big\{I_k=[a_k,b_k]\times[c_k,d_k]\Big\}_{k=1}^n$$

is a set of mutually disjoint domains $I_k\subset\mathcal{D}$. If for $\forall\,\varepsilon>0$, $\exists\,\delta>0$, such that $\sum\limits_{i=1}^n|f(I_i)|<\varepsilon$ for $\sum\limits_{i=1}^n|I_i|<\delta$, where

$$f(I_i)=f(b_i,d_i)-f(b_i,c_i)-f(a_i,d_i)+f(a_i,c_i)$$

and $f(x,c)$, $g(a,y)$ are absolutely continuous functions with respect to x,y respectively, then the function $f(x,y)$ is said to completely continuous on domain $\mathcal{D}$.

Proposition 1.5.1. *If function $f(x,y)$ is a completely continuous on $\mathcal{D}$, then for each fixed $y\in[c,d]$, $f(x,y)$ as a function of x is absolutely continuous on $[a,b]$. similarly, for each fixed $x\in[a,b]$, $f(x,y)$ as a function of y is absolutely continuous on $[c,d]$.*

Proof. Without generality, we only prove the first part. For arbitrary $\varepsilon > 0$, since $f(x, y)$ is complete continuous and $f(x, c)$ is absolutely continuous, there is a $\delta_1 > 0$, so that $\sum\limits_{k=1}^{n} |I_k| < \delta_1$ and $\sum\limits_{k=1}^{n} (b_k - a_k) < \delta_1$, where $\{I_k = [a_k, b_k] \times [c, y]\}_{k=1}^{n}$ is a set of mutually disjoint domains, yield

$$\sum_{k=1}^{n} |f(I_k)| < \varepsilon/2 \ \text{ and } \ \sum_{k=1}^{n} f(b_k, c) - f(a_k, c)| < \varepsilon/2.$$

Choose $\delta = \min\{\frac{\delta_1}{d-c}, \delta_1\}$. Suppose $\sum\limits_{i=1}^{n} (b_i - a_i) < \delta$, then

$$\sum_{k=1}^{n} |I_k| < \sum_{k=1}^{n} (b_k - a_k)(d - c) < \delta(d - c) = \delta_1.$$

Therefore

$$\begin{aligned}
\sum_{k=1}^{n} \big|f(b_k, y) - f(a_k, y)\big| &= \sum_{k=1}^{n} \big|f(a_k, c) + f(b_k, y) - f(a_k, y) - f(b_k, c)\big| \\
&+ \sum_{k=1}^{n} \big|f(b_k, c) - f(a_k, c)\big| \\
&= \sum_{k=1}^{n} |f(I_k)| + \sum_{k=1}^{n} \big|f(b_k, c) - f(a_k, c)\big| \\
&< \varepsilon/2 + \varepsilon/2 = \varepsilon.
\end{aligned}$$

So for each fixed $y \in [c, d]$, the function of one variable $f(x, y)$ is the absolutely continuous on $[a, b]$. □

Following we will give some propositions without proof.

Proposition 1.5.2. *If a function $f(x, y)$ is completely continuous on $\mathcal{D}$, then $f(x, y)$ is continuous on $\mathcal{D}$.*

Proposition 1.5.3. *If a function $f(x, y)$ is integrable on domain $\mathcal{D}$, then the function $\int\limits_a^x \int\limits_c^y f(t, s)\, dt\, ds$, $(x, y) \in \mathcal{D}$ is completely continuous on $\mathcal{D}$.*

Proposition 1.5.4. *If a function $f(x)$ is absolutely continuous on $[a, b]$, then $f(x)$ is completely continuous on $\mathcal{D}$.*

Proposition 1.5.5. *If functions $f(x)$, $g(y)$ are absolutely continuous on $[a, b], [c, d]$ respectively, then $f(x)g(y)$ is completely continuous on $\mathcal{D}$.*

1.5.2 The Binary Function Space $W_2^{(m,n)}(\mathcal{D})$ is a Hilbert space

Definition 1.5.2. The binary function space is defined as

$$W_2^{(m,n)}(\mathcal{D}) = \Big\{ f(x,y) \big| \ \frac{\partial^{m+n-2} f(x,y)}{\partial x^{m-1} \partial y^{n-1}} \text{ is completely continuous in } \mathcal{D}, \\ \frac{\partial^{m+n} f(x,y)}{\partial x^m \partial y^n} \in L^2(\mathcal{D}) \Big\}. \quad (1.5.1)$$

The inner product in $W_2^{(m,n)}(\mathcal{D})$ is defined as follows:

$$\begin{aligned} \langle f(x,y), g(x,y) \rangle_{W_2^{(m,n)}(\mathcal{D})} &= \sum_{i=0}^{m-1} \int_c^d \left[\frac{\partial^n}{\partial y^n} \frac{\partial^i}{\partial x^i} f(a,y) \frac{\partial^n}{\partial y^n} \frac{\partial^i}{\partial x^i} g(a,y) \right] dy \\ &+ \sum_{j=0}^{n-1} \left\langle \frac{\partial^j}{\partial y^j} f(x,c), \frac{\partial^j}{\partial y^j} g(x,c) \right\rangle_{W_2^m} \\ &+ \iint_{\mathcal{D}} \frac{\partial^m}{\partial x^m} \frac{\partial^n}{\partial y^n} f(x,y) \frac{\partial^m}{\partial x^m} \frac{\partial^n}{\partial y^n} g(x,y) \, dx \, dy. \quad (1.5.2) \end{aligned}$$

It is easy to prove that $W_2^{(m,n)}(\mathcal{D})$ is an inner space.

Theorem 1.5.1. *Function space $W_2^{(m,n)}(\mathcal{D})$ is a Hilbert space.*

Proof. Suppose $f_k(x,y)$ is a Cauchy sequence in $W_2^{(m,n)}(\mathcal{D})$, by (1.5.2), we can obtain

$$\frac{\partial^m}{\partial x^i} \frac{\partial^n}{\partial y^n} f_k(a,y), \quad \frac{\partial^j}{\partial y^j} f_k(x,c), \quad \frac{\partial^m}{\partial x^m} \frac{\partial^n}{\partial y^n} f_k(x,y)$$

are Cauchy sequences in $L^2[c,d]$, $W_2^m[a,b]$, $L^2(\mathcal{D})$, respectively. $i = 0, 1, \ldots, m-1$, $j = 0, 1, \ldots, n-1$. So, there exist $F_i(a,y) \in L^2[c,d]$, $G_j(x,c) \in W_2^m[a,b]$, $H(x,y) \in L^2(\mathcal{D})$, satisfying

$$\int_c^d \left[\frac{\partial^i}{\partial x^i} \frac{\partial^n}{\partial y^n} f_k(a,y) - F_i(a,y) \right]^2 dy \underset{k\to\infty}{\longrightarrow} 0, \quad i = 0, 1, \ldots, m-1; \quad (1.5.3)$$

$$\left\| \frac{\partial^j}{\partial y^j} f_k(x,c) - G_j(x,c) \right\|_{W_2^m[a,b]} \underset{k\to\infty}{\longrightarrow} 0, \quad j = 0, 1, \ldots, n-1; \quad (1.5.4)$$

$$\iint_{\mathcal{D}} \left[\frac{\partial^m}{\partial x^m} \frac{\partial^n}{\partial y^n} f_k(x,y) - H(x,y) \right]^2 dx \, dy \underset{k\to\infty}{\longrightarrow} 0. \quad (1.5.5)$$

Suppose

$$\begin{aligned} f(x,y) &= \sum_{i=0}^{m-1}\left[\frac{(x-a)^i}{i!}\overbrace{\int_c^y\cdots\int_c^y}^{n}F_i(a,y)(dy)^n\right] \\ &+ \sum_{j=0}^{n-1}\left[\frac{(y-c)^j}{j!}G_j(x,c)\right] \\ &+ \overbrace{\int_a^x\cdots\int_a^x}^{m}\overbrace{\int_c^y\cdots\int_c^y}^{n}H(x,y)(dy)^m(dy)^n. \end{aligned}$$

(1) Since

$$\begin{aligned} \frac{\partial^{m+n-2}}{\partial x^{m-1}\partial y^{n-1}}f(x,y) &= \int_c^y F_{m-1}(a,y)dy + G_{n-1}(x,c) \\ &+ \int_a^x\int_c^y H(x,y)dxdy, \end{aligned}$$

hence $\frac{\partial^{m+n-2}}{\partial x^{m-1}\partial y^{n-1}}f(x,y)$ is completely continuous in $\mathcal{D}$, and $\frac{\partial^{m+n}}{\partial x^m\partial y^n}f(x,y)=H(x,y)$ is true almost everywhere in $\mathcal{D}$. So, $f(x,y)\in W_2^{(m,n)}(\mathcal{D})$.

(2) Note that

$$\begin{aligned} &\left\|f_k(x,y)-f(x,y)\right\|^2_{W_2^{(m,n)}(\mathcal{D})} \\ &\quad=\sum_{i=0}^{m-1}\int_c^d\left[\frac{\partial^n}{\partial y^n}\frac{\partial^i}{\partial x^i}f_k(a,y)-\frac{\partial^n}{\partial y^n}\frac{\partial^i}{\partial x^i}f(a,y)\right]^2dy \\ &\quad+\sum_{j=0}^{n-1}\left\|\frac{\partial^j}{\partial y^j}f_k(x,c)-\frac{\partial^j}{\partial y^j}f(x,c)\right\|^2_{W_2^m} \\ &\quad+\int_{\mathcal{D}}\left[\frac{\partial^m}{\partial x^m}\frac{\partial^n}{\partial y^n}f_k(x,y)-\frac{\partial^m}{\partial x^m}\frac{\partial^n}{\partial y^n}f(x,y)\right]^2dx\,dy \\ &\quad=\sum_{i=0}^{m-1}\int_c^d\left[\frac{\partial^n}{\partial y^n}\frac{\partial^i}{\partial x^i}f_k(a,y)-F_i(a,y)\right]^2dy \\ &\quad+\sum_{j=0}^{n-1}\left\|\frac{\partial^j}{\partial y^j}f_k(x,c)-G_j(x,c)\right\|^2_{W_2^m} \end{aligned}$$

$$+\iint\limits_{\mathcal{D}}\left[\frac{\partial^m}{\partial x^m}\,\frac{\partial^n}{\partial y^n}f_k(x,y)-H(x,y)\right]^2\,dx\,dy\overset{(1.5.3)-(1.5.5)}{\underset{k\to\infty}{\longrightarrow}}0.$$

Hence, function space $W_2^{(m,n)}(\mathcal{D})$ is a Hilbert Space. □

1.5.3 The Binary Function Space $W_2^{(m,n)}(\mathcal{D})$ is a Reproducing Kernel Space

Theorem 1.5.2. *Let $f(x,y)$, $g_1(x)g_2(y)\in W_2^{(m,n)}(\mathcal{D})$, then*

$$\langle f(x,y),g_1(x)g_2(y)\rangle_{W_2^{(m,n)}}=\big\langle\langle f(x,y),g_1(x)\rangle_{W_2^m},g_2(y)\big\rangle_{W_2^n}.$$

Proof.

$$\begin{aligned}
&\big\langle f(x,y),g_1(x)g_2(y)\big\rangle_{W_2^{(m,n)}}\\
&=\sum_{i=0}^{m-1}\int_c^d\left[\frac{\partial^n}{\partial y^n}\,\frac{\partial^i}{\partial x^i}f(a,y)\,\frac{\partial^n}{\partial y^n}g_2(y)\,\frac{\partial^i}{\partial x^i}g_1(a)\right]dy\\
&\quad+\sum_{j=0}^{n-1}\left\langle\frac{\partial^j}{\partial y^j}f(x,c),g_1(x)\frac{\partial^j}{\partial y^j}g_2(c)\right\rangle_{W_2^m}\\
&\quad+\iint\limits_{\mathcal{D}}\left[\frac{\partial^m}{\partial x^m}\,\frac{\partial^n}{\partial y^n}f(x,y)\,\frac{\partial^m}{\partial x^m}g_1(x)\,\frac{\partial^n}{\partial y^n}g_2(y)\right]dx\,dy\\
&=\int_c^d\frac{\partial^n}{\partial y^n}g_2(y)\,\frac{\partial^n}{\partial y^n}\left[\int_a^b\frac{\partial^m}{\partial x^m}f(x,y)\,\frac{\partial^m}{\partial x^m}g_1(x)\,dx+\sum_{i=0}^{m-1}\frac{\partial^i}{\partial x^i}f(a,y)\,\frac{\partial^i}{\partial x^i}g_1(a)\right]dy\\
&\quad+\sum_{i=0}^{n-1}\left\langle\frac{\partial^i}{\partial y^i}f(x,c),g_1(x)\right\rangle_{W_2^m}\frac{\partial^i}{\partial y^i}g_2(c)\\
&=\int_c^d\frac{\partial^n}{\partial y^n}g_2(y)\,\frac{\partial^n}{\partial y^n}\langle f(x,y),g_1(x)\rangle_{W_2^m}\,dy\\
&\quad+\sum_{i=0}^{n-1}\left\langle\frac{\partial^i}{\partial y^i}\,f(x,c),g_1(x)\right\rangle_{W_2^m}\frac{\partial^i}{\partial y^i}g_2(c)\\
&=\sum_{i=0}^{n-1}\frac{\partial^i}{\partial y^i}\langle f(x,c),g_1(x)\rangle_{W_2^m}\frac{\partial^i}{\partial y^i}g_2(c)\\
&\quad+\int_c^d\frac{\partial^n}{\partial y^n}\langle f(x,y),g_1(x)\rangle_{W_2^m}\frac{\partial^n}{\partial y^n}g_2(y)\,dy
\end{aligned}$$

$$= \left\langle \langle f(x,y), g_1(x)\rangle_{W_2^m}, g_2(y)\right\rangle_{W_2^n}. \qquad \square$$

Corollary 1.5.1. *Let $f_1(x)f_2(y)$, $g_1(x)g_2(y) \in W_2^{(m,n)}(\mathcal{D})$, then*

$$\left\langle f_1(x)f_2(y), g_1(x)g_2(y)\right\rangle_{W_2^{(m,n)}} = \langle f_1(x), g_1(x)\rangle_{W_2^m}\langle f_2(y)g_2(y)\rangle_{W_2^n}.$$

Corollary 1.5.2. *If $f(x) \in W_2^m[a,b]$, then for $n \in \mathbb{N}$, we have*

$$\|f(x)\|_{W_2^m} = \|f(x)\|_{W_2^{(m,n)}}. \tag{1.5.6}$$

Proof. By Proposition 1.5.4 and Corollary 1.5.1, one gets $f(x) \in W_2^{(m,n)}(\mathcal{D})$ and obtains immediately

$$\|f(x)\|^2_{W_2^{(m,n)}} = \langle f(x), f(x)\rangle_{W_2^m} = \|f(x)\|^2_{W_2^m}. \qquad \square$$

By Theorem 1.5.2, we can obtain the next theorem immediately.

Theorem 1.5.3. *$W_2^{(m,n)}(\mathcal{D})$ is a reproducing kernel space which has reproducing kernel*

$$K_{(\xi,\eta)}(x,y) = R_\xi(x)Q_\eta(y), \tag{1.5.7}$$

where $R_\xi(x)$, $Q_\eta(y)$ are the reproducing kernels of $W_2^m[a,b]$ and $W_2^n[c,d]$, respectively.

Similar to Section 1.3.4, we also impose several homogeneous conditions on the space to obtain obtaining the subspace ${}^oW_2^{(m,n)}(\mathcal{D})$ of $W_2^{(m,n)}(\mathcal{D})$.

1.6 The Reproducing Kernel Space $W_2^1(\mathbb{R})$

Sometimes, problems need variable $x \in (-\infty, +\infty)$, so we need construct such reproducing kernel space.

For instance, we define a function space

$$W_2^1(\mathbb{R}) = \Big\{u(x) | u(x) \text{ is absolutely continuous, } u'(x) \in L^2(\mathbb{R})\Big\}.$$

The inner product and norm of $W_2^1(\mathbb{R})$ are defined by

$$\begin{aligned} \langle u(x), v(x)\rangle_{W_2^1(\mathbb{R})} &= \int_{-\infty}^{\infty} (u(x)v(x) + u'(x)v'(x))dx, \\ \|u(x)\|_{W_\infty} &= \sqrt{\langle u(x), u(x)\rangle_{W_2^1(\mathbb{R})}}. \end{aligned} \tag{1.6.1}$$

It is easy to prove that $W_2^1(\mathbb{R})$ is an inner space.

Now, let's find out the expression form of the reproducing kernel function $R_y(x)$ in $W_2^1(\mathbb{R})$.

Suppose $R_y(x)$ is the reproducing kernel function of $W_2^1(\mathbb{R})$, then for any fixed $y \in (-\infty, +\infty)$ and any $u(x) \in W_2^1(\mathbb{R})$, $R_y(x)$ must satisfy

$$\langle u(x), R_y(x) \rangle_{W_2^1(\mathbb{R})} = u(y).$$

Based on (1.6.1), we have

$$\begin{aligned} \langle u(x), R_y(x) \rangle_{W_2^1(\mathbb{R})} &= \int_{-\infty}^{+\infty} \left(u(x) R_y(x) + u'(x) \frac{\partial R_y(x)}{\partial x} \right) dx \\ &= u(x) \left. \frac{\partial R_y(x)}{\partial x} \right|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} u(x) \left(v_y(x) - \frac{\partial^2 R_y(x)}{\partial x^2} \right) dx. \end{aligned}$$

Therefore, $R_y(x)$ is the solution of the following generalized differential equation

$$\begin{cases} R_y(x) - \dfrac{\partial^2 R_y(x)}{\partial x^2} = \delta(x - y), \\ \lim\limits_{x \to -\infty} \dfrac{\partial R_y(x)}{\partial x} = 0, \\ \lim\limits_{x \to +\infty} \dfrac{\partial R_y(x)}{\partial x} = 0. \end{cases} \tag{1.6.2}$$

While $x \neq y$, it is easy to know that $r_y(x)$ is the solution of the following constant linear homogeneous differential equation with $2m$ orders, i.e.,

$$R_y(x) - \frac{\partial^2 R_y(x)}{\partial x^2} = 0, \tag{1.6.3}$$

with the boundary conditions:

$$\begin{cases} \lim\limits_{x \to -\infty} \dfrac{\partial R_y(x)}{\partial x} = 0, \\ \lim\limits_{x \to +\infty} \dfrac{\partial R_y(x)}{\partial x} = 0. \end{cases} \tag{1.6.4}$$

We know that Eq. (1.6.3) has characteristic equation $1 - \lambda^2 = 0$, and the eigenvalues $\lambda = \pm 1$. Therefore, the general solution of Eq. (1.6.2) is as follows

$$r_y(x) = \begin{cases} c_1(y) e^x + c_2(y) e^{-x}, & x < y, \\ d_1(y) e^x + d_2(y) e^{-x}, & x > y. \end{cases} \tag{1.6.5}$$

We can obtain the coefficients $c_1(y) = \frac{e^{-y}}{2}$, $c_2(y) = 0$, $d_1(y) = 0$, $d_2(y) = \frac{e^y}{2}$. So that

$$R_y(x) = \frac{e^{-|x-y|}}{2}.$$

For a variable $x \in [0, +\infty)$, the reproducing kernel spaces can also be constructed. (see Section 2.4.2).

Chapter 2

Some Linear Problems

This book mainly discusses the solution method of the nonlinear problems, and the solution of the linear problems is the base. Although the solution methods of ordinary linear problems are quite mature, many new linear problems are raised in recent years, such as the singularly perturbation problem, obstacle problem, singular boundary value problem, linear boundary value problem, non-local boundary value problem, and cycle problem. These are the hot topics for the moment. This chapter will introduce a new solution method for these problems. This method features advanced theory and simple and effective algorithm.

2.1 Solving Singular Boundary Value Problems

2.1.1 Introduction

In applied mathematics many problems lead to singular boundary value problems given by

$$p(x)u''(x) + \frac{1}{q(x)}u'(x) + \frac{1}{r(x)}u(x) = f(x), \quad 0 < x < 1 \tag{2.1.1}$$

subject to boundary conditions

$$u(0) = \alpha, \quad u(1) = \beta,$$

where $p(x)$, $q(x) \in C(0,1)$, $q(0)q(1) = 0$, $r(0)r(1) = 0$, $r(x)$, $f(x)$ are sufficiently regular given functions and α, β are finite constants. Singular perturbed impulsive boundary value problems arise very frequently in the fields of fluid mechanics, fluid dynamics, elasticity, reaction-diffusion processes, chemical kinetics and other branches of applied mathematics, which have become an important area of investigation in recent years. And singular boundary value problems for ordinary differential equations arise very frequently in the theory of thermal explosions and in the study of Electro-hydrodynamics. Such problems also occur in the study of generalized

axially symmetric potentials after separation of variables has been employed. There is considerable interest on numerical methods on singular boundary value problems. Mohanty and Jha [33] reported numerical techniques for a class of singularly perturbed two point singular boundary value problems on a non-uniform mesh using spline in compression. Mohanty and Arora [34] present tension spline numerical methods and a non-uniform mesh for the efficient numerical integration of singularly perturbed two point singular boundary value problems. Recently, there has been a great deal of interest in radial basis functions ($RBFs$) for interpolation problems, and as a tool for numerically solving differential equations. Ling and Trummer [35] employ RBFs to solve boundary value problems with very thin layers in order to demonstrate that RBF methods are capable of achieving high accuracy and robustness through the collocation. All above methods are discussed for a class of singularly perturbed two point singular boundary value problems as follows

$$\begin{cases} \varepsilon u''(x) + p(x)u'(x) + q(x)u(x) = f(x), \ \ \forall\, x \in [a,b], \\ u(a) = \alpha, \quad u(b) = \beta, \end{cases}$$

where $0 < \varepsilon \ll 1$ is a small positive parameter, $p(x)$ and $q(x)$ are sufficiently smooth real-valued functions. This problem has been well studied and the results can be found in much of the literature[38]–[41]. Wang and Jiang [36] present a general method for solving Singularly perturbed impulsive two-point singular boundary value problems with boundary layer at one end ($left\ or\ right$) point. They divide the defined interval into a series of subintervals and obtain the solution on each subinterval.A class of superlinear semipositone singular second order Dirichlet boundary value problems are discussed by Zhang and Liu [37]. They first approximate the singular semipositone problem to the singular positone problem by a substitution, then using the fixed point index and the Arzela–Ascoli theorem.

Here, we will give a new algorithm for a class of singular two-point boundary value problem Eq. (2.1.1) on the assumption that solution is unique. We describe a significant method based on the theory of numerical analysis in reproducing kernel space. Some model problems are solved, and the numerical results are compared with exact solution. It is demonstrated that this method is of high precision and it is significant. As is evident from the numerical results, this method reaches at least $O(h^5)$ accuracy. The results obtained using this method are better than using the stated existing methods with the same no. of knots

2.1.2 The Reproducing Kernel Spaces

The reproducing kernel space ${}^oW_2^3[0,1])$ is defined as

$${}^oW_2^3[0,1] = \Big\{ u(x) |\ u(x) \in W_2^3[0,1] \text{ by (1.3.2)},\ u(0) = u(1) = 0 \Big\}.$$

Its reproducing kernel is :

$$\begin{aligned}R_y(x) = \frac{1}{11520}\Big(&96y^3(10x^2-5xy+y^2)\\&+\frac{8}{13}x(1-y)\Big(120x^4-4x^4y^3+x^4y^4-5(x^3-24)y^4\\&+6(x^4-5(x^3-24)y+10x(3+x)y^2)+2y(15x(-120-40x+x^3)\\&+3x^4y+10(x^3-24)y^2+5x(3+x)y^3)\\&-2\big(-15x^4-3(600-25x^3+x^4)y+15(x^3-24)y^2+20x(3+x)y^3\big)\Big)\\&-480x^2y^2(x+y+|y-x|)+60xy(x+y+|y-x|)^3\\&-3(x+y+|y-x|)^5\Big).\end{aligned}$$

The reproducing kernel space $W_2^1[0,1]$) is defined as (1.3.2), its reproducing kernel $r_y(x)$ by is given(1.4.1).

2.1.3 Primary Theorem and the Method of Solving Eq. (2.1.1)

The problems are considered under the assumptions that it has a unique solution.

After Multiplicative both sides of Eq. (2.1.1) by $q(x)r(x)$, we obtain

$$p(x)q(x)r(x)u''(x)+r(x)u'(x)+q(x)u(x)=q(x)r(x)f(x).$$

In order to solve Eq. (2.1.1), we introduce a linear operator $\mathbb{L}: {}^oW_2^3[0,1] \longrightarrow W_2^1[0,1]$.

$$(\mathbb{L}u)(x) \overset{def}{=} p(x)q(x)r(x)u''(x)+r(x)u'(x)+q(x)u(x). \tag{2.1.2}$$

It is easy to prove that $L: {}^oW_2^3[0,1] \longrightarrow W_2^1[0,1]$ is a bounded linear operator. On the other hand, the method will be given completely matched in the case of $p(x)=\epsilon$. After homogenization of boundary condition, Eq. (2.1.1) can be transformed into the following form:

$$\begin{cases}(\mathbb{L}u)(x)=f(x),\\ u(0)=0, \quad u(1)=0.\end{cases} \tag{2.1.3}$$

Put

$$\varphi_i(x)=r_{x_i}(y), \quad x_i\in[0,1]$$

and

$$\psi_i(x)=\mathbb{L}^*\varphi_i(x),$$

where $\mathbb{L}^*: W_2^1[0,1] \longrightarrow {}^oW_2^3[0,1]$ is the conjugate operator of $\mathbb{L}$.

Lemma 2.1.1. *Let $\{x_i\}_{i=1}^{\infty}$ is a dense set in $[0,1]$, then $\{\psi_i(x)\}_{i=1}^{\infty}$ is the complete system of ${}^oW_2^3[0,1]$.*

Using Gram–Schmidt process of $\{\psi_i(x)\}_{i=1}^{\infty}$, we obtain a normal orthogonal basis $\{\overline{\psi}_i(x)\}_{i=1}^{\infty}$ of ${}^oW_2^3[0,1]$, which satisfies

$$\overline{\psi}_i(x) = \sum_{k=1}^{i} \beta_{ik}\psi_k(x), \tag{2.1.4}$$

β_{ik} are orthogonalization coefficients.

2.1.4 The Structure of Solution to Operator Eq. (2.1.3)

Theorem 2.1.1.

$$u(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} f(x_k)\overline{\psi}_i(x) \tag{2.1.5}$$

is exact representation of the solution of Eq. (2.1.3)*, where* $\{x_i\}_{i=1}^{\infty}$ *is the dense set in* $[0,1]$.

Proof. Let $u \in {}^oW_2^3[0,1]$ is the unique solution of $Eq(2.1.3)$, thus

$$\begin{aligned} u(x) &= \sum_{i=1}^{\infty}\langle u(x), \overline{\psi}_i(x)\rangle_{{}^oW_2^3}\overline{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle u(x), \mathbb{L}^*\varphi_k(x)\rangle_{{}^oW_2^3}\overline{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle \mathbb{L}u(x), \varphi_k(x)\rangle_{W_2^1}\overline{\psi}_i(x). \end{aligned}$$

Note that $\mathbb{L}u = f$, we have

$$u(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle f(x), \varphi_k(x)\rangle_{W_2^1}\overline{\psi}_i(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} f(x_k)\overline{\psi}_i(x). \qquad \square$$

We denote the approximate solution of u by

$$u_n(x) = \sum_{i=1}^{n}\sum_{k=1}^{i} \beta_{ik} f(x_k)\overline{\psi}_i(x).$$

2.1.5 Numerical experiments

Example 2.1.1. Consider the following singular boundary values problem [34]:

$$\varepsilon u''(x) + \frac{1}{x}u'(x) + (1+x^2)u(x) = \frac{2e^{x^2}x(15371+514x^2) - e^x(1024+1025x+1024x^2)}{1024x}$$

Table 2.1:

h	RMS errors in [34]	RMS1 errors with new algorithm
1/8		3.18851E-03
1/16	8.381E-02	3.7987E-04
1/32	8.014E-02	4.10187E-05
1/64	7.772E-02	4.02599E-06
1/128	7.221E-02	3.47918E-07
1/256	6.655E-02	2.99306E-08
1/512	5.876E-02	3.9648E-09

Table 2.2:

Node	Approximate solution $u_{200}(x)$	True solution $u(x)$	Absolute error	Relative error
1/200	0.0201008	0.0201	8.3482E-07	4.15148E-05
17/200	0.368761	0.368748	1.27754E-05	3.4644E-05
33/200	0.766585	0.766563	2.22816E-05	2.9066E-05
49/200	1.20778	1.20775	2.92765E-05	2.42399E-05
65/200	1.68092	1.68089	3.36899E-05	2.00425E-05
81/200	2.16679	2.16676	3.54846E-05	1.63765E-05
97/200	2.6356	2.63557	3.46806E-05	1.31585E-05
113/200	3.04337	3.04334	3.13725E-05	1.03085E-05
129/200	3.32729	3.32727	2.57478E-05	7.73837E-06
145/200	3.39985	3.39983	1.81138E-05	5.32781E-06
161/200	3.14127	3.14127	8.9354E-06	2.844524E-06
177/200	2.39002	2.39002	1.10766E-06	4.63453E-07
193/200	0.930762	0.930773	1.0023E-05	1.18207E-05

with the boundary conditions $u(0) = u(1) = 0$, $\varepsilon = 2^{-10}$. The exact solution is given by $u(x) = e^{x^2} - e^x$, $x \in [0, 1]$. We choose k ($k = 2^3, 2^4, \ldots, 2^9$) points in $[0, 1]$ and obtain approximate solution $u_k(x)$. The results obtained using this method are better than [34]. We denote the root mean square errors in [34] the RMS errors and RMS1 errors is our new algorithm Table 2.1.

Example 2.1.2. Consider the singular boundary values problem:

$$xe^{\frac{1}{x}}u''(x) + u'(x) + xu(x) = -4e^{2x}\left[-1 + x^2 + x^3 + 2e^{\frac{1}{x}}x(-1 + 2x + 2x^2)\right],$$

with the boundary conditions $u(0) = u(1) = 0$. The exact solution is $u(x) = 4x(1-x)e^{2x}$, $x \in [0, 1]$. Using our method, we choose 200 points in $[0, 1]$ and obtain approximate solution $u_{200}(x)$. The numerical results are presented in Table 2.2.

2.2 Solving the third-order obstacle problems

2.2.1 Introduction

A system of third-order boundary-value problems of the type

$$y''' = \begin{cases} f(x), & a \le x \le c, \\ g(x)y(x) + f(x) + r(x), & c < x \le d, \\ f(x), & d < x \le b, \end{cases} \tag{2.2.1}$$

with the boundary conditions

$$y(a) = \alpha_1, \quad y'(a) = \beta_1, \ \text{ and } \ y'(b) = \beta_2$$

and the continuity conditions of y, y', and y'', at c and d. Here, f, g and r are continuous functions on $[a, b]$ and $[c, d]$, respectively. The α_1, β_1 and β_2, are real finite constants.

Such type of systems arise in the study of obstacle, unilateral, moving and free boundary-value problems and has important applications in other branches of pure and applied sciences, for instance, the physical oceanography and the framework of variational inequality theory, etc. see for examples in [43]–[46] and the references there in. In general it is not possible to obtain the analytical solution of (2.2.1) for arbitrary choices of $f(x)$, $g(x)$ and $r(x)$. In last few decades, many researchers devoted to finite-difference, and every kind of spline methods for obtaining an approximate solution of (2.2.1) [47]–[55]. In this section, we give a new method based on the reproducing kernel space. Comparing with other methods in references, our method is of higher accuracy. Also, the derivatives y_n', y_n'' of y_n uniformly converge to the derivatives y', y'' of y respectively.

In order to put boundary-value conditions of Eq. (2.2.1) into the reproducing kernel space $W_2^3[a, b]$ which will be constructed, it is must to homogenize the boundary-value conditions. Put

$$\widetilde{y}(x) = y(x) - A(x),$$

where

$$A(x) = \frac{2\alpha_1 b - 2a(\alpha_1 + b\beta_1) + a^2(\beta_1 + \beta_2)}{2(b-a)} + \frac{b\beta_1 - a\beta_2}{b-a}x + \frac{\beta_1 - \beta_2}{2(b-a)}x^2\,,$$

then we can obtain homogeneous boundary-value conditions of Eq. (2.2.2). Immediately, we get

$$\widetilde{y}''' = \begin{cases} f(x), & a \le x \le c \\ g(x)\widetilde{y}(x) + f(x) + \widetilde{r}(x), & c < x \le d \\ f(x), & d < x \le b \end{cases} \tag{2.2.2}$$

with the boundary conditions

$$\widetilde{y}(a) = 0, \quad \widetilde{y}'(a) = 0, \ \text{ and } \ \widetilde{y}'(b) = 0,$$

where $\widetilde{r}(x) = r(x) - g(x)A(x)$. Following we will always replace $\widetilde{y}(x)$ and $\widetilde{r}(x)$ with $y(x)$ and $r(x)$ in Eq. (2.2.2), respectively.

2.2.2 Reproducing Kernel Space ${}^{o}W_2^3[0,1]$

The reproducing kernel space

$${}^{o}W_2^3[0,1] = \Big\{ y(x) |\ y(x) \in W_2^3[0,1](1.3.2),\ y(0) = y'(0) = y'(1) = 0 \Big\},$$

Similar to Section 1.3.2, we can obtain the reproducing kernel $R_\xi(x)$ of the space ${}^{o}W_2^3[0,1]$.

$$\begin{aligned} R_\xi(x) &= \frac{1}{3840}\Big(32\xi^3(10x^2 - 5x\xi + \xi^2) + x^2\big(32x^3 + 60(\xi-2)^2\xi^2 \\ &+ 20x(\xi-2)^2\xi^2 - 5x^2\xi(32 - 12\xi - 4\xi^2 + \xi^3)\big) \\ &- 160x^2\xi^2(x + \xi + |x-\xi|) + 20x\xi(x + \xi + |x - v\xi|)^3 \\ &- (x + \xi + |x-\xi)^5\Big). \end{aligned}$$

2.2.3 A bounded linear operator on ${}^{o}W_2^3[0,1]$

Rewrite Eq. (2.2.2) as

$$y'''(x) - G(x)y(x) = F(x), \ \ 0 \le x \le 1, \tag{2.2.3}$$

where

$$\begin{aligned} G(x) &= \begin{cases} 0, & 0 \le x \le c, \\ g(x), & c < x \le d, \\ 0, & d < x \le 1, \end{cases} \\ F(x) &= \begin{cases} f(x), & 0 \le x \le c, \\ f(x) + r(x), & c < x \le d, \\ f(x), & d < x \le 1. \end{cases} \end{aligned}$$

Define a bounded linear operator $\mathbb{L} : {}^{o}W_2^3[0,1] \longrightarrow L^2[0,1]$, for each $y(x) \in {}^{o}W_2^3[0,1]$,

$$(\mathbb{L}y)(x) = y'''(x) - G(x)y(x). \tag{2.2.4}$$

Lemma 2.2.1. *$\mathbb{L}$ is a bounded linear operator.*

Proof.

$$\begin{aligned}\|\mathbb{L}y\|_{L^2}^2 &= \int_0^1 \left[y'''(x) - G(x)y(x)\right]^2 dx \\ &\leq \int_0^1 [y'''(x)]^2 dx + \int_0^1 [G(z)y(z)]^2 dx + 2\left(\int_0^1 |y'''(x)||G(x)||y(x)|dx\right).\end{aligned}$$

Note that

$$|y(x)| = \left|\langle y(\xi), R_\xi(x)\rangle_{{}^oW_2^3}\right| \leq M_1\|y\|_{{}^oW_2^3},$$

where M_1 is a positive real number and $G(x)$ is the bounded function, let $|G(x)| \leq M_2$. This follows that

$$\begin{aligned}\|\mathbb{L}y\|_{L^2}^2 &\leq \int_0^1 [y'''(x)]^2 dx + M_2^2 M_1^2 \|y\|_{{}^oW_2^3}^2 \\ &+ 2M_2\left[\int_0^1 |y'''(x)|^2 dx \int_0^1 |y(x)|^2 dx\right]^{1/2} \\ &\leq \int_0^1 [y'''(x)]^2 dx + M_2^2 M_1^2 \|y\|_{{}^oW_2^3}^2 \\ &+ 2M_2 M_1 \|y\|_{{}^oW_2^3} \sqrt{\int_0^1 |y'''(x)|^2 dx} \\ &\leq \|y\|_{{}^oW_2^3}^2 + M_2^2 M_1^2 \|y\|_{{}^oW_2^3}^2 \\ &+ 2M_2 M_1 \|y\|_{{}^oW_2^3} \sqrt{\|y\|_{{}^oW_2^3}^2} \leq M\|y\|_{{}^oW_2^3}^2.\end{aligned}$$

The proof is complete. □

We rewrite (2.2.2) as the following operator equation:

$$\begin{cases}(\mathbb{L}y)(x) = F(x), \ \ 0 \leq x \leq 1, \\ y(0) = y'(0) = y'(1) = 0.\end{cases} \tag{2.2.5}$$

2.2.4 To Solve Eq. (2.2.5)

We will choose and fix a countable dense subset $\mathrm{S} = \{x_1, x_2, \ldots\} \subset [0,1]$ and define $\psi_i(x)$ by

$$\begin{aligned}\psi_i(x) &\overset{def}{=} \int_0^{x_i} (\mathbb{L}_\xi R_\xi(x))d\xi \\ &= \partial_{x_i}^2 R_{x_i}(x) - \partial_\xi^2 R_\xi(x)|_{\xi=0} - \int_0^{x_i} G(\xi)R_\xi(x)d\xi, \quad i = 1, 2, \ldots.\end{aligned}$$

By Theory 1.3.3, it is easy to get the following lemmas.

Lemma 2.2.2. $\psi_i(x) \in {}^oW_2^3[0,1], \quad i = 1, 2, \ldots$.

Lemma 2.2.3. *The function system* $\{\psi_i(x)\}_{i=1}^\infty$ *is a complete system of the space* ${}^oW_2^3[0,1]$.

Proof. For every i, we have

$$\begin{aligned}0 = \langle y(x), \psi_i(x)\rangle_{{}^oW_2^3} &= \left\langle y(x), \int_0^{x_i} \mathbb{L}_\xi R_\xi(x)d\xi \right\rangle_{{}^oW_2^3} \\ &= \int_0^{x_i} \mathbb{L}_\xi \left(\langle y(x), R_\xi(x)\rangle_{{}^oW_2^3}\right) d\xi = \int_0^{x_i} \mathbb{L}y(\xi)d\xi.\end{aligned}$$

Note that $x_i \in S$ is a dense set. Therefore $\int_0^x \mathbb{L}y(\xi)d\xi = 0, x \in [0,1]$. It follows that $(\mathbb{L}y)(x) \equiv 0$. So $y(x) \equiv 0$ from the existence and uniqueness of the solution of Eq. (2.2.2), namely, the existence of $\mathbb{L}^{-1}$. □

By Gram–Schmidt process, we obtain an orthogonal basis $\{\widetilde{\psi}_i(x)\}_{i=1}^\infty$ of ${}^oW_2^3[0,1]$, such that

$$\widetilde{\psi}_i(x) = \sum_{k=1}^i \beta_{ik}\psi_k(x),$$

where β_{ik} are orthogonal coefficients.

Theorem 2.2.1. *If* $y(x)$ *is the solution of Eq.* (2.2.5)*, then*

$$y(x) = \sum_{i=1}^\infty \sum_{k=1}^i \beta_{ik} \int_0^{x_k} F(\xi)d\xi\widetilde{\psi}_i(x). \tag{2.2.6}$$

Proof. $y(x)$ can be expanded to Fourier series in terms of normal orthogonal basis $\overline{\psi}_i(x)$ in ${}^oW_2^3[0,1]$,

$$\begin{aligned} y(x) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\langle y(x),\widetilde{\psi}_i(x)\rangle_{{}^oW_2^3}\widetilde{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\langle y(x),\psi_k(x)\rangle_{{}^oW_2^3}\widetilde{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\left\langle y(x),\int_0^{x_k}\mathbb{L}_\xi R_\xi(x)d\xi\right\rangle_{{}^oW_2^3}\widetilde{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\int_0^{x_k}\mathbb{L}_\xi\left(\langle y(x),R_\xi(x)\rangle_{{}^oW_2^3}\right)d\xi\widetilde{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\int_0^{x_k}\mathbb{L}_\xi y(\xi)d\xi\widetilde{\psi}_i(x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\int_0^{x_k}F(\xi)d\xi\widetilde{\psi}_i(x). \end{aligned} \tag{2.2.7}$$

The proof is complete. □

By truncating the series of the left-hand side of (2.2.6), we obtain the approximate solution of Eq. (2.2.5)

$$y_n(x)=\sum_{i=1}^{n}\sum_{k=1}^{i}\beta_{ik}F(x_k)\widetilde{\psi}_i(x). \tag{2.2.8}$$

$y_n(x)$ in (2.2.8) is the n-term intercept of $y(x)$ in (2.2.6), so $y_n(x)\longrightarrow y(x)$ in $W_2^3[0,1]$ as $n\longrightarrow\infty$.

Theorem 2.2.2. *The approximate solution y_n and its derivatives y_n', y_n'' uniformly converge to exact solution y and its derivatives y', y'' respectively.*

Proof. By Theory 1.3.4, one can get its proof immediately. □

2.2.5 Numerical Experiments

Example 2.2.1.

$$y'''=\begin{cases}0, & 0\le x\le 1/4,\\ y(x)-1, & 1/4<x\le 3/4,\\ 0, & 3/4<x\le 1,\end{cases}$$

Table 2.3: Observed maximum errors $|y - y_n|$

Our mother	[47]	[48]	[49]	[50]	[51]	[52]	[53]	[54]	[55]
9.63E-6	1.27 E-4	1.61E-4	1.26E-4	1.15E-4	2.35 E-5	1.15 E-4	1.27E-4	7.36E-3	1.23E-4

Table 2.4: The root mean square deviations for the derivatives using our mother

$\left[\sum_{i=1}^{n}(y_n'(x_i) - y'(x_i))^2/n\right]^{1/2}$	$\left[\sum_{i=1}^{n}(y_n''(x_i) - y''(x_i))^2/n\right]^{1/2}$
3.5E-5	2.4E-4

with the boundary conditions

$$y(0) = 0, \quad y'(0) = 0, \ \text{ and } \ y'(1) = 0.$$

True solution is

$$y(x) = \begin{cases} \dfrac{1}{2}a_1 x^2, & 0 \le x \le 1/4, \\ 1 + a_2 e^x + e^{-\frac{x}{2}}\left[a_3 \cos\dfrac{\sqrt{3}}{2}x + a_4 \sin\dfrac{\sqrt{3}}{2}x\right], & 1/4 < x \le 3/4, \\ \dfrac{1}{2}a_5 x(x-2) + a_6, & 3/4 < x \le 1, \end{cases}$$

where

$$a_1 = 0.24391096222648; \ \ a_2 = -0.17847234452746; \ \ a_3 = -0.81893573565615;$$
$$a_4 = -0.30266818001866; \ \ a_5 = -0.24213890868443; \ \ a_6 = -0.06537630092111.$$

By using Mathematica 5.0 and the present method (where $n = 1000$, in (2.2.8)), we calculate the approximate solution $y_n(x)$. Table 2.3 gives the maximum error using our method and the maximum errors by other methods. The root mean square errors for the derivatives of approximate solution $y_n(x)$ and the derivatives of the true solution $y(x)$ are given in Table 2.4. From Figure 2.1–2.4, it is clear that the approximate solution and its derivatives can approximate the true solution and its derivatives well.

2.3 Solving Third-Order Singularly Perturbed Problems

2.3.1 Introduction

Singularly perturbed problems arise frequently in applications including geophysical fluid dynamics, oceanic and atmospheric circulation, chemical reactions, optimal control, etc. It is well known that the solution of singularly perturbed boundary

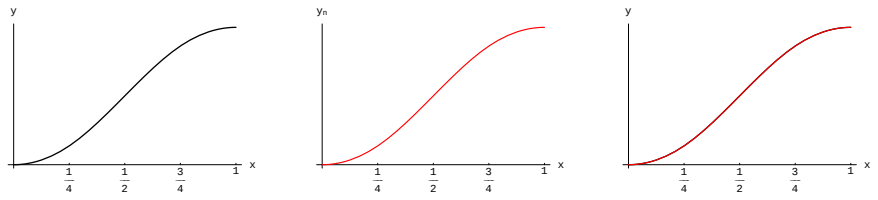


Figure 2.1: The red is curve of y, the black is curve of y_n.

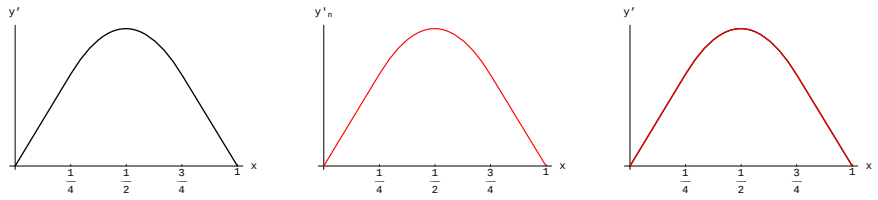


Figure 2.2: The red is curve of y', the black is curve of y_n'.

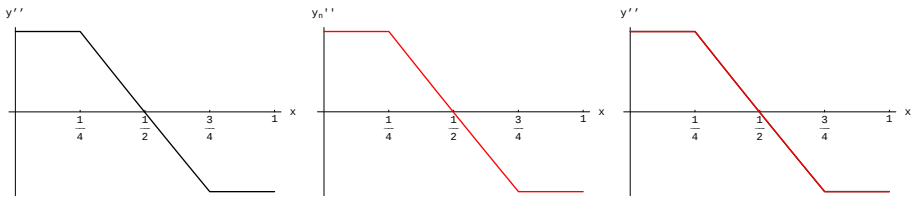


Figure 2.3: The red is curve of y'', the black is curve of y_n''.

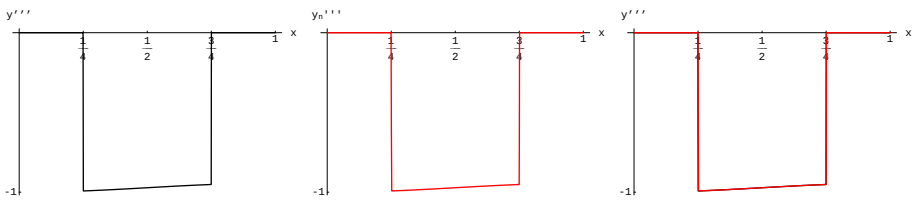


Figure 2.4: The red is curve of y''', the black is curve of y_n'''.

value problem has a multiscale character; that is, there are thin transition layers where the solution varies rapidly, while away from the layers the solution behaves regularly and varies slowly.

Such problems have been investigated by many researchers. The existence and uniqueness of such problems are discussed in [56]. The numerical treatment of singularly perturbed problems present some major computational difficulties, and in recent years a large number of special-purpose methods have been proposed to provide accurate numerical solutions. For the past two decades, many numerical method have appeared in the literature. Most notable among these are collocation methods [58], [59], finite-difference methods [60], [61], finite-element methods [62], boundary-value techniques [63], initial-value techniques [64], [65], Spline techniques [66], [67], and so on. As far as authors' knowledge goes, only few results are reported in the case of third-order differential equations. In fact, Howws has considered the problems and discussed the existence, uniqueness and asymptotic estimates of the solutions [68], [69]. Zhao [70] has also discussed the existence, uniqueness and obtained asymptotic estimates of the solution using the theory of differential inequalities. Roberts, Valarmathi and Ramanujam [71], [72] have suggested methods of finding approximate solutions for third-order singularly perturbed boundary value problems.

Here, a new computational method is suggested to solve the following third-order singularly perturbed boundary value problem in the reproducing kernel space

$$\begin{cases} \varepsilon u'''(x) + a(x)u''(x) + b(x)u'(x) + c(x)u(x) = f(x), \ \ 0 \le x \le 1, \\ u'(0) = q, \quad u(1) = 0, \quad u'(1) = 0, \end{cases} \tag{2.3.1}$$

where $0 < \varepsilon \ll 1$, $f(x)$, $a(x)$, $b(x)$, $c(x)$ are sufficiently smooth functions and $a(x) \ge -\alpha$, $\alpha > 0$.

Remark 2.3.1. Problem (2.3.1) has a boundary layer(at x=0) which is less severe because the boundary conditions are the Neumann type [56].

If put $u(x) = u_1(x)$, $u'(x) = u_2(x)$, then the problem (2.3.1) can be converted into an equivalent problem of the form

$$\begin{cases} u_1'(x) - u_2(x) = 0, \ \ 0 \le x \le 1, \\ \varepsilon u_2''(x) + a(x)u_2'(x) + b(x)u_2(x) + c(x)u_1(x) = f(x), \\ u_1(1) = 0, \quad u_2(0) = q, \quad u_2(1) = 0. \end{cases} \tag{2.3.2}$$

2.3.2 Asymptotic Expansion Approximation

One can look for the asymptotic expansion solution of (2.3.2) in the form

$$\mathbf{u}(x, \varepsilon) = (\mathbf{u}_0 + \mathbf{v}_0) + \varepsilon(\mathbf{u}_1 + \mathbf{v}_1) + \cdots .$$

By the method of stretching variable [18], one can obtain the zeroth-order asymptotic expansion $\mathbf{u}_{as} = \mathbf{u}_0 + \mathbf{v}_0$ where $\mathbf{u}_0$ is a solution of the reduced problem (2.3.2) given by

$$\begin{cases} u_{01}'(x) - u_{02}(x) = 0, \\ a(x)u_{02}'(x) + b(x)u_{02}(x) + c(x)u_{01}(x) = f(x), \ \ 0 \le x \le 1, \\ u_{01}(1) = 0, \quad u_{02}(1) = 0, \end{cases} \tag{2.3.3}$$

where $\mathbf{u}_0 = (u_{01}, u_{02})$ and $\mathbf{v}_0 = (v_{01}, v_{02})$ is the layer correction

$$v_{01} = -\frac{\varepsilon}{a(0)}[q - u_{02}(0)]e^{-\frac{a(0)x}{\varepsilon}}, \quad v_{02} = [q - u_{02}(0)]e^{-\frac{a(0)x}{\varepsilon}}. \tag{2.3.4}$$

Theorem 2.3.1. *The zeroth order asymptotic expansion approximation u_{as} satisfies the inequality*

$$|\mathbf{u}(x) - \mathbf{u}_{as}(x)| \le C\varepsilon,$$

where $\mathbf{u}(x) = (u_1, u_2)$ is the solution of (2.3.2), C is a positive real number.

The proof see for [75].

In order to obtain the approximate solution $\mathbf{u}_{as}$, It suffices to obtain the solution $\mathbf{u}_0(x)$ to the terminal value problem (2.3.3).

2.3.3 Several Reproducing Kernel Spaces and Lemmas

The reproducing kernel space

$${}^oW_2^2[0,1] = \Big\{u(x)|\ u(x) \in W_2^2[0,1] \text{ by (1.3.2)},\ u(1) = 0\Big\}$$

possesses the reproducing kernel function

$$R_y(x)$$
$$= \begin{cases} -\dfrac{1}{84}(y–1)\Big(x^3(y–4)(2+y)–6(y–4)(2+y)–6x(6+(y–2)y)–3x^2(6+(y–2)y)\Big), & x \le y, \\ -\dfrac{1}{84}(x–1)\Big(6(8+2x–x^2)–6(6–2x+x^2)y–3(6–2x+x^2)y^2–(8+2x–x^2)y^3\Big), & x > y. \end{cases}$$

Given the reproducing kernel space $W_2^1[0,1]$ by (1.3.2), the reproducing kernel is $r_y(x)$.

Define the operators $\mathbb{A}_{ij} : {}^oW_2^2[0,1] \longrightarrow W_2^1[0,1], \ \ (i,j = 1,2)$ by

$$\begin{aligned} (\mathbb{A}_{11}v)(x) &= v'(x), \\ (\mathbb{A}_{12}v) &= -v(x), \\ (\mathbb{A}_{21}v)(x) &= c(x)v(x), \\ (\mathbb{A}_{22}v)(x) &= a(x)v'(x) + b(x)v(x), \end{aligned}$$

$a(x)$, $b(x)$, $c(x)$ are given by (2.3.1).

It is easy to prove the operators $\mathbb{A}_{ij}$ is bounded. For instance,

$$\begin{aligned}\|(\mathbb{A}_{11}v)(x)\|^2_{W_2^1} &= \|v'(x)\|^2_{W_2^1} = (v'(0))^2 + \int_0^1 (v''(x))^2 dx \\ &\leq (v(0))^2 + (v'(0))^2 + \int_0^1 (v''(x))^2 dx \\ &= \|v(x)\|^2_{{}^oW_2^2}.\end{aligned}$$

Put $\mathbb{A} = \begin{pmatrix} \mathbb{A}_{11} & \mathbb{A}_{12} \\ \mathbb{A}_{21} & \mathbb{A}_{22} \end{pmatrix}$, then $\mathbb{A}$ is the operator from $\mathbf{W}_2 = {}^oW_2^2[0,1] \bigoplus {}^oW_2^2[0,1]$ to $\mathbf{W}_1 = W_2^1[0,1] \bigoplus W_2^1[0,1]$. Note that the inner product of $\mathbf{W}_i$, $i = 1, 2$ is defined as

$$\begin{aligned}\langle \mathbf{u}(x), \mathbf{v}(x)\rangle_{\mathbf{W_1}} &= \langle u_1(x), v_1(x)\rangle_{W_2^1} + \langle u_2(x), v_2(x)\rangle_{W_2^1}, \\ \langle \mathbf{u}(x), \mathbf{v}(x)\rangle_{\mathbf{W_2}} &= \langle u_1(x), v_1(x)\rangle_{{}^oW_2^2} + \langle u_2(x), v_2(x)\rangle_{{}^oW_2^2}.\end{aligned} \tag{2.3.5}$$

If write out $\mathbf{u}_0 = (u_{01}, u_{02})^\top$ and $\mathbf{F}(x) = (0, f(x))^\top$. Then TVP (2.3.3) can be converted into the following form

$$\begin{cases} \mathbb{A}\mathbf{u}_0(x) = \mathbf{F}(x), \ \ 0 \leq x \leq 1, \\ \mathbf{u}_0(1) = \mathbf{0}, \end{cases} \tag{2.3.6}$$

where $\mathbf{u}_0 \in \mathbf{W}_2$, $\mathbf{F}(x) \in \mathbf{W}_1$.

Lemma 2.3.1. *If $\mathbb{A}_{ij} : {}^oW_2^2[0,1] \longrightarrow W_2^1[0,1], i, j = 1, 2$ are bounded linear operators, then $\mathbb{A} : \mathbf{W}_2 \longrightarrow \mathbf{W}_1$ is a bounded linear operator.*

Proof. Clearly, $\mathbb{A}$ is a linear operator. For $\forall\, \mathbf{u} \in \mathbf{W}_2$,

$$\begin{aligned}\|\mathbb{A}\mathbf{u}\|_{W_2^1} &= \left(\sum_{i=1}^2 \left\| \sum_{j=1}^2 \mathbb{A}_{ij} u_j \right\|^2_{W_2^1} \right)^{\frac{1}{2}} \\ &\leq \left[\sum_{i=1}^2 \left(\sum_{j=1}^2 \|\mathbb{A}_{ij}\| \, \|u_j\|_{{}^oW_2^2} \right)^2 \right]^{\frac{1}{2}} \\ &\leq \left[\sum_{i=1}^2 \left(\sum_{j=1}^2 \|\mathbb{A}_{ij}\|^2 \right) \left(\sum_{j=1}^2 \|u_j\|^2_{{}^oW_2^2} \right) \right]^{\frac{1}{2}} \\ &= \left(\sum_{i=1}^2 \sum_{j=1}^2 \|\mathbb{A}_{ij}\|^2 \right)^{\frac{1}{2}} \|\mathbf{u}\|_{\mathbf{W}_2}.\end{aligned}$$

The boundedness of $\mathbb{A}_{ij}$ implies that $\mathbb{A}$ is bounded. The proof is complete. □

It is easy to see that the adjoint operator of $\mathbb{A}$ is $\mathbb{A}^*= \begin{pmatrix} \mathbb{A}_{11}^* & \mathbb{A}_{21}^* \\ \mathbb{A}_{12}^* & \mathbb{A}_{22}^* \end{pmatrix}$, where $\mathbb{A}_{ij}^*$ is the adjoint operator of $\mathbb{A}_{ij}$.

2.3.4 The Representation of Solution of TVP (2.3.6)

Let $r_y(x)$ is the producing kernel of $W_2^1[0,1]$, $\mathbf{e}_1 = (1,0)^\top$, $\mathbf{e}_2 = (0,1)^\top$.

Lemma 2.3.2. *If* $\mathrm{S} = \{x_1, x_2, \ldots\}$ *is a dense set of* $[0,1]$*, put*

$$\Phi_{ij}(x) = r_{x_i}(x)\mathbf{e}_j, \quad \Psi_{ij}(x) = (\mathbb{A}^*\Phi_{ij})(x), \ \ j = 1,2,$$

then $\{\Psi_{ij}(x)_{j=1,2}\}_{i=1}^{\infty}$ *is the complete system of* $\mathbf{W}_2$*.*

Proof. Given $\mathbf{u}(x) \in \mathbf{W}_2$, if

$$\langle \mathbf{u}(x), \Psi_{ij}(x)\rangle_{\mathbf{W}_2} = 0 \ \ (i = 1,2,\ldots; \ \ j = 1,2),$$

which means that,

$$\langle (\mathbb{A}\mathbf{u})(x), \Phi_{ij}(x)\rangle_{\mathbf{W}_1} = 0 \ \ (i = 1,2,\ldots; \ \ j = 1,2). \tag{2.3.7}$$

Note that

$$\mathbb{A}\mathbf{u} = \big(\mathbb{A}_{11}u_1 + \mathbb{A}_{12}u_2, \mathbb{A}_{21}u_1 + \mathbb{A}_{22}u_2\big)^\top, \quad \Phi_{ij} = R_{x_i}\mathbf{e}_j.$$

Hence

$$\langle (\mathbb{A}\mathbf{u})(x), \Phi_{ij}(x)\rangle_{\mathbf{W}_1} = \langle \mathbb{A}_{j1}u_1 + \mathbb{A}_{j2}u_2, R_{x_i}\rangle_{W_2^1} = [\mathbb{A}_{j1}u_1 + \mathbb{A}_{j2}u_2](x_i),$$

that is $(\mathbb{A}\mathbf{u})(x_i) = \mathbf{0}$ Since S is dense on $[0,1]$, we must have $(\mathbb{A}\mathbf{u})(x) = 0$. It follows that $\mathbf{u} \equiv 0$ from the existence of $\mathbb{A}^{-1}$. Therefore, $\{\Psi_{ij}(x)_{j=1,2}\}_{i=1}^{\infty}$ is the complete system of $\mathbf{W}_2$. So the proof is complete. □

By Gram–Schmidt process, we obtain an orthogonal basis $\{\overline{\Psi}_{ij}(x)_{j=1,2}\}_{i=1}^{\infty}$ of $\mathbf{W}_2$, such that

$$\overline{\Psi}_{ij}(x) = \sum_{l=1}^{i}\sum_{k=1}^{j} \beta_{lk}^{ij}\Psi_{lk}(x),$$

where β_{lk}^{ij} are orthogonal coefficients.

Theorem 2.3.2. *If* $\{x_i\}_{i=1}^{\infty}$ *is dense on* $[0,1]$ *and the solution of* TVP (2.3.6) *is unique, then the solution of* TVP (2.3.6) *is*

$$\mathbf{u}(x) = \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j} \beta_{lk}^{ij}F_k(x_l)\overline{\Psi}_{ij}(x), \tag{2.3.8}$$

Proof. Note that $\langle \mathbf{v}(x), \Phi_{ij}(x)\rangle_{\mathbf{W}_2} = v_j(x_i)$ for each $\mathbf{v}(x) \in \mathbf{W}_2$. Hence, we have

$$\begin{aligned}
\mathbf{u}(x) &= \sum_{i=1}^{\infty}\sum_{j=1}^{2}\langle \mathbf{u}(x), \overline{\Psi}_{ij}(x)\rangle_{\mathbf{W}_2}\overline{\Psi}_{ij}(x) \\
&= \sum_{i=1}^{\infty}\sum_{j=1}^{2}\left\langle \mathbf{u}(x), \sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}\Psi_{lk}(x)\right\rangle_{\mathbf{W}_2}\overline{\Psi}_{ij}(x) \\
&= \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}\big\langle \mathbf{u}(x), \mathbb{A}^*\Phi_{lk}(x)\big\rangle_{\mathbf{W}_2}\overline{\Psi}_{ij}(x) \\
&= \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}\big\langle \mathbb{A}\mathbf{u}(x), \Phi_{lk}(x)\big\rangle_{\mathbf{W}_1}\overline{\Psi}_{ij}(x) \\
&= \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}(\mathbb{A}\mathbf{u})_k(x_l)\overline{\Psi}_{ij}(x) \\
&= \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}F_k(x_l)\overline{\Psi}_{ij}(x). \qquad (2.3.9)
\end{aligned}$$

So, the proof of the theorem is complete. □

Now the zeroth order asymptotic expansion approximation u_{as} can be obtained

$$\mathbf{u}_{as} = \mathbf{u}_0(x) + \mathbf{v}_0(x) = \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}F_k(x_l)\overline{\Psi}_{ij}(x) + \mathbf{v}_0(x) \qquad (2.3.10)$$

and the approximate solution of problem (1.2) can be obtained by the n-term intercept of (2.3.10) and

$$\mathbf{u}_{as}^{n} = \sum_{i=1}^{n}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}F_k(x_l)\overline{\Psi}_{ij}(x) + \mathbf{v}_0(x),$$

where $\mathbf{v}_0(x)$ is given in (2.3.4).

2.3.5 Numerical Experiments

Example 2.3.1. Consider the nonlinear singularly perturbed two-point boundary value problem

$$\begin{cases} \varepsilon u'''(x) + (1-x)u''(x) - u'(x) + xu(x) = f(x), \;\; 0 \le x \le 1, \\ u'(0) = \dfrac{1}{e^{-\frac{1}{\varepsilon}} - \varepsilon} - 2, \quad u(1) = 0, \;\; u'(1) = 0 \end{cases}$$

Table 2.5: Numerical results for Example 2.3.1 ($\varepsilon = 10^{-4}$).

x	True solution u(x)	Approximate solution u_{54}	relative error
$1.0E-06$	1.99005	1.989450	$3.0E-04$
$1.0E-05$	1.90482	1.904220	$3.1E-04$
$9.0E-05$	1.40639	1.405790	$4.2E-04$
$1.2E-04$	1.30095	1.300360	$4.5E-04$
0.08	0.84640	0.846260	$1.6E-04$
0.32	0.46240	0.462360	$8.5E-05$
0.48	0.27040	0.270407	$3.9E-05$
0.64	0.12960	0.129631	$2.4E-04$
0.80	0.04000	0.040030	$7.6E-04$
0.96	0.00160	0.001602	$1.3E-03$
0.999	$1.00E-6$	$1.002E-6$	$2.9E-03$

Table 2.6: Numerical results for Example 2.3.1 ($\varepsilon = 10^{-4}$).

x	True solution $u'(x)$	Approximate solution u'_{54}	relative error
1.0E-06	-9902.50	-9902.50	2.0E-10
1.0E-05	-9050.37	-9050.37	2.1E-09
9.0E-05	-4067.70	-4067.70	2.9E-08
1.2E-04	-3013.94	-3013.94	4.7E-08
0.08	-1.84	-1.83983	9.1E-05
0.32	-1.36	-1.35988	8.5E-05
0.48	-1.04	-1.03992	7.9E-05
0.80	-0.40	-0.39997	5.7E-05
0.96	-0.08	-0.07999	1.0E-05
0.999	-0.002	-0.002	5.7E-07

Table 2.7: Numerical results for Example 2.3.1 ($\varepsilon = 10^{-6}$).

x	True solution$u(x)$	Approximate solution u_{54}	relative error
1.0E-08	1.99005	1.98945	3.0E-04
1.0E-07	1.90484	1.90424	3.1E-04
9.0E-07	1.40657	1.40597	4.2E-04
3.6E-06	1.02732	1.02672	5.8E-04
0.00001	1.00003	0.99942	5.9E-04
0.08	0.8464	0.84626	1.6E-04
0.32	0.4624	0.46236	8.5E-05
0.48	0.2704	0.27040	2.4E-05
0.64	0.1296	0.12963	2.4E-04
0.80	0.0400	0.04003	7.6E-04
0.96	0.0016	0.00160	1.3E-03
0.999	1.0E-6	1.002E-6	2.9E-03

Table 2.8: Numerical results for Example 2.3.1 ($\varepsilon = 10^{-6}$).

x	True solution $u'(x)$	Approximate solution u'_{54}	relative error
1.0E-08	-951231	-951231	1.0E-11
1.0E-07	-904839	-904839	2.1E-11
9.0E-07	-406572	-406572	2.9E-10
3.6E-06	-27325.7	-27325.7	7.2E-09
0.00001	-47.3999	-47.3997	4.3E-06
0.08	-1.84	-1.83983	8.8E-05
0.32	-1.36	-1.35988	8.5E-05
0.48	-1.04	-1.03999	7.9E-05
0.64	-0.72	-0.71994	7.1E-05
0.80	-0.40	-0.39997	5.7E-05
0.96	-0.08	-0.07999	1.0E-05
0.999	-0.002	-0.0020	5.7E-07

The true solution

$$u(x) = 1 + \frac{1 - e^{-(\frac{x}{\epsilon})}}{-1 + e^{-(\frac{1}{\epsilon})}} + (-1 + x)^2.$$

Using our method, we choose 54 points on $[0, 1]$ and take $\varepsilon = 10^{-4}$, $\varepsilon = 10^{-6}$ respectively. The numerical results are given in the following Table 2.5–2.8.

2.4 Solving a Class of Variable Delay Integro-Differential Equations

2.4.1 Introduction

The optimal control problem in population can be changed into solving the functional extreme value problem [76]: $\min J(g)$, where

$$J(g) = \int_0^1 \left[u^2(t) + g^2(t)\right]\, dt,$$

and $u(t)$ satisfies

$$\begin{cases} u'(t) = a_1 u(t) + a_2 u(t-\tau) + \displaystyle\int_{-\tau}^{0} c(\theta) u(t+\theta)\, d\theta + bg(t), \\ u(t) = h(t), \quad -\tau \le t < 0, \quad \tau \ge 0. \end{cases} \tag{2.4.1}$$

Ref [77] studies the behavior of solution of linear integro-differential equation

$$\begin{cases} x'(t) = ax(t) + \int\limits_{-\infty}^{t} k(t-s)x(s)\,ds, & t \geq 0, \\ x(t) = \phi(t), & t < 0. \end{cases} \tag{2.4.2}$$

with unbounded delay, where a is a real number, and$k$ is a continuous real-valued function on the interval $[0, \infty)$ which is assumed to be of sign and not identically zero on $[0, \infty)$. [78] study stability of Runge-Kutta (RK) methods for delay integro-differential equations with a constant delay on the basis of the linear equation

$$u'(t) = Lu(t) + Mu(t-\tau) + \int\limits_{t-\tau}^{t} u(\theta)\,d\theta, \tag{2.4.3}$$

where L, M, K are constant Complex matrices. [79] studies the solvability of the Cauchy problem for functional differential equations with infinite delay in a Banach space X. [80] investigates explicit and implicit continuous Runge–Kutta methods for solving neutral Volterra integro-differential equation with delay.

However, the representation of the exact solution for the Delayed Integro-Differential Equation (DIDEs)has not been discussed. In this section, the representations of solutions to Eq. (2.4.1), Eq. (2.4.2) and Eq. (2.4.3) are given in a special Hilbert space – Reproducing kernel space.

In this section, We discusses Eq. (2.4.1) with more general form

$$\begin{cases} u'(x) + au(x) + bu(x-\tau(x)) + \int\limits_{\lambda(x)}^{x} c(s,x)u(s)\,ds = f(x), \\ u(x) = h(x), \quad -\tau(x) \leq x < 0, \quad \tau(x) \geq 0, \end{cases} \tag{2.4.4}$$

where $\lambda(x) = \max\{x - \tau(x), 0\}$, $\tau(x)$ is a delayed function.

And this method is completely suitable to Eq. (2.4.2), (2.4.3).

2.4.2 The Reproducing Kernel Spaces

Homogenize Eq. (2.4.4), one obtains $u(0) = 0$. Therefore we consider to construct the following reproducing kernel space ${}^{o}W_2^2[0, \infty)$.

(1)

$${}^{o}W_2^2[0,\infty) = \Big\{u(x) |\ u,\ u' \text{ are absolutely continuous functions}$$
$$u,\ u',\ u'' \in L^2[0,\infty),\ u(0) = 0\Big\},$$

and endowed with the inner product as follows:

$$\langle u(x), v(x)\rangle_{W_2^2} = \int_0^\infty (4uv + 5u'v' + u''v'')\,dx.$$

Similar to the Section 1.6, one gets the reproducing kernel:

$$R_t(x) = \frac{e^{-2(t+x)} - 2e^{-(t+x)} - e^{-2|x-t|} + 2e^{-|x-t|}}{12}, \quad t,\, x \in [0,\infty). \tag{2.4.5}$$

(2) The reproducing kernel space $W_2^1[0,\infty)$

$$W_2^1[0,\infty) = \Big\{u(x)|\ u \text{ is absolutely continuous function},\ u,\, u' \in L^2[0,\infty)\Big\},$$

the inner product is given as follows:

$$\langle u(x), v(x)\rangle_{W_2^1} = \int_0^\infty (uv + u'v')\,dx.$$

The reproducing kernel:

$$Q_t(x) = \frac{1}{2}\big(e^{-(x+t)} + e^{-|x-t|}\big). \tag{2.4.6}$$

2.4.3 Linear Operator L on ${}^oW_2^2[0,\infty)$

Let

$$(\mathbb{L}u)(x) \equiv \begin{cases} u'(x) + a(x)u(x) + b(x)u(x-\tau(x)) \\ \qquad + \displaystyle\int_{\lambda(x)}^{x} c(s,x)u(s)\,ds, & \tau(x) \leqslant x < \infty, \\ u'(x) + a(x)u(x) + \displaystyle\int_{\lambda(x)}^{x} c(s,x)u(s)\,ds, & 0 \leqslant x < \tau(x), \end{cases} \tag{2.4.7}$$

where $\lambda(x) = \max\{x - \tau(x), 0\}$.

Now, Eq. (2.4.4) is rewritten with the transformation as follows:

$$\begin{cases} (\mathbb{L}u)(x) = F(x), & 0 \leqslant x < \infty, \\ u(x) = h(x), & -\tau(x) < x < 0, \end{cases} \tag{2.4.8}$$

where

$$F(x) = \begin{cases} f(x), & \tau(x) \leq x < \infty, \\ f(x) - b(x)h(x-\tau(x)), & 0 \leq x < \tau(x). \end{cases}$$

Lemma 2.4.1. *If $a(x)$, $b(x)$ are bounded on interval $[0,\infty)$, $c(x,s)$ is bounded on interval $[0,\infty)\times[0,\infty)$, then operator $\mathbb{L}: W_2^2[0,\infty)\longrightarrow W_2^1[0,\infty)$ is a bounded linear operator.*

Proof. For $u(x)\in W_2^2[0,\infty)$, note that

$$(\mathbb{L}u)(x)=u'(x)+a(x)u(x)+w(x)+\int\limits_{\lambda(x)}^{x}c(s,x)u(s)\,ds,$$

where

$$\begin{aligned} w(x) &= \begin{cases} b(x)u(x-\tau(x)), & \tau(x)\le x<\infty,\\ 0, & 0\le x<\tau(x),\end{cases}\\ \|\mathbb{L}u\|_{W_2^1} &\le \|u'(x)\|_{W_2^1}+\|a(x)u(x)\|_{W_2^1}+\|w(x)\|_{W_2^1}\\ &+ \left\|\int\limits_{\lambda(x)}^{x}c(s,x)u(s)\,ds\right\|_{W_2^1}, \end{aligned}$$

(1)

$$\|a(x)u\|_{W_2^1}^2\le A^2\|u\|_{W_2^2}^2,\quad \|w(x)\|_{W_2^1}^2\le B^2\|u\|_{W_2^2}^2,$$
$$\|u'\|_{W_2^1}^2=\int\limits_0^{\infty}\left[(u')^2+(u'')^2\right]\,dx\le\|u\|_{W_2^2}^2,$$

where A, B are the upper bounds of $|a(x)|$, $|b(x)|$, respectively.

(2) i. When $x-\tau<0$, $\lambda(x)=0$,

$$\left\|\int\limits_0^{x}c(s,x)u(s)ds\right\|_{W_2^1}^2$$
$$=\int\limits_0^{\infty}\left[\left(\int\limits_0^{x}c(s,x)u(s)\,ds\right)^2+\left(\frac{d}{dx}\int\limits_0^{x}c(s,x)u(s)\,ds\right)^2\right]\,dx$$
$$=\int\limits_0^{\infty}\left[\left(\int\limits_0^{x}c(s,x)u(s)\,ds\right)^2+c^2(x,x)u^2(x)+\left(\int\limits_0^{x}c'(s,x)u(s)\,ds\right)^2\right.$$
$$\left.+2c(x,x)u(x)\int\limits_0^{x}c'(s,x)u(s)\,ds\right]\,dx$$

and

$$\int_0^\infty \left(\int_0^x c(s,x)u(s)\,ds \right)^2 dx \le \int_0^\infty \left(\int_0^\infty c(s,x)u(s)\,ds \right)^2 dx$$
$$\le \left(\int_0^\infty c^2(s,x)\,ds \int_0^\infty u^2(s)\,ds \right) dx \le M_1 \|u\|^2_{{}_oW_2^2}.$$

In the same way

$$\int_0^\infty \left(\int_0^x c'_x(s,x)u(s)\,ds \right)^2 dx \le M_2 \|u\|^2_{{}_oW_2^2}.$$

On the other hand,

$$\int_0^\infty \left(c(x,x)u(x) \int_0^x c'_x(s,x)u(s)\,ds \right) dx$$
$$\le \left[\int_0^\infty c^2(x,x)u^2(x)\,dx \int_0^\infty \left(\int_0^x c'_x(s,x)u(s)\,ds \right)^2 dx \right]^{1/2}$$
$$\le \left[M_3\|u\|^2_{W_2^2} \cdot M_4\|u\|^2_{W_2^2} \right]^{1/2} \le M_5\|u\|^2_{W_2^2},$$

where M_1, M_2, M_3, M_4, $M_5 > 0$ are constants. So that

$$\left\| \int_0^x c(s,x)u(s)ds \right\|_{W_2^1} \le M\|u\|_{W_2^2}.$$

ii. When $x - \tau \ge 0$, similar to i), we have

$$\left\| \int_{x-\tau}^x c(s,x)u(s)ds \right\|_{W_2^1} \le M\|u\|_{W_2^2}.$$

Therefore, according to (1), (2), $\mathbb{L}$ is a bounded linear operator. □

Lemma 2.4.2. *If take $t_i \in [0,\infty)$, put*

$$\begin{aligned} \varphi_i(x) &= Q_{t_i}(x), \\ \psi_i(x) &= \mathbb{L}^*\varphi_i(x), \end{aligned}$$

then $\|\psi_i(x)\|_{{}_oW_2^2} \le \|\mathbb{L}\|$, $i = 0,1,2,\ldots$, where $\mathbb{L}^*$ denotes the conjugated operator of $\mathbb{L}$.

Proof. Note that

$$\|\psi_i(x)\|_{{}^oW_2^2} = \left\langle \mathbb{L}^*\varphi_i(x), L^*\varphi_i(x)\right\rangle_{{}^oW_2^2} \le \|\mathbb{L}\|^2\|\varphi_i(x)\|^2_{W_2^1}.$$

Since

$$\|\varphi_i(x)\|^2_{W_2^1} = \left\langle Q_{t_i}(x), Q_{t_i}(x)\right\rangle_{W_2^1} = Q_{t_i}(t_i) = \frac{1}{2}\left(1+e^{-2t_i}\right) \le 1,$$

we have

$$\|\psi_i(x)\|_{W_2^2} \le \|\mathbb{L}\|. \tag{2.4.9}$$

The proof is complete. □

2.4.4 Two Function Sequences: $r_n(x)$, $\rho_n(x)$

For $u(x) \in {}^oW_2^2$, put

$$r_0(x) = u(x), \quad \rho_0(x) = (\mathbb{L}u)(x) = F(x). \tag{2.4.10}$$

If let $t_0 \in [0,+\infty)$ be the maximum point of $|\rho_0(x)|$, and r_n, ρ_n are defined respectively by recursion method as follows:

$$r_n(x) = r_{n-1}(x) - B^2_{n-1}\rho_{n-1}(t_{n-1})\psi_{n-1}(x), \ \ n = 1,2,\ldots, \tag{2.4.11}$$

$$\rho_n(x) = \rho_{n-1}(x) - B^2_{n-1}\rho_{n-1}(t_{n-1})(\mathbb{L}\psi_{n-1})(x), \ \ n = 1,2,\ldots, \tag{2.4.12}$$

where $B_k \overset{def}{=} \frac{1}{\|\psi_k\|}$, $t_k \in [0,+\infty)$ is the maximum point of $|\rho_k(x)|$, $k = 0,1,\ldots$. From Lemma 2.4.2, $B_k \ge \frac{1}{\|\mathbb{L}\|}$. Obviously,

$$\rho_n(x) = (\mathbb{L}r_n)(x), \ \ n = 0,1,2,\ldots.$$

By (2.4.11) and (2.4.12), we can obtain directly,

$$\begin{aligned} u(x) &= r_0(x) = \sum_{k=0}^{n-1} B_k^2\rho_k(t_k)\psi_k(x) + r_n(x), \\ (\mathbb{L}u)(x) &= \rho_0(x) = \sum_{k=0}^{n-1} B_k^2\rho_k(t_k)(\mathbb{L}\psi_k)(x) + \rho_n(x). \end{aligned} \tag{2.4.13}$$

Theorem 2.4.1. *$\|r_n(x)\|_{{}^oW_2^2}$ is a decreasing sequence of numbers.*

Proof.

$$\begin{aligned} \|r_n(x)\|^2_{{}^oW_2^2} &= \langle r_n, r_n\rangle_{{}^oW_2^2} \\ &= \left\langle r_{n-1} - B^2_{n-1}\rho_{n-1}(t_{n-1})\psi_{n-1}, r_{n-1} - B^2_{n-1}\rho_{n-1}(t_{n-1})\psi_{n-1}\right\rangle_{{}^oW_2^2} \end{aligned}$$

$$
\begin{aligned}
&= \|r_{n-1}\|^2_{{}_oW_2^2} - 2B^2_{n-1}\rho_{n-1}(t_{n-1})\langle r_{n-1}, \psi_{n-1}\rangle_{{}_oW_2^2} \\
&+ B^4_{n-1}\rho^2_{n-1}(t_{n-1})\|\psi_{n-1}\|^2_{{}_oW_2^2}.
\end{aligned}
$$

Noting that

$$
\begin{aligned}
\langle r_{n-1}, \psi_{n-1}\rangle_{{}_oW_2^2} &= \langle r_{n-1}, \mathbb{L}^* R_{t_{n-1}}\rangle_{{}_oW_2^2} \\
&= \langle \mathbb{L} r_{n-1}, R_{t_{n-1}}\rangle_{W_2^1} = \langle \rho_{n-1}, R_{t_{n-1}}\rangle_{W_2^1} = \rho_{n-1}(t_{n-1})
\end{aligned}
$$

and $\|\psi_{n-1}\|^{-1} = B_{n-1}$, we have

$$
\begin{aligned}
\|r_n(x)\|^2_{{}_oW_2^2} &= \|r_{n-1}\|^2_{{}_oW_2^2} - 2B^2_{n-1}\rho^2_{n-1}(t_{n-1}) + B^2_{n-1}\rho^2_{n-1}(t_{n-1}) \\
&= \|r_{n-1}\|^2_{{}_oW_2^2} - B^2_{n-1}\rho^2_{n-1}(t_{n-1}). \qquad (2.4.14)
\end{aligned}
$$

Therefore,

$$
\|r_n(x)\|^2_{{}_oW_2^2} \le \|r_{n-1}(x)\|^2_{{}_oW_2^2}. \qquad \square
$$

Corollary 2.4.1. *$\|r_n(x)\|_{{}_oW_2^2}$ is a convergent sequence of numbers.*

Theorem 2.4.2. *The sequence of functions $\rho_n(x) \longrightarrow 0$ (uniformly) on $[0, \infty)$.*

Proof. Noting that (2.4.14) and $\|r_{n+1}\|^2_{{}_oW_2^2}$ is convergent, we have

$$
\|u(x)\|^2_{W_2^2} \equiv \|r_0(x)\|^2_{{}_oW_2^2} = \|r_{n+1}\|^2_{{}_oW_2^2} + \sum_{k=0}^{n} B^2_k\rho^2_k(t_k). \qquad (2.4.15)
$$

Hence $\sum\limits_{k=0}^{\infty} B^2_k\rho^2_k(t_k)$ is a convergent series. This follows that $B^2_n\rho^2_n(t_n) \longrightarrow 0$ $(n \longrightarrow \infty)$. But with $B_n \ge \frac{1}{\|\mathbb{L}\|}$, the above formula implies that $\rho_n(t_n) \longrightarrow 0$ $(n \longrightarrow \infty)$. We note that t_n is the maximum point of $|\rho_n(x)|$, and hence $\rho_n(x) \longrightarrow 0$ $(uniformly)$ on $[0, \infty)$. $\square$

2.4.5 The Representation of Solution of Eq. (2.4.4)

Here, assume that the solution of Eq. (2.4.4) is existent and unique.

Lemma 2.4.3. *Assume that Eq. (2.4.4) has a unique solution and $a(x)$, $b(x)$, $c(x, s)$ satisfy the condition of Lemma 2.4.1. If operator $\mathbb{L} : W_2^2[0, +\infty) \longrightarrow W_2^1[0, +\infty)$ is defined by (2.4.7), then $\mathbb{L} : C^1[0, \infty) \longrightarrow C[0, +\infty)$ is a linear bounded and invertible operator, and its inverse operator $\mathbb{L}^{-1}$ is also linear bounded.*

Proof. Note only that the norms of $C^1[0, \infty)$ and $C[0, \infty)$ are of the following forms

$$
\|u\|_c = \max_{x\in[0,\infty)} |u|, \quad \|u\|_{c^1} = \sup_{x\in[0,\infty)} (|u| + |u'|).
$$

It is easy to prove that $\mathbb{L}$ is a bounded linear operator from $C^1[0,\infty)$ to $C[0,\infty)$ under the conditions of Lemma 2.4.1. On the other hand, $C^1[0,\infty)$ and $C[0,\infty)$ are Banach spaces. Applying the theorem of inverse operator, $\mathbb{L}^{-1}$ is also a bounded linear operator. □

Finally, we give the main theorems of this Section.

Theorem 2.4.3. *Assume that the solution $u(t)$ of Eq. (2.4.4) is unique, then it can be represented as follows*

$$u(x) = \sum_{k=0}^{\infty} B_k^2 \rho_k(t_k)\psi_k(x), \tag{2.4.16}$$

where t_k is the maximum point of $|\rho_k(x)|$, $x \in [0,\infty)$.

Proof. By (2.4.13), we have

$$\mathbb{L}^{-1}((\mathbb{L}u))(x) = \mathbb{L}^{-1}\left(\sum_{k=0}^{n-1} B_k^2 \rho_k(t_k)(\mathbb{L}\psi_k)\right)(x) + (\mathbb{L}^{-1}\rho_n)(x).$$

Hence

$$u(x) = \sum_{k=0}^{n-1} B_k^2 \rho_k(t_k)\psi_k(x) + r_n(x), \tag{2.4.17}$$

where

$$r_n(x) = \mathbb{L}^{-1}\rho_n(x).$$

Since inverse operator $\mathbb{L}^{-1} : W_2^1[0,\infty) \longrightarrow W_2^2[0,\infty)$ is linear bounded, then taking the limit of both sides on $C^1[0,\infty)$, we have $r_n(x) \longrightarrow 0$ $(n \longrightarrow \infty)$. We have proved that $\rho_n(x)$ is uniformly convergent on $[0,\infty)$, thus $r_n(x) \longrightarrow 0$ uniformly holds on $[0,\infty)$. □

Truncating the series of (2.4.16), we get an approximate formula

$$u_m(x) = \sum_{k=0}^{m} B_k^2 \rho_k(t_k)\psi_k(x). \tag{2.4.18}$$

2.4.6 Numerical Experiments

Example 2.4.1. Let $\tau(x) = 0.2$, $x \in [0,\infty)$; $h(x) = x$, $x \in [-0.2, 0)$; $a(x) = b(x) = c(t,x) = 1$,

$$f(x) = \begin{cases} \dfrac{x-0.2}{1+(x-0.2)^2} - \dfrac{2x^2}{(1+x^2)^2} + \dfrac{1}{1+x^2} + \dfrac{x}{1+x^2} \\ \qquad -\dfrac{1}{2}\log\dfrac{1+x^2}{x^2-0.4t+1.04}, & x \geq 0.2, \\ \dfrac{0.8+2x-1.4x^2+3x^3-0.2x^4+x^5}{(1+x^2)^2} + \dfrac{1}{2}\log(1+x^2), & 0 \leq x < 0.2. \end{cases} \tag{2.4.19}$$

Table 2.9: The error of Example 2.4.1

x	$u(x)$	$u_m(x)$	absolute error	relative error
0	0	0.00074495	0.00074495	
$\frac{1}{30}$	0.0332963	0.0339363	0.000639958	0.0192201
$\frac{2}{30}$	0.0663717	0.0668982	0.000526531	0.00793306
$\frac{3}{30}$	0.0990099	0.0993934	0.000383521	0.00387356
$\frac{4}{30}$	0.131004	0.131372	0.000367445	0.00280483
$\frac{5}{30}$	0.162162	0.162461	0.000298705	0.00184201
$\frac{6}{30}$	0.192308	0.1925	0.000192516	0.00100108
$\frac{7}{30}$	0.221286	0.221473	0.000187153	0.000845752
$\frac{8}{30}$	0.248963	0.249141	0.000178519	0.000717052
$\frac{9}{30}$	0.275229	0.27537	0.000140538	0.000510622
$\frac{10}{30}$	0.3	0.300137	0.000137166	0.000457221
$\frac{11}{30}$	0.3232137	0.323315	0.000102774	0.000317978
$\frac{12}{30}$	0.369247	0.36812	0.00112662	0.00305114
$\frac{13}{30}$	0.364827	0.364832	5.143383×10^{-6}	0.000014
$\frac{14}{30}$	0.383212	0.38319	0.0000215783	0.0000563091
$\frac{15}{30}$	0.4	0.400023	0.0000234348	0.000058587
$\frac{16}{30}$	0.415225	0.415224	5.15865×10^{-7}	1.24238×10^{-6}
$\frac{17}{30}$	0.428932	0.428903	0.0000289846	0.0000675739
$\frac{18}{30}$	0.441176	0.441133	0.0000438552	0.000099405
$\frac{19}{30}$	0.452022	0.451976	0.0000465446	0.00010297
$\frac{20}{30}$	0.461538	0.461498	0.0000402762	0.0000872651
$\frac{21}{30}$	0.469799	0.469751	0.0000472465	0.000100567
$\frac{22}{30}$	0.476879	0.476826	0.0000523272	0.000109729
$\frac{23}{30}$	0.482855	0.482832	0.0000230806	0.0000478002
$\frac{24}{30}$	0.487805	0.487791	0.0000135895	0.0000278585
$\frac{25}{30}$	0.491803	0.491777	0.0000260489	0.000052966
$\frac{26}{30}$	0.494924	0.494889	0.000034368	0.000069441
$\frac{27}{30}$	0.497238	0.497236	1.9061×10^{-6}	3.8333×10^{-6}
$\frac{28}{30}$	0.498812	0.498909	0.0000965414	0.000193542
$\frac{29}{30}$	0.499713	0.49986	0.000146988	0.000294144
$\frac{30}{30}$	0.5	0.500144	0.000144167	0.000288334

The true solution of Eq. (2.4.19) is $u(x) = \frac{x}{1+x^2}$, $x \in [0,1]$, Applying (2.4.18) ($m = 200$) and Mathematica 4.2, we calculate the approximate solution $u_m(x)$. The numerical results are given in Table 2.7.

Example 2.4.2. Let $\tau(x) = \frac{5x+1}{10}$, $x \in [0,\infty)$; $h(x) = \frac{5x-x^2}{5}$, $x \in [-\tau(x), 0)$; $a(x) = b(x) = 1$, $c(t,x) = e^{x-t}$,

$$f(x) = \begin{cases} e^{-\frac{7}{10}x}\left(e^{\frac{3}{5}x}(-0.10202 + 0.510101x) - 0.904837x + e^{\frac{x}{2}}(1 + 1.8x)\right), \\ \qquad\qquad x \geq \tau(x), \\ e^{-\frac{6}{5}x}(e^x(2 + 0.8x) - 1) - 0.05(x - 10.2)(x - 0.2), \\ \qquad\qquad 0 \leq x < \tau(x). \end{cases}$$

The true solution of Eq. is $u(x) = xe^{-\frac{x}{5}}$, $x \in [0,1]$. Applying (2.4.18) ($m = 200$)

Table 2.10: The error of Example 4.2.2

x	$u(x)$	$u_m(x)$	absolute error	relative error
0	0	-0.000157968	0.000157968	
$\frac{1}{30}$	0.0331119	0.033048	0.0000638077	0.00192703
$\frac{2}{30}$	0.0657837	0.0657009	0.0000827993	0.00125866
$\frac{3}{30}$	0.0980199	0.0977678	0.000252111	0.00257204
$\frac{4}{30}$	0.129825	0.129263	0.000562132	0.00432993
$\frac{5}{30}$	0.161203	0.160157	0.00104558	0.00648611
$\frac{6}{30}$	0.192158	0.190651	0.0015065	0.0078399
$\frac{7}{30}$	0.222695	0.220752	0.00194306	0.00872522
$\frac{8}{30}$	0.252817	0.250483	0.00233421	0.00923279
$\frac{9}{30}$	0.282529	0.279852	0.00267775	0.00947776
$\frac{10}{30}$	0.311836	0.308865	0.00297045	0.00952569
$\frac{11}{30}$	0.34074	0.3375	0.00324014	0.00950912
$\frac{12}{30}$	0.369247	0.365741	0.00350507	0.00949248
$\frac{13}{30}$	0.397359	0.393613	0.00374645	0.00942837
$\frac{14}{30}$	0.425082	0.421163	0.0039185	0.00921822
$\frac{15}{30}$	0.452419	0.448398	0.00402107	0.00888795
$\frac{16}{30}$	0.479373	0.475319	0.00405412	0.00845713
$\frac{17}{30}$	0.50595	0.501875	0.00407517	0.0080545
$\frac{18}{30}$	0.532152	0.52815	0.0040025	0.00752134
$\frac{19}{30}$	0.557984	0.5541	0.00388445	0.00696158
$\frac{20}{30}$	0.583449	0.579645	0.00380393	0.00651974
$\frac{21}{30}$	0.608551	0.604938	0.00361243	0.00593611
$\frac{22}{30}$	0.633293	0.629855	0.00343788	0.00542857
$\frac{23}{30}$	0.65768	0.654394	0.00328647	0.00499706
$\frac{24}{30}$	0.681715	0.678651	0.00306397	0.0044945
$\frac{25}{30}$	0.705401	0.702542	0.00285902	0.00405304
$\frac{26}{30}$	0.728743	0.726101	0.00264165	0.00362494
$\frac{27}{30}$	0.751743	0.749279	0.00246432	0.00327814
$\frac{28}{30}$	0.774406	0.772081	0.00232423	0.00300131
$\frac{29}{30}$	0.796734	0.794506	0.00222803	0.00279646
$\frac{30}{30}$	0.818731	0.816633	0.00209816	0.00256269

and Mathematica 4.2, we calculate the approximate solution $u_m(x)$. The numerical results are given in Table 2.8.

Chapter 3

Some Algebras Problems

Many mathematics and physics problems will be equivalently converted into the infinite algebraic problems. However, there are few research papers on infinite system of algebraic equation.

This chapter will introduce a new solution method for the infinite system of linear equation and infinite system of quadratic equation.

3.1 Solving Infinite System of Linear Equations

3.1.1 Introduction

In 1962, the convergency of iteration method for regular infinite system of equations

$$X - \lambda AX = b$$

was discussed by Kontorovich in [81], where $A = (a_{ij})_{i,j=1}^{\infty}$ is an infinite matrix satisfying condition

$$\sum_{i,j=1}^{\infty} a_{ij}^2 < \infty \tag{3.1.1}$$

and $\mathbf{x} = (x_1, x_2, \dots)^\top \in \ell^2$, $\mathbf{b} = (b_1, b_2, \dots)^\top \in \ell^2$, and

$$1 - \sum_{i,j=1}^{\infty} |a_{ij}| \geq 0 \ \text{ (regular condition).}$$

In [82], the author demonstrated the convergency of the intercept equation

$$\mathbf{X}_n - \lambda A_n \mathbf{X}_n = b_n$$

when condition (3.1.1) holds, where $A_n = (a_{ij})_{i,j=1}^{n}$, $\mathbf{X}_n = (x_{n1}, x_{n2}, \dots, x_{nn})^\top$, and $\mathbf{b}_n = (b_{n1}, b_{n2}, \dots, b_{nn})^\top$. Ever since then, one has seldom seen any discussions

about the infinite system of linear equations in correlative literatures for a long time. Up until the year 2002, we found the paper [83], but it merely discussed some special issues on structured and infinite systems of linear equations.

This section will present a representation of the exact solution for the infinite system of linear equations

$$A\mathbf{y} = \mathbf{b}. \tag{3.1.2}$$

The reproducing kernel space $W_2^1[0,1]$ is defined as (1.3.2), its reproducing kernel $R_y(x)$ is the form (1.4.1).

Let

$$\Omega = \left\{\frac{1}{2}, \frac{1}{2^2}, \frac{3}{2^2}, \frac{1}{2^3}, \frac{3}{2^3}, \frac{5}{2^3}, \frac{7}{2^3}, \frac{1}{2^4}, \frac{3}{2^4}, \ldots\right\} = \{p_1, p_2, p_3, \ldots\} \tag{3.1.3}$$

obviously Ω is dense in $[0,1]$.

Put

$$R_{p_i}(x) = \varphi_i(x) \in W_2^1[0,1].$$

It is easy to prove $\{\varphi_i(x)\}_{i=1}^{\infty}$ is a complete basis for $W_2^1[0,1]$.

Orthonormalize $\{\varphi_i(x)\}_{i=1}^{\infty}$ by well-known Gram–Schmidt orthonormalization process. We obtain

$$\begin{aligned} \overline{\varphi}_i &= \sum_{k=1}^{i} \beta_{ik}\varphi_k, \\ \varphi_i &= \sum_{k=1}^{i} \alpha_{ik}\overline{\varphi}_k, \end{aligned}$$

where β_{ik} are the orthogonal coefficients, and $\{\overline{\varphi}_i(x)\}_{i=1}^{\infty}$ are normal orthogonal basis. Then for every $u(x) \in W_2^1[0,1]$, we have

$$u(x) = \sum_{i=1}^{\infty} \langle u(x), \overline{\varphi}_i(x)\rangle_{W_2^1} \overline{\varphi}_i(x).$$

3.1.2 A Norm-Preserving Operator ρ from ℓ^2 onto $W_2^1[0,1]$

Definition 3.1.1. Operator $\rho : \ell^2 \longrightarrow W_2^1[0,1]$, defined as

$$\rho\mathbf{y} = \sum_{i=1}^{\infty} y_i \overline{\varphi}_i(x) \tag{3.1.4}$$

for every $\mathbf{y} = (y_1, y_2, \ldots) \in \ell^2$.

Theorem 3.1.1. *Operator $\rho : \ell^2 \longrightarrow W_2^1[0,1]$ has the following properties:*

(1) *ρ is a one-to-one mapping.*

(2) *ρ is a norm-preserving operator.*

(3) *ρ is a bounded linear operator.*

(4) *$\rho^* = \rho^{-1}$, where ρ^* is the conjugate operator of ρ.*

Proof. (1) For any $\mathbf{y} \in \ell^2$, there corresponds $\rho\mathbf{y} = \sum\limits_{i=1}^{\infty} y_i\overline{\varphi}_i(x) \in W_2^1[0,1]$. If there exist $\mathbf{y}, \mathbf{z} \in \ell^2$ such that $\rho\mathbf{y} = \rho\mathbf{z}$, namely $\sum\limits_{i=1}^{\infty}(y_i - z_i)\overline{\varphi}_i(x) = 0$, since $\{\overline{\varphi}_i(x)\}_{i=1}^{\infty}$ is complete, then $y_i = z_i$, $i = 1, 2, \ldots$. Hence ρ is a monomorphism. For any $u \in W_2^1[0,1]$, since $\{\overline{\varphi}_i(x)\}_{i=1}^{\infty}$ is normal orthogonal basis for $W_2^1[0,1]$, we have $u(x) = \sum\limits_{i=1}^{\infty} u_i\overline{\varphi}_i(x)$ and $(u_1, u_2, \ldots) \in \ell^2$.

(2)
$$\|\rho\mathbf{y}\|_{W_2^1}^2 = \left\langle \sum_{i=1}^{\infty} y_i\overline{\varphi}_i(x), \sum_{i=1}^{\infty} y_i\overline{\varphi}_i(x) \right\rangle_{W_2^1} = \sum_{i=1}^{\infty} y_i^2 = \|\mathbf{y}\|_{W_2^1}^2.$$

(3) ρ is norm-preserving, thus is bounded. For any α, $\beta \in \mathbb{R}$, and any $\mathbf{y}, \mathbf{z} \in \ell^2$, since

$$\begin{aligned} \rho(\alpha\mathbf{y} + \beta\mathbf{z}) &= \sum_{i=1}^{\infty}(\alpha y_i + \beta z_i)\overline{\varphi}_i(x) \\ &= \alpha\sum_{i=1}^{\infty} y_i\overline{\varphi}_i(x) + \beta\sum_{i=1}^{\infty} z_i\overline{\varphi}_i(x) = \alpha\rho(\mathbf{y}) + \beta\rho(\mathbf{z}), \end{aligned}$$

then ρ is linear.

(4) Bijective property of ρ guarantees that the inverse mapping $\rho^{-1} : W_2^1[0,1] \longrightarrow \ell^2$ exists. For every $\mathbf{y} \in \ell^2$ and $v(x) = \rho\mathbf{z} \in W_2^1[0,1]$, we have

$$\langle \rho\mathbf{y}, v\rangle_{W_2^1} = \left\langle \sum_{i=1}^{\infty} y_i\overline{\varphi}_i(x), \sum_{i=1}^{\infty} z_i\overline{\varphi}_i(x) \right\rangle_{W_2^1} = \sum_{i=1}^{\infty} y_i z_i = \langle \mathbf{y}, \mathbf{z}\rangle_{\ell^2} = \langle \mathbf{y}, \rho^{-1}v\rangle_{\ell^2},$$

therefore $\rho^* = \rho^{-1}$. □

3.1.3 Transform Infinite System of Linear Equation $A\mathbf{y} = \mathbf{b}$ into Operator Equation $\mathbb{K}u = f$ on $W_2^1[0,1]$

$$\begin{cases} a_{11}y_1 + a_{12}y_2 + \cdots = b_1, \\ a_{21}y_1 + a_{22}y_2 + \cdots = b_2, \\ \vdots \end{cases}$$

is customary to write

$$A\mathbf{y} = \mathbf{b}, \tag{3.1.5}$$

where $A = (a_{ij})_{i,j=1}^{\infty}$ is an infinite matrix, $\mathbf{y} = (y_1, y_2, \ldots) \in \ell^2$ is an unknown vector, $\mathbf{b} = (b_1, b_2, \ldots) \in \ell^2$ is a known vector. Then Eq. (3.1.5) can be viewed as an operator equation on ℓ^2. Because of (3.1.4), $\mathbb{K} = \rho A \rho^{-1}$ is an operator from the space $W_2^1[0,1]$ onto $W_2^1[0,1]$. It follows that to every operator equation $A\mathbf{y} = \mathbf{b}$ on ℓ^2, there corresponds a unique operator equation

$$\mathbb{K}u = f \tag{3.1.6}$$

on $W_2^1[0,1]$, where $u = \rho\mathbf{y}$, $f(x) = \rho\mathbf{b}$.

Put

$$\psi_i = \mathbb{K}^*\varphi_i = \rho A^{\top}\rho^{-1}\varphi_i,$$

where K^* is the conjugate operator of $\mathbb{K}$. Now orthonormalize $\{\psi_i\}_{i=1}^{\infty}$ by Gram–Schmidt process, write

$$\begin{aligned} \overline{\psi}_i &= \sum_{k=1}^{i} \beta_{ik}^* \psi_k, \\ \psi_i &= \sum_{k=1}^{i} \alpha_{ik}^* \overline{\psi}_k, \end{aligned} \tag{3.1.7}$$

where β_{ik}^* are orthogonal coefficients.

3.1.4 Representation of the Solution for Infinite System of Linear Equations $A\mathbf{y} = \mathbf{b}$

Theorem 3.1.2. *Let $Ay = b$ be an operator equation on ℓ^2 (an infinite system of linear equations) given by* (3.1.5). *Let $\mathbb{K}u = f$ be the corresponding operator equation on $W_2^1[0,1]$ given by* (3.1.6). *$\widetilde{y}$ is a solution of equation $Ay = b$ if and only if $\rho\widetilde{y}$ is a solution of equation $\mathbb{K}u = f$.*

Proof. The conclusion of Theorem 3.1.2 follows from the definition of mapping ρ. □

Definition 3.1.2. Let

$$\Psi = \left\{ u(x) |\ u(x) = \sum_{i=1}^{\infty} c_i \overline{\psi}_i,\ (c_i)_{i=1}^{\infty} \in \ell^2 \right\},$$

where $\overline{\psi}_i$ are given by (3.1.7), then projective operator $\mathbb{P}_\Psi : W_2^1[0,1] \longrightarrow \Psi$ is defined by

$$\mathbb{P}_\Psi u = \sum_{i=1}^{\infty} \langle u, \overline{\psi}_i \rangle_{W_2^1} \overline{\psi}_i, \quad u \in W_2^1[0,1].$$

Theorem 3.1.3. *Let $\mathbb{K}u = f$ be an operator equation on $W_2^1[0,1]$ given by* (3.1.6)*, then u is a solution if and only if $\mathbb{P}_\Psi u$ is a solution.*

Proof. Since any function in $W_2^1[0,1]$ is continuous, and $\mathbb{K}$ is an operator from $W_2^1[0,1]$ onto $W_2^1[0,1]$, we only need to show the following equalities hold for the dense set Ω given by (3.1.3),

$$\begin{aligned}
\mathbb{K}u(p_i) &= \big\langle \mathbb{K}u(y), \varphi_i(y)\big\rangle_{W_2^1} = \big\langle u(y), \mathbb{K}^*\varphi_i(y)\big\rangle_{W_2^1} = \big\langle u(y), \psi_i\big\rangle_{W_2^1} \\
&= \left\langle u(y), \sum_{k=1}^{i} \alpha_{ik}^* \overline{\psi}_k \right\rangle_{W_2^1} = \left\langle u(y), \mathbb{P}_\Psi \left(\sum_{k=1}^{i} \alpha_{ik}^* \overline{\psi}_k \right) \right\rangle >_{W_2^1} \\
&= \left\langle \mathbb{P}_\Psi u(y), \sum_{k=1}^{i} \alpha_{ik}^* \overline{\psi}_k \right\rangle_{W_2^1} = \big\langle \mathbb{P}_\Psi u(y), \psi_i\big\rangle_{W_2^1} \\
&= \big\langle \mathbb{P}_\Psi u(y), \mathbb{K}^*\varphi_i(y)\big\rangle_{W_2^1} = \big\langle \mathbb{K}\mathbb{P}_\Psi u(y), \varphi_i(y)\big\rangle_{W_2^1} \\
&= (\mathbb{K}\mathbb{P}_\Psi u)(p_i).
\end{aligned}$$

The proof is complete. □

Theorem 3.1.4. *If $\sum\limits_{i=1}^{\infty}\sum\limits_{k=1}^{i} \beta_{ik}^* f(p_k)\overline{\psi}_i$ is convergent in $W_2^1[0,1]$, write*

$$\widetilde{u} = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}^* f(p_k)\overline{\psi}_i, \tag{3.1.8}$$

then $\widetilde{u}$ is a solution of $\mathbb{K}u = f$.

Proof. Since $\widetilde{u}(x), f(x)$ are continuous in $W_2^1[0,1]$, and (3.1.3) is dense in $W_2^1[0,1]$, we only need to show that for any $p_n \in \Omega$, there holds

$$\mathbb{K}\widetilde{u}(p_n) = f(p_n), \quad n = 1, 2, \ldots . \tag{3.1.9}$$

We have

$$\begin{aligned}
\mathbb{K}\widetilde{u}(p_n) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}^* f(p_k)\mathbb{K}\overline{\psi}_i(p_n) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}^* f(p_k)\langle \mathbb{K}\overline{\psi}_i(x), \varphi_n(x)\rangle_{W_2^1} \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}^* f(p_k)\langle \overline{\psi}_i(x), \mathbb{K}^*\varphi_n(x)\rangle_{W_2^1} \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}^* f(p_k)\langle \overline{\psi}_i(x), \psi_n(x)\rangle_{W_2^1}.
\end{aligned} \tag{3.1.10}$$

Next, we will use mathematical induction to prove (3.1.9).
While $n = 1$,

$$\begin{aligned}
\mathbb{K}\widetilde{u}(p_1) &= \frac{1}{\beta_{11}^*}\beta_{11}^*\mathbb{K}\widetilde{u}(p_1) \\
&= \frac{1}{\beta_{11}^*}\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^* f(p_k)\langle\overline{\psi}_i(x), \beta_{11}^*\psi_1(x)\rangle_{W_2^1} \\
&= \frac{1}{\beta_{11}^*}\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^* f(p_k)\langle\overline{\psi}_i(x), \overline{\psi}_1(x)\rangle_{W_2^1} \\
&= \frac{1}{\beta_{11}^*}\beta_{11}^* f(p_1) = f(p_1).
\end{aligned}$$

Suppose $\mathbb{K}\widetilde{u}(p_n) = f(p_n)$ for $n = 1, 2, \ldots, m$, then applying (3.1.10), we have

$$\begin{aligned}
\beta_{m+1,m+1}^*\mathbb{K}\widetilde{u}(p_{m+1}) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^* f(p_k)\langle\overline{\psi}_i(x), \beta_{m+1,m+1}^*\psi_{m+1}\rangle_{W_2^1}, \\
\beta_{m+1,l}^*\mathbb{K}\widetilde{u}(p_l) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^* f(p_k)\langle\overline{\psi}_i(x), \beta_{m+1,l}^*\psi_l\rangle_{W_2^1}, \quad l = 1, 2, \ldots, m.
\end{aligned}$$

Summing both sides separately and applying the orthogonality of $\{\overline{\psi}_i\}_{i=1}^{\infty}$, we obtain

$$\begin{aligned}
left\ \ side &= \beta_{m+1,m+1}^*\mathbb{K}\widetilde{u}(p_{m+1}) + \sum_{l=1}^{m}\beta_{m+1,l}^* f(p_l), \\
right\ \ side &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^* f(p_k)\langle\overline{\psi}_i(x), \overline{\psi}_{m+1}(x)\rangle_{W_2^1} \\
&= \beta_{m+1,m+1}^* f(p_{m+1}) + \sum_{l=1}^{m}\beta_{m+1,l}^* f(p_l).
\end{aligned}$$

Since left side is equal to right side, and $\beta_{m+1,m+1}^* \neq 0$, we yield $\mathbb{K}\widetilde{u}(p_{m+1}) = f(p_{m+1})$. This completes the proof. □

Theorem 3.1.5. *Operator equation $\mathbb{K}u = f$ has solutions, then* (3.1.8) *is convergent in $W_2^1[0, 1]$, and it is a solution of $\mathbb{K}u = f$.*

Proof. Assume that u is a solution of $\mathbb{K}u = f$, because of Theorem 3.1.3, $\mathbb{P}_\Psi u$ is a solution.

$$\mathbb{P}_\Psi u = \sum_{i=1}^{\infty}\langle u, \overline{\psi}_i\rangle_{W_2^1}\overline{\psi}_i = \sum_{i=1}^{\infty}\left\langle u, \sum_{k=1}^{i}\beta_{ik}^*\psi_k\right\rangle_{W_2^1}\overline{\psi}_i$$

$$
\begin{aligned}
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}\langle u,\psi_k\rangle_{W_2^1}\overline{\psi}_i = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}\langle u,\mathbb{K}^{*}\varphi_k\rangle_{W_2^1}\overline{\psi}_i \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}\langle \mathbb{K}u,\varphi_k\rangle_{W_2^1}\overline{\psi}_i = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}\langle f,\varphi_k\rangle_{W_2^1}\overline{\psi}_i \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\overline{\psi}_i,
\end{aligned}
$$

thus (3.1.8) is convergent in $W_2^1[0,1]$. By Theorem 3.1.4, (3.1.8) is a solution of $\mathbb{K}u = f$. □

Theorem 3.1.6. *Operator equation $\mathbb{K}u = f$ has solutions if and only if*

$$\sum_{i=1}^{\infty}\left[\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\right]^2 < \infty.$$

Proof. Suppose u is a solution of $\mathbb{K}u = f$, by Theorem 3.1.3,

$$\mathbb{P}_{\Psi}u = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\overline{\psi}_i$$

is a solution too. Since $\{\overline{\psi_i}\}_{i=1}^{\infty}$ is a normal orthogonal basis for $W_2^1[0,1]$, the square sum of the Fourier coefficients of $\mathbb{P}_{\Psi}u$ is convergent, namely

$$\sum_{i=1}^{\infty}\left[\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\right]^2 < \infty.$$

Inversely, if

$$\sum_{i=1}^{\infty}\left[\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\right]^2 < \infty,$$

applying Theorem 3.1.4, the function denoted by (3.1.8) is a solution of $\mathbb{K}u = f$, then $\mathbb{K}u = f$ has one solution at least. □

Theorem 3.1.7. *If operator equation $\mathbb{K}u = f$ has solutions, then* (3.1.8) *is its minimum norm solution.*

Proof. Let $u \in W_2^1[0,1]$ is one solution of $\mathbb{K}u = f$, and $\widetilde{u}$ is another solution given by (3.1.8), $\|u\|_{W_2^1} \geq \|\mathbb{P}_{\Psi}u\|_{W_2^1} = \|\widetilde{u}\|_{W_2^1}$, thus we finish the proof. □

Corollary 3.1.1. *If $\mathbb{K}u = f$ has a unique solution, then* (3.1.8) *is its unique solution.*

Since by the application of operator ρ, the correspondence of infinite system of equations $Ay = b$ and operator equation $\mathbb{K}u = f$ is one-to-one. It follows the corresponding theorems:

Theorem 3.1.8. *For infinite system of linear equations $A\mathbf{y} = \mathbf{b}$, $\mathbf{y} \in \ell^2$, $\mathbf{b} \in \ell^2$, if* (3.1.8) *is convergent, then*

$$\mathbf{y} = \rho^{-1}\widetilde{u} = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\rho^{-1}\overline{\psi}_i \tag{3.1.11}$$

is the minimum norm solution of $A\mathbf{y} = \mathbf{b}$, where $f = \rho\mathbf{b}$, and β_{ik}^{*} is given by (3.1.7).

Theorem 3.1.9. *Infinite system of linear equations $A\mathbf{y} = \mathbf{b}$ has solutions, where $\mathbf{b}, \mathbf{y} \in \ell^2$ if and only if*

$$\sum_{i=1}^{\infty}\left[\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\right]^2 < \infty,$$

where $f = \rho\mathbf{b}$, and β_{ik}^{*} is given by (3.1.7).

We will particularly express the right side of (3.1.11) in order that it can be achieved easily by computer in the course of solving Eq. (3.1.2).

$$\begin{aligned}\mathbf{y} &= \rho^{-1}\widetilde{u} = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\rho^{-1}\overline{\psi}_i = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\rho^{-1}\sum_{l=1}^{i}\beta_{il}^{*}\psi_l \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\sum_{l=1}^{i}\beta_{ik}^{*}\beta_{il}^{*}f(p_k)\rho^{-1}K^{*}\varphi_l = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\sum_{l=1}^{i}\beta_{ik}^{*}\beta_{il}^{*}f(p_k)A^{\top}\rho^{-1}\varphi_l.\end{aligned}$$

Since

$$\rho^{-1}\varphi_l = \rho^{-1}\left(\sum_{s=1}^{l}\alpha_{ls}\overline{\varphi}_s\right) = (\alpha_{l1}, \alpha_{l2}, \ldots, \alpha_{ll}, 0, \ldots)^{\top},$$

we yield

$$\mathbf{y} = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\sum_{l=1}^{i}\beta_{ik}^{*}\beta_{il}^{*}f(p_k)A^{\top}(\alpha_{l1}, \alpha_{l2}, \ldots, \alpha_{ll}, 0, \ldots)^{\top}.$$

3.1.5 Recursion Relation

Basing on above theories, we solve the infinite system of linear equations. First, we truncate the series on the right side of Eq. (3.1.11) to n-order

$${}_n\mathbf{y} = \sum_{i=1}^{n}\left[\sum_{k=1}^{i}\beta_{ik}^{*}f(p_k)\right]\rho^{-1}\overline{\psi_i}$$

and calculate the approximate solution of Eq. (3.1.2). By this time, many corresponding data have been calculated by computer, such as:

$$\varphi_1, \varphi_2, \ldots, \varphi_n \quad \overline{\varphi}_1, \overline{\varphi}_2, \ldots, \overline{\varphi}_n, \quad \psi_1, \psi_2, \ldots, \psi_n \quad \overline{\psi}_1, \overline{\psi}_2, \ldots, \overline{\psi}_n .$$

Observing the representation (3.1.11), we found it is the key to calculate the orthogonal coefficients β_{ij}^* of ψ_i.

Let n-dimensional vector $\lambda_i = (\lambda_{i1}, \lambda_{i2}, \ldots, \lambda_{in}) = \rho^{-1}\psi_i$ $(1 \le i \le n)$, since ρ is norm-preserving, so that the orthogonal coefficients of λ_i are that of ψ_i.

Go step further, when solving

$$_{n+1}\mathbf{y} = \sum_{i=1}^{n+1} \left[\sum_{k=1}^{i} \beta_{ik}^* f(p_k) \right] \rho^{-1}\overline{\psi_i}, \tag{3.1.12}$$

we can utilize efficiently calculated data and only need to compute:

(1) $\overline{\varphi}_{n+1}(x)$, $\varphi_{n+1}(x) = R_{p_{n+1}}(x)$.

(2) Express φ_{n+1} by $\overline{\varphi}_1, \overline{\varphi}_2, \ldots, \overline{\varphi}_{n+1}$, that is $\varphi_{n+1} = \sum\limits_{j=1}^{n+1} \alpha_{n+1,j}\overline{\varphi}_j$, where

$$\alpha_{n+1,j} = \begin{cases} (\varphi_{n+1}, \overline{\varphi}_j), & 1 \le j \le n, \\ \left(\|\varphi_{n+1}\|^2 - \sum\limits_{k=1}^{n} (\varphi_{n+1}, \overline{\varphi}_k)^2 \right)^{1/2}, & j = n+1. \end{cases}$$

Observe that α_{kj} $(1 \le k \le n,\ 1 \le j \le k)$ are known data which have been calculated by computer.

(3) $$\begin{cases} \lambda_i^* = \left(\lambda_{i1}, \lambda_{i2}, \ldots, \lambda_{in}, \sum\limits_{k=1}^{i} a_{k,n+1}\alpha_{ik} \right), & 1 \le i \le n, \\ \lambda_{n+1}^* = \left(\sum\limits_{k=1}^{n+1} a_{k1}\alpha_{n+1k}, \sum\limits_{k=1}^{n+1} a_{k2}\alpha_{n+1k}, \ldots, \sum\limits_{k=1}^{n+1} a_{kn+1}\alpha_{n+1k} \right). \end{cases}$$

(4) Orthonormalizing $\lambda_1^*, \lambda_2^*, \ldots, \lambda_{n+1}^*$, we obtain renewed β_{ij}^*, $1 \le i \le n+1$, $1 \le j \le i$. Finally, insert them into (3.1.12).

In fact, we establish a recursion relation from $_n\mathbf{y}$ to $_{n+1}\mathbf{y}$, where $_n\mathbf{y}$ and $_{n+1}\mathbf{y}$ are the representations of the approximate solution. Because of this recursion relation, we can decrease the computational quantity and improve the calculating rate.

3.1.6 Numerical Experiments

Example 3.1.1. Consider infinite system of linear equations $A\mathbf{y} = \mathbf{b}$

$$\begin{cases} y_1 + \frac{1}{2}y_2 + \frac{1}{3}y_3 + \cdots + \frac{1}{k}y_k + \cdots = \sum_{k=1}^{\infty} \frac{1}{k^2}, \\ \frac{1}{2}y_1 + \frac{1}{3}y_2 + \frac{1}{4}y_3 + \cdots + \frac{1}{k+1}y_k + \cdots = \sum_{k=1}^{\infty} \frac{1}{(k+1)k}, \\ \frac{1}{3}y_1 + \frac{1}{4}y_2 + \frac{1}{5}y_3 + \cdots + \frac{1}{k+2}y_k + \cdots = \sum_{k=1}^{\infty} \frac{1}{(k+2)k}, \\ \vdots \\ \frac{1}{i}y_1 + \frac{1}{i+1}y_2 + \frac{1}{i+2}y_3 + \cdots + \frac{1}{k+i-1}y_k + \cdots = \sum_{k=1}^{\infty} \frac{1}{(k+i-1)k}, \\ \vdots \end{cases}$$

where A is the Hilbert matrix

$$A^{\top} = \begin{pmatrix} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \cdots \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}^{\top}$$

and

$$\mathbf{b} = (1.64493, 1, 0.75, 0.611111, \ldots)^{\top}.$$

(3.1.11) is the representation of exact solution for Eq. (3.1.2). Truncate the series of the right side of it into

$${}_{40}\mathbf{y} = \sum_{i=1}^{40} \left[\sum_{k=1}^{i} \beta_{ik}^{*} f(p_k) \right] \rho^{-1} \overline{\psi_i}.$$

And calculate the approximate solution ${}_{40}\mathbf{y}$ of y, the results are listed in Table 3.1.

Then we use truncation method (by Mathematic 4.2):

$$A_{40}\mathbf{y} = \mathbf{b}_{40}, \quad A_{40} = \{a_{ij}\}_{i,j=1}^{40}, \quad \mathbf{b}_{40} = (b_1, b_2, \ldots, b_{40})^{\top} \tag{3.1.13}$$

to obtain another approximate solution ${}^{40}\mathbf{y}$ which is exhibited in Table 3.1 also. The calculation shows that Hilbert matrix is highly ill-conditioned, and such equations

can not be solved by truncated Eq. (3.1.13). However, our method can dispose these problems. Even for such highly ill-conditioned system of equations, our computed results are satisfactory.

Table 3.1:Results comparison of Example 3.1.1

	true value	${}^{40}\mathbf{y}$	${}_{40}\mathbf{y}$	relative error for ${}_{40}\mathbf{y}$
1	1	2.047030×10^{13}	1.000000	4.89124×10^{-9}
2	1/2	-1.96554×10^{16}	0.500000	3.96063×10^{-8}
3	1/3	1.955890×10^{19}	0.333333	1.54487×10^{-6}
4	1/4	-8.03550×10^{20}	0.250004	1.40162×10^{-5}
5	1/5	1.019960×10^{23}	0.199999	5.51890×10^{-6}
6	1/6	1.381770×10^{25}	0.166665	9.80502×10^{-6}
7	1/7	1.232920×10^{26}	0.142858	2.66426×10^{-6}
8	1/8	-8.26035×10^{27}	0.125000	1.65623×10^{-6}
9	1/9	7.110850×10^{29}	0.111112	1.10451×10^{-5}
10	1/10	-7.61771×10^{30}	0.100001	9.57183×10^{-6}
11	1/11	7.998140×10^{31}	0.0909099	8.46755×10^{-6}
12	1/12	-1.30578×10^{32}	0.0833330	4.11262×10^{-6}
13	1/13	1.343850×10^{31}	0.0769221	1.20842×10^{-5}
14	1/14	-1.04097×10^{34}	0.0714275	1.45617×10^{-5}
15	1/15	3.257810×10^{35}	0.0666658	1.24155×10^{-5}
16	1/16	1.105250×10^{36}	0.0624995	7.77679×10^{-6}
17	1/17	-4.27379×10^{36}	0.0588225	1.67575×10^{-5}
18	1/18	-8.58541×10^{36}	0.0555547	1.62441×10^{-5}
19	1/19	-1.34486×10^{38}	0.0526309	1.34229×10^{-5}
20	1/20	-1.86373×10^{39}	0.0499993	1.38695×10^{-5}
21	1/21	-2.35635×10^{39}	0.0476188	5.42959×10^{-6}
22	1/22	5.155600×10^{39}	0.0454543	5.51872×10^{-6}
23	1/23	8.893650×10^{39}	0.0434785	5.93369×10^{-6}
24	1/24	9.225140×10^{39}	0.0416671	9.48900×10^{-6}
25	1/25	2.990560×10^{40}	0.0400008	1.97706×10^{-5}
26	1/26	-1.23654×10^{41}	0.0384616	1.15103×10^{-6}
27	1/27	-2.63304×10^{38}	0.0370377	1.82324×10^{-5}
28	1/28	-2.85190×10^{40}	0.0357150	1.97581×10^{-5}
29	1/29	-1.50677×10^{40}	0.0344839	3.35827×10^{-5}
30	1/30	-6.48704×10^{40}	0.0333342	2.68295×10^{-5}
31	1/31	2.040300×10^{40}	0.0322592	3.54792×10^{-5}
32	1/32	-2.96692×10^{39}	0.0312508	2.68350×10^{-5}
33	1/33	8.331960×10^{39}	0.0303039	2.72509×10^{-5}
34	1/34	-1.46123×10^{40}	0.0294127	3.29089×10^{-5}
35	1/35	3.301260×10^{39}	0.0285727	4.33587×10^{-5}
36	1/36	-9.92557×10^{38}	0.0277786	3.02898×10^{-5}
37	1/37	3.623730×10^{37}	0.0270275	1.91543×10^{-5}
38	1/38	-3.84128×10^{37}	0.0263164	2.14527×10^{-5}
39	1/39	-1.55332×10^{36}	0.0256417	2.72266×10^{-5}
40	1/40	1.990750×10^{35}	0.0250005	2.02794×10^{-5}

Example 3.1.2. 1. Sampling Theorem:

The Whittaker–Shannon–Kotel'nikov (WSK) sampling theorem states that if $f(t)$ is a bandlimited function with bandwidth σ, then it is completely determined by its values at the points $t_n = n\pi/\sigma, n \in Z$, and can be reconstructed

by means of the formula

$$f(t) = \sum_{n\in Z} f(t_n)\frac{\sin\sigma(t-t_n)}{\sigma(t-t_n)},$$

where the series is absolutely and uniformly convergent on compact sets [5]. While $\sigma = 1, t_n = n\pi$, we have

$$f(t) = \sum_{n\in Z} f(n\pi)\frac{\sin(t-n\pi)}{(t-n\pi)}.$$

2. Discussion on solving integral equation

$$u(s) + \int_R h(s,t)u(t)dt = f(s),$$

where

$$\begin{aligned} h(s,t) &= \left[\frac{(1+e^2)-2e\cos s}{\sqrt{\pi/2}(1+s^2)}\cdot\frac{(1+e^2)-2e\cos t}{\sqrt{\pi/2}(1+t^2)}\right]^{1/2}, \\ f(s) &= \left[1+(e^2-1)\sqrt{2\pi}\right]\cdot\left[\frac{(1+e^2)-2e\cos s}{\sqrt{\pi/2}(1+s^2)}\right]^{1/2}. \end{aligned}$$

It can be tested that the true solution of the integral equation is

$$u(s) = \left[\frac{(1+e^2)-2e\cos s}{\sqrt{\pi/2}(1+s^2)}\right]^{1/2}.$$

3. It can be proved that the support of $\widehat{h}(\omega,t), \widehat{f}(\omega)$ is respectively compact, where $\widehat{h}, \widehat{f}$ is separately the Fourier transform of h, f, then $h(s,t)u(t)$ satisfies the condition of Shannon sampling theorem. Then we discrete this integral equation into an infinite system of linear equations $A\mathbf{y} = \mathbf{b}$, its specific form as follows:

$$u(x_j) + \pi g(x_j)\sum_{k\in Z} g(x_k)u(x_k) = f(x_j), \quad j \in Z, \tag{3.1.14}$$

where

$$\begin{aligned} g(x) &= \left[\frac{(1+e^2)-2e\cos x}{\sqrt{\pi/2}(1+x^2)}\right]^{1/2}, \\ x_j &= j\pi, j \in Z. \end{aligned}$$

By the way, if write

$$\begin{cases} a_{jj} = \pi g(x_j)g(x_j) + 1, & j = k, \\ a_{jk} = \pi g(x_j)g(x_k), & j \neq k, \end{cases}$$

then

$$A = (a_{ij})_{i,j=1}^{\infty}.$$

Now solving discreted infinite system of linear equations (3.1.14). First truncate the series on the right side of (3.1.11) into

$${}_{81}\mathbf{y} = \sum_{i=1}^{81} \left[\sum_{k=1}^{i} \beta_{ik}^{*} f(p_k) \right] \rho^{-1} \overline{\psi_i}.$$

Calculate the approximate solution ${}_{81}\mathbf{y}$. Then compared with the true value as Table 3.2.

Table 3.2: Results comparison of Example 3.1.2

	true value	${}_{81}\mathbf{y}$	relative error for ${}_{81}\mathbf{y}$
1	0.012213	0.012218	3.687386×10^{-4}
2	0.027107	0.027117	3.687386×10^{-4}
3	0.012856	0.012861	3.687385×10^{-4}
4	0.028572	0.028583	3.687385×10^{-4}
5	0.013570	0.013575	3.687385×10^{-4}
6	0.030205	0.030216	3.687386×10^{-4}
7	0.014369	0.014374	3.687385×10^{-4}
8	0.032035	0.032047	3.687385×10^{-4}
9	0.015267	0.015272	3.687385×10^{-4}
10	0.034102	0.034114	3.687386×10^{-4}
11	0.016284	0.016290	3.687385×10^{-4}
12	0.036453	0.036467	3.687385×10^{-4}
13	0.017447	0.017454	3.687385×10^{-4}
14	0.039153	0.039168	3.687386×10^{-4}
15	0.018789	0.018796	3.687386×10^{-4}
16	0.042285	0.042301	3.687386×10^{-4}
17	0.020355	0.020362	3.687387×10^{-4}
18	0.045961	0.045978	3.687386×10^{-4}
19	0.022205	0.022213	3.687385×10^{-4}
20	0.050338	0.050356	3.687386×10^{-4}
21	0.024425	0.024434	3.687385×10^{-4}
22	0.055635	0.055655	3.687386×10^{-4}
23	0.027138	0.027148	3.687386×10^{-4}
24	0.062178	0.062201	3.687386×10^{-4}
25	0.030529	0.030540	3.687386×10^{-4}
26	0.070465	0.070491	3.687386×10^{-4}
27	0.034888	0.034901	3.687386×10^{-4}

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	true value	${}_{81}\mathbf{y}$	relative error for ${}_{81}\mathbf{y}$
28	0.081300	0.081330	3.687386×10^{-4}
29	0.040699	0.040714	3.687386×10^{-4}
30	0.096070	0.096105	3.687386×10^{-4}
31	0.048831	0.048849	3.687386×10^{-4}
32	0.117395	0.117438	3.687386×10^{-4}
33	0.061021	0.061044	3.687386×10^{-4}
34	0.150875	0.150930	3.687385×10^{-4}
35	0.081312	0.081342	3.687385×10^{-4}
36	0.211015	0.211093	3.687386×10^{-4}
37	0.121754	0.121799	3.687386×10^{-4}
38	0.350437	0.350566	3.687386×10^{-4}
39	0.241242	0.241331	3.687386×10^{-4}
40	1.007408	1.007780	3.687386×10^{-4}
41	1.534845	1.535411	3.687386×10^{-4}
42	1.007408	1.007780	3.687386×10^{-4}
43	0.241242	0.241331	3.687386×10^{-4}
44	0.350437	0.350566	3.687386×10^{-4}
45	0.121754	0.121799	3.687386×10^{-4}
46	0.211015	0.211093	3.687386×10^{-4}
47	0.081312	0.081342	3.687385×10^{-4}
48	0.150875	0.150930	3.687386×10^{-4}
49	0.061021	0.061044	3.687386×10^{-4}
50	0.117395	0.117438	3.687386×10^{-4}
51	0.048831	0.048849	3.687386×10^{-4}
52	0.096070	0.096105	3.687385×10^{-4}
53	0.040699	0.040714	3.687386×10^{-4}
54	0.081300	0.081330	3.687386×10^{-4}
55	0.034888	0.034901	3.687386×10^{-4}
56	0.070465	0.070491	3.687386×10^{-4}
57	0.030529	0.030540	3.687386×10^{-4}
58	0.062178	0.062201	3.687385×10^{-4}
59	0.027138	0.027148	3.687385×10^{-4}
60	0.055635	0.055655	3.687386×10^{-4}
61	0.024424	0.024434	3.687385×10^{-4}
62	0.050338	0.050356	3.687386×10^{-4}
63	0.022205	0.022213	3.687386×10^{-4}
64	0.045961	0.045978	3.687386×10^{-4}
65	0.020355	0.020362	3.687386×10^{-4}
66	0.042285	0.042301	3.687386×10^{-4}
67	0.018789	0.018796	3.687386×10^{-4}
68	0.039153	0.039168	3.687385×10^{-4}
69	0.017447	0.017454	3.687385×10^{-4}
70	0.036453	0.036467	3.687385×10^{-4}
71	0.016284	0.016290	3.687386×10^{-4}
72	0.034102	0.034114	3.687385×10^{-4}
73	0.015267	0.015272	3.687386×10^{-4}
74	0.032035	0.032047	3.687385×10^{-4}
75	0.014369	0.014374	3.687384×10^{-4}

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	true value	${}_{81}\mathbf{y}$	relative error for ${}_{81}\mathbf{y}$
76	0.030205	0.030216	3.687387×10^{-4}
77	0.013570	0.013575	3.687388×10^{-4}
78	0.028572	0.028583	3.687387×10^{-4}
79	0.012856	0.012861	3.687384×10^{-4}
80	0.027107	0.027117	3.687385×10^{-4}
81	0.012213	0.012218	3.687382×10^{-4}

3.2 A Solution of Infinite System of Quadratic Equations

3.2.1 Introduction

Current applications from the sciences abound with non-linear problems. To obtain the mathematical solution for these problems, one may often convert them into a related system of non-linear algebraic equations, typically in an infinite number of unknowns; and of these system of quadratic equations are the most basic ones. We begin with a review of some examples from the theory of integral equations, difference equations, and partial differential equations. For example,

$$\begin{cases} u''(x) + a(x)[u'(x)]^2 + b(x)[u(x)]^2 = f(x), \\ u(0) = u_0, \quad u(1) = u_1, \end{cases} \tag{3.2.1}$$

whose nonlinear term is quadratic form. Using difference method, (3.2.1) can be written as follows

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h_i^2} + a_i \left(\frac{u_{i+1} - u_i}{h_i} \right)^2 + b_i u_i^2 = f_i, \quad i = 1, 2, \ldots, n. \tag{3.2.2}$$

which is a system of quadratic equations with respect to unknown u_i. Another example,

$$u(x) + \int_0^1 k(x,t)[u(t)]^2 dt = f(x), \tag{3.2.3}$$

using projection interpolation method, (3.2.3) can be transformed into the following form

$$u_j = \int_0^1 k(x_j, t) \left[\sum_{i=1}^{n} u_i e_i(t) \right]^2 dt = f_j, \quad j = 1, 2, \ldots, n, \tag{3.2.4}$$

which is also a system of quadratic equations with respect u_j, where $e_i(t)$ $(i = 1, 2, \ldots, n)$ are the interpolation basis functions. If $n \longrightarrow \infty$, then (3.2.2) and (3.2.4) will be an infinite system of quadratic equations. Therefore, it is important that

solving a system of quadratic equations in the research of non-linear problems. Few papers focus on solving algebraic equations [85], [86]. In this section, an algorithm is proposed for solving infinite system of quadratic equations

$$\begin{cases} \sum\limits_{i=1}^{\infty} a_{ii,k}x_i^2 + 2\sum\limits_{i<j}^{\infty} a_{ij,k}x_ix_j = b_k, \quad k = 1, 2, \ldots, \\ x_1 = x_0, \end{cases} \tag{3.2.5}$$

where x_0 is a constant and $\sum\limits_{ij}^{\infty}(a_{ij,k})^2 < \infty$.

If we introduce symmetric matrix $A_k = (a_{ij,k})_{\infty\times\infty}$, $(k \in \mathbb{N})$, then (3.2.5) can be converted as follows

$$\begin{cases} \langle \mathbf{x}, A_k\mathbf{x}^\top\rangle_{\ell^2} = b_k, \quad k = 1, 2, \ldots, \\ x_1 = x_0, \end{cases} \tag{3.2.6}$$

where $\mathbf{x} = (x_1, x_2, \ldots) \in \ell^2$-space, $(b_1, b_2, \ldots) \in \ell^2$, $\langle \cdot, \cdot \rangle_{\ell^2}$ denotes the inner product in ℓ^2.

3.2.2 Linear Operators in Reproducing Kernel Space

The reproducing kernel space $W_2^1[0, 1]$ is defined as (1.3.2), its reproducing kernel is $R_\eta(t)$ by (1.4.1). The reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$ is defined as (1.5.1), its reproducing kernel is $K_{(\xi,\eta)}(s, t) = R_\xi(s)R_\eta(t)$.

Choose a countable dense subset

$$\mathrm{T} = \{t_1, t_2, \ldots\}$$

of $[0, 1]$, and put

$$\phi_i(t) = R_{t_i}(t), \quad i \in \mathbb{N}.$$

The normal orthogonal basis $\{\overline{\phi}_i(t)\}_{i=1}^{\infty}$ can be derived from Gram–Schmidt orthonormalization process of $\{\phi_i(t)\}_{i=1}^{\infty}$,

$$\overline{\phi}_i(t) = \sum_{l=1}^{i} \alpha_{il}\phi_l(t) \quad (\alpha_{ii} > 0, \quad i = 1, 2, \ldots).$$

Lemma 3.2.1. *If* $\mathrm{T} = \{t_1, t_2, \ldots\}$ *is dense in* $[0, 1]$, *then* $\{\phi_i(t)\}_{i=1}^{\infty}$ *is a complete system in* $W_2^1[0, 1]$.

Proof. Let $u \in W_2^1[0, 1]$ such that $\langle u(\cdot), \phi_i(\cdot)\rangle_{W_2^1} = 0$ for all $i \in \mathbb{N}$. Since $\langle u(\cdot), \phi_i(\cdot)\rangle_{W_2^1} = u(t_i)$, then $u(t_i) = 0$ for all $t_i \in \mathrm{T}$. Note that T is dense in $[0, 1]$, hence $u(t) = 0$. So the proof is complete. □

Now, three operators are defined.

(1) A linear operator $\rho : \ell^2 \longrightarrow W_2^1[0,1]$ is defined by

$$\rho(\mathbf{x}) = \sum_{i=1}^{\infty} x_i \overline{\phi}_i(t). \tag{3.2.7}$$

It is clear that ρ is a one-to-one and norm-preserving mapping. Using ρ, Eq. (3.2.6) can be converted as follows

$$\big\langle u(\cdot\,), (\rho A_k \rho^{-1} u)(\cdot\,) \big\rangle_{W_2^1} = b_k, \quad k = 1, 2, \ldots,$$

where $u(t) = \rho(\mathbf{x})$.

(2) We define an operator $\widetilde{\mathbb{A}}_k : W_2^1[0,1] \longrightarrow W_2^1[0,1]$ by

$$(\widetilde{\mathbb{A}}_k u)(t) = (\rho A_k \rho^{-1} u)(t), \quad k \in \mathbb{N}$$

then we convert (3.2.6) into the following form

$$\langle u(\cdot\,), (\widetilde{\mathbb{A}}_k u)(\cdot\,) \rangle_{W_2^1} = b_k, \quad k = 1, 2, \ldots,$$

and for each $u(t) \in W_2^1[0,1]$, we have

$$\big\langle u(\cdot\,), (\widetilde{\mathbb{A}}_k u)(\cdot\,) \big\rangle_{W_2^1} = u(0)(\widetilde{\mathbb{A}}_k u)(0) + \int_0^1 u'(t)(\widetilde{\mathbb{A}}_k u)'(t)\, dt.$$

(3) Let ∂ and I be the differential operator and identical operator in $W_2^1[0,1]$, respectively. For any $k \in \mathbb{N}$, we define an operator $\mathbb{H}_k : W_2^{(1,1)}(\mathcal{D}) \longrightarrow \mathbb{R}$ by

$$\mathbb{H}_k v = \Big[\widetilde{\mathbb{A}}_k^{(\diamond)} I^{(*)} v(*, \diamond)\Big](0) + \int_0^1 \Big[\partial_* \partial_\diamond \widetilde{\mathbb{A}}_k^{(\diamond)} v(*, \diamond)\Big](t)\, dt, \quad v \in W_2^{(1,1)}(\mathcal{D}), \tag{3.2.8}$$

where "$\diamond$" and "$*$" denote the variables corresponding to a function.

An operator $\mathbb{L} : W_2^{(1,1)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$ is defined by

$$(\mathbb{L}v)(t) = \rho((\mathbb{H}_1 v, \mathbb{H}_2 v, \ldots)) = \sum_{k=1}^{\infty} (\mathbb{H}_k v)\overline{\phi}_k(t), \quad v \in W_2^{(1,1)}(\mathcal{D}), \tag{3.2.9}$$

where $\mathbb{H}_k$ and ρ are given by (3.2.8) and (3.2.7) respectively. Note that, $\widetilde{\mathbb{A}}_k$, ∂ and ρ are bounded linear operators, hence, $\mathbb{L}$ is also a bounded linear operator.

Lemma 3.2.2. *Let $\rho(\mathbf{x}) = u(t)$, $\mathbf{x} = (x_1, x_2, \ldots) \in \ell^2$, then*

$$u(t_1) = x_1 \|\phi_1\|_{W_2^1}.$$

Proof. Since $\{\overline{\phi}_i(t)\}_{i=1}^{\infty}$ is the normal orthogonal basis in $W_2^1[0,1]$, then we have, by mean of (1.1.1) and (3.2.7)

$$\begin{aligned} u(t_1) &= \big\langle u(\cdot\,), \phi_1(\cdot\,)\big\rangle_{W_2^1} = \|\phi_1\|\big\langle u(\cdot\,), \overline{\phi}_1(\cdot\,)\big\rangle_{W_2^1} \\ &= \|\phi_1\|_{W_2^1}\left\langle \sum_{i=1}^{\infty} x_i\overline{\phi}_i(\cdot\,), \overline{\phi}_1(\cdot\,)\right\rangle_{W_2^1} = x_1\|\phi_1\|_{W_2^1}. \end{aligned}$$

The proof is complete. □

Theorem 3.2.1. *Let $\rho(\mathbf{x}) = u(t)$, $\mathbf{x} = (x_1, x_2, \ldots) \in \ell^2$, then $\mathbf{x}$ is a solution of Eq. (3.2.6) if and only if $u(s)u(t) \in W_2^{(1,1)}(\mathcal{D})$ is a solution of*

$$(\mathbb{L}v)(t) = f(t), \tag{3.2.10}$$

and $u(t_1) = x_0\|\phi_1\|_{W_2^1}$, where $f(t) = \sum\limits_{k=1}^{\infty} b_k\overline{\phi}_k(t)$.

Proof. If $\mathbf{x}$ is a solution of Eq. (3.2.6), in view of (3.2.9), (3.2.8) and (3.2.6) then we have

$$\begin{aligned} (\mathbb{L}u(*)u(\cdot\,))(t) &= \rho\big(\mathbb{H}_1u(*)u(\cdot\,), \ldots\big) \\ &= \rho\left(u(0)(\rho A_1\rho^{-1}u)(0) + \int_0^1 \left[u'(t)(\rho A_1\rho^{-1}u)'(t)\right]\,dt, \ldots\right) \\ &= \rho\left(\big\langle u(\cdot\,), (\rho A_1\rho^{-1}u)(\cdot\,)\big\rangle_{W_2^1}, \ldots\right) \\ &= \rho\left(\langle \mathbf{x}, A_1\mathbf{x}^T\rangle_{\ell^2}, \ldots\right) \\ &= \rho(b_1, \ldots) \\ &= f(t). \end{aligned}$$

From Lemma 3.2.2, one gets $u(t_1) = x_0\|\phi_1\|$.

Conversely, if $u(s)u(t)$ is a solution of Eq. (3.2.10), then we get

$$\rho\left(\big\langle u(t), (\rho A_1\rho^{-1}u)(t)\big\rangle_{W_2^1}, \ldots\right) = \rho(b_1, \ldots).$$

Since ρ is one-to-one, we obtain $(\langle \mathbf{x}, A_1\mathbf{x}^{\top}\rangle_{\ell^2}, \ldots) = (b_1, \ldots)$, thus $\mathbf{x}$ is a solution of Eq. (3.2.5).

Note that

$$\begin{aligned} x_0|\phi_1| &= u(t_1) = \sum_{k=1}^{\infty} x_i\overline{\phi}_i(t_1) = \sum_{k=1}^{\infty} x_i\langle\overline{\phi}_i, \phi_1\rangle_{W_2^1} \\ &= \sum_{k=1}^{\infty} x_i\big\langle\overline{\phi}_i, \|\phi_1\|\overline{\phi_1}\big\rangle_{W_2^1} = x_1\|\phi_1\|_{W_2^1}. \end{aligned}$$

Thus $x_1 = x_0$. □

Using the adjoint operator $\mathbb{L}^*$ of $\mathbb{L}$ defined by (3.2.9), we define $\psi_i(s,t)$ in $W(\mathcal{D})$ by

$$\psi_i(s,t) = (\mathbb{L}^*\phi_i)(s,t) = (\mathbb{L}^*R_{t_i})(s,t), \quad i \in \mathbb{N}. \tag{3.2.11}$$

Lemma 3.2.3. *If $\psi_i(s,t)$ is given by* (3.2.3)*, it holds that*

$$\psi_i(s,t) = \sum_{k=1}^{\infty}\left[\sum_{i=1}^{\infty}\sum_{j=1}^{\infty}\alpha_{ij,k}\overline{\phi}_i(s)\overline{\phi}_j(t)\right]\overline{\phi}_k(t_i), \quad i \in \mathbb{N}. \tag{3.2.12}$$

Proof. By virtue of (1.1.1) and (3.2.9), we have

$$\begin{aligned}\psi_i(s,t) &= \left\langle (\mathbb{L}^*\phi_i)(*,\cdot), R_s(*)R_t(\cdot)\right\rangle_{W_2^{(1,1)}} = \left\langle \phi_i(\diamond), (\mathbb{L}R_s(*)R_t(\cdot))(\diamond)\right\rangle_{W_2^1}\\ &= (\mathbb{L}R_s(*)R_t(\cdot))(t_i) = \sum_{k=1}^{\infty}\left\langle R_s(\cdot), (\widetilde{\mathbb{A}}_kR_t)(\cdot)\right\rangle_{W_2^1}\overline{\phi}_k(t_i)\\ &= \sum_{k=1}^{\infty}\left(\rho A_k\rho^{-1}R_t\right)(s)\overline{\phi}_k(t_i). \qquad (*)\end{aligned}$$

On the other hand, since

$$R_t(s) = \sum_{k=1}^{\infty}\left\langle R_t(\cdot), \overline{\phi}_k(\cdot)\right\rangle_{W_2^1}\overline{\phi}_k(s) = \sum_{k=1}^{\infty}\overline{\phi}_k(t)\overline{\phi}_k(s)$$

we have

$$\rho^{-1}R_t(s) = (\overline{\phi}_1(t), \overline{\phi}_2(t), \cdots) \in \ell^2.$$

Hence

$$(\rho A_k\rho^{-1}R_t)(s) = \sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty}\alpha_{ij,k}\overline{\phi}_j(t)\right)\overline{\phi}_i(s). \qquad (**)$$

Substituting $(**)$ into $(*)$, we obtain the desired result. □

Again the normal orthogonal basis $\{\overline{\psi}_i(s,t)\}_{i=1}^{\infty}$ in $W_2^{(1,1)}(\mathcal{D})$ can be derived from Gram–Schmidt orthonormalization process of $\{\psi_i(s,t)\}_{i=1}^{\infty}$,

$$\overline{\psi}_i(s,t) = \sum_{k=1}^{i}\beta_{ik}\psi_k(s,t).$$

Let $\mathrm{S} = \overline{span\{\overline{\psi}_i(s,t)\}_{i=1}^{\infty}}$ and $S^{\perp}$ be the orthogonal complement of S in $W_2^{(1,1)}(\mathcal{D})$, thus, $W_2^{(1,1)}(\mathcal{D}) = \mathrm{S}\oplus\mathrm{S}^{\perp}$. Choose a countable dense subset $\mathrm{B} = \{(s_1,t_1),(s_2,t_2),\ldots\}$ of $\mathcal{D}$. Take

$$\sigma_j(s,t) = R_{s_j}(s)R_{t_j}(t), \quad j \in \mathbb{N},$$

which is a basis of $W_2^{(1,1)}(\mathcal{D})$. Orthonormalizing $\{\overline{\psi}_1, \overline{\psi}_2, \ldots, \sigma_1, \sigma_2, \ldots\}$, we obtain, for each $j \in \mathbb{N}$,

$$\overline{\sigma}_j(s,t) = \sum_{k=1}^{\infty} \beta_{jk}^* \psi_k(s,t) + \sum_{m=1}^{j} \beta_{jm}^* \sigma_m(s,t), \quad j \in \mathbb{N}.$$

Hence, $\{\overline{\psi}_1, \overline{\psi}_2, \ldots, \overline{\sigma_1}, \overline{\sigma_2}, \ldots\}$ is a normal orthogonal basis of $W_2^{(1,1)}(\mathcal{D})$.

3.2.3 Separated Solution of (3.2.10)

By Theorem 3.2.1, we know that obtaining a solution of Eq. (3.2.5) is equivalent to obtaining the separated solution of Eq. (3.2.10).

Theorem 3.2.2. *Let $\lambda = (\lambda_1, \lambda_2, \ldots)$ be an arbitrary constant in ℓ^2, then*

$$v(s,t) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} f(t_k) \overline{\psi}_k(s,t) + \sum_{j=1}^{\infty} \lambda_j \overline{\sigma}_j(s,t) \tag{3.2.13}$$

is a solution of (3.2.10)*, where $t_k \in \mathrm{T}$.*

Proof. Applying $\mathbb{L}$ to both sides of (3.2.13), one gets

$$(\mathbb{L}v)(t) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} f(t_k) (\mathbb{L}\overline{\psi}_i)(t) + \sum_{j=1}^{\infty} \lambda_j (\mathbb{L}\overline{\sigma}_j)(t).$$

Since

$$(\mathbb{L}\overline{\sigma}_j)(t_l) = \langle \mathbb{L}\overline{\sigma}_j, \phi_l \rangle_{W_2^1} = \langle \overline{\sigma}_j, \mathbb{L}^* \phi_l \rangle_{W_2^{(1,1)}} = \langle \overline{\sigma}_j, \psi_l \rangle_{W_2^{(1,1)}} = 0$$

for all $t_l \in \mathrm{T}$, we have $(\mathbb{L}\overline{\sigma}_j)(t) = 0$. Thus

$$\begin{aligned} (\mathbb{L}v)(t_l) &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} f(t_k) (\mathbb{L}\overline{\psi}_i)(t_l) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} f(t_k) \langle \mathbb{L}\overline{\psi}_i, \phi_l \rangle_{W_2^1} \\ &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} f(t_k) \langle \overline{\psi}_i, \psi_l \rangle_{W_2^{(1,1)}}. \end{aligned}$$

Multiplying both sides of the above equality by β_{nl} and summing with respect to l $(1 \le l \le n)$, then

$$\sum_{l=1}^{n} \beta_{nl} (\mathbb{L}v)(t_l) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} f(t_k) \langle \overline{\psi}_i, \overline{\psi}_n \rangle_{W_2^{(1,1)}} = \sum_{k=1}^{n} \beta_{nk} f(t_k).$$

Hence,for all $n \in \mathbb{N}$, we obtain $(\mathbb{L}v)(t_n) = f(t_n)$. Note that T is dense in $[0,1]$, $(\mathbb{L}v)(t) = f(t)$ holds for all $t \in [0,1]$. So the proof is complete. □

If $v(s,t) = u(s)u(t)$ is given by (3.2.13), from Lemma 3.2.2 we get

$$v(t_1,t) = x_1\|\phi_1\|_{W_2^1}u(t), \quad v(t,t) = u^2(t). \tag{3.2.14}$$

Theorem 3.2.3. *If $v(s,t) = u(s)u(t)$ is given by* (3.2.13)*, then*

$$u(t) = \frac{1}{x_1\|\phi_1\|_{W_2^1}}\left[\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(t_k)\overline{\psi}_i(t_1,t) + \sum_{j=1}^{\infty}\lambda_j\overline{\sigma}_j(t_1,t)\right]. \tag{3.2.15}$$

Furthermore $\rho^{-1}(u) = (x_1, x_2, \ldots)$ is the solution of (3.2.5)*.*

Proof. In view of (3.2.13), we have

$$u(s)u(t) = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(t_k)\overline{\psi}_i(s,t) + \sum_{j=1}^{\infty}\lambda_j\overline{\sigma}_j(s,t).$$

Putting $s = t_1$ and dividing both sides of the above equality by $u(t_1)$, (3.2.15) holds from (3.2.14) and Theorem 3.2.1. □

Remark 3.2.1. Considering the numerical computation, we define the approximation to $u(t)$ by $u_{n,m}(t)$,

$$u_{nm}(t) = \frac{1}{x_1\|\phi_1\|_{W_2^1}}\left[\sum_{i=1}^{n}\sum_{k=1}^{i}\beta_{ik}f(t_k)\overline{\psi}_i(t_1,t) + \sum_{j=1}^{m}\lambda_j\overline{\sigma}_j(t_1,t)\right], \tag{3.2.16}$$

and $\rho^{-1}(u_{nm}) = (x_1, x_2, \ldots)$ is the approximation solution of Eq. (3.2.5).

In order to obtain u_{nm} in (3.2.16), we have to determine the values of $\lambda_1, \ldots, \lambda_m$. According to Theorem 3.2.3, if $v(s,t) = u(s)u(t)$ is a solution of (3.2.10), then $\rho^{-1}(u) = (x_1, x_2, \ldots)$ is the solution of Eq. (3.2.5). Thus, $\lambda_1, \lambda_2, \ldots$ should satisfy

$$\|v(t,t) - u^2(t)\|_{W_2^1}^2 = 0.$$

Therefore, we can get u_{nm} by finding $(\lambda_1, \ldots, \lambda_m)$ making

$$G(\lambda_1, \lambda_2, \ldots, \lambda_m) = \left\|v_{nm}(t,t) - u_{nm}^2(t)\right\|_{W_2^1}^2$$

minimal, where $v_{nm}(t,t)$ is a partial sum of (3.2.13), then we can get u_{nm}. Fortunately $G(\lambda_1, \ldots, \lambda_m)$ is a polynomial with respect to $\lambda_1, \ldots, \lambda_m$ whose optimization problem is familiar to us.

Example 3.2.1.

$$\begin{cases}\mathbf{x}^{\top}\cdot A_k\mathbf{x} = b_k, \;\; k = 1, 2, \ldots, \\ x_1 = 1,\end{cases} \tag{3.2.17}$$

where

$$b_1 = \sum_{n,m=1}^{\infty} \frac{1}{nm2^{n+m-2}}, \quad b_2 = \sum_{n,m=1}^{\infty} \frac{1}{nm3^{n+m-2}}, \quad b_3 = \sum_{n,m=1}^{\infty} \frac{1}{nm4^{n+m-2}},$$
$$b_4 = \sum_{n,m=1}^{\infty} \frac{1}{nm5^{n+m-2}}, \quad b_5 = \sum_{n,m=1}^{\infty} \frac{1}{nm6^{n+m-2}}, \quad \ldots,$$

the infinite matrixes are given by

$$A_1 = (a_{ij})_{i,j=1}^{\infty}, \ a_{ij} = \frac{1}{2^{i+j-2}}, \quad A_2 = (a_{ij})_{i,j=1}^{\infty}, \ a_{ij} = \frac{1}{3^{i+j-2}},$$
$$A_3 = (a_{ij})_{i,j=1}^{\infty}, \ a_{ij} = \frac{1}{4^{i+j-2}}, \quad A_4 = (a_{ij})_{i,j=1}^{\infty}, \ a_{ij} = \frac{1}{5^{i+j-2}},$$
$$A_5 = (a_{ij})_{i,j=1}^{\infty}, \ a_{ij} = \frac{1}{6^{i+j-2}}, \ \ldots$$

Using our method, we obtain an approximate solution $\widetilde{\mathbf{x}}$ of Eq. (3.2.17), its true solution is $\mathbf{x}$. The numerical results are given in the following Table 3.3.

Table 3.3: Error of Example 3.2.1

$\mathbf{x}$	$\widetilde{\mathbf{x}}$	error	$\mathbf{x}$	$\widetilde{\mathbf{x}}$	error	$\mathbf{x}$	$\widetilde{\mathbf{x}}$	error
1	1.0008	0.000803595	$\frac{1}{2}$	0.493129	0.00687093	$\frac{1}{3}$	0.37745	0.0441171
$\frac{1}{4}$	0.2298	0.0202002	$\frac{1}{5}$	0.203195	0.00319469	$\vdots$	$\vdots$	$\vdots$

Part II

2

Chapter 4

Integral equations

The research of nonlinear integral equations was started in 1920's to 1930's. People mainly focused on the qualitative research of the theory. As to the numerical solution method, in most cases, the nonlinear integral equations would be converted into nonlinear algebraic equations. The solution of the nonlinear algebraic equations seems to rely on Newton iteration method or its derivative method. The defects of these methods are well known to us.

In this chapter, we will introduce a new solution method for the typical nonlinear integral equation, nonlinear integral-differential equation.

4.1 Solving Fredholm Integral Equations of the First Kind and A Stability Analysis

4.1.1 Introduction

In many fields of science, one is interested in continuous functions u which are not directly accessible by experiment, but are related to an experimentally measurable quantity f according to

$$(\mathbb{A}u)(x) = f(x) + \varepsilon(x), \tag{4.1.1}$$

where $\mathbb{A}$ denotes the operator which maps the function u on the experiment data f and $\varepsilon(x)$ means the measurement errors.

In order to determine u one has to invert Eq. (4.1.1). In many cases $\mathbb{A}$ is linear, i.e. it may correspond to a Fredholm integral equation of the first kind. Some examples of linear inverse problems are the analysis of heat conduction data [91], the analysis of deep level transient spectroscopy(DLTS) data [92], the analysis of nuclear magnetic resonance(NMR) data [93], the analysis of static light scattering(SLS) data [94] and so on. All the cited inverse problems are known to be ill-posed (for a strict definition of ill-posed [95]), since the function u does not depend continuously upon the data f. The measurement errors of the experimental data f will result in unbounded errors of the estimate u. In order to solve ill-posed problems, it is essential

to find the approximate solution of the ill-posed problems, which are called "the regularization method". Tikhonov provides this method in 1960s [96]. The basic idea of regularization is to replace the original equation by a "close" equation involving a small parameter, so that the changed equation can be solved in a stable way and its solution is close to the solution of the original equation when the parameter is small. To construct the regularization operator and choose appropriate regularization parameter, *prior information* [97] is imposed on the solution u. A specific prior information is the smoothness constraint taking into account the phenomenological fact. The smoothness constraint is used by the well-known Tikhonov regularization. There exist the programs to construct the regularization operator and choose appropriate regularization parameter [98], [99]. These programs are restricted to one specific linear regularization term which models the smoothness constraint.

In this section, a new method is proposed in order to solve an ill-posed problem on Fredholm integral equation of the first kind. The representation of the exact solution is obtained and a stability analysis of the solution on Fredholm integral equation of the first kind is discussed in the reproducing kernel space. If the integral equation has solutions, the discussion of the stability of any a solution is transformed into the discussion of the stability of the minimal norm solution. The conclusion is obtained that the stability problem on the first kind of Fredholm integral equation is a well-posed problem in the reproducing kernel space $W_2^1[a,b]$. In fact, the analytic condition of the functions is easy to be satisfied on the reproducing kernel space $W_2^1[a,b]$. The measurement errors of the experimental data can not result in large errors of the exact solution. It will be propitious to know the true solution. The numerical experiments are illustrated in the last part. It is shown the method given is valid.

4.1.2 Representation of Exact Solution for Fredholm Integral Equation of the First Kind

The reproducing kernel space $W_2^1[a,b]$ is given by (1.3.2), its reproducing kernel is

$$R_y(x) = \begin{cases} 1-a+x, & x \le y, \\ 1-a+y, & x > y. \end{cases} \tag{4.1.2}$$

Here, let us consider Fredholm integral equation of the first kind

$$(\mathbb{A}u)(x) \equiv \int_a^b K(x,s)u(s)\,ds = f(x), \quad u,\, f \in W_2^1[a,b], \tag{4.1.3}$$

where $u(x)$ is determined function, $f(x)$ is a given function and the kernel $K(x,s)$ satisfies the condition

$$\iint_{[a,b]\times[a,b]} (k(x,s))^2\,dx\,ds \le M, \quad M \in \mathbb{R}^+. \tag{4.1.4}$$

It is easy to prove $\mathbb{A}$ is a bounded linear operator from $W_2^1[a,b]$ to $W_2^1[a,b]$ under the condition (4.1.4). And let $\mathbb{A}^*$ is the conjugate operator of $\mathbb{A}$. In order to obtain the representation of the exact solution of Eq. (4.1.3), let

$$\varphi_i(x) = R_{x_i}(x), \quad \psi_i(x) = \mathbb{A}^*\varphi_i(x),$$

where $\{x_i\}_{i=1}^{\infty}$ is dense in the interval $[a,b]$.

Define

$$\overline{\psi}_i(x) = \sum_{k=1}^{i} \beta_{ik}\psi_k(x),$$

where β_{ik} are coefficients of Gram–Schmidt orthonomalization.

Moreover, let

$$\Psi = \left\{ u |\ u = \sum_{i=1}^{\infty} c_i\overline{\psi}_i \ \text{ for } \ \{c_i\}_{i=1}^{\infty} \in \ell^2 \right\}$$

and $\mathbb{P}$ be the projection operator from $W_2^1[a,b]$ to Ψ, that is

$$\mathbb{P}u = \sum_{i=1}^{\infty} \langle u, \overline{\psi}_i\rangle_{W_2^1}\overline{\psi}_i,$$

($\mathbb{P} = \mathbb{P}^*$, $\mathbb{P}^2 = \mathbb{P}$). Then a solution of (4.1.3) could be obtained.

Theorem 4.1.1. *Let $\{x_i\}_{i=1}^{\infty}$ be dense in the interval $[a,b]$. If the equation* (4.1.3) *has unique solution and define*

$$u_{min}(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} f(x_k)\overline{\psi}_i(x), \tag{4.1.5}$$

then $u_{min}(x)$ is the solution of (4.1.3).

Proof. Let $u(x) \in W_2^1[a,b]$ be the solution of (4.1.3). From the definition of $\mathbb{P}$, $u(x)$ could be expressed as $\mathbb{P}u(x) = \sum\limits_{i=1}^{\infty} \langle u, \overline{\psi}_i\rangle_{W_2^1}\overline{\psi}_i(x)$. Hence, it has

$$\begin{aligned}
\mathbb{P}u(x) &= \sum_{i=1}^{\infty} \left\langle u, \sum_{k=1}^{i} \beta_{ik}\mathbb{A}^*\varphi_k \right\rangle_{W_2^1} \overline{\psi}_i(x) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle \mathbb{A}u, \varphi_k\rangle_{W_2^1}\overline{\psi}_i(x) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} f(x_k)\overline{\psi}_i(x) = u_{min}(x).
\end{aligned}$$

Subsequently, it need prove the form (4.1.5) is the solution of (4.1.3). Since $\{x_i\}_{i=1}^{\infty}$ is dense in the interval $[a,b]$, it only prove $(\mathbb{A}u_{\min})(x_i) = f(x_i)$.

$$\begin{aligned}(\mathbb{A}u_{\min})(x_i) &= \big\langle(\mathbb{A}u_{\min})(x), \varphi_i(x)\big\rangle_{W_2^1} \\ &= \big\langle u_{\min}(s), \mathbb{A}^*\varphi_i(s)\big\rangle_{W_2^1} = \big\langle(\mathbb{P}u)(s), \psi_i(s)\big\rangle_{W_2^1} \\ &= \big\langle u(s), \mathbb{P}(\psi_i(s))\big\rangle_{W_2^1} = \big\langle u(s), \psi_i(s)\big\rangle_{W_2^1} \\ &= \big\langle(Au)(s), \varphi_i(s)\big\rangle_{W_2^1} = \langle f(s), \varphi_i(s)\rangle_{W_2^1} = f(x_i). \qquad (4.1.6)\end{aligned}$$

The proof is complete. □

Furthermore, the solutions of (4.1.3) are not unique, the representation of all the exact solutions for (4.1.3) are obtained.

Let $\Psi^{\perp} = \{u|\ \langle u, \Psi\rangle_{W_2^1} = 0\}$ and define the null space $\mathrm{N}(\mathbb{A}) = \{u|\ \mathbb{A}u = 0\}$ of $\mathbb{A}$. It is obvious $\mathrm{N}(\mathbb{A}) = \Psi^{\perp}$. Assume the sequence

$$\psi_1, \psi_2, \ldots, \psi_n, \ldots, r_1, r_2, \ldots, r_m, \ldots$$

is a complete system in $W_2^1[a,b]$ and

$$\overline{\psi}_1, \overline{\psi}_2, \ldots, \overline{\psi}_n, \ldots, \overline{r}_1, \overline{r}_2, \ldots, \overline{r}_m, \ldots$$

be the normal orthogonal basis in $W_2^1[a,b]$ obtained from $\{\psi_i\}_{i=1}^{\infty}$ and $\{r_i\}_{i=1}^{\infty}$. Namely,

$$\begin{aligned}\overline{\psi}_1 &= \frac{\psi_1}{\|\psi_1\|}, \ldots, \\ \overline{\psi}_{n+1} &= \frac{\psi_{n+1} - \sum\limits_{i=1}^{\infty}\langle\psi_{n+1}, \overline{\psi}_i\rangle\overline{\psi}_i}{\big\|\psi_{n+1} - \sum\limits_{i=1}^{\infty}\langle\psi_{n+1}, \overline{\psi}_i\rangle\overline{\psi}_i\big\|}, \quad n = 1, 2, \ldots, \\ \overline{r}_1 &= \frac{r_1 - \sum\limits_{i=1}^{\infty}\langle r_1, \overline{\psi}_i\rangle\overline{\psi}_i}{\big\|r_1 - \sum\limits_{i=1}^{\infty}\langle r_1, \overline{\psi}_i\rangle\overline{\psi}_i\big\|}, \ldots, \\ \overline{r}_{m+1} &= \frac{r_{m+1} - \sum\limits_{i=1}^{\infty}\langle r_{m+1}, \overline{\psi}_i\rangle\overline{\psi}_i - \sum\limits_{i=1}^{m}\langle r_{m+1}, \overline{r}_i\rangle\overline{r}_i}{\big\|\overline{r}_{m+1} - \sum\limits_{i=1}^{\infty}\langle r_{m+1}, \overline{\psi}_i\rangle\overline{\psi}_i - \sum\limits_{i=1}^{m}\langle r_{m+1}, \overline{r}_i\rangle\overline{r}_i\big\|}, \quad m = 1, 2, \ldots.\end{aligned}$$

From the process of Schmidt orthnormalization, it holds $\{r_i\}_{i=1}^{\infty} \perp \Psi$, i.e., $\Psi^{\perp} = \{\overline{r}_1, \overline{r}_2, \ldots, \overline{r}_m, \ldots\}$. Therefore, the following theorem is proposed.

Theorem 4.1.2. *If the equation* (4.1.3) *has solutions, the representation of all exact solutions is given by*

$$u(x) = u_{\min}(x) + \sum_{i=1}^{\infty} \alpha_i \overline{r}_i(x), \tag{4.1.7}$$

where real sequence $\{\alpha_i\}_{i=1}^{\infty} \in \ell^2$ *and*

$$u_{min}(x) = \sum_{i=1}^{\infty} \left[\sum_{k=1}^{i} \beta_{ik} f(x_k) \right] \overline{\psi}_i(x)$$

is the minimal norm solution of (4.1.3).

Proof. It is obvious that the form (4.1.7) is a solution of (4.1.3) because of $\Psi^{\perp} = \mathrm{N}(\mathbb{A})$. Here, it will prove the form (4.1.5) is the minimal norm solution of (4.1.3). Let u be any a solution of (4.1.3). Then $u = u_{\min} + v$, where $u_{\min} \in \Psi$ and $v \in \Psi^{\perp}$. It holds that

$$\|u\|^2 = \|u_{\min}\|^2 + \|v\|^2 \geq \|u_{\min}\|^2.$$

It shows that $u_{\min}(x)$ is the minimal norm solution of (4.1.3). □

4.1.3 The Stability of the Solution on the Eq. (4.1.3)

It is well known the problem on the stability of the solution for Eq. (4.1.3) is an ill-posed problem in the space $C[a, b]$ or $L^2[a, b]$. Here, it would be discussed in the reproducing kernel space $W_2^1[a, b]$.

Firstly, stability of the solution for Eq. (4.1.3) is defined in the reproducing kernel space $W_2^1[a, b]$.

Let $u(x)$ be a solution of (4.1.3). It is called that the approximate method on solution $u(x)$ from $u_n(x)$ with the right-hand side $f_n(x)$ is stable in $W_2^1[a, b]$, if $\lim\limits_{n\to\infty} \|f - f_n\|_{W_2^1} = 0$, then $\lim\limits_{n\to\infty} \|u - u_n\|_{W_2^1} = 0$.

From Section 4.1.2, any a solution of (4.1.3) has the form $u = u_{min} + v$, where $u \in \Psi$ and $v \in \Psi^{\perp}$. Then it holds that

$$\mathbb{A}u = \mathbb{A}u_{min} + \mathbb{A}v = \mathbb{A}u_{min} = f.$$

Therefore, the discussion of the stability of any a solution for (4.1.3) is equivalent to the stability of the minimal norm solution for (4.1.3). So, thereinafter for this Section, the operator $\mathbb{A}$ is restricted on Ψ, hence $\mathbb{A}^{-1}$ is existent.

Theorem 4.1.3. *If the equation* (4.1.3) *has solutions and let* $u_{\min}(x)$ *is the minimal norm solution, then the approximate method on the minimal norm solution* $u_{\min}(x)$ *from* $u_{\min}^{(n)}(x)$ *is stable in the reproducing kernel space* $W_2^1[a, b]$.

Proof. Let

$$\mathbb{A}u_{\min}^{(n)}(x) = f_n(x) \text{ and } f(x) = f_n(x) + \varepsilon_n(x),$$

where $\varepsilon_n(x)$ is a perturbation and $\varepsilon_n(x) \xrightarrow{W_2^1} 0$ $(n \longrightarrow \infty)$. From the form (4.1.5), note that

$$u_{\min}(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} f(x_k)\overline{\psi}_i(x)$$

and

$$u_{\min}^{(n)}(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} f_n(x_k)\overline{\psi}_i(x)$$

for $f,\ f_n \in W_2^1[a,b]$. It has

$$u_{\min}(x) - u_{\min}^{(n)}(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\varepsilon_n(x_k)\overline{\psi}_i(x) = \mathbb{P}\mathbb{A}^{-1}\varepsilon_n(x).$$

From the continuity of $\mathbb{P}$, $\mathbb{A}^{-1}$ and $\varepsilon_n(x) \xrightarrow{W_2^1} 0$ $(n \longrightarrow \infty)$, it follows

$$\lim_{n\to\infty} \left\|u_{\min}(x) - u_{\min}^{(n)}(x)\right\|_{W_2^1} = \|\mathbb{P}\|\,\|\mathbb{A}^{-1}\| \lim_{n\to\infty} \|\varepsilon_n(x)\|_{W_2^1} = 0.$$

Furthermore, a corollary is obtained according to a property (see [21])

$$\|u\|_{W_2^1} \geq M\|u\|_C. \tag{4.1.8}$$

The following important fact follows immediately from Theorem 4.1.3 and the form (4.1.8). □

Corollary 4.1.1. *Assume the conditions in Theorem* 4.1.3 *are satisfied. If* $\varepsilon_n(x) \xrightarrow{W_2^1} 0$ *as* $n \longrightarrow \infty$, *then*

$$u_{\min}^{(n)}(x) - u_{\min}(x) \xrightarrow{C} 0 \ \text{ as } \ n \longrightarrow \infty.$$

4.1.4 Numerical Experiments

For the Eq. (4.1.3), let

$$u_n(s) = u(s) + n\sin n^2 s$$

be approximate solutions of (4.1.3) of which the right-hand side is perturbed by

$$f_n(x) = f(x) + n\int_a^b K(x,s)\sin n^2 s\, ds.$$

Here, we define the error estimate expressions of the right-hand side of (4.1.3) and the solution of (4.1.3), respectively,

$$\begin{aligned} \|f_n - f\|_C &= \max_{x\in[a,b]} |f_n(x) - f(x)|, \\ \|u_n - u\|_C &= \max_{x\in[a,b]} |u_n(x) - u(x)|. \end{aligned}$$

Note that

$$\begin{aligned} |f_n(x) - f(x)| &= \left| \int_a^b K(x,s) n \sin n^2 s \, ds \right| \\ &= \left| -K(x,s)\frac{1}{n}\cos n^2 s \Big|_a^b + \frac{1}{n}\int_a^b K_s(x,s)\cos n^2 s \, ds \right| \\ &\leq \frac{L}{n}, \end{aligned}$$

where L is a constant independent of n.In the space $C[a,b]$, it holds $\|u_n - u\|_C \longrightarrow \infty$ as $n \longrightarrow \infty$ as soon as $\|f_n - f\|_C$ tends to 0 as $n \longrightarrow \infty$. It is shown that a small perturbation being put on the right-hand side of (4.1.3) may cause dramatically large errors in the solution in the space $C[a,b]$, namely, Fredholm integral equation of the first kind is an ill-posed problem in $C[a,b]$. But it is a well-posed problem in $W_2^1[a,b]$ from above discussions. Here, we give an example as follows:

Example 4.1.1. Let us consider Fredholm integral equation of the first kind

$$\int_0^{\frac{8\pi}{3}} K(x,s)u(s)\, ds = f(x), \tag{4.1.9}$$

where $K(x,s) = \cos(3x+s)$. For simplicity, we take $f(x) = 0$. Then $u(x) = 0$. We give the right-hand side of (4.1.9) a perturbation

$$\varepsilon_n(x) = \int_0^{\frac{8\pi}{3}} K(x,s) n \sin n^2 s \, ds.$$

Table 4.1: $n = 40$

Node	perturbations ε_n	Error $u_{\min}^{(n)}$ in $W_2^1[0, \frac{8\pi}{3}]$	Error u_n in $C[0, \frac{8\pi}{3}]$
0	0.0187	5.519E-03	0
$2\pi/3$	0.0187	-8.425E-04	34.641
π	-0.0187	-4.158E-03	0
$4\pi/3$	0.0187	3.143E-03	-34.641
$7\pi/3$	-0.0187	4.741E-03	-34.641
$8\pi/3$	0.0187	3.301E-03	34.641

Table 4.2: $n = 200$

Node	perturbations ε_n	Error $u_{\min}^{(n)}$ in $W_2^1[0, \frac{8\pi}{3}]$	Error u_n in $C[0, \frac{8\pi}{3}]$
0	0.00375	1.104E-03	0
$2\pi/3$	0.00375	1.686E-04	173.205
π	-0.00375	8.321E-04	0
$4\pi/3$	0.00375	6.289E-04	-173.205
$7\pi/3$	-0.00375	9.487E-04	-173.205
$8\pi/3$	0.00375	6.670E-04	173.205

Example 4.1.2. Let us consider a 1D heat conduction equation with initial and boundary condition [91], [100]

$$\begin{cases} u_t(x,t) = u_{xx}(x,t), & 0 < x < \pi, \ \ t > 0, \\ u(0,t) = u(\pi,t) = 0, & t > 0, \\ u(x,0) = f(x), & 0 < x < \pi. \end{cases} \tag{4.1.10}$$

The solution of Eq. (4.1.10) is

$$u(x,t) = \sum_{n=1}^{\infty} a_n e^{-n^2 t} \sin(nx), \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(y) \sin(ny)\, dy.$$

The inverse heat conduction problem considered involves determining $f(x)$ from the given data $u(\cdot, T)$. Then the problem is come down to solve the solution of Fredholm integral equation of the first kind

$$\int_0^{\pi} k(x,y) f(y)\, dy = u(x,T), \ \ 0 \le x \le \pi, \tag{4.1.11}$$

where $k(x,y) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-n^2 T} \sin(nx) \sin(ny)$ at $t = T$. While the inverse problem is an ill-posed problem in the space $L^2[0,\pi]$, namely, a small perturbation of the right-hand side $u(\cdot, T)$ can cause a large perturbation in the solution $f(x)$.

Take $T = 1$,

$$u(x,1) = \sum_{n=1}^{\infty} \frac{2\sin(n\pi)}{\pi - n^2\pi} e^{-n^2} \sin(nx),$$

the true solution is $f(x) = \sin x$. We calculate the approximate solution $\widehat{f}(x)$. All computations are performed by the Mathematica software package. We present the numerical results in Table 4.3 and Table 4.4 when the right-hand side of Eq. (4.1.11) is put on perturbations $\varepsilon = 0.05$ and $\varepsilon = 0.005$, respectively, in the space $W_2^1[0,\pi]$.

Table 4.3: $\varepsilon = 0.05$

Node	$f(x)$	$\widehat{f}(x)$	absolute error
$\pi/12$	0.258819	0.227559	3.126E-02
$3\pi/12$	0.707107	0.671077	3.603E-02
$5\pi/12$	0.965926	0.924806	4.112E-02
$7\pi/12$	0.965926	0.923126	4.28E-02
$9\pi/12$	0.707107	0.667107	4E-02
$11\pi/12$	0.258819	0.223359	3.546E-02

Table 4.4: $\varepsilon = 0.005$

Node	$f(x)$	$\widehat{f}(x)$	absolute error
$\pi/12$	0.258819	0.255693	3.126E-03
$3\pi/12$	0.707107	0.0703504	3.603E-03
$5\pi/12$	0.965926	0.961814	4.112E-03
$7\pi/12$	0.965926	0.961646	4.28E-03
$9\pi/12$	0.707107	0.703107	4E-03
$11\pi/12$	0.258819	0.255273	3.546E-03

From the above numerical examples, it shows that numerical solution is stable when the right-hand side is put on a small perturbations.

4.2 Solving Nonlinear Volterra–Fredholm Integral Equations

4.2.1 Introduction

We are concerned with the nonlinear Volterra–Fredholm integral equations of the form

$$u(x) = f(x) + Gu(x), \tag{4.2.1}$$

where

$$Gu(x) = \lambda_1 \int_a^x K_1(x,\xi) N_1(u(\xi))\, d\xi + \lambda_2 \int_a^b K_2(x,\xi) N_2(u(\xi))\, d\xi,$$

$u(x)$ is determined function and $u(x), f(x) \in W_2^1[a,b]$, $N_1(\cdot), N_2(\cdot)$ the continuous nonlinear terms in $W_2^1[a,b]$, $W_2^1[a,b]$ is a reproducing kernel space.

The literatures of nonlinear integral equations contain few numerical methods [101]. The commonly used methods are the projection method, time collocation method [102], [103], trapezoidal Nystrom method [104] and the Adomian decomposition method [101], [105], [106]. The existing techniques encountered difficulties in terms of computational work used and approximate solutions were obtained for numerical purposes. A Taylor expansion approach for solving Fredholm integral

equations has been presented by Kanwal and liu [107] and then this has been extended by Sezer to Volterra integral equation [108]. A Taylor expansion approach for the nonlinear Volterra–Fredholm integral equations has been given in [109]. But it only handle a special form of (4.2.1).

In this study, the representation of the exact solution of the nonlinear Volterra–Fredholm integral equations will be obtained in the reproducing kernel space. The exact solution is given by the form of series. Its approximate solution is obtained by truncating the series and a new numerical approximate method is obtained. The error of the approximate solution is monotone deceasing in the sense of $\|\cdot\|_{W_2^1}$. The intrinsic merit of the method given in the paper lies in its speedy convergence. In addition, it must be pointed out that additional conditions as linearization or unjustified assumptions are not required for nonlinear cases.

4.2.2 Theoretic Basis of the Method

Given the reproducing kernel space $W_2^1[a,b]$ by (1.3.2), its reproducing kernel is $R_y(x)$ given by (4.1.2).

Here, representation of the solution of (4.2.1) is given in the reproducing kernel space $W_2^1[a,b]$.

Let $\psi_i(x) = R_{x_i}(x)$, where $\{x_i\}_{i=1}^{\infty}$ is dense in the interval $[a,b]$. From the property of the reproducing kernel, it holds

$$\langle u(x), \psi_i(x)\rangle_{W_2^1} = u(x_i). \tag{4.2.2}$$

Theorem 4.2.1. *If $\{x_i\}_{i=1}^{\infty}$ is dense in the interval $[a,b]$, then $\{\psi_i(x)\}_{i=1}^{\infty}$ is the complete function system of $W_2^1[a,b]$.*

Proof. Note that $\{x_i\}_{i=1}^{\infty}$ is dense in the interval $[a,b]$. For $u(x) \in W_2^1[a,b]$, if

$$\langle u(x), \psi_i(x)\rangle_{W_2^1} = u(x_i) = 0 \ \ (i = 1, 2, \ldots),$$

then we have $u(x) \equiv 0$ from the density of $\{x_i\}_{i=1}^{\infty}$ and continuity of $u(x)$. □

Practise Gram–Schmidt orthonomalization for $\{\psi_i(x)\}_{i=1}^{\infty}$

$$\overline{\psi}_i(x) = \sum_{k=1}^{i} \beta_{ik}\psi_k(x),$$

where β_{ik} are coefficients of Gram–Schmidt orthonomalization and $\{\overline{\psi}_i(x)\}_{i=1}^{\infty}$ is a normal orthogonal basis of $W_2^1[a,b]$.

Subsequently, the following theorem is obtained.

Theorem 4.2.2. *Let $\{x_i\}_{i=1}^{\infty}$ be dense in the interval $[a,b]$. If the equation* (4.2.1) *has a unique solution, then the solution satisfies the form*

$$u(x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}(f(x_k) + Gu(x_k))\overline{\psi}_i(x). \tag{4.2.3}$$

Proof. Assume $u(x)$ be the solution of Eq. (4.2.1). From Theorem (4.2.1), $u(x)$ is expanded in Fourier series, it has

$$\begin{aligned} u(x) &= \sum_{i=1}^{\infty} \big\langle u(x), \overline{\psi}_i(x) \big\rangle_{W_2^1} \overline{\psi}_i(x) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} \langle u(x), \psi_k(x) \rangle_{W_2^1} \overline{\psi}_i(x) \\ &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} \big\langle f(x) + Gu(x), \psi_k(x) \big\rangle_{W_2^1} \overline{\psi}_i(x) \\ &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} \big(f(x_k) + Gu(x_k) \big) \overline{\psi}_i(x). \end{aligned}$$

The proof is complete. □

4.2.3 Implementations of the Method

Here, a method of solving (4.2.3) of (4.2.1) is given in the reproducing kernel space. Rewrite (4.2.3) as

$$u(x) = \sum_{i=1}^{\infty} A_i \overline{\psi}_i(x),$$

where

$$A_i = \sum_{k=1}^{i} \beta_{ik} (f(x_k) + Gu(x_k)).$$

In fact, A_i is unknown. A_i is approximated by known B_i.

For a numerical computation, let initial function $u_1(x) = f(x)$ and the n-term approximation to $u(x)$ is defined by

$$u_{n+1}(x) = \sum_{i=1}^{n} B_i \overline{\psi}_i(x), \tag{4.2.4}$$

where

$$\begin{aligned} B_1 &= \beta_{11} (G u_1(x_1) + f(x_1)), \\ u_2(x) &= B_1 \overline{\psi}_1(x), \\ B_2 &= \sum_{k=1}^{2} \beta_{2k} (G u_2(x_k) + f(x_k)), \\ u_3(x) &= B_1 \overline{\psi}_1(x) + B_2 \overline{\psi}_2(x), \\ &\cdots \\ B_n &= \sum_{k=1}^{n} \beta_{nk} (G u_n(x_k) + f(x_k)), \\ u_{n+1}(x) &= \sum_{k=1}^{n} B_k \overline{\psi}_k(x). \end{aligned} \tag{4.2.5}$$

Lemma 4.2.1. *If* $u_n(x) \xrightarrow{\|\cdot\|_{W_2^1}} \overline{u}(x)$ $(n \longrightarrow \infty)$ *and* $x_n \longrightarrow y$, *then*

$$u_n(x_n) \longrightarrow \overline{u}(y) \ \ (n \longrightarrow \infty),$$

Proof. Since

$$\begin{aligned} \big|u_n(x_n) - \overline{u}(y)\big| &= \big|u_n(x_n) - u_n(y) + u_n(y) - \overline{u}(y)\big| \\ &\leq |u_n(x_n) - u_n(y)| + \big|u_n(y) - \overline{u}(y)\big|. \end{aligned}$$

It follows that

$$\begin{aligned} |u_n(x_n) - u_n(y)| &= \big|\langle u_n(x), R_{x_n}(x) - R_y(x)\rangle_{W_2^1}\big| \\ &\leq \|u_n(x)\|_{W_2^1}\big\|R_{x_n}(x) - R_y(x)\big\|_{W_2^1}. \end{aligned}$$

From the convergence of $u_n(x)$, there exists a constant N, such that

$$\|u_n(x)\|_{W_2^1} \leq 2\|\overline{u}(x)\|_{W_2^1}$$

as soon as $n \geq N$. At the same time, we can prove

$$\|R_{x_n}(x) - R_y(x)\|_{W_2^1} \longrightarrow 0$$

as soon as $n \longrightarrow \infty$. Hence $|u_n(x_n) - u_n(y)| \longrightarrow 0$ as soon as $x_n \longrightarrow y$ from $\|u\|_C \leq M\|u\|_{W_2^1}$.

On the other hand, for any $y \in [a, b]$, it holds that

$$\big\|u_n(y) - \overline{u}(y)\big\|_C \longrightarrow 0 \ \ (n \longrightarrow \infty)$$

as soon as

$$\big\|u_n(y) - \overline{u}(y)\big\|_{W_2^1} \longrightarrow 0 \ \ (n \longrightarrow \infty).$$

Hence, it follows that

$$u_n(x_n) \longrightarrow \overline{u}(y) \ \ (n \longrightarrow \infty)$$

as soon as $x_n \longrightarrow y$. □

By means of the continuation of $N_1(\cdot)$, $N_2(\cdot)$, it is obtained that $N_i(u_n(x_n)) \longrightarrow N_i(\overline{u}(y))$ $(n \longrightarrow \infty)$, $i = 1, 2$. It shows that

$$Gu_n(x_n) \longrightarrow G\overline{u}(y) \ \ (n \longrightarrow \infty).$$

The convergence theorem is established for the method mentioned above.

Theorem 4.2.3. *Assume* $\|u_n(x)\|_{W_2^1}$ *is bounded in* (4.2.4), *if* $\{x_i\}_{i=1}^{\infty}$ *is dense in* $[a,b]$, *then the* n*-term approximate solution* $u_n(x)$ *converges to the exact solution* $u(x)$ *of Eq.* (4.2.1) *and the exact solution is expressed as*

$$u(x) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(x), \tag{4.2.6}$$

where B_i *is given by* (4.2.5).

Proof. The proof will be divided into three steps.

(i) The convergence of (4.2.4) will be proved. From the form (4.2.5), one gets

$$u_{n+1}(x) = u_n(x) + B_n \overline{\psi}_n(x). \tag{4.2.7}$$

Using the orthogonality of $\{\overline{\psi}_i\}_{i=1}^{\infty}$, it follows that

$$\|u_{n+1}(x)\|_{W_2^1}^2 = \|u_n(x)\|_{W_2^1}^2 + B_n^2.$$

The sequence $\|u_n(x)\|_{W_2^1}$ is monotone increasing. Due to $\|u_n(x)\|_{W_2^1}$ being bounded, $\{\|u_n(x)\|_{W_2^1}\}$ is convergent as soon as $n \longrightarrow \infty$. Then there exists a constant c such that

$$\sum_{i=1}^{\infty} B_i^2 = c. \tag{4.2.8}$$

It implies that

$$B_i = \sum_{k=1}^{i} \beta_{ik}\big(f(x_k) + Gu_i(x_k)\big) \in \ell^2, \quad i = 1, 2, \ldots. \tag{4.2.9}$$

If $m > n$, using the orthogonality of $u_{n+1}(x) - u_n(x)$, $n = 2, 3, \ldots$, it holds

$$\begin{aligned} &\|u_m(x) - u_n(x)\|_{W_2^1}^2 \\ &= \big\|u_m(x) - u_{m-1}(x) + u_{m-2}(x) + \cdots + u_{n+1}(x) - u_n(x)\big\|_{W_2^1}^2 \\ &= \|u_m(x) - u_{m-1}(x)\|_{W_2^1}^2 + \cdots + \|u_{n+1}(x) - u_n(x)\|_{W_2^1}^2 \\ &= \sum_{i=n+1}^{m} B_i^2 \longrightarrow 0 \ \ (n \longrightarrow \infty). \end{aligned} \tag{4.2.10}$$

Considering the completeness of $W_2^1[a,b]$, it has

$$u_n(x) \xrightarrow{\|\cdot\|_{W_2^1}} \overline{u}(x) \ \ (n \longrightarrow \infty).$$

Hence

$$\overline{u}(x) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(x). \tag{4.2.11}$$

(ii) Define the projection operator

$$\mathbb{P}_n\overline{u}(x) = \sum_{i=1}^{n} B_i\overline{\psi}_i(x).$$

Then

$$u_{n+1}(x) = \mathbb{P}_n\overline{u}(x).$$

We can prove

$$u_{n+1}(x_k) = \overline{u}(x_k), \quad k \le n.$$

In fact,

$$\begin{aligned} u_{n+1}(x_k) &= \langle u_{n+1}(x), \psi_k(x)\rangle_{W_2^1} = \langle P_n\overline{u}(x), \psi_k(x)\rangle_{W_2^1} \\ &= \langle \overline{u}(x), P_n\psi_k(x)\rangle_{W_2^1} = \langle \overline{u}(x), \psi_k(x)\rangle_{W_2^1} = \overline{u}(x_k). \end{aligned}$$

So

$$Gu_{n+1}(x_k) = G\overline{u}(x_k), \quad k \le n.$$

(iii) It is proved that $\overline{u}(x)$ is the solution of Eq. (4.2.1).

From (4.2.11), it follows

$$\overline{u}(x_j) = \sum_{i=1}^{\infty} B_i\langle\overline{\psi}_i(x), \psi_j(x)\rangle_{W_2^1}. \tag{4.2.12}$$

Multiplying both sides of (4.2.12) by β_{nj} and summing for j from 1 to n, it has

$$\begin{aligned} \sum_{j=1}^{n}\beta_{nj}\overline{u}(x_j) &= \sum_{i=1}^{\infty} B_i\left\langle\overline{\psi}_i(x), \sum_{j=1}^{n}\beta_{nj}\psi_j\right\rangle_{W_2^1} \\ &= \sum_{i=1}^{\infty} B_i\langle\overline{\psi}_i(x), \overline{\psi}_n\rangle_{W_2^1} = B_n. \end{aligned} \tag{4.2.13}$$

From (4.2.13) and (4.2.4), if $n = 1$, then

$$\overline{u}(x_1) = B_1 = f(x_1) + Gu_1(x_1).$$

If $n = 2$, then

$$\begin{aligned} \beta_{21}\overline{u}(x_1) + \beta_{22}\overline{u}(x_2) &= B_2 \\ &= \beta_{21}(f(x_1) + Gu_2(x_1)) + \beta_{22}(f(x_2) + Gu_2(x_2)) \\ &= \beta_{21}(f(x_1) + G\overline{u}(x_1)) + \beta_{22}(f(x_2) + Gu_2(x_2)) \\ &= \beta_{21}\overline{u}(x_1) + \beta_{22}(f(x_2) + Gu_2(x_2)). \end{aligned} \tag{4.2.14}$$

Namely,

$$\overline{u}(x_2) = f(x_2) + Gu_2(x_2). \tag{4.2.15}$$

In the same way, it has

$$\overline{u}(x_n) = f(x_n) + Gu_n(x_n). \tag{4.2.16}$$

For any $y \in [a, b]$, there exists a subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ converging to y since $\{x_i\}_{i=1}^{\infty}$ is dense in $[a, b]$. From Lemma (4.2.1) and the above form, it holds that

$$\overline{u}(y) = f(y) + G\overline{u}(y). \tag{4.2.17}$$

So $\overline{u}(x)$ is the solution of Eq. (4.2.1) and

$$u(x) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(x). \qquad \square$$

Theorem 4.2.4. *Assume $u(x)$ is the solution of Eq.* (4.2.1) *and $r_n(x)$ is the error in the approximate solution $u_{n+1}(x)$, where $u_{n+1}(x)$ is given by* (4.2.4)*. Then the error r_n is monotone decreasing in the sense of $\|\cdot\|_{W_2^1}$.*

Proof. Suppose that $u(x)$ and $u_{n+1}(x)$ are given by (4.2.6) and (4.2.4) respectively. It holds

$$\begin{aligned} \|r_n(x)\|_{W_2^1}^2 &= \|u(x) - u_{n+1}(x)\|_{W_2^1}^2 \\ &= \left\| \sum_{i=n+1}^{\infty} B_i \overline{\psi}_i(x) \right\|_{W_2^1}^2 = \sum_{i=n+1}^{\infty} B_i^2. \end{aligned} \tag{4.2.18}$$

It shows that the error r_n is monotone decreasing in the sense of $\|\cdot\|_{W_2^1}$. $\square$

4.2.4 Numerical Experiment

Above method will be applied to many numerical examples. All computations are performed by the Mathematica 5.0 software package. The approximate solution $u_{n+1}(x)$ are calculated by(4.2.4). The numerical results are presented in Table 4.5–4.6.

Example 4.2.1. We consider the nonlinear Volterra integral equation

$$u(x) = f(x) + \int_0^x F(x, \xi) N_1(u(\xi))\, d\xi, \tag{4.2.19}$$

where $x \in [0,1]$, $u(x)$, $f(x) \in W_2^1[0,1]$. The existence and uniqueness of solution for Eq. (4.2.19) have be established in [110], [102]. If

$$\begin{aligned} N_1(u(x)) &= u^2(x), \\ F(x,\xi) &= \sin(x-\xi), \\ f(x) &= \frac{1}{6}(-3+8\cos x+\cos 2x), \end{aligned}$$

then $u(x) = \cos x$ is the true solution. In calculation, define n-term approximate solution can be solved by the process (4.2.5) and $u_1(x) = f(x)$. The approximate solution $u_{n+1}(x)$ with $n = 25$ and error are shown in Table 4.5.

Table 4.5: The absolute error of exact solution of Example 4.2.1

Node	True solution u	Approximate solution u_{25}	Absolute error
0.08	0.996802	0.9968	1.70553E-6
0.16	0.987227	0.98722	6.81112E-6
0.24	0.971338	0.971323	1.52827E-5
0.32	0.949235	0.949208	2.8514E-5
0.4	0.921061	0.921019	4.20859E-5
0.48	0.886995	0.886935	6.02055E-5
0.56	0.847255	0.847172	8.27812E-5
0.64	0.802096	0.801991	1.05202E-4
0.72	0.751806	0.751674	1.318E-4
0.8	0.696707	0.696546	1.60689E-4
0.88	0.637151	0.636959	1.92082E-4
0.96	0.547352	0.573306	2.14353E-4

Example 4.2.2. A nonlinear Volterra–Fredholm integral equation

$$u(x) = f(x) + \int_0^x F_1(x,\xi)N_1(u(\xi))\,d\xi + \int_0^1 F_2(x,\xi)N_2(u(\xi))\,d\xi,$$

where $x \in [0,1]$, $u(x)$, $f(x) \in W_2^1[0,1]$,

$$\begin{gathered} F_1(x,\xi) = \sin(x-\xi), \quad F_2(x,\xi) = x-\xi, \\ N_1(u(x)) = \cos(u(x)), \quad N_2(u(x)) = 1+u^2(x), \\ f(x) = \frac{1}{12}\left(19-28x+6x\sin 1\cos x - 6x\cos 1\sin x - 6\sin 1\sin x\right). \end{gathered}$$

Then $u(x) = 1-x$ is the true solution.The approximate solution $u_{n+1}(x)$ with $n = 25$ and error are shown in Table 4.6.

Table 4.6: The absolute error of exact solution of Example 4.2.2

Node	True solution u	Approximate solution u_{25}	Absolute error
0.08	0.92	0.919882	1.18148E-4
0.16	0.84	0.839856	1.44021E-4
0.24	0.76	0.759834	1.65987E-4
0.32	0.68	0.679815	1.84716E-4
0.4	0.6	0.599799	2.00596E-4
0.48	0.52	0.519786	2.14138E-4
0.56	0.44	0.439775	2.25485E-4
0.64	0.36	0.369764	2.36015E-4
0.72	0.28	0.279755	2.45419E-4
0.8	0.2	0.199747	2.52873E-4
0.88	0.12	0.119737	2.63029E-4
0.96	0.04	0.039724	2.75999E-4

4.3 Solving a Class of Nonlinear Volterra–Fredholm Integral Equations

4.3.1 Introduction

We shall consider the general nonlinear mixed Volterra–Fredholm integral equation of the form

$$u(t,x) = f(t,x) + \int_0^t \int_\Omega F\big(t,x,\tau,\xi,u(\tau,\xi)\big)\, d\xi\, d\tau, \quad (t,x) \in [0,T] \times \Omega, \qquad (4.3.1)$$

where $u(t,x)$ is determined function, $f(t,x)$ and $F(t,x,\tau,\xi,u(\tau,\xi))$ are analytic functions on $D = [0,T] \times \Omega$, where Ω is a closed subset of R^n, $n = 1,2,3$.

Equations of this type arise in the theory of nonlinear parabolic boundary value problems, the mathematical model of the spatiotemporal development of an epidemic, and various physical, mechanical, and biological problems [111], [112].

The existence and uniqueness of the solution for the equation (4.2.1) are discussed in [6], [7]. Significant progress has been made in numerical analysis linear and nonlinear version of the Eq. (4.3.1). For the linear case, some methods for numerical treatment are given in [6], [7], [113]. For nonlinear case, the literature of integral equations contains few numerical methods [9] for handling the equation Eq. (4.3.1). In recent years, there has been renewed interest in Eq. (4.3.1), such as the time collocation and time discretization methods [7], [8], the particular trapezoidal Nystrom method [21], the Adomian decomposition method [9], [22], [23] and so on.

4.3.2 Solving Eq. (4.3.1) in the Reproducing Kernel Space

The reproducing kernel spaces $W_2^{(2,1)}(\mathcal{D})$, $\mathcal{D} = [0,T] \times [a,b]$, ($m = 2$, $n = 1$) and $W_2^1[a,b]$ by the Section 1.5.2 and the Section 1.3.2, respectively, their reproducing kernels are $K_{(\eta,\xi)}(t,x)$ and $r_\xi(x)$, respectively.

We will discuss on solving the Eq. (4.3.1) in the reproducing kernel space $W_2^{(2,1)}(\mathcal{D})$. Applying partial derivative of two sides of (4.3.1) with respect to t, it has

$$\begin{aligned}\frac{\partial u(t,x)}{\partial t} &= \frac{\partial f(t,x)}{\partial t} \\ &+ \int_a^b F(t,x,t,\xi,u(t,\xi))d\xi + \int_0^t \int_a^b \frac{\partial F(t,x,\tau,\xi,u(\tau,\xi))}{\partial t}\, d\xi\, d\tau, \quad (4.3.2)\end{aligned}$$

with initial values

$$u(0,x) = f(0,x). \quad (4.3.3)$$

Obviously, Eq. (4.3.2) is equivalent to (4.3.1).

The operator $\mathbb{L} : W_2^{(2,1)}(\mathcal{D}) \longrightarrow W_2^1[a,b]$ is defined by

$$(\mathbb{L}u)(x) \triangleq \left.\frac{\partial u(t,x)}{\partial t}\right|_{t=0}. \quad (4.3.4)$$

If let $t = 0$ in Eq. (4.3.2), operator equation becomes

$$(\mathbb{L}u)(x) = G(x), \quad (4.3.5)$$

where

$$G(x) = \left.\frac{\partial f(t,x)}{\partial t}\right|_{t=0} + \int_a^b F\big(0,x,0,\xi,u(0,\xi)\big)d\xi. \quad (4.3.6)$$

It is easy to prove the operator $\mathbb{L}$ is bounded linear operator.

Obviously, the solution of Eq. (4.3.2) satisfies (4.3.5). But it must be pointed out Eq. (4.3.5) is not equivalent to Eq. (4.3.2). Here, we will solve Eq. (4.3.2) in virtue of the definition of the operator $\mathbb{L}$.

In order to obtain the representation of the exact solution of Eq. (4.3.2), let

$$\varphi_i(x) = R_{x_i}(x),$$

where $\{x_i\}_{i=1}^\infty$ is dense in the interval $[a,b]$. Let $\mathbb{L}^*$ denotes the conjugated operator of $\mathbb{L}$ from $W_2^{(2,1)}(\mathcal{D})$ to $W_2^1[a,b]$.

Write

$$\psi_i(t,x) = (\mathbb{L}^*\varphi_i)(t,x)$$

and

$$\overline{\psi}_i(t,x)=\sum_{k=1}^{i}\beta_{ik}\psi_k(t,x),$$

where β_{ik} are coefficients of Gram–Schmidt orthonomalization. Then the span for $\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}$ is a subspace of $W_2^{(2,1)}(\mathcal{D})$

$$span\big(\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}\big)=\left\{u(t,x)|\ u(t,x)=\sum_{i=1}^{n}c_i\overline{\psi}_i(t,x),\ c_i\in\mathbb{R},\ n\in\mathbb{N}\right\}. \quad (4.3.7)$$

Let S be the closure of this subspace

$$\mathrm{S}=\overline{span}\big(\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}\big)$$

and $\mathrm{S}^{\perp}$ denote orthcomplement space of S in $W_2^{(2,1)}(\mathcal{D})$. Using the following method, a normal orthogonal basis $\{\overline{\rho}_j(t,x)\}_{j=1}^{\infty}$ of $\mathrm{S}^{\perp}$ could be obtained.

Take a set of points $\mathrm{B}=\{p_j(t_j,x_j)\}_{j=1}^{\infty}$ as a dense set of region $\mathcal{D}=[0,T]\times[a,b]$ and put

$$\rho_j(t,x)=K_{(t_j,x_j)}(t,x),\ \ j=1,2,\ldots. \quad (4.3.8)$$

Apply generalized Schimidt orthonormalization for the function system $\{\rho_j(t,x)\}_{j=1}^{\infty}$ about orthonormal system $\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}$, that is

$$\overline{\rho}_j(t,x)=\frac{\rho_j(t,x)-\sum\limits_{i=1}^{\infty}\langle\rho_j(t,x),\overline{\psi}_i(t,x)\rangle\overline{\psi}_i(t,x)-\sum\limits_{i=1}^{j-1}\langle\rho_j(t,x),\overline{\rho}_i(t,x)\rangle\overline{\rho}_i(t,x)}{\big\|\rho_j(t,x)-\sum\limits_{i=1}^{\infty}\langle\rho_j(t,x),\overline{\psi}_i(t,x)\rangle\overline{\psi}_i(t,x)-\sum\limits_{i=1}^{j-1}\langle\rho_j(t,x),\overline{\rho}_i(t,x)\rangle\overline{\rho}_i(t,x)\big\|},$$

$j=1,2,\ldots$, where $\sum\limits_{i=1}^{j-1}=0$ as soon as $j=1$. And put

$$\overline{\rho}_j(t,x)=\sum_{k=1}^{\infty}\beta_{jk}\psi_k(t,x)+\sum_{m=1}^{j}\beta^*_{jm}\rho_m(t,x),\ \ j=1,2,\ldots. \quad (4.3.9)$$

Since $\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}$ and $\{\overline{\rho}_j(t,x)\}_{j=1}^{\infty}$ are normal orthogonal basis of S and $\mathrm{S}^{\perp}$, respectively. $\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}\cup\{\overline{\rho}_j(t,x)\}_{j=1}^{\infty}$ is the normal orthogonal basis of $W_2^{(1,1)}(\mathcal{D})$.

Theorem 4.3.1. *Take $\{s_k\}_{k=1}^{\infty}$ as a dense set of interval $[a,b]$. Assume $u(t,x)$ is the solution of Eq.* (4.3.5)*, then the solution is expressed as*

$$u(t,x)=\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}G(s_k)\overline{\psi}_i(t,x)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(t,x), \quad (4.3.10)$$

where $G(x)$ by (4.3.6), α_j satisfy the infinite linear system

$$\begin{aligned} u(0,x_n) &= f(0,x_n) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}G(s_k)\overline{\psi}_i(0,x_n)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(0,x_n), \quad n=1,2,\ldots. \end{aligned} \tag{4.3.11}$$

Proof. Assume $u(t,x)$ be the solution of Eq. (4.3.5) and $u(t,x)\in W_2^{(2,1)}(\mathcal{D})$. $u(t,x)$ be expanded in Fourier series

$$u(t,x)=\sum_{i=1}^{\infty}\big\langle u(t,x),\overline{\psi}_i(t,x)\big\rangle\overline{\psi}_i(t,x)+\sum_{j=1}^{\infty}\big\langle u(t,x),\overline{\rho}_j(t,x)\big\rangle\overline{\rho}_j(t,x)$$

by the normal orthogonal basis $\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty}\cup\{\overline{\rho}_j(t,x)\}_{j=1}^{\infty}$ of $W_2^{(2,1)}(\mathcal{D})$.

Let $\alpha_j=\langle u(t,x),\overline{\rho}_j(t,x)\rangle$, it follows

$$\begin{aligned} u(t,x) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\big\langle u(t,x),(\mathbb{L}^*\varphi_k)(t,x)\big\rangle_{W_2^{(1,1)}}\overline{\psi}_i(t,x)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(t,x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\big\langle(\mathbb{L}u)(x),\varphi_k(x)\big\rangle_{W_2^1}\overline{\psi}_i(t,x)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(t,x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\langle G(x),\varphi_k(x)\rangle_{W_2^1}\overline{\psi}_i(t,x)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(t,x) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}G(s_k)\overline{\psi}_i(t,x)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(t,x). \end{aligned}$$

Specially, it holds that

$$u(0,x)=\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}G(s_k)\overline{\psi}_i(0,x)+\sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(0,x).$$

From the uniqueness of the solution and the initial values (4.3.3), α_j can be obtained by solving infinite system of linear equations (4.3.11). □

The approximate solution is obtained by truncating the series

$$u_{n,m}(t,x)=\sum_{i=1}^{n}\sum_{k=1}^{i}\beta_{ik}G(s_k)\overline{\psi}_i(t,x)+\sum_{j=1}^{m}\alpha_j\overline{\rho}_j(t,x), \tag{4.3.12}$$

and α_j satisfy the linear system

$$f(0,x_l)=\sum_{i=1}^{n}\sum_{k=1}^{i}\beta_{ik}G(s_k)\overline{\psi}_i(0,x_l)+\sum_{j=1}^{m}\alpha_j\overline{\rho}_j(0,x_l), \quad l=1,2,\ldots,m. \tag{4.3.13}$$

In calculation, α_j could be obtained by solving Eq. (4.3.13).

Theorem 4.3.2. *Assume $u(t,x)$ is the solution of Eq. (4.3.2) and $r_{n,m}(t,x)$ are the error in the approximate solution $u_{n,m}(t,x)$, where $u(t,x)$, $u_{n,m}(t,x)$ are given by (4.3.10), (4.3.12) respectively. Then the following conclusions hold.*

(i) *The approximate solution $u_{n,m}(t,x)$ converge to the exact solution $u(t,x)$ in the sense of $\|\cdot\|_{W_2^{(2,1)}}$.*

(ii) *The error $r_{n,m}(t,x)$ is monotone decreasing in the sense of $\|\cdot\|_{W_2^{(2,1)}}$,*

$$\lim_{n,m\to\infty} \|r_{n,m}(t,x)\|_{W_2^{(2,1)}} = 0.$$

Proof. (i) Since $\{\overline{\psi}_i(t,x)\}_{i=1}^{\infty} \cup \{\overline{\rho}_j(t,x)\}_{j=1}^{\infty}$ is the normal orthogonal basis of $W_2^{(2,1)}(\mathcal{D})$, the form (4.3.10) is the Fourier series, so $u_{n,m}(t,x)$ converge to $u(t,x)$.

(ii) From

$$\begin{aligned}
&\|r_{n+1,m+1}(t,x)\|^2_{W_2^{(2,1)}} \\
&\quad = \sum_{i=n+1}^{\infty}\left[\sum_{k=1}^{i}\beta_{ik}G(s_k)\right]^2 + \sum_{j=m+1}^{\infty}\alpha_j^2 - \left[\sum_{k=1}^{n+1}\beta_{(n+1)k}G(s_k)\right]^2 - \alpha_{m+1}^2 \\
&\quad = \|r_{n,m}(t,x)\|^2_{W_2^{(2,1)}} - \left[\sum_{k=1}^{n+1}\beta_{(n+1)k}G(s_k)\right]^2 - \alpha_{m+1}^2.
\end{aligned}$$

So

$$\left\|r_{n+1,m+1}(t,x)\right\|^2_{W_2^{(2,1)}} \le \|r_{n,m}(t,x)\|^2_{W_2^{(2,1)}}.$$

By (i), of course

$$\lim_{n,m\to\infty} \|r_{n,m}(t,x)\|_{W_2^{(2,1)}} = 0. \qquad \square$$

4.3.3 Numerical Experiments

Some examples would be operated using the method proposed in the paper. All computations are performed by the Mathematica 4.0 software package. The approximate solution $u_{n,m}(t,x)$ is calculated by (4.3.12) and absolute error.

Example 4.3.1. Firstly, let us consider a linear Volterra–Fredholm integral equation

$$u(t,x) = f(t,x) + \int_0^t \int_0^1 F(x,\xi,\tau)u(\xi,\tau)\,d\xi\,d\tau, \quad t \in [0,1], \tag{4.3.14}$$

where

$$\begin{aligned} F(x,\xi,\tau) &= -\tau\cos(x-\xi), \\ f(t,x) &= \frac{1}{8}\,e^{-2t}\Big(8\cos x+(e^{-2t}-2t-1)\big(\cos x+\sin 1\cos(1-x)\big)\Big). \end{aligned}$$

Then $u(t,x)=\cos x e^{-2t}$ is the true solution of Eq. (4.3.14). The approximate solution with $n=150$, $m=80$ in the form (4.3.12) and error are shown in Table 4.7.

Table 4.7: The error of Example 4.3.1

Node	True solution u	Approximate solution $u_{150,80}$	Absolute error
$(\frac{1}{10},\frac{1}{15})$	0.816912	0.818221	0.0013089
$(\frac{1}{10},\frac{2}{5})$	0.754101	0.755338	0.00123737
$(\frac{1}{10},\frac{11}{15})$	0.608274	0.609004	0.000730235
$(\frac{3}{10},\frac{1}{15})$	0.547593	0.554697	0.0071047
$(\frac{3}{10},\frac{2}{5})$	0.505489	0.512153	0.00666431
$(\frac{3}{10},\frac{11}{15})$	0.407738	0.410699	0.00296107
$(\frac{7}{10},\frac{1}{15})$	0.246049	0.248285	0.00223576
$(\frac{7}{10},\frac{2}{5})$	0.227131	0.227568	0.000437002
$(\frac{7}{10},\frac{11}{15})$	0.183209	0.169654	0.0135549
$(\frac{9}{10},\frac{1}{15})$	0.164932	0.14778	0.0171516
$(\frac{9}{10},\frac{2}{5})$	0.15225	0.15013	0.00212
$(\frac{9}{10},\frac{11}{15})$	0.122808	0.111906	0.010902

Example 4.3.2. A nonlinear Volterra–Fredholm integral equation is of form

$$u(t,x)=f(t,x)+\int_0^t\int_0^1 F(x,\xi,\tau)(1+u^2(\xi,\tau))\,d\xi\,d\tau,\quad t\in[0,1], \tag{4.3.15}$$

where

$$\begin{aligned} F(x,\xi,\tau) &= -e^{\tau}\cos(x-\xi), \\ f(t,x) &= e^{-2t}\left(1-\frac{x^2t^2}{2}\right) \\ &- \frac{1}{12500}\Big(e^{-5t}\Big(20\big(-44+33e^{5t}-220t-550t^2 \\ &\quad -500t^3-625t^3\big)\cos(1-x)+\big(3012+12500e^{4t}-15512e^{5t} \\ &\quad +2560t+6400t^2+6500t^3+8125t^4\big)\sin(1-x) \\ &\quad -4(-869-3125e^4t+3994e^{5t}-1220t-3050t^2 \\ &\quad -3000t^3-3750t^4)\sin x\Big)\Big). \end{aligned}$$

Then $u(t,x)=(1-\frac{x^2t^2}{2})e^{-2t}$ is the true solution of Eq. (4.3.15). The approximate solution with $n=150$, $m=100$ and error are shown in Table 4.8.

Table 4.8: The error of Example 4.3.2

Node	True solution u	Approximate solution $u_{150,100}$	Absolute error
$(\frac{1}{10}, \frac{1}{15})$	0.408456	0.409073	0.000616843
$(\frac{1}{10}, \frac{2}{5})$	0.376616	0.377193	0.000577055
$(\frac{1}{10}, \frac{11}{15})$	0.299292	0.299591	0.000299591
$(\frac{3}{10}, \frac{1}{15})$	0.273796	0.276999	0.00320283
$(\frac{3}{10}, \frac{2}{5})$	0.252453	0.255402	0.00294834
$(\frac{3}{10}, \frac{11}{15})$	0.200621	0.20153	0.00090935
$(\frac{1}{2}, \frac{1}{15})$	0.183531	0.187178	0.00364687
$(\frac{1}{2}, \frac{2}{5})$	0.169225	0.172311	0.00308633
$(\frac{1}{2}, \frac{11}{15})$	0.13448	0.132859	0.0016215
$(\frac{7}{10}, \frac{1}{15})$	0.123024	0.121958	0.00106683
$(\frac{7}{10}, \frac{2}{5})$	0.113435	0.111281	0.00215332
$(\frac{7}{10}, \frac{11}{15})$	0.183531	0.187178	0.00364687
$(\frac{9}{10}, \frac{1}{15})$	0.0824658	0.0799499	0.0025159
$(\frac{9}{10}, \frac{2}{5})$	0.0760375	0.0615089	0.0145286
$(\frac{9}{10}, \frac{11}{15})$	0.0604259	0.0582806	0.021453

Example 4.3.3. A nonlinear Volterra–Fredholm integral equation is

$$u(t,x) = f(t,x) + \int_0^t \int_0^1 F(x,\xi,\tau)\log(u(\xi,\tau))\,d\xi\,d\tau, \quad t \in [0,1], \tag{4.3.16}$$

where

$$\begin{aligned} F(x,\xi,\tau) &= -\tau(x-\xi), \\ f(t,x) &= -\frac{1}{4}\,e^{-2t}(x^2-2) - \frac{1}{12}\,t^2(2x-1)\left(4t - 3\log\left(\frac{2-x^2}{4}\right)\right). \end{aligned}$$

Then $u(t,x) = (\frac{2-x^2}{4})e^{-2t}$ is the true solution of the Eq. (4.3.16). The approximate solution with $n = 150, m = 80$ and error are shown in Table 4.9.

Table 4.9: The error of Example 4.3.3

Node	True solution u	Approximate solution $u_{150,80}$	Absolute error
$(\frac{1}{10}, \frac{1}{15})$	0.818713	0.818914	0.000201784
$(\frac{1}{10}, \frac{2}{5})$	0.818076	0.81893	0.000853889
$(\frac{1}{10}, \frac{11}{15})$	0.816529	0.818949	0.00241938
$(\frac{3}{10}, \frac{1}{15})$	0.5448702	0.549319	0.00061675
$(\frac{3}{10}, \frac{2}{5})$	0.54486	0.549357	0.00825307
$(\frac{3}{10}, \frac{11}{15})$	0.53553	0.549417	0.0138862
$(\frac{1}{2}, \frac{1}{15})$	0.367675	0.368534	0.000858552
$(\frac{1}{2}, \frac{2}{5})$	0.360522	0.368582	0.00806044
$(\frac{1}{2}, \frac{11}{15})$	0.34315	0.368661	0.025511
$(\frac{9}{10}, \frac{1}{15})$	0.165001	0.165976	0.000974688
$(\frac{9}{10}, \frac{2}{5})$	0.154588	0.166026	0.0114384
$(\frac{9}{10}, \frac{11}{15})$	0.129297	0.121906	0.007391

4.4 New Algorithm for Nonlinear Integro-Differential Equations

4.4.1 Introduction

Although the literature of integro-differential equation is now extensive, more research works tend to center around the global stability of steady states [115], the global existence of solutions [116], [117], [118]. However, our purpose is to study the exact solution for a class of nonlinear integro-differential equations of the form

$$\begin{cases} \mathbb{A}v^2(x,t) + v_t(x,t) + \int\limits_a^b k(x,\xi)v(\xi,t)d\xi + v(x,t) = f(x,t), \\ \qquad\qquad 0 \le x \le 1, \;\; 0 \le t \le T \\ v(x,0) = 0 \end{cases} \tag{4.4.1}$$

where $\mathbb{A}$ is a bounded linear operator, the kernel k might be unsymmetric.

4.4.2 Solving the Nonlinear Operator Equation

The reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$, $(\mathcal{D} = [a,b] \times [0,T], T > 0)$ has been given by (1.5.1), its reproducing kernel is $R_{(\xi,\eta)}(x,t) = r_{\mathbf{p}}(\mathbf{s})$. Define the reproducing kernel space

$${}^oW_2^{(1,2)}(\mathcal{D}) = \Big\{u(x,t)|\; u(x,t) \in W_2^{(1,2)}(\mathcal{D}) \text{ by (1.5.1)},\; u(x,0) = 0\Big\}.$$

One can obtain its reproducing kernel and writes $R_{(\xi,\eta)}(x,t) = R_{\mathbf{p}}(\mathbf{s})$, (Readers may get its representation similar to the Section 1.5.2).

We construct the direct product $\widetilde{W}(\mathcal{D}^2)$ of the reproducing kernel spaces ${}^oW_2^{(1,2)}(\mathcal{D})$ and ${}^oW_2^{(1,2)}(\mathcal{D})$, that is,

$$\widetilde{W}(\mathcal{D}^2) = {}^oW_2^{(1,2)}(\mathcal{D}) \otimes {}^oW_2^{(1,2)}(\mathcal{D}).$$

Theorem 4.4.1. *$\widetilde{W}(\mathcal{D}^2)$ is a reproducing kernel and its reproducing kernel is*

$$K_{(\mathbf{p}_1,\mathbf{p}_2)}(\mathbf{q}_1,\mathbf{q}_2) = R_{\mathbf{p}_1}(\mathbf{q}_1)R_{\mathbf{p}_2}(\mathbf{q}_2), \;\; \mathbf{p}_1,\, \mathbf{q}_1,\, \mathbf{p}_2,\, \mathbf{q}_2 \in \mathcal{D}.$$

This proof is omitted.

We shall give the exact solution of the nonlinear operator equation of the form

$$\begin{cases} \mathbb{A}v^2 + \mathbb{P}v + \mathbb{K}v + \mathbb{E}v = f, \\ v(x,0) = 0, \end{cases} \tag{4.4.2}$$

where $v(x,y) \in {}^{o}W_2^{(1,2)}(\mathcal{D})$, the operator $\mathbb{A} : {}^{o}W_2^{(1,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ is a bounded linear operator, and put

$$\mathbb{C} = \mathbb{P} + \mathbb{K} + \mathbb{E},$$

then Eq. (4.4.2) can be written

$$\mathbb{A}v^2 + \mathbb{C}v = f.$$

$\mathbb{P} : {}^{o}W_2^{(1,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ denotes partial differential operator, namely,

$$(\mathbb{P}v)(x,t) = \frac{\partial v(x,t)}{\partial t},$$

$\mathbb{K} : {}^{o}W_2^{(1,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ denotes integral operator, that is,

$$(\mathbb{K}u)(x,t) = \int_b^a k(x,z)u(z,t)dz,$$

and $\mathbb{E}$ denotes identical operator.

Obviously, identical operator $\mathbb{E}$ and partial differential operator $\mathbb{P}$ are bounded linear from ${}^{o}W_2^{(1,2)}(\mathcal{D})$ to $W_2^{(1,1)}(\mathcal{D})$. If $k(x,y) \in W_2^{(1,1)}(\mathcal{D})$ one can prove that integral operator $\mathbb{K}$ is bounded linear from ${}^{o}W_2^{(1,2)}(\mathcal{D})$ to $W_2^{(1,1)}(\mathcal{D})$.

(1) To solve the Eq. (4.4.2), we will construct the following operator $\mathbb{T} : W(\mathcal{D}^2) \longrightarrow W_2^{(1,1)}(\mathcal{D})$, which is defined by

$$[\mathbb{T}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}) = [\mathbb{L}_1 u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}) + [\mathbb{L}_2 u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}), \tag{4.4.3}$$

with

$$\begin{aligned}
[\mathbb{L}_1 u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}) &= \left[\mathbb{A}_{\mathbf{s}'}[\mathbb{B}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}')\right](\mathbf{s}),\\
[\mathbb{B}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}') &\stackrel{def}{=} u(\mathbf{s}',\mathbf{s}'),\\
[\mathbb{L}_2 u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}) &= \left\langle u(\mathbf{q}_1,\mathbf{q}_2), R_{\mathbf{s}_0}(\mathbf{q}_1)[\mathbb{C}^* R_{\mathbf{s}}(\mathbf{s}')](\mathbf{q}_2)\right\rangle_{\widetilde{W}},
\end{aligned}$$

where the fixed $\mathbf{s}_0 \in \mathcal{D}$.

Theorem 4.4.2. *$\mathbb{T} : W((\mathcal{D}^2)) \longrightarrow W(\mathcal{D})$, is a bounded linear operator.*

[119] gives the similar proof in details.

Our main motivation is that the nonlinear operator Eq. (4.4.2) is solved through transforming into linear operator Eq. (4.4.4). Now in the following, we will give the relation of solutions on two equations.

(2) Let a operator equation

$$[\mathbb{T}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}) = f(\mathbf{s}), \tag{4.4.4}$$

where f is given by (4.4.2).

Theorem 4.4.3. *If there exists a $\mathbf{s}_0 \in \mathcal{D}$, so that $v(\mathbf{s}_0) = 1$ then $v(\mathbf{s}) \in {}^oW_2^{(1,2)}(\mathcal{D})$ is the solution of Eq.* (4.4.2) *iff $v(\mathbf{q}_1)v(\mathbf{q}_2)$ is the solution of the Eq.* (4.4.4).

Proof. $\Longrightarrow$ Since $v(\mathbf{s}) \in {}^oW_2^{(1,2)}(\mathcal{D})$ is a solution of the Eq. (4.4.2), and there exists an $\mathbf{s}_0 \in \mathcal{D}$ such that $v(\mathbf{s}_0) = 1$, it follows that

$$\begin{aligned}
[\mathbb{T}(v(\mathbf{q}_1)v(\mathbf{q}_2))](\mathbf{s}) &= [\mathbb{L}_1 v(\mathbf{q}_1)v(\mathbf{q}_2)](\mathbf{s}) + [\mathbb{L}_2 v(\mathbf{q}_1)v(\mathbf{q}_2)](\mathbf{s}) \\
&= [\mathbb{A}_{\mathbf{s}'}[\mathbb{B}v(\mathbf{q}_1)v(\mathbf{q}_2)](\mathbf{s}')](\mathbf{s}) \\
&+ \langle v(\mathbf{q}_1)v(\mathbf{q}_2), R_{\mathbf{s}_0}(\mathbf{q}_1)[\mathbb{C}^* R_{\mathbf{s}}(\mathbf{s}')](\mathbf{q}_2)\rangle_{\widetilde{W}} \\
&= \left[\mathbb{A}_{\mathbf{s}'}[v(\mathbf{s}')v(\mathbf{s}')]\right](\mathbf{s}) \\
&+ \langle v(\mathbf{q}_1), R_{\mathbf{s}_0}(\mathbf{q}_1)\rangle_{{}_oW_2^{(1,2)}} \langle v(\mathbf{q}_2), [\mathbb{C}^* R_{\mathbf{s}}(\mathbf{s}')](\mathbf{q}_2)\rangle_{{}_oW_2^{(1,2)}} \\
&= (\mathbb{A}v^2(\mathbf{s}'))(\mathbf{s}) + v(\mathbf{s}_0)\langle [\mathbb{C}v(\mathbf{q}_2)](\mathbf{s}'), R_{\mathbf{s}}(\mathbf{s}')\rangle_{{}_oW_2^{(1,2)}} \\
&= (\mathbb{A}v^2(\mathbf{s}'))(\mathbf{s}) + v(\mathbf{s}_0)[Cv(\mathbf{q}_2)](\mathbf{s}) \\
&= (\mathbb{A}v^2)(\mathbf{s}) + (\mathbb{C}v)(\mathbf{s}) = f(\mathbf{s}).
\end{aligned}$$

Consequently, $v(\mathbf{q}_1)v(\mathbf{q}_2)$ is a solution of Eq. (4.4.4).

$\Longleftarrow$ By the proof of the above necessity, we have

$$\left[\mathbb{T}(v(\mathbf{q}_1)v(\mathbf{q}_2))\right](\mathbf{s}) = (\mathbb{A}v^2(\mathbf{s}'))(\mathbf{s}) + v(\mathbf{s}_0)\cdot[Cv(\mathbf{q}_2)](\mathbf{s}).$$

Because $v(\mathbf{q}_1)v(\mathbf{q}_2)$ is a solution of Eq. (4.4.4) and $v(\mathbf{s}_0) = 1$ the above formula is simplified to

$$\left[\mathbb{T}(v(\mathbf{q}_1)v(\mathbf{q}_2))\right](\mathbf{s}) = (\mathbb{A}v^2)(\mathbf{s}) + (\mathbb{C}v)(\mathbf{s}) = f(\mathbf{s}).$$

These complete the proof. □

The solution of Eq. (4.4.4) with the form $v(\mathbf{q}_1)v(\mathbf{q}_2)$ is called the separable solution. As a result, we can obtain the solution of the nonlinear operator Eq. (4.4.2) by looking for the separable solution of the linear operator Eq. (4.4.4).

(3) Let $\{\mathbf{s}_i = (x_i, t_i)\}_{i=1}^{\infty}$ be any dense subset of $\mathcal{D}$, put

$$\varphi_i(x,t) = r_{\mathbf{s}_i}(\mathbf{s}) = r_{(x_i,t_i)}(x,t),$$

$$\psi_i(\mathbf{q}_1,\mathbf{q}_2) \;=\; (\mathbb{T}^*\varphi_i)(\mathbf{q}_1,\mathbf{q}_2),$$

where $\mathbb{T}^*$ is conjugate operator of $\mathbb{T}$. Then $\{\varphi_i\}_{i=1}^{\infty}$ constitutes a complete system of space $W_2^{(1,1)}(\mathcal{D})$, and $\{\overline{\psi}_i\}_{i=1}^{\infty}$ is a complete orthonormal system of the subspace

$$\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})} = \overline{span[\{\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2)\}_{i=1}^{n}]} \subset \widetilde{W}(\mathcal{D}^2),$$

where $\{\overline{\psi}_i\}_{i=1}^{\infty}$ derives from Gram–Schmidt othoganolization process of $\{\psi_i\}_{i=1}^{\infty}$, that is,

$$\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2) = \sum_{k=1}^{i}\beta_{ik}\psi_k(\mathbf{q}_1,\mathbf{q}_2).$$

Lemma 4.4.1. *Suppose that $\mathbb{P}_n$ is a projection operator of space $\widetilde{W}(\mathcal{D}^2)$ into the subspace $\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}$, that is*

$$(\mathbb{P}_n u)(\mathbf{q}_1,\mathbf{q}_2) = \sum_{i=1}^{n}\langle u,\overline{\psi}_i\rangle_{\widetilde{W}}\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2).$$

If $u(\mathbf{q}_1,\mathbf{q}_2)$ is a solution of Eq. (4.4.4)*, then*

$$u_n(\mathbf{q}_1,\mathbf{q}_2) \overset{def}{=} \sum_{i=1}^{n}\left(\sum_{k=1}^{i}\beta_{ik}f(\mathbf{s}_k)\right)\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2) = (\mathbb{P}_n u)(\mathbf{q}_1,\mathbf{q}_2),$$

and $u_n(\mathbf{q}_1,\mathbf{q}_2)$ satisfies the Eq. (4.4.4) *on $\{\mathbf{s}_i\}_{i=1}^{n}$, that is, $[\mathbb{T}u_n(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}_i) = f(\mathbf{s}_i)$.*

Proof. Noting that $\mathbb{P}_n$ is a self-conjugate operator and $\varphi_i(\mathbf{s})$ is the reproducing kernel of space $W_2^{(1,1)}(\mathcal{D})$, it follows that

$$\begin{aligned}
[\mathbb{T}u_n(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}_i) &= \big\langle[\mathbb{T}u_n(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}),\varphi_i(\mathbf{s})\big\rangle_{W_2^{(1,1)}}\\
&= \big\langle[\mathbb{T}(\mathbb{P}_n u)(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}),\varphi_i(\mathbf{s})\big\rangle_{W_2^{(1,1)}}\\
&= \Big\langle[\mathbb{P}_n u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}_3,\mathbf{s}_4),[\mathbb{T}^*\varphi_i(\mathbf{s})](\mathbf{s}_3,\mathbf{s}_4)\Big\rangle_{\widetilde{W}}\\
&= \Big\langle u(\mathbf{q}_1,\mathbf{q}_2),\mathbb{P}_n\big[(\mathbb{T}^*\varphi_i(\mathbf{s}))(\mathbf{s}_3,\mathbf{s}_4)\big](\mathbf{q}_1,\mathbf{q}_2)\Big\rangle_{\widetilde{W}}\\
&= \Big\langle u(\mathbf{q}_1,\mathbf{q}_2),(\mathbb{T}^*\varphi_i(\mathbf{s}))(\mathbf{q}_1,\mathbf{q}_2)\Big\rangle_{\widetilde{W}}\\
&= \big\langle[\mathbb{T}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}),\varphi_i(\mathbf{s})\big\rangle_{W_2^{(1,1)}}\\
&= [\mathbb{T}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}_i) = f(\mathbf{s}_i).
\end{aligned}$$

$u_n(\mathbf{q}_1,\mathbf{q}_2) \longrightarrow u(\mathbf{q}_1,\mathbf{q}_2)$ by Lemma 4.4.1, so that

$$u_n(\mathbf{q}_1,\mathbf{q}_2)=\sum_{i=1}^{n}\left(\sum_{k=1}^{i}\beta_{ik}f(\mathbf{s}_k)\right)\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2)$$

is an approximate solution of Eq. (4.4.4). □

Corollary 4.4.1. *If the series* $\sum\limits_{i=1}^{n}\big(\sum\limits_{k=1}^{i}\beta_{ik}f(\mathbf{s}_k)\big)\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2)$ *converges, then* $u(\mathbf{q}_1,\mathbf{q}_2)=\sum\limits_{i=1}^{\infty}\big(\sum\limits_{k=1}^{i}\beta_{ik}f(\mathbf{s}_k)\big)\overline{\psi}_i(\mathbf{q}_1,\mathbf{q}_2)$ *is an exact solution of Eq.* (4.4.4).

Let $\mathrm{N}(\mathbb{T})$ be the null space of Eq. (4.4.4), that is,

$$\mathrm{N}(\mathbb{T})=\Big\{u(\mathbf{q}_1,\mathbf{q}_2)|\ [\mathbb{T}u(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s})=0\Big\}.$$

For the null space $\mathrm{N}(\mathbb{T})$, the following lemma holds.

Lemma 4.4.2. $\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}=\mathrm{N}(\mathbb{T})$, *that is,*

$$\mathrm{N}(\mathbb{T})\oplus\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}=\widetilde{W}(\mathcal{D}^2).$$

Proof. For any $h(\mathbf{q}_1,\mathbf{q}_2)\in\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}$,

$$\begin{aligned}[\mathbb{T}h(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}) &= \big\langle[\mathbb{T}h(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s}_3),r_M(\mathbf{s}_3)\big\rangle_{W_2^{(1,1)}}\\ &= \Big\langle h(\mathbf{q}_1,\mathbf{q}_2),[\mathbb{T}^*r_M(\mathbf{s}_3)](\mathbf{q}_1,\mathbf{q}_2)\Big\rangle_{\widetilde{W}}=0.\end{aligned}$$

Hence, $h(\mathbf{q}_1,\mathbf{q}_2)\in\mathrm{N}(\mathbb{T})$,

$$\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}\subset\mathrm{N}(\mathbb{T}).$$

On the other hand, for any $g(\mathbf{q}_1,\mathbf{q}_2)\in\mathrm{N}(\mathbb{T})$ and $u(\mathbf{s})\in W_2^{(1,1)}(\mathcal{D})$, $\mathbb{T}^*u\in\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}$, and

$$\big\langle(\mathbb{T}^*u)(\mathbf{q}_1,\mathbf{q}_2),g(\mathbf{q}_1,\mathbf{q}_2)\big\rangle_{\widetilde{W}}=\big\langle u(\mathbf{s}),[\mathbb{T}g(\mathbf{q}_1,\mathbf{q}_2)](\mathbf{s})\big\rangle_{W_2^{(1,1)}}=0.$$

So, $g(\mathbf{q}_1,\mathbf{q}_2)\in\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}$, that is,

$$\mathrm{N}(\mathbb{T})\subset\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}.$$

Summing up the above, $\overline{\mathbb{T}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}=\mathrm{N}(\mathbb{T})$. □

Applying Lemma 4.4.1 and Lemma 4.4.2, the following theorem is obtained.

Theorem 4.4.4. *Put* $\{\mathbf{s}_i\}_{i=1}^{\infty}$ *be any dense subset of* $\mathcal{D}$*, if Eq.* (4.4.4) *has a solution, then its solution* $u(\mathbf{q}_1, \mathbf{q}_2)$ *can be represented as follows,*

$$u(\mathbf{q}_1, \mathbf{q}_2) = \sum_{i=1}^{\infty} \left(\sum_{k=1}^{i} \beta_{ik} f(\mathbf{s}_k) \right) \overline{\psi}_i(\mathbf{q}_1, \mathbf{q}_2) + \omega(\mathbf{q}_1, \mathbf{q}_2),$$

where $\omega(\mathbf{q}_1, \mathbf{q}_2) \in \mathrm{N}(\mathbb{T})$.

This proof is omitted since it is obvious.

Assume $\{\rho_k\}_{k=1}^{\infty}$ is a normal orthogonal basis of space $\mathrm{N}(\mathbb{T})$, then $u(\mathbf{q}_1, \mathbf{q}_2)$ is written as

$$u(\mathbf{q}_1, \mathbf{q}_2) = \sum_{i=1}^{\infty} \left(\sum_{k=1}^{i} \beta_{ik} f(\mathbf{s}_k) \right) \overline{\psi}_i(\mathbf{q}_1, \mathbf{q}_2) + \sum_{i=1}^{\infty} c_i \rho_i(\mathbf{q}_1, \mathbf{q}_2),$$

where $\{c_i\}_{i=1}^{\infty} \in \ell^2$.

As a result, we can find the separable solution from all solutions of Eq. (4.4.4) to give the solution of Eq. (4.4.2).

4.4.3 The Algorithm of Finding the Separable Solution

According to Theorem 4.4.3, we must obtain the separable solution of Eq. (4.4.4) for solving the nonlinear operator Eq. (4.4.2). In the following, we give the numerical algorithm of finding the separable solution, and the following notations used are of the same meanings as before.

Let $\{\mathbf{s}_i = (x_i, y_i)\}_{i=1}^{\infty}$ be any dense subset of $\mathcal{D}$, $\{\overline{\psi}_i\}_{i=1}^{\infty}$ constitutes the normal orthogonal basis of space $\mathbb{T}^* W_2^{(1,1)}(\mathcal{D})$,

$$\{R_k(\mathbf{s}', \mathbf{s}'')\}_{k=1}^{\infty} = \left\{ R_{\mathbf{s}_i}(\mathbf{s}') R_{\mathbf{s}_j}(\mathbf{s}'') \right\}_{i,j=1}^{\infty} \subset \widetilde{W}(\mathcal{D}^2),$$

where the subscript k corresponds to certain permutation of (i, j). Hence, through the Gram–Schmidt process, the sequence $\{\overline{\psi}_i\} \cup \{\widetilde{r}_k\}$ is the normal orthogonal basis defined on $\widetilde{W}(\mathcal{D}^2)$, derived from the sequence $\{\psi_i\} \cup \{r_k\}$, where

$$\begin{aligned}
\overline{\psi}_1 &= \frac{\psi_1}{\|\psi_1\|}, \ldots, \\
\overline{\psi}_{i+1} &= \frac{\psi_{i+1} - \sum_{k=1}^{i} \langle \psi_{i+1}, \overline{\psi}_k \rangle \overline{\psi}_k}{\left\| \psi_{i+1} - \sum_{k=1}^{i} \langle \psi_{i+1}, \overline{\psi}_k \rangle \overline{\psi}_k \right\|}, \ldots, \quad i = 2, 3, \ldots
\end{aligned}$$

$$\widetilde{r}_1 = \frac{r_1 - \sum_{k=1}^{\infty} \langle r_1, \overline{\psi}_k \rangle \overline{\psi}_k}{\left\| r_1 - \sum_{k=1}^{\infty} \langle r_1, \overline{\psi}_k \rangle \overline{\psi}_k \right\|}, \ldots,$$

$$\widetilde{r}_i = \frac{r_i - \sum_{k=1}^{\infty} \langle r_i, \overline{\psi}_k \rangle \overline{\psi}_k - \sum_{k=1}^{i-1} \langle r_i, \widetilde{r}_k \rangle \widetilde{r}_k}{\left\| r_i - \sum_{k=1}^{\infty} \langle r_i, \overline{\psi}_k \rangle \overline{\psi}_k - \sum_{k=1}^{i-1} \langle r_i, \widetilde{\rho}_k \rangle \widetilde{r}_k \right\|}, \ldots, \quad i = 2, 3, \ldots.$$

It is easy to see that

$$\overline{\psi}_i = \sum_{k=1}^{i} \beta_{ik} \psi_k, \quad \widetilde{r}_i = \sum_{k=1}^{\infty} \beta_{ik} \psi_k + \sum_{k=1}^{i} c_{ik} r_k,$$

$\{\widetilde{r}_i\}_{i=1}^{\infty}$ is a complete orthonormal system in $\mathrm{N}(\mathbb{T})$.

We denote the separable solution of Eq. (4.4.4) by $u(\mathbf{s}', \mathbf{s}'')$, so it follows that $u(\mathbf{s}', \mathbf{s}'') = u(\mathbf{s}', \mathbf{s}_0) u(\mathbf{s}'', \mathbf{s}_0)$, and $u(\mathbf{s}', \mathbf{s}_0)$ is the solution of Eq. (4.4.2). Applying Corollary 4.4.1, we have

$$\begin{aligned} u(\mathbf{s}', \mathbf{s}'') &= \sum_{i=1}^{\infty} \left(\sum_{k=1}^{i} \beta_{ik} f(\mathbf{s}_k) \right) \overline{\psi}_i(\mathbf{s}', \mathbf{s}'') + \omega \\ &= \sum_{i=1}^{\infty} \left(\sum_{k=1}^{i} \beta_{ik} f(\mathbf{s}_k) \right) \overline{\psi}_i(\mathbf{s}', \mathbf{s}'') + \sum_{k=1}^{\infty} \alpha_k \widetilde{r}_k(\mathbf{s}', \mathbf{s}''), \end{aligned}$$

where $\alpha_k = \langle u(\mathbf{s}', \mathbf{s}''), \widetilde{r}_k(\mathbf{s}', \mathbf{s}'') \rangle_{\widetilde{W}}$. Put $\gamma_k = u(\mathbf{s}_k, \mathbf{s}_0)$, simplify α_k as follows,

$$\begin{aligned} \alpha_k &= \left\langle u(\mathbf{s}', \mathbf{s}''), \widetilde{r}_k(\mathbf{s}', \mathbf{s}'') \right\rangle_{\widetilde{W}} = \left\langle u(\mathbf{s}', \mathbf{s}''), \sum_{i=1}^{\infty} \beta_{ki} \psi_i + \sum_{i=1}^{k} c_{ki} r_i \right\rangle_{\widetilde{W}} \\ &= \sum_{i=1}^{\infty} \beta_{ki} \left\langle u(\mathbf{s}', \mathbf{s}''), \mathbb{T}^* \varphi_i \right\rangle_{\widetilde{W}} + \sum_{i=1}^{k} c_{ki} \left\langle u(\mathbf{s}', \mathbf{s}''), K_{\mathbf{s}_{i_1}}(\mathbf{s}') K_{\mathbf{s}_{i_2}}(\mathbf{s}'') \right\rangle_{\widetilde{W}} \\ &= \sum_{i=1}^{\infty} \beta_{ki} \left\langle \mathbb{T} u(\mathbf{s}', \mathbf{s}''), \varphi_i \right\rangle_{W_2^{(1,1)}} + \sum_{i=1}^{k} c_{ki} u(\mathbf{s}_{i_1}, \mathbf{s}_{i_2}) \\ &= \sum_{i=1}^{\infty} \beta_{ki} f(\mathbf{s}_i) + \sum_{i=1}^{k} c_{ki} \gamma_{i_1} \gamma_{i_2}, \end{aligned}$$

where i corresponds to certain permutation of (i_1, i_2). Observing that the above derivation, $\{\alpha_k\}_{k=1}^{\infty}$ has been substituted for the unknown function $u(\mathbf{s}', \mathbf{s}'')$. So, employing the relation of α_k and γ_k, we can look for γ_k such that for any $\varepsilon > 0$,

$$\min \left\| u(\mathbf{s}', \mathbf{s}'') - u(\mathbf{s}', \mathbf{s}_0) u(\mathbf{s}'', \mathbf{s}_0) \right\|_{\widetilde{W}} < \varepsilon.$$

Running this algorithm with Mathematica 4.2 yields $\{\alpha_k\}_{k=1}^{\infty}$ to give the solution of Eq. (4.4.4).

4.4.4 Numerical Experiments

We employ the method introduced in Section 4.4.2 and Section 4.4.3 to solve the following two special integro-differential equations, in which different forms of the kernel function $k(x, y)$ are considered. And the error and images of true solution and approximated solution for every example are given through symbolic and numerical computations performed by using Mathematica 4.2. It shows that the algorithm is valid.

Example 4.4.1. Integro-differential equation

$$\begin{cases} \mathbb{A}v^2 + (\mathbb{P} + \mathbb{K} + \mathbb{E})v = f, \\ v(x, 0) = 0, \end{cases}$$

where $\mathbf{s}_0 = (\frac{\pi}{2}, 1), \mathbb{A} = \mathbb{E}$,

$$\begin{aligned} [\mathbb{P}v(t,\tau)](x,y) &= \left.\frac{\partial v(t,\tau)}{\partial t}\right|_{(x,y)}, \\ [\mathbb{K}v(t,\tau)](x,y) &= \int_0^2 (x - 6z)v(z,y)dz, \end{aligned}$$

the true solution is $v(x, y) = y \sin x$. The numerical results and images are given by the following Table 4.10 and Figure 4.2. Moreover, we choose arbitrarily some t, such as $t = 0.5$ and $t = 1.0$, to demonstrate that the approximate solution coincides with the exact solution through the image. See for Figure 4.1.

Table 4.10: Numerical results of Example 4.4.1

Nodes (x, t)	$v(x, t)$	$u_{appro}(x, t)$	Relative error
(12/19,1/7)	0.0843457	0.0843315	1.68E-04
(14/19,2/7)	0.191986	0.191961	1.32E-04
(26/19,2/7)	0.279883	0.279749	4.79E-04
(16/19,3/7)	0.319734	0.319825	2.83E-04
(30/19,3/7)	0.428557	0.428455	2.39E-04
(10/19,4/7)	0.287058	0.287195	4.85E-04
(18/19,4/7)	0.463933	0.46364	6.31E-04
(24/19,5/7)	0.412572	0.40996	6.65E-03
(36/19,5/7)	0.219386	0.219641	1.17E-03
(6/19,6/7)	1.47286	1.46833	3.07E-03
(18/19,6/7)	0.783193	0.789973	8.66E-03
(0.0,1.0)	2.71828	2.71828	3.74E-12
(8/19,1.0)	1.78416	1.77881	3.0E-03
(28/19,1.0)	0.622704	0.622404	4.81E-04

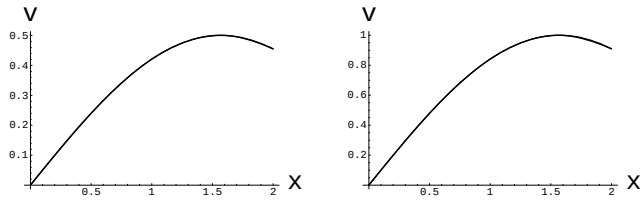


Figure 4.1: Left are curves of $v(x, 0.5)$ and $u_{appro}(x, 0.5)$, Right are curves of $v(x, 1.0)$ and $u_{appro}(x, 1.0)$.

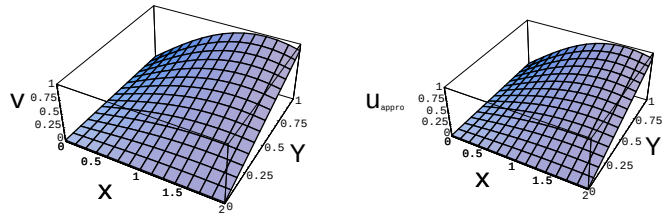


Figure 4.2: Left is the true solution $v(x, t)$, Right is the approximate solution $u_{appro}(x, t)$.

Example 4.4.2. Integro-differential equation

$$\begin{cases} \mathbb{A}v^2 + (\mathbb{P} + \mathbb{K} + \mathbb{E})v = f \\ v(x, 0) = 0 \end{cases}$$

where $M_0 = (1, 1)$, $\mathbb{A} = \mathbb{E}$,

$$\begin{aligned} [\mathbb{P}v(\eta, \tau)](x, t) &= (-7)\ \frac{\partial v(\eta, \tau)}{\partial \tau}\bigg|_{(x,t)}, \\ [\mathbb{K}v(\eta, \tau)](x, t) &= \int_0^2 xzv(z, t)dz, \end{aligned}$$

the true solution is $v(x, t) = te^{(t-x)}$. The numerical results and images are given by the Table 4.11 and Figure 4.4. Moreover, we choose arbitrarily some t, such as $t = 0.5$ and $t = 1.0$, to demonstrate that the approximate solution coincides with the exact solution through the image. See for Figure 4.3.

Table 4.11: Numerical results of Example 4.4.2

Nodes (x,t)	$v(x,t)$	$u_{appro}(x,t)$	Relative error
(0.0,2/7)	0.380203	0.38077	1.5E-03
(26/19,2/7)	0.096765	0.0967429	2.3E-04
(8/19,3/7)	0.431806	0.429352	5.7E-03
(28/19,3/7)	0.150708	0.150755	3.1E-04
(20/19,4/7)	0.353165	0.35348	8.9E-04
(26/19,4/7)	0.257533	0.257567	1.3E-04
(12/19,5/7)	0.775874	0.775421	5.8E-04
(36/19,5/7)	0.219386	0.219641	1.2E-03
(6/19,6/7)	1.47286	1.46833	3.1E-03
(22/19,6/7)	0.63451	0.639626	8.1E-03
(10/19,1.0)	1.6059	1.60779	1.2E-03
(2.0,1.0)	0.367879	0.367879	6.9E-12

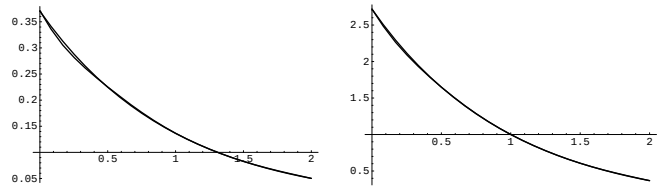


Figure 4.3: Left are curves of $v(x,0.5)$ and $u_{appro}(x,0.5)$, Right are curves of the approximate solution $v(x,1.0)$ and $u_{appro}(x,1.0)$.

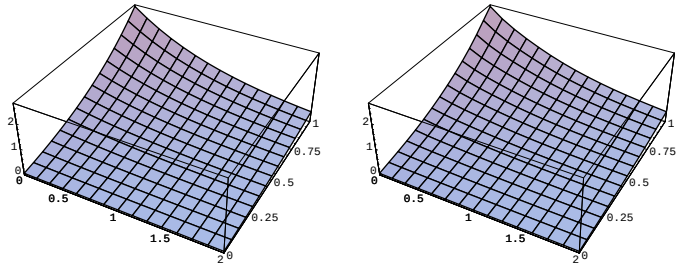


Figure 4.4: Left is the true solution $v(x,t)$, Right is the approximate solution $u_{appro}(x,t)$.

Chapter 5

Differential Equations

Many mathematics models in science, technology and engineering can be described with differential equation. The basic equations for the modern natural science are differential equations. And in recent years, many differential equations have been established to reflect various objective phenomena. However, people could hardly obtain the precise solution of these equations. With the high-speed development of modern computer, the numerical method of the differential equations has witnessed unprecedented development.

This chapter is to introduce a new numerical method with obvious advantages to the readers. The method is very simple and easy to realize in the computer, and the derivative of the approximate solution will uniformly converge to the derivative of precise solution.

5.1 Solving Variable-Coefficient Burgers Equation

5.1.1 Introduction

Systems of partial differential equations have attracted much attention in studying evolution equations describing wave propagation, investigating the shallow water waves [120], [121], and in examining the chemical reaction-diffusion model of Brusselator [122]. While Burgers equation has been found to describe various phenomena such as a mathematic model of turbulence [123]. Burgers equation is a very important fluid dynamical model both for the conceptual understanding of physical flow and testing various new solution approaches. Moreover, simulation of Burgers equation is a natural first step towards developing methods for computations of complex flows. The existence and uniqueness of solutions to Burgers equation have been proved [124], [125]. An analytic solution of STDBE was given by Fletcher using the Hopf-cole transformation [126] and the exact solution of one-dimensional Burgers equation have been surveyed by Beton and Platzman [127]. In the past few years, a great deal of effort have been expended to compute the solution of burgers equation.

In recent years, many authors have used a variety of numerical techniques based on finite-difference, finite-element, cubic spline function and boundary element method in attempting to solve the equation [128]–[132].

New exact solutions of one-dimensional inhomogeneous Burgers equation are proposed [133]. The transformation properties of a variable-coefficient Burgers equation was discussed by Cnristocloulos Sophocleous [134]. Here, we are concerned with the following one-dimensional inhomogeneous variable-coefficient Burgers equations with initial and boundary conditions

$$\begin{cases} U_t + a_1(x,t)UU_x + a_2(x,t)U_{xx} = f(x,t), \ \ 0 \le x \le 1, \ \ 0 \le t \le 1, \\ U(0,x) = g_1(x), \\ U(t,0) = g_2(t), \quad U(t,1) = g_3(t) \end{cases} \tag{5.1.1}$$

where $a_1(x,t)$, $a_2(x,t)$ and $f(x,t)$ are continuous. Through transformation of function, Eq. (5.1.1) can be converted into the following equivalent form:

$$\begin{cases} u_t + b_1(x,t)u_{xx} + b_2(x,t)u + b_3(x,t)u_x - b_4(x,t)uu_x - f(x,t) = 0, \\ u(0,x) = 0, \\ u(t,0) = 0, \quad u(t,1) = 0, \end{cases} \tag{5.1.2}$$

where $0 \le x \le 1$, $0 \le t \le 1$. In order to solve Eq. (5.1.1), it therefore remains only to solve Eq. (5.1.2).

Put

$$\mathbb{L}u \equiv u_t + b_1(x,t)u_{xx} + b_2(x,t)u + b_3(x,t)u_x,$$

then Eq. (5.1.2) can further be converted into the following form:

$$\begin{cases} \mathbb{L}u = f(x,t) + b_4(x,t)uu_x, \\ u(0,x) = 0, \\ u(t,0) = 0, \quad u(t,1) = 0. \end{cases} \tag{5.1.3}$$

5.1.2 The Solution of Eq. (5.1.3)

Let $\mathcal{D} = [0,1] \times [0,1]$. The reproducing kernel space

$${}^0W_2^{(2,3)}(\mathcal{D}) = \left\{ u(t,x) |\ u(t,x) \in W_2^{(2,3)}(\mathcal{D}),\ u(t,0) = u(t,1) = u(0,x) = 0 \right\} \tag{5.1.4}$$

and their reproducing kernel is $K_{(\xi,\eta)}(t,x) = Q_\xi(t)R_\eta(x)$, where

$$Q_\xi(t) \ = \ \begin{cases} -\dfrac{1}{6}\, t(t^2 - 3\xi(2+t)), & t \le \xi, \\ -\dfrac{\xi^3}{6} + \dfrac{1}{2}\, \xi(2+\xi)t, & t > \xi, \end{cases} \tag{5.1.5}$$

$$R_\eta(x) = \begin{cases} \frac{1}{18720}\Big(x(\eta-1)\big(-120\eta(2+\eta)(18+(\eta-6)\eta) \\ \quad +5x^3\eta(2+\eta)(18+(\eta-6)\eta) \\ \quad -x^4\big(156+\eta(2+\eta)(18+(\eta-6)\eta)\big) \\ \quad -30x\eta(-120+\eta(6+(\eta-4)\eta)) \\ \quad -10x^2\eta(-120+\eta(6+(\eta-4)\eta))\Big)\Big), & x \le \eta, \\ \frac{1}{18720}\Big((x-1)\eta\big(-120x(2+x)(18+(x-6)x) \\ \quad +30x\big(120-x(6+(x-4)x)\eta \\ \quad +10x(120-x(6+(x-4)x))\eta^2 \\ \quad +5x(2+x)\big(18+(x-6)x\eta^3 \\ \quad -(156+x(2+x)(18+(x-6)x))\eta^4\big)\big)\Big), & x > \eta. \end{cases} \tag{5.1.6}$$

The reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$ is define by (1.5.1) and their reproducing kernel is

$$r_{(\xi,\eta)}(t,x) = \widetilde{r}_\xi(t)\widetilde{r}_\eta(x),$$

where $\widetilde{r}_\xi(t)$ by (1.4.1).

Next, the solution of Eq. (5.1.3) is given in the reproducing kernel space ${}^oW_2^{(2,3)}(\mathcal{D})$.

Theorem 5.1.1. *For Eq.* (5.1.3)*, if* $\{\mathbf{p}_i\}_{i=1}^\infty$ *is dense on* $\mathcal{D}$*, then* $\{\psi_i(\mathbf{p})\}_{i=1}^\infty$ *is the complete system of* ${}^oW_2^{(2,3)}(\mathcal{D})$*.*

The proof is omitted.

It is clear that $\mathbb{L} : {}^oW_2^{(2,3)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ is a bounded linear operator. Put

$$\mathbf{p} = (t,x), \quad \mathbf{p}_i = (t_i, x_i),$$
$$\varphi_i(\mathbf{p}) = r_{\mathbf{p}_i}(\mathbf{p}) \ \text{ and } \ \psi_i(\mathbf{p}) = \mathbb{L}^*\varphi_i(\mathbf{p}),$$

where $\mathbb{L}^*$ is the conjugate operator of $\mathbb{L}$. The normal orthogonal basis $\{\overline{\psi}_i(\mathbf{p})\}_{i=1}^\infty$ of ${}^oW_2^{(2,3)}(\mathcal{D})$ can be derived from Gram–Schmidt orthogonalization process of $\{\psi_i(\mathbf{p})\}_{i=1}^\infty$,

$$\overline{\psi}_i(\mathbf{p}) = \sum_{k=1}^{i} \beta_{ik}\psi_k(\mathbf{p}) \ \ (\beta_{ii} > 0, \ \ i = 1, 2, \ldots). \tag{5.1.7}$$

Theorem 5.1.2. *If* $\{\mathbf{p}_i\}_{i=1}^\infty$ *is dense on* $\mathcal{D}$ *and the solution of Eq.* (5.1.3) *is unique, then the solution of Eq.* (5.1.3) *satisfies the form*

$$u(\mathbf{p}) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}F(\mathbf{p}_k, u(\mathbf{p}_k)u_x(\mathbf{p}_k))\overline{\psi}_i(\mathbf{p}), \tag{5.1.8}$$

where

$$F(\mathbf{p}, u(\mathbf{p})u_x(\mathbf{p})) = f(\mathbf{p}) + b_4(\mathbf{p})uu_x(\mathbf{p}).$$

Proof. Because $\{\overline{\psi}_i(\mathbf{p})\}_{i=1}^{\infty}$ is the complete normal orthogonal basis of ${}^oW_2^{(2,3)}(\mathcal{D})$ and

$$\langle v(\mathbf{p}), \varphi_i(\mathbf{p})\rangle_{{}_oW_2^{(2,3)}} = v(\mathbf{p}_i),$$

for each $v(\mathbf{p}) \in {}^oW_2^{(2,3)}(\mathcal{D})$, we have

$$\begin{aligned}
u(\mathbf{p}) &= \sum_{i=1}^{\infty}\langle u(\mathbf{p}), \overline{\psi}_i(\mathbf{p})\rangle_{{}_oW_2^{(2,3)}}\overline{\psi}_i(\mathbf{p}) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\langle u(\mathbf{p}), L^*\varphi_k(\mathbf{p})\rangle_{{}_oW_2^{(2,3)}}\overline{\psi}_i(\mathbf{p}) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\langle \mathbb{L}u(\mathbf{p}), \varphi_k(\mathbf{p})\rangle_{W_2^{(1,1)}}\overline{\psi}_i(\mathbf{p}) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\left\langle F(\mathbf{p}, u(\mathbf{p})u_x(\mathbf{p})), \varphi_k(\mathbf{p})\right\rangle_{W_2^{(1,1)}}\overline{\psi}_i(\mathbf{p}) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}F(\mathbf{p}_k, u(\mathbf{p}_k)u_x(\mathbf{p}_k))\overline{\psi}_i(\mathbf{p}) \qquad (5.1.9)
\end{aligned}$$

and the proof of the theorem is complete. □

5.1.3 The Implementation Method

In (5.1.8), denotes

$$u(\mathbf{p}) = \sum_{i=1}^{\infty} A_i\overline{\psi}_i(\mathbf{p}), \qquad (5.1.10)$$

where

$$A_i = \sum_{k=1}^{i}\beta_{ik}F\big(\mathbf{p}_k, u(\mathbf{p}_k)u_x(\mathbf{p}_k)\big).$$

Let $\mathbf{p}_1 = (0, 0)$, it follows that $u(\mathbf{p}_1) = 0$ and $u(\mathbf{p}_1)u_x(\mathbf{p}_1) = 0$ from the initial and boundary conditions of Eq. (5.1.3), then $F(\mathbf{p}_1, u(\mathbf{p}_1)u_x(\mathbf{p}_1))$ is known. Considering the numerical computation, we put $u_0(\mathbf{p}_1) = u(\mathbf{p}_1)$ and define the n-term approximation to $u(\mathbf{p})$ by

$$u_n(\mathbf{p}) = \sum_{i=1}^{n} B_i\overline{\psi}_i(\mathbf{p}), \qquad (5.1.11)$$

where

$$\begin{aligned} B_1 &= \beta_{11} F\big(\mathbf{p}_1, u_0(\mathbf{p}_1)\partial_x u_0(\mathbf{p}_1)\big), \\ u_1(\mathbf{p}) &= B_1 \overline{\psi}_1(\mathbf{p}), \\ B_2 &= \sum_{k=1}^{2} \beta_{2k} F\big(\mathbf{p}_k, u_{k-1}(\mathbf{p}_k)(u_{k-1})_x(\mathbf{p}_k)\big), \\ u_2(\mathbf{p}) &= \sum_{i=1}^{2} B_i \overline{\psi}_i(\mathbf{p}), \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ B_n &= \sum_{k=1}^{n} \beta_{nk} F\big(\mathbf{p}_k, u_{k-1}(\mathbf{p}_k)(u_{k-1})_x(\mathbf{p}_k)\big). \end{aligned} \tag{5.1.12}$$

Next, the convergence of $u_n(\mathbf{p})$ will be proved. Now, two Lemmas are given first.

Lemma 5.1.1. *If $u(t,x) \in {}^{o}W_2^{(2,3)}(\mathcal{D})$, then there exists a constant $C > 0$ such that*

$$|u(t,x)| \le C\|u(t,x)\|_{{}^{o}W_2^{(2,3)}}, \quad |u_t(t,x)| \le C\|u(t,x)\|_{{}^{o}W_2^{(2,3)}},$$
$$|u_x(t,x)| \le C\|u(t,x)\|_{{}^{o}W_2^{(2,3)}}, \quad |u_{xt}(t,x)| \le C\|u(t,x)\|_{{}^{o}W_2^{(2,3)}},$$
$$|u_{xx}(t,x)| \le C\|u(t,x)\|_{{}^{o}W_2^{(2,3)}}.$$

Proof. For any $(t,x) \in \mathcal{D}$,

$$|u(t,x)| = \big|\langle u(\xi,\eta), K_{(t,x)}(\xi,\eta)\rangle_{{}^{o}W_2^{(2,3)}}\big| \le \|u(\xi,\eta)\|_{{}^{o}W_2^{(2,3)}} \, \|K_{t,x}(\xi,\eta)\|_{{}^{o}W_2^{(2,3)}},$$

then there exists a constant $C_1 > 0$ such that

$$|u(t,x)| \le C_1\|u(\xi,\eta)\|_{{}^{o}W_2^{(2,3)}}.$$

Note that

$$\begin{aligned} |u_x(t,x)| &= \big|\langle u(\xi,\eta), \partial_x K_{(t,x)}(\xi,\eta)\rangle_{{}^{o}W_2^{(2,3)}}\big| \\ &\le \|u(\xi,\eta)\|_{{}^{o}W_2^{(2,3)}} \big\|\partial_x K_{(t,x)}(\xi,\eta)\big\|_{W_3}. \end{aligned}$$

Hence there exists a constant $C_2 > 0$ such that

$$|u_x(t,x)| \le C_2\|u(\xi,\eta)\|_{{}^{o}W_2^{(2,3)}}.$$

In like manner, there exist constants $C_3 > 0, C_4 > 0, C_5 > 0$ such that

$$|u_t(t,x)| \le C_3\|u(t,x)\|_{{}^{o}W_2^{(2,3)}}, \quad |u_{xt}(t,x)| \le C_4\|u(t,x)\|_{{}^{o}W_2^{(2,3)}}$$

and

$$|u_{xx}(t,x)| \le C_5\|u(t,x)\|_{{}^{o}W_2^{(2,3)}}.$$

Let $C = \max\{C_1, C_2, C_3, C_4, C_5\}$, then the proof is complete. □

Lemma 5.1.2. *If* $u_n \overset{\|\cdot\|_{{}_oW_2^{(2,3)}}}{\longrightarrow} \overline{u}$ $(n \longrightarrow \infty)$, $\mathbf{p}_n = (t_n, x_n) \longrightarrow M = (t, x)$ $(n \longrightarrow \infty)$, $\|u_n\|_{{}_oW_2^{(2,3)}}$ *is bounded and* $F(\mathbf{p}, u(\mathbf{p})u_x(\mathbf{p}))$ *is continuous, then*

$$F(\mathbf{p}_n, u_{n-1}(\mathbf{p}_n)\partial_x u_{n-1}(\mathbf{p}_n)) \longrightarrow F\big(\mathbf{p}, \overline{u}(\mathbf{p})\overline{u}_x(\mathbf{p})\big) \ \text{ as } \ n \longrightarrow \infty.$$

Proof. From the given condition $u_n \overset{\|\cdot\|}{\longrightarrow} \overline{u}$ $(n \longrightarrow \infty)$ and Lemma 5.1.1, it follows that, for any $\mathbf{p} \in \mathcal{D}$

$$\begin{gathered}
\big|u_{n-1}(\mathbf{p}) - \overline{u}(\mathbf{p})\big| \longrightarrow 0 \ \ (n \longrightarrow \infty), \\
\big|\partial_x u_{n-1}(\mathbf{p}) - \overline{u}_x(\mathbf{p})\big| \longrightarrow 0 \ \ (n \longrightarrow \infty) \ \ \text{(uniformly)}, \\
|\partial_t u_{n-1}| \le C\|u_{n-1}\|_{{}_oW_2^{(2,3)}}, \quad |\partial_x u_{n-1}| \le C\|u_{n-1}\|_{{}_oW_2^{(2,3)}}, \\
|\partial_{xt} u_{n-1}| \le C\|u_{n-1}\|_{{}_oW_2^{(2,3)}} \ \text{ and } \ |\partial_{x^2} u_{n-1}| \le C\|u_{n-1}\|_{{}_oW_2^{(2,3)}}.
\end{gathered}$$

Observing that

$$\begin{aligned}
\big|u_{n-1}(\mathbf{p}_n) - \overline{u}(\mathbf{p})\big| &= \big|u_{n-1}(\mathbf{p}_n) - u_{n-1}(\mathbf{p}) + u_{n-1}(\mathbf{p}) - \overline{u}(\mathbf{p})\big| \\
&\le |\partial_t u_{n-1}|\,|t_n - t| + |\partial_x u_{n-1}|\,|x_n - x| + \big|u_{n-1}(\mathbf{p}) - \overline{u}(\mathbf{p})\big|
\end{aligned}$$

and

$$\begin{aligned}
\big|\partial_x u_{n-1}(\mathbf{p}_n) - \overline{u}_x(\mathbf{p})\big| &= \big|\partial_x u_{n-1}(\mathbf{p}_n) - \partial_x u_{n-1}(\mathbf{p}) + \partial_x u_{n-1}(\mathbf{p}) - \overline{u}_x(\mathbf{p})\big| \\
&\le |\partial_{xt} u_{n-1}|\,|t_n - t| + |\partial_{x^2} u_{n-1}|\,|x_n - x| \\
&+ \big|\partial_x u_{n-1}(\mathbf{p}) - \overline{u}_x(\mathbf{p})\big|,
\end{aligned}$$

by the boundedness of $\|u_n\|_{{}_oW_2^{(2,3)}}$, we get

$$\big|u_{n-1}(\mathbf{p}_n) - \overline{u}(\mathbf{p})\big| \longrightarrow 0 \ \text{ and } \ \big|\partial_x u_{n-1}(\mathbf{p}_n) - \overline{u}_x(\mathbf{p})\big| \longrightarrow 0 \ \text{ as } \ n \longrightarrow \infty.$$

The continuation of $F(\mathbf{p}, u(\mathbf{p})u_x(\mathbf{p}))$ implies that

$$F\big(\mathbf{p}_n, u_{n-1}(\mathbf{p}_n)\partial_x u_{n-1}(\mathbf{p}_n)\big) \longrightarrow F\big(\mathbf{p}, \overline{u}(\mathbf{p})\overline{u}_x(\mathbf{p})\big) \ \text{ as } \ n \longrightarrow \infty. \qquad \square$$

Theorem 5.1.3. *Suppose that* $\|u_n\|$ *is bounded in* (5.1.11)*, if* $\{\mathbf{p}_i\}_{i=1}^{\infty}$ *is dense on* $\mathcal{D}$*, then the n-term approximate solution* $u_n(\mathbf{p})$ *derived from the above method converges to the exact solution* $u(\mathbf{p})$ *of Eq.* (5.1.3) *and*

$$u(\mathbf{p}) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(\mathbf{p}), \tag{5.1.13}$$

where B_i *is given by* (5.1.12)*.*

Proof. First, we will prove the convergence of u_n.

By (5.1.11), we infer that

$$u_{n+1}(\mathbf{p}) = u_n(\mathbf{p}) + B_{n+1}\overline{\psi}_{n+1}(\mathbf{p}). \tag{5.1.14}$$

From the orthogonality of $\{\overline{\psi}_i\}_{i=1}^{\infty}$, it follows that

$$\|u_{n+1}\|^2_{{}^oW_2^{(2,3)}} = \|u_n\|^2_{{}^oW_2^{(2,3)}} + B^2_{n+1}. \tag{5.1.15}$$

From (5.1.15), it holds that $\|u_{n+1}\|_{{}^oW_2^{(2,3)}} \geq \|u_n\|_{{}^oW_2^{(2,3)}}$. Due to the condition that $\|u_n\|_{{}^oW_2^{(2,3)}}$ is bounded, $\|u_n\|_{{}^oW_2^{(2,3)}}$ is convergent and there exists a constant c such that

$$\sum_{i=1}^{\infty} B_i^2 = c.$$

This implies that

$$B_i \in \ell^2, \;\; i = 1, 2, \ldots .$$

If $m > n$, then

$$\|u_m - u_n\|^2_{{}^oW_2^{(2,3)}} = \big\|u_m - u_{m-1} + u_{m-1} - u_{m-2} + \cdots + u_{n+1} - u_n\big\|^2_{{}^oW_2^{(2,3)}}.$$

In view of $(u_m - u_{m-1}) \perp (u_{m-1} - u_{m-2}) \perp \cdots \perp (u_{n+1} - u_n)$, it follows that

$$\|u_m - u_n\|^2_{{}^oW_2^{(2,3)}} = \|u_m - u_{m-1}\|^2_{{}^oW_2^{(2,3)}} + \cdots + \|u_{n+1} - u_n\|^2_{{}^oW_2^{(2,3)}}.$$

Furthermore

$$\|u_m - u_{m-1}\|^2_{{}^oW_2^{(2,3)}} = B_m^2.$$

Consequently,

$$\|u_m - u_n\|^2_{{}^oW_2^{(2,3)}} = \sum_{l=n+1}^{m} B_l^2 \longrightarrow 0 \;\text{ as }\; n \longrightarrow \infty.$$

The completeness of ${}^oW_2^{(2,3)}(\mathcal{D})$ shows that

$$u_m \xrightarrow{\|\cdot\|} \overline{u} \;\text{ as }\; m \longrightarrow \infty.$$

Second, we will prove that $\overline{u}$ is the solution of Eq. (5.1.3).

It follows that, on taking limits in (5.1.11)

$$\overline{u}(\mathbf{p}) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(\mathbf{p}).$$

Since

$$\begin{aligned}(\mathbb{L}\overline{u})(\mathbf{p}_n) &= \sum_{i=1}^{\infty} B_i \langle \mathbb{L}\overline{\psi}_i, \varphi_n \rangle_{W_2^{(1,1)}} \\ &= \sum_{i=1}^{\infty} B_i \langle \overline{\psi}_i, \mathbb{L}^*\varphi_n \rangle_{{}^oW_2^{(2,3)}} = \sum_{i=1}^{\infty} B_i \langle \overline{\psi}_i, \psi_n \rangle_{{}^oW_2^{(2,3)}},\end{aligned}$$

it follows that

$$\begin{aligned}\sum_{j=1}^{n} \beta_{nj}(\mathbb{L}\overline{u})(\mathbf{p}_j) &= \sum_{i=1}^{\infty} B_i \left\langle \overline{\psi}_i, \sum_{j=1}^{n} \beta_{nj}\psi_j \right\rangle_{{}^oW_2^{(2,3)}} \\ &= \sum_{i=1}^{\infty} B_i \langle \overline{\psi}_i, \overline{\psi}_n \rangle_{{}^oW_2^{(2,3)}} = B_n.\end{aligned}$$

If $n = 1$, then

$$(\mathbb{L}\overline{u})(\mathbf{p}_1) = F\big(\mathbf{p}_1, u_0(\mathbf{p}_1)\partial_x u_0(\mathbf{p}_1)\big).$$

If $n = 2$, then

$$\begin{aligned}&\beta_{21}(\mathbb{L}\overline{u})(\mathbf{p}_1) + \beta_{22}(L\overline{u})(\mathbf{p}_2) \\ &\qquad = \beta_{21}F\big(\mathbf{p}_1, u_0(\mathbf{p}_1)\partial_x u_0(\mathbf{p}_1)\big) + \beta_{22}F\big(\mathbf{p}_2, u_1(\mathbf{p}_2)\partial_x u_1(\mathbf{p}_2)\big).\end{aligned}$$

It is clear that

$$(L\overline{u})(\mathbf{p}_2) = F\big(\mathbf{p}_2, u_1(\mathbf{p}_2)\partial_x u_1(\mathbf{p}_2)\big).$$

Moreover, it is easy to see by induction that

$$(\mathbb{L}\overline{u})(\mathbf{p}_j) = F\big(\mathbf{p}_j, u_{j-1}(\mathbf{p}_j)\partial_x u_{j-1}(\mathbf{p}_j)\big), \quad j = 1, 2, \ldots. \tag{5.1.16}$$

Since $\{\mathbf{p}_i\}_{i=1}^{\infty}$ is dense on $\mathcal{D}$, for $\forall\, \mathbf{z} \in \mathcal{D}$, there exists a subsequence $\{\mathbf{p}_{n_j}\}$ such that

$$\mathbf{p}_{n_j} \longrightarrow \mathbf{z} \ \text{ as } \ j \longrightarrow \infty.$$

Hence, let $j \longrightarrow \infty$ in (5.1.16), by the convergence of u_n and Lemma 5.1.2, we have

$$(\mathbb{L}\overline{u})(\mathbf{z}) = F\big(\mathbf{z}, \overline{u}(\mathbf{z})\overline{u}_x(\mathbf{z})\big). \tag{5.1.17}$$

From (5.1.17), it follows that $\overline{u}(\mathbf{p})$ satisfies Eq. (5.1.3).

Since $\overline{\psi}_i(\mathbf{p}) \in {}^oW_2^{(2,3)}(\mathcal{D})$, clearly, $\overline{u}(\mathbf{p})$ satisfies the initial and boundary conditions of Eq. (5.1.3).

That is, $\overline{u}(\mathbf{p})$ is the solution of Eq. (5.1.3). The application of the uniqueness of solution to Eq. (5.1.3) then yields that

$$u(\mathbf{p}) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(\mathbf{p}). \tag{5.1.18}$$

The proof is complete. □

Theorem 5.1.4. *Assume $u(\mathbf{p})$ is the solution of Eq. (5.1.3) and $r_n(\mathbf{p})$ is the approximate error between $u_n(\mathbf{p})$ and $u(\mathbf{p})$, where $u_n(\mathbf{p})$ is given by (5.1.11). Then the error $r_n(\mathbf{p})$ is monotone decreasing in the sense of $\|\cdot\|_{{}_oW_2^{(2,3)}}$.*

Proof. From (5.1.11), (5.1.13), it follows that

$$\|r_n\|^2_{{}_oW_2^{(2,3)}} = \left\| \sum_{i=n+1}^{\infty} B_i \overline{\psi}_i(\mathbf{p}) \right\|^2_{{}_oW_2^{(2,3)}} = \sum_{i=n+1}^{\infty} B_i^2. \tag{5.1.19}$$

(5.1.19) shows that the error r_n is monotone decreasing in the sense of $\|\cdot\|_{{}_oW_2^{(2,3)}}$. □

Table 5.1: Comparison of results at $t = 0.08$

x	True solution $u(x,t)$	Approximate solution u_{100}	Absolute error
0.08	-0.00588	-0.00579	0.00008
0.16	-0.01075	-0.01059	0.00016
0.24	-0.01459	-0.01437	0.00021
0.32	-0.01740	-0.01715	0.00025
0.40	-0.01920	-0.01892	0.00027
0.48	-0.01996	-0.01969	0.00027
0.56	-0.01971	-0.01944	0.00026
0.64	-0.01843	-0.01818	0.00024
0.72	-0.01612	-0.01591	0.00020
0.80	-0.01280	-0.01263	0.00016
0.88	-0.00844	-0.00834	0.00010
0.96	-0.00307	-0.00302	0.00004

5.1.4 Numerical Experiments

Some numerical examples are studied to demonstrate the accuracy of the present method. In the process of computation, all the symbolic and numerical computations performed by using Mathematica 4.2. Results obtained by the method are compared with the exact solution of each example and are found to be in good agreement with each other.

Example 5.1.1. Consider the equation

$$\begin{cases} u_t + (xt+1)u_{xx} + x^2 u + (x+1)u_x - t\sin(x)uu_x = f(x,t), \\ u(0,x) = 0, \\ u(t,0) = 0, \quad u(t,1) = 0, \end{cases}$$

Table 5.2: Comparison of results

(t,x)	True solution $u(x,t)$	Approximate solution u_{100}	Absolute error
(0.08,0.08)	-0.00588	-0.00579	0.00008
(0.16.0.16)	-0.02150	-0.02121	0.00028
(0.24.0.24)	-0.04377	-0.04321	0.00055
(0.32,0.32)	-0.06963	-0.06877	0.00086
(0.40,0.40)	-0.09600	-0.09483	0.00116
(0.48,0.48)	-0.11980	-0.11834	0.00146
(0.56,0.56)	-0.13798	-0.13628	0.00170
(0.64,0.64)	-0.14745	-0.14558	0.00186
(0.72,0.72)	-0.14515	-0.14322	0.00192
(0.80,0.80)	-0.12800	-0.12611	0.00188
(0.88,0.88)	-0.09292	-0.09124	0.00168
(0.96,0.96)	-0.03686	-0.03587	0.00098

where $0 \le t \le 1$, $0 \le x \le 1$ and

$$f(t,x)=2t(xt+1)+x^2-x+t(2x-1)(x^2-x)t^2\sin(x)+x^3(x-1)t+(x+1)(2x-1)t.$$

The true solution is $(x^2 - x)t$. Using our method, we choose 100 points on $\mathcal{D} = [0,1] \times [0,1]$ and obtain the approximate solution $u_{100}(t,x)$ on $\mathcal{D}$. The numerical results are given in the following Table 5.1–5.2.

Table 5.3: Comparison of results at $t = 0.48$

x	True solution $u(x,t)$	Approximate solution u_{121}	Absolute error
0.08	-0.03395	-0.03347	0.00047
0.16	-0.06179	-0.06095	0.00084
0.24	-0.08342	-0.08230	0.00112
0.32	-0.09877	-0.09747	0.00130
0.40	-0.10789	-0.10648	0.00140
0.48	-0.11088	-0.10945	0.00142
0.56	-0.10792	-0.10655	0.00137
0.64	-0.09927	-0.09803	0.00124
0.72	-0.08525	-0.08420	0.00104
0.80	-0.06625	-0.06545	0.00080
0.88	-0.04270	-0.04220	0.00050
0.96	-0.01513	-0.01495	0.00017

Table 5.4: Comparison of results

(t,x)	True solution $u(x,t)$	Approximate solution u_{121}	absolute error
(0.08,0.08)	-0.00587	-0.00578	0.00009
(0.16.0.16)	-0.02132	-0.02101	0.00030
(0.24.0.24)	-0.04294	-0.04236	0.00058
(0.32,0.32)	-0.06728	-0.06640	0.00088
(0.40,0.40)	-0.09098	-0.08981	0.00142
(0.48,0.48)	-0.11088	-0.10945	0.00146
(0.56,0.56)	-0.12415	-0.12254	0.00160
(0.64,0.64)	-0.12839	-0.12672	0.00166
(0.72,0.72)	-0.12174	-0.12015	0.00158
(0.80,0.80)	-0.10292	-0.10158	0.00133
(0.88,0.88)	-0.07128	-0.07039	0.00089
(0.96,0.96)	-0.02684	-0.02653	0.00030

Example 5.1.2. Consider the equation

$$\begin{cases} u_t + (xt+1)u_{xx} + x^2u + (x+1)u_x - t\sin(x)uu_x = f(x,t), \\ u(0,x) = 0, \\ u(t,0) = 0, \quad u(t,1) = 0 \end{cases}$$

where $0 \le t \le 1$, $0 \le x \le 1$ and

$$\begin{aligned} f(t,x) &= (x-1)\cos(t)\sin(t) + x^2(x-1)\sin(x)\sin(t) \\ &+ (1+xt)\big(2\cos(x) - (x-1)\sin(x)\big)\sin(t) \\ &+ (x+1)\big((x-1)\cos(x)\sin(t) + \sin(x)\sin(t)\big) \\ &+ (x-1)t(\sin(x))^2\sin(t)\big((x-1)\cos(x)\sin(t) + \sin(x)\sin(t)\big). \end{aligned}$$

The true solution is $(x-1)\sin(x)\sin(t)$. Using our method, we choose 121 points on $\mathcal{D} = [0,1] \times [0,1]$ and obtain the approximate solution $u_{121}(t,x)$ on $\mathcal{D}$. The numerical results are given in the following Table 5.3–5.4.

5.2 The Nonlinear Age-Structured Population Model

5.2.1 Numerical Experiments

We will discuss how to solve nonlinear age-structured population model [138]

$$\text{(I)}\begin{cases}\dfrac{\partial p(t,x)}{\partial t}+\dfrac{\partial p(t,x)}{\partial x}=-(d_1(x)+d_2(x)P(t))p(t,x), & t\geq 0,\ \ 0\leq x<A,\\ t=0:\ p(0,x)=p_0(x), & 0\leq x<A,\\ x=0:\ p(t,0)=\displaystyle\int_a^A[b_1(\xi)-b_2(\xi)P(t)]p(t,\xi)\,d\xi, & t\geq 0,\\ P(t)=\displaystyle\int_0^A p(t,x)\,dx, & t\geq 0,\end{cases}$$

where t, x denote time and age respectively, $P(t)$ denotes the total population number at time t, $p(t,x)$ is the age-specific density of individuals of age x at time t, which means $\int_a^{a+\Delta a} p(t,x)dx$ gives the number of individuals that have age between a and $a+\Delta a$ at time t, $d_1(x)$ is the natural death rate(without considering competition), $d_2(x)P(t)$ is the increase of death rate when considering competition; $b_1(x)$ is the natural fertility(without considering competition), $b_2(x)P(t)$ is the decrease of fertility when considering competition, a denotes the lowest age that an individual can bear, and A is the maximum age that an individual of the population may reach.

In general, model (I) can be written as

$$\begin{cases}\dfrac{\partial p(t,x)}{\partial t}+\dfrac{\partial p(t,x)}{\partial x}=-\mu(x,I_\mu(t),t)p(t,x), & t\geq 0,\ \ 0\leq x<A,\\ t=0:\ p(0,x)=p_0(x), & 0\leq x<A,\\ x=0:\ p(t,0)=\displaystyle\int_0^A\beta(\xi,I_\beta(t),t)p(t,\xi)\,d\xi & t\geq 0,\end{cases}\tag{5.2.1}$$

where μ, β denote fertility and death rate respectively,

$$\begin{aligned}I_\mu(t) &= \int_0^A\gamma_\mu(x)p(t,x)dx,\\ I_\beta(t) &= \int_0^A\gamma_\beta(x)p(t,x)dx,\end{aligned}$$

and $\gamma_\mu(x)$,$\gamma_\beta(x)$ are weight functions. If let $\gamma_\mu(x)\equiv\gamma_\beta(x)\equiv 1$,

$$\mu(x,z,t) = d_1(x)+d_2(x)z,$$

$$\beta(x,z,t) \quad = \quad b_1(x) - b_2(x)z,$$

then Eq. (5.2.1) will become population model (I).

In 1995, for the case $\gamma_\mu(x) = \gamma_\beta(x) \equiv 1$, based on Runge-Kutta method, Abia and López-Marcos brought forward a difference schemes for solving Eq. (5.2.1) [135]. In 1997, for the case $\gamma_\mu(x) = \gamma_\beta(x) \equiv 1$ and β, μ are independent of t, Iannelli and Kim introduced a spline algorithm for solving Eq. (5.2.1) [136]. Further, in 1999, Abia and López-Marcos formulated a second-order finite difference schemes for solving Eq. (5.2.1) [137].

5.2.2 Solving Population Model can be Turned into Solving Operator Equation (IV)

We assume $\mathcal{D} = [a,b] \times [c,d]$. The reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$ is defined by (1.5.1), its reproducing kernel is $R_\xi^{ab}(t)R_\eta^{cd}(x)$, where

$$R_\xi^{ab}(t) = \begin{cases} 1 - a + t, & t \le \xi, \\ 1 - a + \xi, & t > \xi. \end{cases} \tag{5.2.2}$$

In addition, similarly, the reproducing kernel space is given by

$${}^oW_2^{(2,2)}(\mathcal{D}) = \Big\{u(t,x) |\ u(t,x) \in W_2^{(2,2)}(\mathcal{D}) \text{ by (1.5.1)},\ u(a,x) = 0\Big\},$$

its reproducing kernel is $\overline{K}_\xi^{ab}(t)K_\eta^{cd}(x)$, where

$$\overline{K}_\xi^{ab}(t) \quad = \quad \begin{cases} -\dfrac{1}{6}\,(a-t)(2a^2 - t^2 + 3\xi(2+t) - a(6+3\xi+t)), & t \le \xi, \\ -\dfrac{1}{6}\,(a-\xi)(2a^2 - \xi^2 + 6t + 3\xi t - a(6+\xi+3t)), & t > \xi, \end{cases} \tag{5.2.3}$$

$$K_\eta^{cd}(x) \quad = \quad \begin{cases} \dfrac{1}{6}\big(6 - 2c^3 - x^3 + 3\eta x(2+x) + 3c^2(2+\eta+x) \\ \qquad -6c(\eta + x + \eta x)\big), & t \le \eta, \\ \dfrac{1}{6}\big(6 - 2c^3 - \eta^3 + 6\eta x + 3\eta^2 x + 3c^2(2+\eta+x) \\ \qquad -6c(\eta + x + \eta x)\big), & t > \eta. \end{cases} \tag{5.2.4}$$

We assume that

(H1) zero-order compatible condition is valid, that is

$$p_0(0) = \int_a^A \big[b_1(\xi) - b_2(\xi)P(0)\big]p_0(\xi)d\xi \tag{5.2.5}$$

(H2) First-order compatible condition holds, that is,

$$\begin{aligned} p_0'(0)+\mu(0,0)p_0(0) &= \int_a^A \big[b_1(\xi)-b_2(\xi)P(0)\big]\left[\frac{\partial p(0,\xi)}{\partial \xi}+\mu(0,\xi)p_0(\xi)\right]d\xi \\ &+ \int_a^A b_2(\xi)p_0(\xi)\,d\xi\cdot P'(t-x)\Big|_{x=t}, \end{aligned}$$

where

$$\begin{aligned} \mu(t,x) &= d_1(x)+d_2(x)P(t), \\ P'(t-x)\big|_{x=t} &= p_0(0)-\int_0^A \big[\mu(0,\eta)p_0(\eta)\big]d\eta, \\ P(0) &= \int_0^A p_0(x)dx = C\ (Constant). \end{aligned}$$

(H3) On interval $[0,A]$, $p_0(x)$, $0\le p_0(x)\le\alpha$, is properly smooth, and when $x\longrightarrow A_{-0}$, $p_0(x)\longrightarrow 0=p_0(A)$.

(H4) On interval $[0,A]$, $0\le d_1(x)$, $0\le d_2(x)\le\beta$, are properly smooth. When $x\longrightarrow A_{-0}$, we have

$$d_1(x)\longrightarrow+\infty \ \text{ and } \ \int_0^x d_1(\xi)\,d\xi\longrightarrow\infty.$$

(H5) On interval $[0,A]$, $0\le b_1(x)\le C_1$, $0\le b_2(x)\le C_2$ and are properly smooth.

(H6) Take B, $A-1\le B<A$, such that $d_1(x)$, $d_2(x)\in C^{(1)}[0,B]$, where $C^{(1)}[0,B]$ denotes the set of all functions with first continuous derivative.

In the above assumptions, α,β,C_1,C_2 are all positive constants.

In [138], upon the assumption (H1)–(H5), the author proved the theorem on the existence and uniqueness of the global solution for Eqs. (I). From the process of its proof we know that Eqs. (I) still has a unique global solution when we restrict $t\in[0,n]$, where $n\in\mathbb{R}^+$.

Here, due to the need of our method, we also assume that (H6) is valid. Fur-

thermore, we assume that under the assumptions (H1)–(H6), Eq.

$$\text{(II)}\begin{cases}\dfrac{\partial p(t,x)}{\partial t}+\dfrac{\partial p(t,x)}{\partial x}=-(d_1(x)+d_2(x)P(t))p(t,x), & 0\le t\le n,\ \ 0\le x<B,\\ p(0,x)=p_0(x), & 0\le x<B,\\ p(t,0)=\displaystyle\int_a^B\big[b_1(\xi)-b_2(\xi)P(t)\big]p(t,\xi)d\xi, & 0\le t\le n,\\ P(t)=\displaystyle\int_0^B p(t,x)dx,\end{cases}$$

still have a unique global solution, where $n\in R^+$.

In the subsequent part of this section, we consider how to solve Eqs. (II).

5.2.3-1 Solving Eq. (II) can turned into solving Eq. (IV)

Let

$$p_1(t,x)=p(t,x)-p_0(x).$$

Write

$$P_1(t)=\int_0^B p_1(t,x)dx,\quad \lambda_0=\int_0^B p_0(x)\,dx. \tag{5.2.6}$$

Then Eqs. (II) can be equivalently turned into Eqs.

$$\text{(III)}\ \frac{\partial p_1}{\partial t}+\frac{\partial p_1}{\partial x}=-(d_1(x)+d_2(x)P(t))(p_1+p_0(x))-p_0'(x), \tag{5.2.7}$$
$$0\le t\le n,\ \ 0\le x<B,$$

$$\begin{aligned}p_1(0,x)&=0,\ \ 0\le x<B,\\ p_1(t,0)&=-p_0(0),\\ &\quad+\int_a^B(b_1(\xi)-b_2(\xi)P(t))(p_1(t,\xi)+p_0(\xi))d\xi,\ \ 0\le t\le n,\\ P(t)&=\int_0^B p(t,x)dx.\end{aligned} \tag{5.2.8}$$

Integrating both sides of (5.2.7) for x from 0 to x_1, we get

$$\int_0^{x_1}\frac{\partial p_1(t,x)}{\partial t}\,dx+p_1(t,x_1)-p_1(t,0)$$

$$= -\int_0^{x_1} \big(d_1(x) + d_2(x)P(t)\big)(p_1 + p_0(x))\,dx - p_0(x_1) + p_0(0). \quad (5.2.9)$$

Put

$$b(x) = b_1(x) - \lambda_0 b_2(x), \;\; \lambda_1 = \int_a^B b(x)p_0(x)\,dx, \;\; \lambda_2 = \int_a^B b_2(x)p_0(x)\,dx \quad (5.2.10)$$

then by (5.2.8), we have

$$\begin{aligned}
p_1(t,0) &= -p_0(0) + \int_a^B \big[b_1(\xi) - b_2(\xi)(P_1(t) + \lambda_0)\big](p_1(t,\xi) + p_0(\xi))\,d\xi \\
&= -p_0(0) + \int_a^B \big[b(\xi) - b_2(\xi)P_1(t)\big](p_1(t,\xi) + p_0(\xi))d\xi \\
&= -p_0(0) + \int_a^B b(\xi)p_1(t,\xi)\,d\xi + \int_a^B b(\xi)p_0(\xi)\,d\xi \\
&- P_1(t)\int_a^B b_2(\xi)p_0(\xi)\,d\xi - P_1(t)\int_a^B b_2(\xi)p_1(t,\xi)\,d\xi \\
&= -p_0(0) + \lambda_1 + \int_a^B b(\xi)p_1(t,\xi)d\xi - \lambda_2 P_1(t) \\
&- P_1(t)\int_a^B b_2(\xi)p_1(t,\xi)\,d\xi. \qquad (5.2.11)
\end{aligned}$$

Set

$$d(x) = d_1(x) + \lambda_0 d_2(x), \;\; f_1(x_1) = \int_0^{x_1} d(x)p_0(x)\,dx \quad (5.2.12)$$

then

$$\begin{aligned}
&\int_0^{x_1} \big[d_1(x) + d_2(x)P(t)\big](p_1 + p_0(x))\,dx \\
&\qquad = \int_0^{x_1} \big[d_1(x) + d_2(x)(P_1(t) + \lambda_0)\big](p_1 + p_0(x))\,dx
\end{aligned}$$

$$= \int_0^{x_1} [d(x) + d_2(x)P_1(t)](p_1 + p_0(x))\,dx$$

$$= f_1(x_1) + \int_0^{x_1} d(x)p_1(t,x)\,dx + P_1(t)\int_0^{x_1} d_2(x)p_1(t,x)\,dx$$

$$+ P_1(t)\int_0^{x_1} d_2(x)p_0(x)\,dx. \tag{5.2.13}$$

Substituting (5.2.11), (5.2.13) into (5.2.9) and rearranging it, we obtain

$$P_1(t)\left[\int_a^B b_2(\xi)p_1(t,\xi)d\xi + \int_0^{x_1} d_2(x)p_1(t,x)dx\right] + \int_0^{x_1}\frac{\partial p_1(t,x)}{\partial t}dx$$

$$+p_1(t,x_1) + P_1(t)\left[\lambda_2 + \int_0^{x_1} d_2(x)p_0(x)dx\right]$$

$$-\int_a^B b(\xi)p_1(t,\xi)d\xi + \int_0^{x_1} d(x)p_1(t,x)dx$$

$$= \lambda_1 - p_0(x_1) - f_1(x_1), \tag{5.2.14}$$

where $P_1(t)$, λ_0, λ_1, λ_2, $b(x)$, $d(x)$, $f_1(x_1)$ are respectively defined by (5.2.6), (5.2.10), (5.2.12) respectively.

Define operators

$$(\mathbb{A}p_1)(t,x_1) = \int_0^{x_1}\frac{\partial p_1(t,x)}{\partial t}dx + p_1(t,x_1), \tag{5.2.15}$$

$$(\mathbb{B}p_1)(t,x_1) = P_1(t) = \int_0^B p_1(t,x)dx, \tag{5.2.16}$$

$$(\mathbb{B}_1p_1)(t,x_1) = (\mathbb{B}p_1)(t,x_1)\left[\lambda_2 + \int_0^{x_1} d_2(x)p_0(x)dx\right], \tag{5.2.17}$$

$$(\mathbb{C}_1p_1)(t,x_1) = \int_a^B b_2(\xi)p_1(t,\xi)d\xi, \tag{5.2.18}$$

$$(\mathbb{C}_2p_1)(t,x_1) = \int_0^{x_1} d_2(x)p_1(t,x)dx, \tag{5.2.19}$$

$$(\mathbb{C}p_1)(t,x_1) = (\mathbb{C}_1p_1)(t,x_1) + (\mathbb{C}_2p_1)(t,x_1), \tag{5.2.20}$$

$$(\mathbb{D}_1p_1)(t,x_1) = \int_a^B b(\xi)p_1(t,\xi)d\xi, \tag{5.2.21}$$

$$(\mathbb{D}_2p_1)(t,x_1) = \int_0^{x_1} d(x)p_1(t,x)dx, \tag{5.2.22}$$

$$(\mathbb{D}p_1)(t,x_1) = -(\mathbb{D}_1p_1)(t,x_1) + (\mathbb{D}_2p_1)(t,x_1), \tag{5.2.23}$$

$$f(t,x_1) = \lambda_1 - f_1(x_1) - p_0(x_1), \tag{5.2.24}$$

then (5.2.15) becomes

$$\mathbb{B}p_1 \cdot \mathbb{C}p_1 + (\mathbb{A} + \mathbb{B}_1 + \mathbb{D})p_1 = f(t,x_1).$$

Furthermore, let

$$(\mathbb{F}p_1)(t,x_1) = (\mathbb{A}p_1)(t,x_1) + (\mathbb{B}_1p_1)(t,x_1) + (\mathbb{D}p_1)(t,x_1), \tag{5.2.25}$$

then (5.2.2) turns into

$$\mathbb{B}p_1 \cdot \mathbb{C}p_1 + \mathbb{F}p_1 = f(t,x_1).$$

Theorem 5.2.1. *Eq.* (II) *equivalent to*

$$\text{(IV)} \quad \begin{cases} \mathbb{B}p_1 \cdot \mathbb{C}p_1 + \mathbb{F}p_1 = f(t,x_1), & 0 \le t \le n, \ \ 0 \le x \le B, \\ p_1(0,x_1) = 0, & 0 \le x \le B. \end{cases} \tag{5.2.26}$$

Proof. Since Eq. (II) is equivalent to Eq. (III), it's enough to show that Eq. (III) is equivalent to Eq. (IV). From the above reasoning we know (III)$\Longrightarrow$(IV). Next we will prove (IV)$\Longrightarrow$(III). In fact, Eq. (5.2.26) is the other form of Eq. (5.2.2). Differentiating on both sides of (5.2.2) for x_1, we get (5.2.7). Subsequently integrating on both sides of (5.2.7) for x from 0 to x_1 we obtain (5.2.9). Comparing (5.2.9) with (5.2.2), one gets (5.2.8). Hence, from (5.2.26), (5.2.7) and (5.2.8) are deduced. This finishes (IV)$\Longrightarrow$(III). □

5.2.3-2 The Boundedness of Operators

We'll prove operators $\mathbb{A}$, $\mathbb{B}$, $\mathbb{B}_1$, $\mathbb{C}_1$, $\mathbb{C}_2$, $\mathbb{C}$, $\mathbb{D}_1$, $\mathbb{D}_2$, $\mathbb{D}$ defined by (5.2.15)–(5.2.23) are all bounded linear operators of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.

Using Hölder inequality, we can easily obtain the following lemma.

Lemma 5.2.1. *Suppose that* $p(t,x) \in L^2(\mathcal{D})$, *then*

$$q(t,x_1) = \int_0^{x_1} p(t,x)dx \in L^2(\mathcal{D}).$$

Theorem 5.2.2. $\mathbb{A}$ *defined by* (5.2.15) *is a bounded linear operator of* ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.

Proof. We first prove that if $p \in {}^oW_2^{(2,2)}(\mathcal{D})$, then $\mathbb{A}p \in W_2^{(1,1)}(\mathcal{D})$. Due to $p \in {}^oW_2^{(2,2)}(\Omega) \subset W_2^{(1,1)}(\mathcal{D})$, it's enough to prove that

$$q(t,x_1) = \int_0^{x_1} \frac{\partial p(t,x)}{\partial t}\,dx \in W_2^{(1,1)}(\mathcal{D}).$$

In fact, from $p \in {}^oW_2^{(2,2)}(\mathcal{D})$ it follows that $\frac{\partial p(t,x)}{\partial t}$ is absolutely continuous on domain $\mathcal{D}$, and $\frac{\partial^2 p(t,x)}{\partial t^2} \in L^2(\mathcal{D})$. We get that $q(t,x_1)$ is absolutely continuous on $\mathcal{D}$. On the other hand, we have

$$\begin{aligned}
\frac{\partial q(t,x_1)}{\partial x_1} &= \frac{\partial p(t,x_1)}{\partial t} \in L^2(\mathcal{D}),\\
\frac{\partial q(t,x_1)}{\partial t} &= \int_0^{x_1} \frac{\partial^2 p(t,x)}{\partial t^2}\,dx_1 \in L^2(\mathcal{D}) \ \text{(obtained by Lemma 2.1)},\\
\frac{\partial q(t,x_1)}{\partial t \partial x_1} &= \frac{\partial^2 p(t,x_1)}{\partial t^2} \in L^2(\mathcal{D}).
\end{aligned}$$

Therefore, $q(t,x_1) \in W_2^{(1,1)}(\mathcal{D})$.

Next, we will prove $\mathbb{A}$ defined by (5.2.15) is a bounded linear operator of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$. Obviously, $\mathbb{A}$ is linear. Next, we'll show that $\mathbb{A}$ is bounded. Denote by I the embedding operator of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ and define operator $\mathbb{A}_1$ as follows

$$(\mathbb{A}_1 p)(t,x_1) = \int_0^{x_1} \frac{\partial p(t,x)}{\partial t}\,dx, \ \ \forall\, p \in {}^oW_2^{(2,2)}(\mathcal{D}),$$

then $\mathbb{A} = I + \mathbb{A}_1$. The boundedness of I can be obtained by

$$\|Ip\|_{W_2^{(1,1)}} = \|p\|_{W_2^{(1,1)}} \le \|p\|_{{}^oW_2^{(2,2)}}, \ \ \forall\, p \in {}^oW_2^{(2,2)}(\mathcal{D}).$$

While the boundedness of $\mathbb{A}_1$ can be proved as follows: for any $p \in {}^oW_2^{(2,2)}(\mathcal{D})$, from Hölder inequality it follows that

$$(\mathbb{A}_1 p)^2(t,x_1) = \left(\int_0^{x_1} p_t'(t,x)dx\right)^2 \le \int_0^{x_1} (p_t'(t,x))^2 dx \int_0^{x_1} 1^2 dx$$

$$\leq \quad B\int_0^B (p_t'(t,x))^2 dx.$$

therefore,

$$\int_0^n\int_0^B (\mathbb{A}_1 p)^2(t,x_1)\,dx_1\,dt \quad \leq \quad B\int_0^B\int_0^n\int_0^B p_t'^2(t,x)\,dx\,dt\,dx_1 \tag{5.2.27}$$

$$\leq \quad B^2\int_0^n\int_0^B (p_t'(t,x))^2\,dx\,dt \leq B^2\|p\|^2_{{}^oW_2^{(2,2)}}. \tag{5.2.28}$$

Similarly,

$$\iint_{\mathcal{D}}\left[\frac{\partial}{\partial t}(\mathbb{A}_1 p)(t,x_1)\right]^2 d\sigma \quad \leq \quad B^2\|p\|^2_{{}^oW_2^{(2,2)}},$$

$$\iint_{\mathcal{D}}\left[\frac{\partial}{\partial x_1}(\mathbb{A}_1 p)(t,x_1)\right]^2 d\sigma \quad \leq \quad \|p\|^2_{{}^oW_2^{(2,2)}},$$

$$\iint_{\mathcal{D}}\left[\frac{\partial^2}{\partial t\partial x_1}(\mathbb{A}_1 p)(t,x_1)\right]^2 d\sigma \quad \leq \quad \|p\|^2_{{}^oW_2^{(2,2)}},$$

$$\|\mathbb{A}_1 p\|^2_{W_2^{(1,1)}} \quad = \quad \iint_{\mathcal{D}}(\mathbb{A}_1 p)^2 + \left(\frac{\partial}{\partial x_1}\mathbb{A}_1 p\right)^2 + \left(\frac{\partial}{\partial t}\mathbb{A}_1 p\right)^2 + \left(\frac{\partial^2}{\partial t\partial x_1}\mathbb{A}_1 p\right)^2 d\sigma$$

$$\leq \quad 2(B^2+1)\|p\|^2_{{}^oW_2^{(2,2)}}. \tag{5.2.29}$$

This completes our proof. □

It can be proved similarly that $\mathbb{C}_2$, $\mathbb{D}_2$ are both bounded linear operators of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.

Imitating the proof of Theory 5.2.2, we can also prove the following two theorems.

Theorem 5.2.3. *Suppose that $b(x) \in C[0,B]$, $\alpha, \beta \in [0,B]$, $\alpha < \beta$. Define operator*

$$(\mathbb{K}p)(t,x) = \int_\alpha^\beta b(\xi)p(t,\xi)d\xi, \ \ \forall\, p \in {}^oW_2^{(2,2)}(\mathcal{D}), \tag{5.2.30}$$

then $\mathbb{K}$ is a bounded linear operator of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.

Corollary 5.2.1. *$\mathbb{B}$, $\mathbb{C}_1$, $\mathbb{D}_1$ defined by* (5.2.16), (5.2.18) *and* (5.2.21) *are all bounded linear operators of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.*

Theorem 5.2.4. *Operator $\mathbb{B}_1$ defined by* (5.2.17) *is a bounded linear operator of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.*

5.2.3 The Exact Solution of Eq.(IV)

From the (5.2.24) it follows that $f(t, x_1)$ can be regarded as a function of $W_2^{(1,1)}(\mathcal{D})$.

By the definition of ${}^oW_2^{(2,2)}(\mathcal{D})$, we know that solving Eq. (IV) is equivalent to solving operator Eq. (5.2.31),

$$\mathbb{B}p_1 \cdot \mathbb{C}p_1 + \mathbb{F}p_1 = f(t, x_1), \;\; p_1 \in {}^oW_2^{(2,2)}(\mathcal{D}), \;\; f(t, x_1) \in W_2^{(1,1)}(\mathcal{D}) \tag{5.2.31}$$

where $\mathcal{D} = [0, n] \times [0, B]$, n, $B \in R^+$, $\mathbb{B}$, $\mathbb{C}$, $\mathbb{F}$ defined by (5.2.16), (5.2.20), (5.2.25), respectively, are all bounded linear operators of ${}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ (which have been proved in Section 5.2.3) and Eq. (5.2.31) has a unique solution (see reference [7]).

For the convenience of discussion, we introduce the following definition.

Definition 5.2.1. Let $\mathbf{s}$, $\mathbf{s}' \in \mathcal{D}$. Then we call that function $u(\mathbf{s}, \mathbf{s}')$ is separable if their exists a function $v(\mathbf{s})$ such that $u(\mathbf{s}, \mathbf{s}') = v(\mathbf{s})v(\mathbf{s}')$.

5.2.4-1 Solving Eq. (5.2.31) can be Turned into Finding the Separable Solution of Eq. (5.2.34)

Assume that we know the value of the exact solution $p_1(t, x_1)$ of Eq. (5.2.31) at some point $\mathbf{s}_0 \in \mathcal{D}$

$$p_1(\mathbf{s}_0) = C_0 \neq 0. \tag{5.2.32}$$

Let $\mathbf{s}$, $\mathbf{s}'$, $\mathbf{z}$, $\mathbf{z}'$, $\mathbf{y}$, $\mathbf{y}' \in \mathcal{D} = [0, n] \times [0, B]$, $\mathbf{s} = (x, y)$, $\mathbf{s}' = (\xi, \eta)$. $W_2^{(1,1)}(\mathcal{D})$, ${}^oW_2^{(2,2)}(\mathcal{D})$ have reproducing kernels $R_{\mathbf{s}}^{nB}(\mathbf{s}'), K_{\mathbf{s}}^{nB}(\mathbf{s}')$ separately and

$$\begin{aligned} R_{\mathbf{s}}^{nB}(\mathbf{s}') &= R_x^{0n}(\xi) R_y^{0B}(\eta), \\ K_{\mathbf{s}}^{nB}(\mathbf{s}') &= \overline{K}_x^{0n}(\xi) K_y^{0n}(\eta), \end{aligned}$$

where $R_\xi^{ab}(x)$, $\overline{K}_\xi^{ab}(x)$, $K_\eta^{cd}(y)$ are given by (5.2.2), (5.2.3), (5.2.4).

Define operator

$$\mathbb{L}: \; \widetilde{W}(\mathcal{D}^2) = {}^oW_2^{(2,2)}(\mathcal{D}) \otimes {}^oW_2^{(2,2)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$$

as follows

$$(\mathbb{L}u)(\mathbf{s}) = (\mathbb{K}u)(\mathbf{s}) + (\mathbb{H}u)(\mathbf{s}), \;\; \forall\, u \in \widetilde{W}(\mathcal{D}^2), \tag{5.2.33}$$

where

$$(\mathbb{K}u)(\mathbf{s}) = \left\langle u(\mathbf{z}, \mathbf{z}'), \left[\mathbb{B}_W^* R_{\mathbf{s}}^{nB}(\mathbf{y})\right](\mathbf{z}) \cdot \left[\mathbb{C}_{\mathbf{s}}^* R_{\mathbf{s}}^{nB}(\mathbf{y}')\right](\mathbf{z}') \right\rangle_{\widetilde{W}} \tag{5.2.34}$$

and

$$(\mathbb{H}u)(\mathbf{s}) = \left\langle u(\mathbf{z},\mathbf{z}'), \frac{1}{C_0} K^{nB}_{\mathbf{s}_0}(\mathbf{z}) \cdot \left[\mathbb{F}^*_{\mathbf{y}'} R^{nB}_{\mathbf{s}}(\mathbf{y}')\right](\mathbf{z}') \right\rangle_{\widetilde{W}}, \tag{5.2.35}$$

where $\mathbb{B}^*$, $\mathbb{C}^*$, $\mathbb{F}^*$ are conjugate operators of $\mathbb{B}$, $\mathbb{C}$, $\mathbb{F}$, respectively, and expression $[\mathbb{B}^*_W R^{nB}_{\mathbf{s}}(\mathbf{y})](\mathbf{z})$ indicates that operator $\mathbb{B}^*$ is applied to $R^{nB}_{\mathbf{s}}(\mathbf{y})$ as function of $\mathbf{y}$ and that the resulting function is considered as function of $\mathbf{z}$.

Next, we give the main theorem.

Theorem 5.2.5. *Suppose that $\mathbf{s}, \mathbf{s}' \in \mathcal{D}$ and $p_1(\mathbf{s}_0) = C_0 \neq 0$, then $p_1(\mathbf{s})$ is one solution of Eq.* (5.2.31) *if and only if $p_1(\mathbf{s}')p_1(\mathbf{s}'')$ is one solution of operator equation*

$$\mathbb{L}u = f, \quad u \in \widetilde{W}(\mathcal{D}^2), \quad f \in W_2^{(1,1)}(\mathcal{D}). \tag{5.2.36}$$

Proof. From (5.2.33)–(5.2.35), $p_1(\mathbf{s}_0) = C_0$ and using the equality

$$\langle u, \mathbb{B}^* v\rangle_{{}_o W_2^{(2,2)}} = \langle \mathbb{B}u, v\rangle_{W_2^{(1,2)}},$$

it follows that

$$\left[\mathbb{L}p_1(\mathbf{s}')p_1(\mathbf{s}'')\right](\mathbf{s}) = (\mathbb{B}p_1)(\mathbf{s}) \cdot (\mathbb{C}p_1)(\mathbf{s}) + (\mathbb{F}p_1)(\mathbf{s}). \tag{5.2.37}$$

Necessity. If $p_1(\mathbf{s})$ is one solution of (5.2.31), then

$$(\mathbb{B}p_1)(\mathbf{s}) \cdot (\mathbb{C}p_1)(\mathbf{s}) + (\mathbb{F}p_1)(\mathbf{s}) = f(\mathbf{s}). \tag{5.2.38}$$

By equalities (5.2.37), (5.2.38), it follows that $p_1(\mathbf{s}')p_1(\mathbf{s}'')$ is one solution of Eq. (5.2.36).

Sufficiency. If $p_1(\mathbf{s}')p_1(\mathbf{s}'')$ is one solution of (5.2.36), by equality (5.2.37), then (5.2.38) is obtained. That means $p_1(\mathbf{s})$ is one solution of Eq. (5.2.31). □

From Theorem 5.2.5 we know that solving Eq. (5.2.31) is equivalent to finding the separable solution of linear operator equation (5.2.36) which satisfies $p_1(\mathbf{s}_0) = C_0$ (which is unique as Eq. (5.2.31) has a unique solution). So in the following, we first give the representation of all solutions of linear operator equation $\mathbb{L}u = f$, then find the separable solution from them.

5.2.4-2 The Analytic Representation of all Solutions of $\mathbb{L}u = f$

Similar to [24], we can prove $\mathbb{L}$ is a bounded linear operator of $\widetilde{W}(\mathcal{D}^2) \longrightarrow W_2^{(1,1)}(\mathcal{D})$.

[139], [159] give the structure of the solution space Λ of $\mathbb{L}u = f$, namely,

$$\Lambda = \left\{u_0 + v |\ v \in (\mathbb{L}^* W_2^{(1,1)}(\mathcal{D}))^{\perp}\right\}.$$

By using equality

$$(\mathbb{L}^* W_2^{(1,1)}(\mathcal{D}))^{\perp} = N(\mathbb{L}) = \left\{v |\ \mathbb{L}v = 0,\ v \in \widetilde{W}(\mathcal{D}^2)\right\},$$

we rephrase the result of [24], [139] as follows.

Theorem 5.2.6. *Let* $\{\mathbf{s}_i\}_{i=1}^{\infty}$ *be a dense subset of* $W_2^{(1,1)}(\mathcal{D})$. *For every* $i = 1, 2, \ldots$, *write*

$$\varphi_i(\mathbf{s}) = R_{\mathbf{s}_i}^{nB}(\mathbf{s}), \quad \psi_i(\mathbf{s}', \mathbf{s}'') = (\mathbb{L}^*\varphi_i)(\mathbf{s}', \mathbf{s}''),$$

where $\mathbb{L}^*$ *denotes the conjugate operator of* $\mathbb{L}$. *Let* $\{\widetilde{\psi}_i\}_{i=1}^{\infty}$ *be the complete orthonormal system on* $\mathcal{D}$ *obtained from* $\{\psi_i\}_{i=1}^{\infty}$ *by Gram–Schmidt process, namely,*

$$\widetilde{\psi}_i(\mathbf{s}', \mathbf{s}'') = \sum_{k=1}^{i} \beta_{ik}\psi_i(\mathbf{s}', \mathbf{s}''), \ \ \beta_{ii} > 0, \ \ i = 1, 2, \ldots,$$

then

$$u_0(\mathbf{s}', \mathbf{s}'') = \sum_{i=1}^{\infty} \left(\sum_{k=1}^{i} \beta_{ik} f(\mathbf{s}_k) \right) \widetilde{\psi}_i(\mathbf{s}', \mathbf{s}'') \tag{5.2.39}$$

is the minimal norm solution and the solution space of $\mathbb{L}u = f$ *can be denoted as*

$$u_0 + \mathrm{N}(\mathbb{L}) = u_0 + \left\{ v | \ \mathbb{L}v = 0, \ v \in {}^{o}W_2^{(2,2)}(\mathcal{D}) \right\}. \tag{5.2.40}$$

Remark 5.2.1. In Refs. [139], [182], the authors did not give the representation of $\mathrm{N}(\mathbb{L})$ or $(\mathbb{L}^*W_2^{(1,1)}(\mathcal{D}))^{\perp}$.

Remark 5.2.2. If we obtain a complete orthonormal system $\widetilde{r}_1, \widetilde{r}_2, \ldots, \widetilde{r}_n, \ldots$ of null space $\mathrm{N}(\mathbb{L})$, then the solution set Λ of $\mathbb{L}u = f$ can be denoted as

$$\left\{ u_0 + \sum_{i=1}^{\infty} \alpha_i \widetilde{r}_i | \text{ real sequence } \{\alpha_i\}_{i=1}^{\infty} \in \ell^2 \right\}. \tag{5.2.41}$$

Note that $\mathrm{N}(\mathbb{L}) = \overline{\mathbb{L}^*W_2^{(1,1)}(\mathcal{D})}^{\perp}$, namely, $\mathrm{N}(\mathbb{L}) \oplus \overline{\mathbb{L}^*W} = \widetilde{W}(\mathcal{D}^2)$, we know: if sequence $r_1, r_2, \ldots, r_n, \ldots$ is a complete set of $\widetilde{W}(\mathcal{D}^2)$, then sequence

$$\psi_1, \psi_2, \ldots, \psi_n, \ldots, r_1, r_2, \ldots, r_m, \ldots$$

is also a complete set of $\widetilde{W}(\mathcal{D}^2)$. Denote by

$$\widetilde{\psi}_1, \widetilde{\psi}_2, \ldots, \widetilde{\psi}_n, \ldots, \widetilde{r}_1, \widetilde{r}_2, \ldots, \widetilde{r}_m, \ldots$$

the result that is obtained from it by Gram–Schmidt process. Then $\{\widetilde{\psi}_i\}_{i=1}^{\infty}$ is a complete orhtonormal set of $\overline{\mathbb{L}^*W_2^{(1,1)}(\mathcal{D})}$ and $\{\widetilde{r}_i\}_{i=1}^{\infty}$ is a complete orhtonormal set of $\mathrm{N}(\mathbb{L})$. From the process of Gram–Schmidt orthonormalization, the following lemma holds obviously.

Lemma 5.2.2. *Let $\{r_i\}_{i=1}^{\infty}$ be a complete set of $\widetilde{W}(\mathcal{D}^2)$, $\{v_i\}_{i=1}^{\infty}$ be a complete set of $\mathbb{L}^*W_2^{(1,1)}(\mathcal{D})$, and $v_1 \neq 0$. If sequence*

$$\widetilde{v}_1, \widetilde{v}_2, \ldots, \widetilde{v}_n, \ldots, \widetilde{r}_1, \widetilde{r}_2, \ldots, \widetilde{r}_m, \ldots$$

is a complete orthonormal system defined on $\widetilde{W}(\mathcal{D}^2)$ obtained from

$$v_1, v_2, \ldots, v_n, \ldots, r_1, r_2, \ldots, r_m, \ldots$$

by Gram–Schmidt process, namely,

$$\begin{aligned}
\widetilde{v}_1 &= \frac{v_1}{\|v_1\|_{\widetilde{W}}}, \\
\widetilde{v}_{k+1} &= \frac{v_{k+1} - \sum_{i=1}^{k} \langle v_{k+1}, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i}{\big\|v_{k+1} - \sum_{i=1}^{k} \langle v_{k+1}, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i\big\|_{\widetilde{W}}}, \quad k = 1, 2, \ldots, \\
\widetilde{r}_1 &= \frac{r_1 - \sum_{i=1}^{\infty} \langle r_1, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i}{\big\|r_1 - \sum_{i=1}^{\infty} \langle r_1, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i\big\|_{\widetilde{W}}}, \\
\widetilde{r}_{k+1} &= \frac{r_{k+1} - \sum_{i=1}^{\infty} \langle r_{k+1}, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i - \sum_{i=1}^{k} \langle r_{k+1}, \widetilde{r}_i\rangle_{\widetilde{W}} \widetilde{r}_i}{\big\|r_{k+1} - \sum_{i=1}^{\infty} \langle r_{k+1}, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i - \sum_{i=1}^{k} \langle r_{k+1}, \widetilde{r}_i\rangle_{\widetilde{W}} \widetilde{r}_i\big\|_{\widetilde{W}}}, \quad k = 1, 2, \ldots,
\end{aligned} \tag{5.2.42}$$

(*To the second formula of* (5.2.42)*, we follow the following rules: if for some positive integer k, $r_{k+1} - \sum_{i=1}^{k}(r_{k+1}, \widetilde{r}_i)\widetilde{r}_i = 0$, then we delete r_{k+1} from the sequence $\widetilde{r}_1, \ldots, \widetilde{r}_k$, $r_{k+1}, r_{k+2}, \ldots, v_1, v_2, \ldots$ and rewrite $r_{k+2}, r_{k+3}, \ldots, v_1, v_2, \ldots$ as $r_{k+1}, r_{k+2}, \ldots, v_1, v_2, \ldots$, after that, we again compute $\widetilde{v}_{k+1}$ by the second formula of* (5.2.42) *from the new sequence $\widetilde{r}_1, \ldots, \widetilde{r}_k, r_{k+1}, r_{k+2}, \ldots, v_1, v_2, \ldots$. We will repeat the same process until $r_{k+1} - \sum_{i=1}^{k}(r_{k+1}, \widetilde{r}_i)\widetilde{r}_i \neq 0$. To the last two formulas of* (5.2.42)*, we follow the same rules.*) *then $\{\widetilde{r}_i\}_{i=1}^{\infty}$ is a completely orthonormal set of $\mathrm{N}(\mathbb{L})$.*

Remark 5.2.3. Write

$$\widetilde{r}_1^{(n)} = \frac{r_1 - \sum_{i=1}^{n} \langle r_1, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i}{\big\|r_1 - \sum_{i=1}^{n} \langle r_1, \widetilde{v}_i\rangle_{\widetilde{W}} \widetilde{v}_i\big\|_{\widetilde{W}}},$$

$$\widetilde{r}_{k+1}^{(n)} = \frac{r_{k+1} - \sum_{i=1}^{k} \langle r_{k+1}, \widetilde{r}_i^{(n)} \rangle_{\widetilde{W}} \widetilde{r}_i^{(n)} - \sum_{i=1}^{n} \langle r_{k+1}, \widetilde{v}_i \rangle_{\widetilde{W}} \widetilde{v}_i}{\left\| r_{k+1} - \sum_{i=1}^{k} \langle r_{k+1}, \widetilde{r}_i^{(n)} \rangle_{\widetilde{W}} \widetilde{r}_i^{(n)} - \sum_{i=1}^{n} \langle r_{k+1}, \widetilde{v}_i \rangle_{\widetilde{W}} \widetilde{v}_i \right\|_{\widetilde{W}}}, \quad k = 1, 2, \ldots,$$

then it is not difficult to prove that for every $i = 1, 2, \ldots$, we have

$$\|\widetilde{r}_i^{(n)} - \widetilde{r}_i\|_{\widetilde{W}} \longrightarrow 0, \quad \text{when} \quad n \longrightarrow \infty.$$

Theorem 5.2.7. *Let $\{\mathbf{s}_i\}_{i=1}^{\infty}$ be a dense subset of $\mathcal{D}$. In Lemma* 5.2.2*, take $v_i = \psi_i$, $\{r_i\}_{i=1}^{\infty} = \{K_{\mathbf{s}_i}(\mathbf{s}') \cdot K_{\mathbf{s}_j}(\mathbf{s}'')\}_{i,j=1}^{\infty}$, namely, arranging $\{K_{\mathbf{s}_i}(\mathbf{s}') \cdot K_{\mathbf{s}_j}(\mathbf{s}'')\}_{i,j=1}^{\infty}$ in proper order and regard it as $\{r_i\}_{i=1}^{\infty}$. Then $\{\widetilde{r}_i\}_{i=1}^{\infty}$ defined by* (5.2.42) *is a completely orthonormal set of* $\mathrm{N}(\mathbb{L})$*, where $K_{\mathbf{s}}(\mathbf{s}')$ denotes the reproducing kernel of ${}^{o}W_2^{(2,2)}(\mathcal{D})$.*

Proof. We only need to show that $\{K_{\mathbf{s}_i}(\mathbf{s}') \cdot K_{\mathbf{s}_j}(\mathbf{s}'')\}_{i,j=1}^{\infty}$ is complete in $\widetilde{W}(\mathcal{D}^2)$. Let $\varphi \in \widetilde{W}(\mathcal{D}^2)$ satisfy that for every $i, j = 1, 2, \ldots$,

$$\left\langle \varphi(\mathbf{s}', \mathbf{s}''), \overline{K}_{\mathbf{s}_i}(\mathbf{s}') \cdot K_{\mathbf{s}_j}(\mathbf{s}'') \right\rangle_{\widetilde{W}} = 0.$$

Note that $K_{\mathbf{s}_i}(\mathbf{s}') \cdot K_{\mathbf{s}_j}(\mathbf{s}'')$ is the reproducing kernel of $\widetilde{W}(\mathcal{D}^2)$, we have $\varphi(\mathbf{s}_i, \mathbf{s}_j) = 0$. From the density of set $\{\mathbf{s}_i\}$ in $\mathcal{D}$, namely, the density of set $\{\mathbf{s}_i, \mathbf{s}_j\}_{i,j=1}^{\infty}$ in $\mathcal{D}^2$, it follows that $\varphi \equiv 0$ on $\mathcal{D}^2$. This completes our proof. □

5.2.4-3 The Representation of the Exact Solution of Eq. (5.2.31)

Here we consider how to find the separable solution in the function set of (5.2.41).

We first give a lemma.

Lemma 5.2.3. *Suppose $q(\mathbf{s}, \mathbf{s}') \in \widetilde{W}(\mathcal{D}^2)$, then the following two propositions are equivalent.*

Proposition A: *There exists a function $v(\mathbf{s})$, such that $q(\mathbf{s}, \mathbf{s}') = v(\mathbf{s})v(\mathbf{s}')$ and $v(\mathbf{s}_0) = C_0 \neq 0$;*

Proposition B: *$q(\mathbf{s}, \mathbf{s}') \neq 0$ and*

$$q(\mathbf{s}, \mathbf{s}') = \frac{1}{C_0^2} q(\mathbf{s}, \mathbf{s}_0) \cdot q(\mathbf{s}', \mathbf{s}_0).$$

Since Eq. (5.2.31) has a unique solution which satisfies $p(\mathbf{s}_0) = C_0$, Eq. (5.2.36) has a unique solution which satisfied $p(\mathbf{s}_0) = C_0$. By Lemma 5.2.3, (5.2.41), we obtain the following method of finding the exact solution of Eq. (5.2.31).

Method: Set

$$q(\mathbf{s}, \mathbf{s}') = u_0 + \sum_{i=1}^{\infty} \alpha_i \widetilde{r}_i, \tag{5.2.41*}$$

where u_0 is defined by (5.2.39), $\widetilde{r}_i$ is defined by Theorem 3.3. The unknowns are real sequence $\{\alpha_i\}_{i=1}^{\infty} \in \ell^2$. Let C_0 be a nonzero known number. Seeking $q(\mathbf{s}, \mathbf{s}')$ such that

$$0 \neq q(\mathbf{s}, \mathbf{s}') = \frac{1}{C_0^2}\, q(\mathbf{s}, \mathbf{s}_0) \cdot q(\mathbf{s}', \mathbf{s}_0). \tag{5.2.43}$$

(Suppose (5.2.43) has a unique solution) or equivalently finding $\{\alpha_i\}_{i=1}^{\infty}$ such that

$$\left\| q(\mathbf{s}, \mathbf{s}') - \frac{1}{C_0^2} q(\mathbf{s}, \mathbf{s}_0) \cdot q(\mathbf{s}', \mathbf{s}_0) \right\|_{\widetilde{W}(\mathcal{D}^2)} = 0,$$

then from Theorem 3.1 it follows that $\frac{q(\mathbf{s}, \mathbf{s}_0)}{C_0}$ is the exact solution of Eq. (5.2.31).

5.2.4-4 The Numerical Algorithm for Solving the ε Approximately Solution of Eq. (5.2.31)

We give a numerical algorithm of finding the approximate solution of (5.2.43). We first introduce the concept of ε-approximate solution.

Definition 5.2.2. Suppose that $\varepsilon > 0$, we call $u(\mathbf{s})$ be the ε-approximate solution of Eq. (5.2.31), if it holds that

$$\|\mathbb{B}u \cdot \mathbb{C}u + \mathbb{F}u - f\|_{W_2^{(1,1)}} \leq \varepsilon.$$

Theorem 5.2.8. *Suppose that $q(\mathbf{s}, \mathbf{s}')$ has the form (5.2.41*) and $q(\mathbf{s}_0, \mathbf{s}_0) = C_0^2$. Put*

$$\varepsilon_1(\mathbf{s}', \mathbf{s}'') = q(\mathbf{s}', \mathbf{s}'') - \frac{1}{C_0^2}\, q(\mathbf{s}', \mathbf{s}_0) \cdot q(\mathbf{s}'', \mathbf{s}_0).$$

If for some $\varepsilon > 0$, $\|\varepsilon_1(\mathbf{s}', \mathbf{s}'')\|_{\widetilde{W}} < \varepsilon$, then $\frac{q(\mathbf{s}', \mathbf{s}_0)}{C_0}$ is a $\varepsilon \cdot \|\mathbb{L}\|$ approximate solution of Eq. (5.2.31).

Proof. Notice that $q(\mathbf{s}', \mathbf{s}'')$ has the form (5.2.41*), we obtain $\mathbb{L}q = f$. Furthermore, from $q(\mathbf{s}_0, \mathbf{s}_0) = C_0^2$ it follows

$$\begin{aligned} \mathbb{L}\varepsilon_1 &= \mathbb{L}q - \frac{1}{C_0^2}\, \mathbb{L}\left[q(\mathbf{s}', \mathbf{s}_0) \cdot q(\mathbf{s}'', \mathbf{s}_0)\right] \\ &= f - \mathbb{B}_{\mathbf{s}'}\left(\frac{q(\mathbf{s}', \mathbf{s}_0)}{C_0}\right) \cdot \mathbb{C}_{\mathbf{s}''}\left(\frac{q(\mathbf{s}'', \mathbf{s}_0)}{C_0}\right) + \mathbb{F}_{\mathbf{s}''}\left(\frac{q(\mathbf{s}'', \mathbf{s}_0)}{C_0}\right). \end{aligned}$$

If put $u(\mathbf{s}) = \frac{q(\mathbf{s}', \mathbf{s}_0)}{C_0}$, then

$$\|\mathbb{L}\varepsilon_1\|_{W_2^{(1,1)}} = \left\|f - (\mathbb{B}u\mathbb{C}u + \mathbb{F}u)\right\|_{W_2^{(1,1)}} \leq \|\mathbb{L}\| \cdot \varepsilon. \tag{5.2.44}$$

That means $u(\mathbf{s}) = \frac{q(\mathbf{s}', \mathbf{s}_0)}{C_0}$ is a $\varepsilon \cdot \|\mathbb{L}\|$-approximate solution of Eq. (5.2.31). □

In order to give the numerical algorithm for solving Eq. (5.2.31), we need a further analysis as follows.

(1) The choice of r_i. Take first n points which belong to $\mathcal{D}$. Put r_{lm} as follows

$$r_{lm} = K_{\mathbf{s}_l}^{nB}(\mathbf{z}') \cdot K_{\mathbf{s}_m}^{nB}(\mathbf{z}''), \quad l, m = 1, 2, \ldots, n,$$

where $K_{\mathbf{s}}^{nB}(\mathbf{z}')$ is the reproducing kernel of ${}^oW_2^{(2,2)}(\mathcal{D})$ and arrange them by the diagonal rule, namely, $r_{11}, r_{12}, r_{21}, r_{31}, r_{22}, r_{13}, r_{14}, r_{23}, \ldots, r_{nn}$. Then regard it as sequence $\{r_i\}_{i=1}^{n^2}$.

(2) The choice of unknowns. Take $\{\gamma_i = \frac{q(\mathbf{s}_i, \mathbf{s}_0)}{C_0}\}_{i=1}^{n}$ as unknown numbers, where $p(\mathbf{s}', \mathbf{s}'')$ has the form (5.2.41*). Then from (5.2.43) it follows that $q(\mathbf{s}_i, \mathbf{s}_j) = \gamma_i\gamma_j$. Moreover, α_i can be denoted by $\{\gamma_i\}_{i=1}^{n}$ as follows

$$\begin{aligned}
\alpha_i &= \big\langle q(\mathbf{s}', \mathbf{s}''), \widetilde{r}_i \big\rangle_{\widetilde{W}} = \left\langle q(\mathbf{s}', \mathbf{s}''), \sum_{k=1}^{\infty} \beta_{ik}\psi_k + \sum_{k=1}^{i} c_{ik} r_k \right\rangle_{\widetilde{W}} \\
&= \sum_{k=1}^{\infty} \beta_{ik} \big\langle q(\mathbf{s}', \mathbf{s}''), L^*\varphi_k \big\rangle_{\widetilde{W}} + \sum_{k=1}^{i} c_{ik} \big\langle q(\mathbf{s}', \mathbf{s}''), r_k \big\rangle_{\widetilde{W}} \\
&= \sum_{k=1}^{\infty} \beta_{ik} \big\langle (\mathbb{L}q)(\mathbf{s}), R_{\mathbf{s}_k}^{nB}(\mathbf{s}) \big\rangle_{W_2^{(1,1)}} \\
&+ \sum_{k=1}^{i} c_{ik} \big\langle q(\mathbf{s}', \mathbf{s}''), K_{\mathbf{s}_{k_1}}^{nB}(\mathbf{s}') K_{\mathbf{s}_{k_2}}^{nB}(\mathbf{s}'') \big\rangle_{\widetilde{W}} \\
&= \sum_{k=1}^{\infty} \beta_{ik} f(\mathbf{s}_k) + \sum_{k=1}^{i} c_{ik} q(\mathbf{s}_{k_1}, \mathbf{s}_{k_2}) \\
&= \sum_{k=1}^{\infty} \beta_{ik} f(\mathbf{s}_k) + \sum_{k=1}^{i} c_{ik} \gamma_{k_1} \gamma_{k_2}, \qquad (5.2.45)
\end{aligned}$$

where k correspond to ordered pair $(\mathbf{s}_{k_1}, \mathbf{s}_{k_2})$.

Summing up the analysis (1), (2), we obtain the following algorithm.

(A1) Take $\{Q_i\}_{i=1}^{N_1}$ which belong to $\mathcal{D}$ and the choice of N_1 follows (A3).

(A2) For every $i = 1, 2, \ldots, N_1$, put

$$\varphi_i(\mathbf{s}) = R_{Q_i}^{nB}(\mathbf{s}),$$

$\psi_i(\mathbf{s}', \mathbf{s}'') = (\mathbb{L}^*\varphi_i)(\mathbf{s}', \mathbf{s}'')$. Let $\{\widetilde{\psi}_i\}_{i=1}^{N_1}$ be the orthonormal system defined on $\mathcal{D}^2$ obtained from $\{\psi_i\}_{i=1}^{N_1}$ by Gram–Schmidt process, namely,

$$\widetilde{\psi}_i(\mathbf{s}', \mathbf{s}'') = \sum_{k=1}^{i} \beta_{ik}\psi_i(\mathbf{s}', \mathbf{s}''), \quad \beta_{ii} > 0, \quad i = 1, 2, \ldots, N_1.$$

(A3) Note that

$$u_{0,N_1}(\mathbf{s}',\mathbf{s}'')=\sum_{i=1}^{N_1}\left(\sum_{k=1}^{i}\beta_{ik}f(\mathbf{s}_k)\right)\widetilde{\psi}_i(\mathbf{s}',\mathbf{s}'').$$

We obtain the approximate solution u_{0,N_1} of u_0 defined by (5.2.39), where N_1 is selected such that

$$\max_{\mathbf{s}',\mathbf{s}''\in\mathcal{D}}\left|u_{0,N_1}(\mathbf{s}',\mathbf{s}'')-u_{0,N_1-1}(\mathbf{s}',\mathbf{s}'')\right|$$

is very small.

(A4) Take $\{\mathbf{s}_i\}_{i=1}^{N_2}$ which belong to $\mathcal{D}$. Put $\mathbf{s}_1=\mathbf{s}_0$. Let $\{r_{lm}=K_{\mathbf{s}_l}^{nB}(\mathbf{z}')K_{\mathbf{s}_m}^{nB}(\mathbf{z}'')\}_{l,m=1}^{N_2}$ and arrange $\{r_{l,m}\}_{l,m=1}^{N_2}$ according to diagonal rule, namely, $r_{11},r_{12},r_{21},r_{31},r_{22}$, $r_{13},r_{14},\ldots,r_{nn}$. Then regard it as sequence $\{r_i\}_{i=1}^{N_2^2}$ and let $\{\widetilde{r}_i\}_{i=1}^{N_2^2}$ be the orthonormal system defined on $\mathcal{D}^2$ obtained from $\widetilde{\psi}_1,\widetilde{\psi}_2,\ldots,\widetilde{\psi}_{N_1},r_1,r_2,\ldots,r_{N_2^2}$ by Gram–Schmidt process, namely,

$$\widetilde{r}_i=\sum_{k=1}^{N_1}\beta_{ik}\psi_k+\sum_{k=1}^{i}c_{ik}r_k.$$

(A5) Take $\{\gamma_i=\frac{q(\mathbf{s}_i,\mathbf{s}_0}{C_0}\}_{i=1}^{N_2}$ the unknown numbers and from (5.2.45), we have

$$\alpha_i=\langle q(\mathbf{s}',\mathbf{s}''),\widetilde{r}_i\rangle_{\widetilde{W}}=\sum_{k=1}^{N_1}\beta_{ik}f(\mathbf{s}_i)+\sum_{k=1}^{i}c_{ik}\gamma_{k_1}\gamma_{k_2},$$

where k correspond to ordered pair $(\mathbf{s}_{k_1},\mathbf{s}_{k_2})$.

(A6) Let

$$\begin{aligned}q(\mathbf{s}',\mathbf{s}'') &= u_{0,N_1}(\mathbf{s}',\mathbf{s}'')+\sum_{i=1}^{N_2^2}\alpha_i\widetilde{r}_i(\mathbf{s}',\mathbf{s}''), \qquad (5.2.46)\\ \varepsilon_1(\mathbf{s}',\mathbf{s}'') &= q(\mathbf{s}',\mathbf{s}'')-\frac{1}{C_0^2}\,q(\mathbf{s}',\mathbf{s}_0)\cdot q(\mathbf{s}'',\mathbf{s}_0)\end{aligned}$$

Solve $\{\gamma_i\}_{i=1}^{N_2}$ by

$$\min\left\{\|\varepsilon_1(\mathbf{s}',\mathbf{s}'')\|_{\widetilde{W}},\ \gamma_i\in\mathbb{R},\ i=1,2,\ldots,N_2,\ \gamma_1=\frac{q(C_0,C_0)}{C_0}=C_0\right\}.$$

(Put $\gamma_1=C_0$ for satisfying Theorem 5.2.8).

(A7) Substituting $\{\gamma_i\}_{i=1}^{N_2}$ obtained by (A6) into (5.2.46). Then by Theorem 5.2.8. We know that $\frac{p(\mathbf{s}',\mathbf{s}_0)}{C_0}$ is the ε-approximate solution of Eq. (5.2.31).

5.2.4 Numerical Experiments

By using the algorithm given by Section 5.2.4-4, we solve Eq. (5.2.47). The following Example 5.2.1 is given by [140] as a numerical experiment.

Example 5.2.1. Suppose that $p = p(t, x)$ denote the age-specific density of individuals of age x at time t. Solve population equation

$$\begin{cases} \dfrac{\partial p}{\partial t} + \dfrac{\partial p}{\partial x} = -P(t) \cdot p(t, x), & t \geq 0, \ \ 0 \leq x < A, \\ p(0, x) = \dfrac{e^{-x}}{2}, & 0 \leq x < A, \\ p(t, 0) = P(t), & t \geq 0, \\ P(t) = \displaystyle\int_0^A p(t, x)dx, & t \geq 0, \end{cases} \tag{5.2.47}$$

where $A = +\infty$. And it's easy to verify that

$$p(t, x) = \frac{e^{-x}}{1 + e^{-t}}, \ \ t \geq 0, \ \ x \geq 0$$

is the exact solution of Eq. (5.2.47).

Take $\mathcal{D} = [0, 1] \times [0, 10]$, which denotes (1 unit time) $\times$ (10 unit age). The numerical results are listed in Table 5.5.

5.3 Solving a Kind of Nonlinear Partial Differential Equations

5.3.1 Introduction

Giving the exact solutions of the nonlinear PDE (Partial Differential Equation) is important to study theory of nonlinear problem and its applications. In [141],using the two-dimensional differential transform method, author solved the following PDE,

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x}\left(D(u)\frac{\partial u}{\partial x}\right).$$

For the nonlinear evolution equation with nonlinear term in the form

$$u_{tt} + au_{xx} + bu + cu^3 = 0,$$

Yang Chen et.al. [142] have also given the exact solution. In this paper, we focus on a kind of nonlinear PDEs with initial-boundary value conditions in the form

$$\frac{\partial}{\partial x}\left(a(x)\frac{\partial u(x, t)}{\partial x}\right) - \frac{\partial^2 u(x, t)}{\partial t^2} + H(u(x, t)) = f(x, t)$$

Table 5.5: The error of Example 5.2.1

nodes		the exact solution	approximate solution	
t	x	$p(t,x)$	$p_1(t,x)$	error $p(t,x)-p_1(t,x)$
0.00	0.00	0.5	0.5	0
0.20	0.00	0.549834	0.523672	2.6162E-02
0.40	0.00	0.598688	0.554367	4.4321E-02
0.60	0.00	0.645656	0.596953	4.4703E-02
0.80	0.00	0.689974	0.653438	3.6536E-02
1.00	0.00	0.731059	0.719825	1.1234E-02
0.00	1.00	0.18394	0.18394	0
0.20	1.00	0.202273	0.195806	6.467E-03
0.40	1.00	0.220245	0.208099	1.2146E-02
0.60	1.00	0.237524	0.22346	1.4064E-02
0.80	1.00	0.253827	0.24594	7.887E-03
1.00	1.00	0.268941	0.274462	-5.552E-03
0.00	2.00	0.0676676	0.0676676	0
0.20	2.00	0.074419	0.0688321	5.5869E-03
0.40	2.00	0.0810236	0.0714055	9.6181E-03
0.60	2.00	0.0873801	0.0770204	1.03597E-02
0.80	2.00	0.0933779	0.087962	5.4159E-03
1.00	2.00	0.0989390	0.0923431	6.5159E-03
0.00	3.00	0.0248935	0.0248935	0
0.20	3.00	0.0273767	0.0229024	4.4722E-03
0.40	3.00	0.0298069	0.0222456	7.3509E-03
0.60	3.00	0.0321453	0.0239349	8.2104E-03
0.80	3.00	0.0343518	0.0279446	6.4072E-03
1.00	3.00	0.0363973	0.0333908	2.9965E-03
0.00	4.00	0.00915782	0.00915782	0
0.20	4.00	0.0100706	0.00824638	1.8242E-03
0.40	4.00	0.0109653	0.0076789	3.2864E-03
0.60	4.00	0.0118256	0.00805066	3.7749E-03
0.80	4.00	0.0126373	0.00947644	3.1609E-03
1.00	4.00	0.0133898	0.0115372	1.8526E-03
0.00	5.00	0.00336897	0.00336897	0
0.20	5.00	0.00370475	0.00403647	-3.3172E-04
0.40	5.00	0.00403393	0.00428807	-2.5414E-04
0.60	5.00	0.0043504	0.00448504	-1.3464E-04
0.80	5.00	0.00464901	0.005010781	-3.5880E-04
1.00	5.00	0.00492583	0.00579783	-8.7200E-04
0.00	0.00	0.00123938	0.00123938	0
0.20	6.00	0.0013629	0.000471366	8.9154E-04
0.40	6.00	0.001484	-0.000839367	2.32337E-03
0.60	6.00	0.00160024	-0.00236794	3.85194E-03
0.80	6.00	0.00171028	-0.00367041	5.38069E-03
1.00	6.00	0.0018211	-0.00478155	6.49366E-03
0.00	8.00	0.000167731	0.000167731	0
0.20	8.00	0.000184449	-0.000274359	4.58849E-03
0.40	8.00	0.000200837	-0.000611792	6.31876E-03
0.60	8.00	0.000216594	-0.00611792	9.81151E-03
0.80	8.00	0.000231461	-0.0129138	1.30453E-02
1.00	8.00	0.000245343	-0.0157958	1.6041E-02
0.00	10.00	0.0000227	0.0000227	0
0.20	10.00	0.0000249642	-0.000189344	2.14306E-04
0.40	10.00	0.0000271804	-0.00111487	1.14205E-03
0.60	10.00	0.0000271804	-0.00249723	2.52654E-03
0.80	10.00	0.0000293128	-0.00386516	3.89648E-03
1.00	10.00	0.00003319	-0.00571415	5.20734E-03

where, $H(x)$ is a nonlinear term with any type, $a(x)$, $f(x,t)$ are sufficiently regular given functions. For the kind of PDEs with general type, we give the representation of exact solution in the reproducing kernel space. Numerical example shows that this method is valid.

5.3.2 Transformation of the Nonlinear Partial Differential Equation

Let $\mathcal{D} = [0,1] \times [0,\infty)$. Given the reproducing kernel spaces

$$ {}^{o}W_2^{(2,3)}(\mathcal{D}) = \Big\{ u(x,t) |\ u(x,t) \in W_2^{(2,3)}(\mathcal{D}),\ u(0,t) = u_x(1,t) = u_t(x,0) = 0 \Big\}, $$

and $W_2^1[0,1]$ (see (1.3.2)). Their reproducing kernels are $K_{(\xi,\eta)}(x,t) = R_\xi(x)G_\eta(t)$, $r_\xi(x)$ respectively, where $r_\xi(x)$ by (1.4.1) and

$$ \begin{aligned} R_\xi(x) &= \begin{cases} -\dfrac{1}{6}x(x^2 - 3\xi(2+x)), & x \le \xi, \\ -\dfrac{1}{6}\xi(\xi^2 - 3x(2+\xi)), & x > \xi, \end{cases} \\ G_\eta(t) &= \begin{cases} \dfrac{1}{8}\, e^{-\eta}\big((3+\eta(3+\eta)+t^2)\cosh t - (3+2\eta)t\sinh t\big), & x \le \eta, \\ \dfrac{1}{8} e^{-t}\big((3+t(3+t)+\eta^2)\cosh \eta - (3+2t)\eta \sinh \eta\big), & x > \eta. \end{cases} \end{aligned} $$

Consider the nonlinear differential equation with initial-boundary value conditions

$$ \begin{cases} \dfrac{\partial}{\partial x}\left[a(x)\,\dfrac{\partial u(x,t)}{\partial x}\right] - \dfrac{\partial^2 u(x,t)}{\partial t^2} + H(u(x,t)) = f(x,t), \\ u(x,0) = h(x), \\ \dfrac{\partial}{\partial t}\,u(x,0) = 0, \\ u(0,t) = 0, \\ \dfrac{\partial}{\partial x}\,u(1,t) = 0, \ \ 0 \le x \le 1, \ \ 0 \le t < +\infty, \end{cases} \tag{5.3.1} $$

where $u(x,t) \in {}^{o}W_2^{(2,3)}(\mathcal{D})$ and $a(x)$, $h(x)$, $f(x,t)$, $H(x)$ sufficiently regular given functions. Then, in the ${}^{o}W_2^{(2,3)}(\mathcal{D})$, Eq. (5.3.1) is equivalent to

$$ \begin{cases} a(x)\,\dfrac{\partial u}{\partial x} - a(0)\,\dfrac{\partial u}{\partial x}\Big|_{x=0} - \displaystyle\int_0^x u_t''(z,t)dz + \int_0^x H(u((z,t))dz = \int_0^x f(z,t)dz \\ u(x,0) = h(x). \end{cases} \tag{5.3.2} $$

5.3.3 The Definition of Operator $\mathbb{L}$

A linear operator $\mathbb{L} : {}^{o}W_2^{(2,3)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$ is defined by

$$(\mathbb{L}u)(x) = \int_0^x u_t''(z,0)dz$$

Then Eq. (5.3.2) is transformed into an equation:

$$\begin{aligned} &(\mathbb{L}u)(x) = F(x), \\ &F(x) = a(x)h'(x) - a(0)h'(0) + \int_0^x (H(h(x)) - f(z,0))dz \end{aligned} \tag{5.3.3}$$

It is easy to prove that the linear operator $\mathbb{L} : {}^{o}W_2^{(2,3)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$ is bounded.

Lemma 5.3.1. *Let $\mathbb{L}^*$ denote the conjugate operator of $\mathbb{L}$, then $\mathbb{L}^*$ is a bounded operator from $W_2^1[0,1]$ to ${}^{o}W_2^{(2,3)}(\mathcal{D})$ and*

$$(\mathbb{L}^* r_\xi(\cdot))(x,t) = \frac{d^2}{d\eta^2} G_t(\eta)\Big|_{\eta=0} \int_0^\xi R_x(s)ds.$$

5.3.4 Decomposition into Direct Sum of ${}^{o}W_2^{(2,3)}(\mathcal{D})$

Take $\mathrm{A} = \{s_1, s_2, \ldots\}$ as a dense set of interval $[0,1]$, put $\varphi_i(x) = r_{s_i}(x)$. And let

$$\psi_i(x,t) = (\mathbb{L}^*\varphi_i)(x,t), \;\; i = 1, 2, \ldots .$$

Now we orthonormalize the function system $\{\psi_i(x,t)\}_{i=1}^{\infty}$, and obtain an orthonormal system $\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty}$, which hold for all i,

$$\overline{\psi}_i(x,t) = \sum_{k=1}^{i} \beta_{ik}\psi_k(x,t), \;\; i = 1, 2, \ldots,$$

where β_{ik} is the coefficient of orthonormalization. Then $span(\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty})$ is a subspace of ${}^{o}W_2^{(2,3)}(\mathcal{D})$:

$$span\big(\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty}\big) = \left\{ u(x,t) |\; u(x,t) = \sum_{i=1}^{n} c_i\overline{\psi}_i(x,t),\; c_i \in \mathbb{R},\; n \in \mathbb{N} \right\}.$$

Let S denote the closure of this subspace, that is

$$\mathrm{S} = \overline{span}\big(\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty}\big)$$

and $\mathrm{S}^{\perp}$ denotes orthcomplement space of S in ${}^{o}W_2^{(2,3)}(\mathcal{D})$. Taking a set of points $\mathrm{B} = \{p_1(\xi_1, \eta_1), p_2(\xi_2, \eta_2), \ldots\}$ as a dense set of region $\mathcal{D} = [0,1] \times [0, +\infty)$, and putting

$$\rho_j(x,t) = K_{(\xi_j,\eta_j)}(x,t), \quad j = 1, 2, \ldots,$$

where $K_{(\xi,\eta)}(x,t)$ is the reproducing kernel of ${}^{o}W_2^{(2,3)}(\mathcal{D})$, we proceed the generalized Schmidt orthonormalization for the function system $\{\rho_j(x,t)\}$ about orthonormal system $\{\overline{\psi}_j(x,t)\}$, that is

$$\begin{aligned}
&\overline{\rho}_j(x,t) \\
&= \frac{\rho_j(x,t) - \sum\limits_{k=1}^{\infty} \langle \rho_j(x,t), \overline{\psi}_k(x,t)\rangle \overline{\psi}_k(x,t) - \sum\limits_{m=1}^{j-1} \langle \rho_j(x,t), \overline{\rho}_m(x,t)\rangle \overline{\rho}_k(x,t)}{\left\| \rho_j(x,t) - \sum\limits_{k=1}^{\infty} \langle \rho_j(x,t), \overline{\psi}_k(x,t)\rangle \overline{\psi}_k(x,t) - \sum\limits_{m=1}^{j-1} \langle \rho_j(x,t), \overline{\rho}_m(x,t)\rangle \overline{\rho}_k(x,t) \right\|}.
\end{aligned}$$

So we have

$$\overline{\rho}_j(x,t) = \sum_{k=1}^{\infty} \beta_{jk} \psi_k(x,t) + \sum_{m=1}^{j} \beta_{jm}^{*} \rho_m(x,t), \quad j = 1, 2, \ldots.$$

Whereas ${}^{o}W_2^{(2,3)}(\mathcal{D}) = \mathrm{S} \oplus \mathrm{S}^{\perp}$ and $\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty} \cup \{\overline{\rho}_j(x,t)\}_{j=1}^{\infty}$ is a normal orthogonal basis of ${}^{o}W_2^{(2,3)}(\mathcal{D})$.

5.3.5 Solving the Nonlinear Partial Differential Equation

Lemma 5.3.2. *Assume that $u(x,t)$ is the solution of Eq.* (5.3.3)*, then it follows that*

$$\begin{aligned}
u(x,t) &= \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} \left(a(s_k) h'(s_k) - k(0) h'(0) + \int_0^{s_k} \big(H(h(z)) - f(z,0) \big) dz \right) \overline{\psi}_i(x,t) \\
&+ \sum_{j=1}^{\infty} \alpha_j \overline{\rho}_j(x,t), \qquad (5.3.4)
\end{aligned}$$

where $a(x)$, $h(x)$, $f(x,t)$, $H(x)$ are given functions in (5.3.1)*, and α_j are coefficients to be determined.*

Proof. Due to $u(x,t) \in {}^{o}W_2^{(2,3)}(\mathcal{D})$, hence

$$\begin{aligned}
u(x,t) &= \sum_{i=1}^{\infty} \big\langle u(x,t), \overline{\psi}_i(x,t) \big\rangle_{{}^{o}W_2^{(2,3)}} \overline{\psi}_i(x,t) \\
&+ \sum_{j=1}^{\infty} \big\langle u(x,t), \overline{\rho}_j(x,t) \big\rangle_{{}^{o}W_2^{(2,3)}} \overline{\rho}_j(x,t).
\end{aligned}$$

Let $\alpha_j = \langle u(x,t), \overline{\rho}_j(x,t)\rangle_W$, then

$$\begin{aligned}
u(x,t) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\big\langle u(x,t), (\mathbb{L}^*\phi_k)(x,t)\big\rangle_{{}_oW_2^{(2,3)}}\overline{\psi}_i(x,t) \\
&+ \sum_{j=1}^{\infty}\big\langle u(x,t), \overline{\rho}_j(x,t)\big\rangle_{{}_oW_2^{(2,3)}}\overline{\rho}_j(x,t) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\langle(\mathbb{L}u)(x), \phi_k(x)\rangle_{W_2^1}\overline{\psi}_i(x,t) \\
&+ \sum_{j=1}^{\infty}\big\langle u(x,t), \overline{\rho}_j(x,t)\big\rangle_{{}_oW_2^{(2,3)}}\overline{\rho}_j(x,t) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\langle F(x), \phi_k(x)\rangle_{W_2^1}\overline{\psi}_i(x,t) \\
&+ \sum_{j=1}^{\infty}\big\langle u(x,t), \overline{\rho}_j(x,t)\big\rangle_{{}_oW_2^{(2,3)}}\overline{\rho}_j(x,t) \\
&= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}F(s_k)\overline{\psi}_i(x,t) + \sum_{j=1}^{\infty}\big\langle u(x,t), \overline{\rho}_j(x,t)\big\rangle_{{}_oW_2^{(2,3)}}\overline{\rho}_j(x,t),
\end{aligned}$$

and from (5.3.3), we can obtain (5.3.4) □

Further we have

$$\begin{aligned}
h(x) &= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\left(a(s_k)h'(s_k) - a(0)h'(0) + \int_0^{s_k}(H(h(z)) - f(z,0))dz\right)\overline{\psi}_i(x,0) \\
&+ \sum_{j=1}^{\infty}\alpha_j\overline{\rho}_j(x,0). \qquad (5.3.5)
\end{aligned}$$

Make inner on the both sides of (5.3.5) with $\overline{\rho}_j(x,0)$, we can get an infinite linear equations about unknown α_j

$$\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}\left(a(s_k)h'(s_k) - a(0)h'(0) + \int_0^{s_k}(H(h(z)) - f(z,0))dz\right)b_{il} + \sum_{j=1}^{\infty}c_{jl}\alpha_j = d_l, \quad (5.3.6)$$

where $a_i,\ b_{il},\ c_{ij},\ d_{jl},\ r_l$ are constants given by formula

$$\begin{cases} b_{il} &= \left\langle \overline{\psi_i}(x,0), \overline{\rho}_l(x,0) \right\rangle_{W_2^1} \\ &= \dfrac{1}{60^2} \displaystyle\sum_{n=1}^{\infty} \sum_{k=1}^{i} \beta_{ln}\beta_{ik} \int_0^{s_k} dx \int_0^{s_n} R_x(z)dz - \frac{1}{60} \sum_{k=1}^{i} \sum_{m=1}^{l} \beta^*_{lm}\beta_{ik} G_{\eta_m}(0) \int_0^{s_k} R_{\xi_m}(z)dz \\ c_{jl} &= \left\langle \overline{\rho_j}(x,0), \overline{\rho}_l(x,0) \right\rangle_{W_2^1} \\ &= \dfrac{1}{60^2} \displaystyle\sum_{k=1}^{\infty} \sum_{p=1}^{\infty} \beta_{jk}\beta_{lp} \int_0^{s_k} dx \int_0^{s_p} R_x(z)dz - \frac{1}{60} \sum_{k=1}^{\infty} \sum_{m=1}^{l} \beta^*_{lm}\beta_{jk} G_{\eta_m}(0) \int_0^{s_k} R_{\xi_m}(z)dz \\ &\quad -\dfrac{1}{60} \displaystyle\sum_{p=1}^{\infty} \sum_{n=1}^{i} \beta^*_{jn}\beta_{lp} G_{\eta_n}(0) \int_0^{s_p} R_{\xi_n}(z)dz + \sum_{k=1}^{j} \sum_{m=1}^{l} \beta^*_{jn}\beta^*_{lm} G_{\eta_m}(0) G_{\eta_n}(0) R_{\xi_m}(\xi_n)dz \\ d_l &= \left\langle h(x), \overline{\rho}_l(x,0) \right\rangle_{W_2^1} \\ &= -\dfrac{1}{60} \displaystyle\sum_{k=1}^{\infty} \beta_{lk} \int_0^{s_k} h(z)dz + \sum_{n=1}^{l} \beta_{jn} G_{\eta_n}(0) h(\xi_n), \quad l = 1, 2, \ldots . \end{cases}$$

Theorem 5.3.1. *Assume that* $\mathrm{A} = \{s_1, s_2, \ldots\}$ *is a dense set of interval* $[0, 1]$, $\alpha_j,\ j = 1, 2, \ldots$ *are the solutions of the infinite Eq.* (5.3.6), *then the solution of Eq.* (5.3.1) *can be given by* (5.3.4).

5.3.6 Numerical Experiments

Example 5.3.1. For the nonlinear partial differential equation:

$$\begin{cases} \dfrac{\partial}{\partial x}\left[(4 - e^{-x})\dfrac{\partial u(x,t)}{\partial x}\right] - \dfrac{\partial^2 u(x,t)}{\partial t^2} + (1 + u(x,t))^2 = f(x,t), \\ u(x,0) = x^2 - 2x, \\ \dfrac{\partial}{\partial t} u(x,0) = 0, \\ u(0,t) = 0, \\ \dfrac{\partial}{\partial x} u(1,t) = 0, \quad 0 \le x \le 1, \quad 0 \le t \le +\infty, \end{cases}$$

where

$$\begin{aligned} f(x,t) &= 1 - e^{-\frac{t^2}{10}} \left(8 - 4e^{-x} - \frac{22}{5}\, x + 2xe^{-x} + \frac{2}{25}\, t^2 x + \frac{11}{5}\, x^2 - \frac{1}{25}\, t^2 x^2\right) \\ &+ x^2 e^{-\frac{t^2}{5}} (4 - 4x + x^2), \end{aligned}$$

the true solution is

$$u(x,t) = (x^2 - 2x)e^{-\frac{t^2}{10}}.$$

Table 5.6: The error of exact solution $u(x,t)$ and approximate solution $u_n(x,t)$

(x,t)	$u(x,t)$	$u_n(x,t)$	absolute error	(x,t)	$u(x,t)$	$u_n(x,t)$	absolute error
$(0,1)$	0	0	0	$(\frac{1}{10},\frac{9}{10})$	0.175217	0.156436	0.0187804
$(\frac{2}{10},\frac{8}{10})$	0.337682	0.309757	0.0279252	$(\frac{3}{10},\frac{7}{10})$	0.485612	0.456851	0.0287618
$(\frac{4}{10},\frac{6}{10})$	0.61737	0.592773	0.0245968	$(\frac{5}{10},\frac{5}{10})$	0.731482	0.713683	0.0177999
$(\frac{6}{10},\frac{4}{10})$	0.826667	0.815818	0.0108493	$(\frac{7}{10},\frac{3}{10})$	0.901847	0.896596	0.00525078
$(\frac{8}{10},\frac{2}{10})$	0.956168	0.954423	0.00174454	$(\frac{9}{10},\frac{1}{10})$	0.98901	0.988771	0.000239182

By using Mathematica 4.2 and applying (5.3.6), we calculate the the approximate solution $u_n(x,t)$ by truncating the series in (5.3.4). The numerical results are given in the Table 5.6.

5.4 Solving the Damped Nonlinear Klein–Gordon Equation

5.4.1 Introduction

Klein–Gordon equation arises in relativistic quantum mechanics and field theory. So it is of a great importance for the high energy physicists. The study of the Klein–Gordon equation have been investigated considerably in the last few years [143]–[148]). Authors have developed numerical methods for solving the Klein–Gordon equation [149]–[153]. However, there are few references in which the numerical methods for solving damped nonlinear Klein–Gordon equation are presented. In [154], [155], authors studied the optimal control of damped Klein–Gordon equations. Identification problem for damped Klein–Gordon equations was discussed in [156]. [157] gave a simulation test, but numerical results. Hence, we give the numerical solution of the model presented in [157]. The results demonstrate that our method is right and is of high accuracy. Meanwhile, we construct a model of damped nonlinear Klein–Gordon equation with a true solution in Section 5. The numerical results of this model also demonstrate the efficiency of the present method. Throughout the study, reproducing kernel space plays an important role in solving damped nonlinear Klein–Gordon equation. In addition, the good convergence of numerical solution has close relations with this space frame. Particularly, the Klein–Gordon equation with damping term can be described by

$$\frac{\partial^2 y}{\partial t^2} + \alpha \frac{\partial y}{\partial t} - \beta \Delta y + \delta |y|^\gamma y = f(t,x), \quad (t,x) \in Q, \tag{5.4.1}$$

with the initial-boundary values conditions

$$y = 0 \ \text{ on } \ \Sigma,$$
$$y(0,x) = y_0(x), \quad \frac{\partial y(0,x)}{\partial t} = y_1(x), \quad x \in \Omega,$$

where α, β, $\gamma > 0$, $\delta \in R$, f is a forcing function and Δ is a Laplacian, Ω is a bounded set in R^n with a piecewise smooth boundary Γ; $\Sigma = (0,T_0) \times \Gamma$, $Q = (0,T_0) \times \Omega$.

Eq. (5.4.1) is known as one of the nonlinear wave equations arising in relativistic quantum mechanics and has been studied extensively. In [1], the existence and uniqueness of the solution of Eq. (5.4.1) has been established.

Here, we will consider the one-dimension form of Eq. (5.4.1) as follows:

$$\begin{cases} \dfrac{\partial^2 y(t,x)}{\partial t^2} + \alpha \dfrac{\partial y(t,x)}{\partial t} - \beta \dfrac{\partial^2 y(t,x)}{\partial x^2} + \delta |y(t,x)|^\gamma y(t,x) = f(t,x), \quad (t,x) \in Q, \\ y(t,0) = y(t,1) = 0, \quad t \in (0,1), \\ y(0,x) = y_0(x), \quad \dfrac{\partial y(0,x)}{\partial t} = y_1(x), \quad x \in [0,1] \end{cases} \tag{5.4.2}$$

where $Q = (0,1) \times [0,1]$.

In order to put initial boundary value conditions of $Eq(1.2)$ into the reproducing kernel space ${}^oW_2^{(3,3)}(\mathcal{D})$ constructed in the following part, it is must to homogenize the initial conditions. Put

$$\widetilde{y}(t,x) = y(t,x) - a(t,x)y_0(x) - b(t,x)y_1(x),$$

where

$$a(t,x) = \begin{cases} e^{-\frac{t^2}{x(1-x)}}, & 0 < x < 1, \\ 0, & x = 0,1, \end{cases} \quad b(t,x) = \begin{cases} te^{-\frac{t}{x(1-x)}}, & 0 < x < 1, \\ 0, & x = 0,1. \end{cases}$$

Denote $a(t,x)y_0(x) + b(t,x)y_1(x)$ by $A(t,x)$, then we can obtain homogeneous initial conditions of Eq. (5.4.2). Immediately, we get

$$\begin{cases} \dfrac{\partial^2 \widetilde{y}(t,x)}{\partial t^2} + \alpha \dfrac{\partial \widetilde{y}(t,x)}{\partial t} - \beta \dfrac{\partial^2 \widetilde{y}(t,x)}{\partial x^2} \\ \qquad + \delta |\widetilde{y}(t,x) - A(t,x)|^\gamma (\widetilde{y}(t,x) - A(t,x)) = \widetilde{f}(t,x), \\ \widetilde{y}(t,0) = \widetilde{y}(t,1) = 0, \quad t \in (0,1), \\ \widetilde{y}(0,x) = \dfrac{\partial \widetilde{y}(0,x)}{\partial t} = 0, \quad x \in [0,1], \end{cases} \tag{5.4.3}$$

where

$$\widetilde{f}(t,x) = f(t,x) + \frac{\partial^2 A(t,x)}{\partial t^2} + \alpha \frac{\partial A(t,x)}{\partial t} - \beta \frac{\partial^2 A(t,x)}{\partial x^2}.$$

Following we will always replace $\widetilde{y}(t,x)$ and $\widetilde{f}(t,x)$ with $y(t,x)$ and $f(t,x)$ in (5.4.3), respectively.

5.4.2 Linear Operator on Reproducing Kernel Spaces

Let $\mathcal{D} = [0,1] \times [0,1]$.

1) The reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$ is defined by (1.5.1) and its reproducing kernel is $h_{(\xi,\eta)}(t,x) = r_\xi(t) r_\eta(x)$, where $r_\xi(t)$ is the form (1.4.1).

2) The reproducing kernel space

$$
\begin{aligned}
&{}^{o}W_2^{(3,3)}(\mathcal{D})\\
&\quad=\Big\{y(t,x)|y\in W_2^{(3,3)}(\mathcal{D}),\ y(t,0)=y(t,1)=y(0,x)=\partial_t y(0,x)=0\Big\}.
\end{aligned}
$$

It possesses the reproducing kernel function

$$
K_{(\xi,\eta)}(t,x)=Q_\xi(t)R_\eta(x), \tag{5.4.4}
$$

where

$$
Q_\xi(t)=\begin{cases}\dfrac{1}{120}\,t^2\big(-5\xi t^2+t^3+10\xi^2(3+t)\big), & t\le\xi,\\[2mm] \dfrac{1}{120}\,\xi^2\big(-5t\xi^2+\xi^3+10t^2(3+\xi)\big), & t<\xi\end{cases} \tag{5.4.5}
$$

$R_\eta(x)$ by (5.1.6).

To solve Eq. (5.4.3), we introduce a linear operator $\mathbb{L}:{}^{o}W_2^{(3,3)}(\mathcal{D})\longrightarrow W_2^{(1,1)}(\mathcal{D})$. For every $y(t,x)\in{}^{o}W_2^{(3,3)}(\mathcal{D})$,

$$
(\mathbb{L}y)(t,x)=\frac{\partial^2 y(t,x)}{\partial t^2}+\alpha\,\frac{\partial y(t,x)}{\partial t}-\beta\,\frac{\partial^2 y(t,x)}{\partial x^2}\,. \tag{5.4.6}
$$

Lemma 5.4.1. *$\mathbb{L}:{}^{o}W_2^{(3,3)}(\mathcal{D})\longrightarrow W_2^{(1,1)}(\mathcal{D})$ is a bounded linear operator.*

Proof. Note that, $y(t,x)=\langle y(s,z),G_t(s)R_x(z)\rangle_{{}^{o}W_2^{(3,3)}}$,

$$
\begin{aligned}
\left|\frac{\partial^2 y(t,x)}{\partial t^2}\right| &= \left|\left\langle y(s,z),\frac{\partial^2 G_t(s)}{\partial t^2}\,R_x(z)\right\rangle_{{}^{o}W_2^{(3,3)}}\right|\\
&\le \left\|\frac{\partial^2 G_t(s)}{\partial t^2}R_x(z)\right\|_{{}^{o}W_2^{(3,3)}}\|y(s,z)\|_{{}^{o}W_2^{(3,3)}}\\
&\le M_1\|y(s,z)\|_{{}^{o}W_2^{(3,3)}},\\
\left|\frac{\partial y(t,x)}{\partial t}\right| &\le \left\|\frac{\partial G_t(s)}{\partial t}R_x(z)\right\|\|y(s,z)\|_{{}^{o}W_2^{(3,3)}}\le M_2\|y(s,z)\|_{{}^{o}W_2^{(3,3)}},\\
\left|\frac{\partial^2 y(t,x)}{\partial x^2}\right| &\le \left\|\frac{\partial^2 R_x(z)}{\partial x^2}G_t(s)\right\|_{{}^{o}W_2^{(3,3)}}\|y(s,z)\|_{{}^{o}W_2^{(3,3)}}\le M_3\|y(s,z)\|_{{}^{o}W_2^{(3,3)}}.
\end{aligned}
$$

So that

$$
\|\mathbb{L}y\|_{W_2^{(1,1)}}\le M\|y\|_{{}^{o}W_2^{(3,3)}},
$$

where M is a positive real number. □

We rewrite Eq. (5.4.3) as the following operator equation:

$$
\begin{cases}(\mathbb{L}y)(t,x)=f(t,x)-\delta\big|y(t,x)-A(t,x)\big|^r(y(t,x)-A(t,x)),\\ y(t,0)=y(t,1)=0,\quad y(0,x)=\dfrac{\partial y}{\partial t}(0,x)=0.\end{cases} \tag{5.4.7}
$$

5.4.3 The Solution of Eq. (5.4.3)

Using the adjoint operator $\mathbb{L}^*$ of $\mathbb{L}$ given in Eq. (5.4.6), we will choose and fix a countable dense subset $\mathrm{T} = \{p_1(t_1, x_1), p_1(t_2, x_2) \ldots\} \subset \mathcal{D}$ and define $\psi_i(t, x)$ by

$$\psi_i(t, x) \stackrel{def}{=} (\mathbb{L}^* h_{(t_i, x_i)})(t, x), \quad i = 1, 2, \ldots .$$

Lemma 5.4.2.

$$\psi_i(t, x) = (\mathbb{L} K_{(t,x)})(t_i, x_i).$$

Proof.

$$\begin{aligned} \psi_i(t, x) &= \big\langle \psi_i(\circ, \star), K_{(t,x)}(\circ, \star) \big\rangle_{{}^oW_2^{(3,3)}} \\ &= \big\langle (\mathbb{L}^* h_{(t_i,x_i)})(\circ, \star), K_{(t,x)}(\circ, \star) \big\rangle_{{}^oW_2^{(3,3)}} \\ &= \big\langle (h_{(t_i,x_i)}), (\mathbb{L} K_{(t,x)}) \big\rangle_{W_2^{(1,1)}} = (\mathbb{L} K_{(t,x)})(t_i, x_i). \end{aligned}$$

The proof is complete. □

Lemma 5.4.3. *The function system $\{\psi_i(t, x)\}_{i=1}^{\infty}$ is a complete system of the space ${}^oW_2^{(3,3)}(\mathcal{D})$.*

The proof is omitted.

By Gram–Schmidt process, we obtain an orthogonal basis $\{\widetilde{\psi}_i(x, y)\}_{i=1}^{\infty}$ of ${}^oW_2^{(3,3)}(\mathcal{D})$, such that

$$\widetilde{\psi}_i(x, y) = \sum_{k=1}^{i} \beta_{ik} \psi_k(x, y),$$

where β_{ik} are orthogonal coefficients.

Theorem 5.4.1. *If $y(t, x)$ is the solution of Eq. (5.4.3), then*

$$y(t, x) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} \big[f(t_k, x_k) - \alpha_k \big] \widetilde{\psi}_i(t, x), \tag{5.4.8}$$

where

$$\alpha_k = \delta \big| y(t_k, x_k) - A(t_k, x_k) \big|^{\gamma} (y(t_k, x_k) - A(t_k, x_k)), \quad k = 1, 2, \ldots . \tag{5.4.9}$$

Proof. $y(t, x)$ can be expanded to Fourier series in terms of normal orthogonal basis $\overline{\psi}_i(t, x)$ in ${}^oW_2^{(3,3)}(\mathcal{D})$,

$$y(t, x) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \langle y(t, x), \widetilde{\psi}_i(t, x) \rangle_{{}^oW_2^{(3,3)}} \widetilde{\psi}_i(t, x)$$

$$= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle y(t,x), \psi_k(t,x)\rangle_{{}^{o}W_2^{(3,3)}} \widetilde{\psi}_i(t,x)$$

$$= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle \mathbb{L}y(t,x), \varphi_k(t,x)\rangle_{W_2^{(1,1)}} \widetilde{\psi}_i(t,x)$$

$$= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\langle f(t,x) - \delta|y(t,x) - A(t,x)|^{\gamma}(y(t,x) - A(t,x)), \varphi_k(t,x)\rangle_{W_2^{(1,1)}} \widetilde{\psi}_i(t,x)$$

$$= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}\Big[f(t_k,x_k) - \delta\big|y(t_k,x_k) - A(t_k,x_k)\big|^{\gamma}(y(t_k,x_k) - A(t_k,x_k))\Big]\widetilde{\psi}_i(t,x).$$

Denote $\delta|y(t_k,x_k) - A(t_k,x_k)|^{\gamma}(y(t_k,x_k) - A(t_k,x_k))$ by α_k, $k = 1, 2, \ldots$, then

$$y(t,x) = \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik}[f(t_k,x_k) - \alpha_k]\widetilde{\psi}_i(t,x). \qquad \square$$

In order to obtain α_k in (5.4.8), put

$$\sum_{k=1}^{\infty}\Big[\delta\big|y(t_k,x_k) - A(t_k,x_k)\big|^{\gamma}(y(t_k,x_k) - A(t_k,x_k)) - \alpha_k\Big]^2 = 0. \tag{5.4.10}$$

By truncating the series of the left-hand side of (5.4.10), we obtain

$$\sum_{k=1}^{n}\Big[\delta\big|y_n(t_k,x_k) - A(t_k,x_k)\big|^{\gamma}(y_n(t_k,x_k) - A(t_k,x_k)) - \alpha_k\Big]^2 = 0, \tag{5.4.11}$$

where

$$y_n(t,x) = \sum_{i=1}^{n}\sum_{k=1}^{i} \beta_{ik}[f(t_k,x_k) - \alpha_k]\widetilde{\psi}_i(t,x).$$

Using Mathematica 5.0, we obtain α_k when

$$J(\alpha_1,\alpha_2,\ldots,\alpha_n) = \sum_{k=1}^{n}\Big[\delta\big|y_n(t_k,x_k) - A(t_k,x_k)\big|^{\gamma}\big(y_n(t_k,x_k) - A(t_k,x_k)\big) - \alpha_k\Big]^2$$

is minimum. Consequently, we obtain the approximate solution of Eq. (5.4.3)

$$y_n(t,x) = \sum_{i=1}^{n}\sum_{k=1}^{i} \beta_{ik}[f(t_k,x_k) - \alpha_k]\widetilde{\psi}_i(t,x). \tag{5.4.12}$$

$y_n(t,x)$ in (5.4.12) is the n-term intercept of $y(t,x)$ in (5.4.8). So $y_n(t,x) \longrightarrow y(t,x)$ in ${}^{o}W_2^{(3,3)}(\mathcal{D})$ as $n \longrightarrow \infty$. Following we shall give the algorithm of obtaining $y_n(t,x)$ in (5.4.12).

(1) Pick initial values α_k^0;

(2) Substitute α_k^0 into (5.4.12) and compute $y_n(t,x)$;

(3) Calculate $J(\alpha_1^0, \alpha_2^0, \ldots, \alpha_n^0)$;

(4) If $J(\alpha_1^0, \alpha_2^0, \ldots, \alpha_n^0) < 10^{-20}$, then computation terminate; otherwise, substitution of $y_n(t,x)$ into (5.4.9) yields new α_k^1;

(5) Calculate $J(\alpha_1^1, \alpha_2^1, \ldots, \alpha_n^1)$;

(6) If $J(\alpha_1^1, \alpha_2^1, \ldots, \alpha_n^1) < J(\alpha_1^0, \alpha_2^0, \ldots, \alpha_n^0)$, then replace α_k^0 with α_k^1 and return to (2); otherwise, give up α_k^1. Taking α_k^0 as initial values, we can obtain the minimal value points of $J(\alpha_1, \alpha_2, \ldots, \alpha_n)$ using Mathematics ,that is new α_k^1 and replace α_k^0 with α_k^1 , return to (2).

Theorem 5.4.2. *The approximate solution y_n and its derivatives $\partial_t y_n$, $\partial_x y_n$, $\partial_{tt} y_n$, $\partial_{xx} y_n$, $\partial_{tx} y_n$ uniformly converge to exact solution y and its derivatives $\partial_t y$, $\partial_x y$, $\partial_{tt} y$, $\partial_{xx} y$, $\partial_{tx} y$, respectively.*

Similar to Theory 1.3.4, the proof can be obtained.

5.4.4 Numerical experiments

Example 5.4.1. For Eq. (5.4.1), put $\alpha = 1$, $\beta = 1$, $\delta = 1$, $\gamma = \frac{1}{2}$, that is

$$\begin{cases} \dfrac{\partial^2 y(t,x)}{\partial t^2} + \dfrac{\partial y(t,x)}{\partial t} - \dfrac{\partial^2 y(t,x)}{\partial x^2} + |y(t,x)|^{\frac{1}{2}} y(t,x) = f(t,x), & (t,x) \in Q, \\ y(t,0) = y(t,1) = 0, & t \in (0,1), \\ y(0,x) = \dfrac{\partial y(0,x)}{\partial t} = 0, & x \in [0,1], \end{cases}$$

where

$$f(t,x) = 2 + 2t + \pi^2 t^2 + t^2 \sqrt{|t^2 \sin \pi x|}\, \sin \pi x.$$

Its true solution $y(t,x) = t^2 \sin \pi x$. By using Mathematica 5.0 and the presented method in Section 5.4.3, where $n = 144$, in (5.4.12), we calculate the approximate solution $y_n(t,x)$. The numerical results are given in Table 5.7. The mean root square deviations for the derivatives of approximate solution $y_n(t,x)$ and the derivatives of the true solution $y(t,x)$ are given in Table 5.8.

Example 5.4.2. In [157], the author gave a simulation to test (5.4.2), where

$$y_0(x) = \sin 3\pi x, \quad y_1(x) = \cos 3\pi x, \quad f(t,x) = 0.$$

We also give the numerical solution of the model. First we homogenize initial conditions of (5.4.2), then (5.4.2) can be converted into (5.4.3). Applying the method presented in Section 4 (where $n = 64$, in (5.4.12)), we can obtain the approximate solution $\widetilde{y}_n(t,x)$ of (5.4.3) and the approximate solution $y_n(t,x)$ of (5.4.2). We shall test approximate solution $y_n(t,x)$ in the following way.

(1) We verify whether $y_n(t,x)$ satisfies the initial-boundary value condition or not (see Figure 5.1–5.2).

(2) Substituting approximate solution $y_n(t,x)$ into the left-hand of origin model (5.4.2), we compute the error of $\left|\frac{\partial_t^2 y_n}{\partial t^2} + \alpha\frac{\partial_t y_n}{\partial t} - \beta\frac{\partial_x^2 y_n}{\partial x^2} + \delta|y_n|^\gamma y_n - f\right|$. The error is satisfying (see Table 5.9).

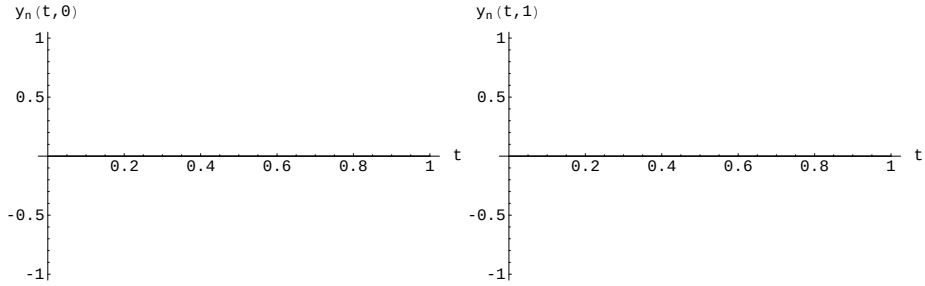


Figure 5.1: Left image is $y_n(t,0)$ and right image is $y_n(t,1)$ in Example 2.

5.4.5 Conclusion

We have presented a method for solving damped nonlinear Klein–Gordon equation. The essence of this method is the Fourier series expansion in reproducing kernel space. Its advantages are:

(1) We use the conjugate operator to construct a basis

$$\psi_i(t,x) = \mathbb{L}^*(h_{(t_i,x_i)}(t,x))(\mathcal{D}),$$

where $h_{(\xi,\eta)}(t,x)$ is the reproducing kernel of $W_2^{(1,1)}$. In general, the conjugate operator L^* acting on a function is difficult to be represented directly.

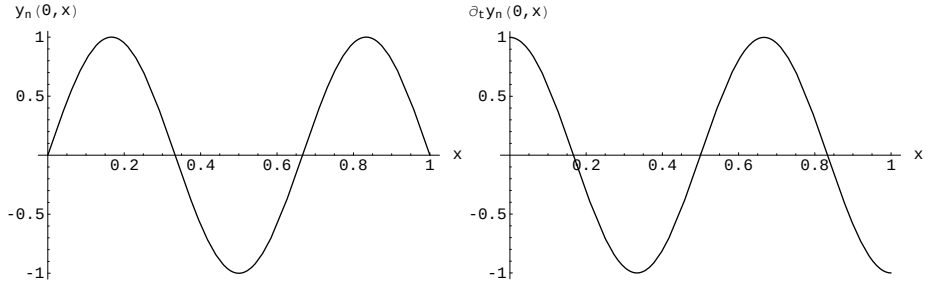


Figure 5.2: Left image is $y_n(0,x)$ and right image is $\partial_t y_n(0,x)$ in Example 2.

Table 5.7: The error of function $y(t,x)$

(t,x)	$y(t,x)$	$y_n(t,x)$	absolute error
$(0,0)$	0	0	0
$(\frac{1}{12},\frac{1}{12})$	0.00179735	0.00169857	0.0000987847
$(\frac{1}{4},\frac{1}{12})$	0.0161762	0.0158423	0.000333869
$(\frac{1}{2},\frac{1}{12})$	0.0647048	0.0642757	0.000429068
$(\frac{3}{4},\frac{1}{12})$	0.145586	0.145455	0.000130918
$(1,\frac{1}{2})$	0.258819	0.259287	0.000468143
$(\frac{1}{4},\frac{1}{6})$	0.03125	0.0305827	0.000667308
$(\frac{1}{2},\frac{1}{6})$	0.125	0.124041	0.000958924
$(\frac{3}{4},\frac{1}{6})$	0.28125	0.280683	0.000566792
$(1,\frac{1}{6})$	0.5	0.50033	0.00033007
$(\frac{1}{4},\frac{1}{4})$	0.0441942	0.0432463	0.000947855
$(\frac{1}{2},\frac{1}{4})$	0.176777	0.175361	0.0014152
$(\frac{3}{4},\frac{1}{4})$	0.397748	0.396793	0.000954693
$(1,\frac{1}{4})$	0.707107	0.707287	0.000179913
$(\frac{1}{4},\frac{1}{3})$	0.0541266	0.0529653	0.00116132
$(\frac{1}{2},\frac{1}{3})$	0.216506	0.214735	0.00177185
$(\frac{3}{4},\frac{1}{3})$	0.487139	0.485863	0.00127588
$(1,\frac{1}{3})$	0.866025	0.866042	0.0000167167
$(\frac{1}{4},\frac{5}{12})$	0.0603704	0.0590765	0.00129385
$(\frac{1}{2},\frac{5}{12})$	0.241481	0.239486	0.00199504
$(\frac{3}{4},\frac{5}{12})$	0.543333	0.54185	0.00148321
$(1,\frac{5}{12})$	0.965926	0.965824	0.000101594
$(\frac{1}{4},\frac{1}{2})$	0.0625	0.0611596	0.00134036
$(\frac{1}{2},\frac{1}{2})$	0.25	0.24792	0.00207973
$(\frac{3}{4},\frac{1}{2})$	0.5625	0.560918	0.00158192
$(1,\frac{1}{2})$	1	0.9998	0.000200106
$(\frac{1}{4},\frac{7}{12})$	0.0603704	0.0590726	0.00129781
$(\frac{1}{2},\frac{7}{12})$	0.241481	0.239461	0.00202072
$(\frac{3}{4},\frac{7}{12})$	0.543333	0.541769	0.00156416
$(1,\frac{7}{12})$	0.965926	0.965657	0.000268917
$(\frac{1}{4},\frac{2}{3})$	0.0541266	0.0529572	0.00116935
$(\frac{1}{2},\frac{2}{3})$	0.216506	0.214679	0.0018273
$(\frac{3}{4},\frac{2}{3})$	0.487139	0.485697	0.00144252
$(1,\frac{2}{3})$	0.866025	0.865703	0.000322126
$(\frac{1}{4},\frac{3}{4})$	0.0441942	0.043232	0.000962198
$(\frac{1}{2},\frac{3}{4})$	0.176777	0.175264	0.00151266
$(\frac{3}{4},\frac{3}{4})$	0.397748	0.396521	0.0012265
$(1,\frac{3}{4})$	0.707107	0.70675	0.000356661
$(\frac{1}{4},\frac{5}{6})$	0.03125	0.0305613	0.000688701
$(\frac{1}{2},\frac{5}{6})$	0.125	0.123903	0.00109662
$(\frac{3}{4},\frac{5}{6})$	0.28125	0.280321	0.000928807
$(1,\frac{5}{6})$	0.5	0.499634	0.000365649
$(\frac{1}{4},\frac{11}{12})$	0.0161762	0.015813	0.000363239
$(\frac{1}{2},\frac{11}{12})$	0.0647048	0.0641123	0.000592424
$(\frac{3}{4},\frac{11}{12})$	0.145586	0.145049	0.00053677
$(1,\frac{11}{12})$	0.258819	0.258529	0.000290169
$(\frac{1}{4},1)$	0	0	0
$(\frac{1}{2},1)$	0	0	0
$(\frac{3}{4},1)$	0	0	0
$(1,1)$	0	0	0

Table 5.8: The mean root square deviations for the derivatives

$\sqrt{\sum_{i=1}^{n}(y_n-y)_t^2/n}$	$\sqrt{\sum_{i=1}^{n}(y_n-y)_x^2/n}$	$\sqrt{\sum_{i=1}^{n}(y_n-y)_{tt}^2/n}$	$\sqrt{\sum_{i=1}^{n}(y_n-y)_{xx}^2/n}$	$\sqrt{\sum_{i=1}^{n}(y_n-y)_{tx}^2/n}$
0.000152208	0.00972659	0.000114178	0.00930846	0.000741472

Table 5.9: The error of $\left|\frac{\partial_t^2 y_n}{\partial t^2}+\alpha\frac{\partial_t y_n}{\partial t}-\beta\frac{\partial_x^2 y_n}{\partial x^2}+\delta|y_n|^\gamma y_n-f\right|$

(t,x)	error	(t,x)	error	(t,x)	error
$(\frac{1}{8},\frac{1}{8})$	1.30109E-8	$(\frac{1}{4},\frac{1}{8})$	1.88352E-8	$(\frac{3}{8},\frac{1}{8})$	2.52278E-8
$(\frac{1}{2},\frac{1}{8})$	3.25739E-8	$(\frac{5}{8},\frac{1}{8})$	4.49226E-8	$(\frac{3}{4},\frac{1}{8})$	5.73209E-8
$(\frac{7}{8},\frac{1}{8})$	6.74096E-8	$(1,\frac{1}{8})$	1.04987E-7	$(\frac{1}{8},\frac{1}{4})$	2.5026E-8
$(\frac{1}{4},\frac{1}{4})$	3.11488E-8	$(\frac{3}{8},\frac{1}{4})$	4.3293E-8	$(\frac{1}{2},\frac{1}{4})$	6.49887E-8
$(\frac{5}{8},\frac{1}{4})$	9.94257E-8	$(\frac{3}{4},\frac{1}{4})$	1.18767E-7	$(\frac{7}{8},\frac{1}{4})$	1.60329E-7
$(1,\frac{1}{4})$	2.0458E-7	$(\frac{1}{8},\frac{3}{8})$	3.40199E-8	$(\frac{1}{4},\frac{3}{8})$	2.95831E-8
$(\frac{3}{8},\frac{3}{8})$	4.01212E-8	$(\frac{1}{2},\frac{3}{8})$	8.61876E-8	$(\frac{5}{8},\frac{3}{8})$	1.56464E-7
$(\frac{3}{4},\frac{3}{8})$	2.03541E-7	$(\frac{7}{8},\frac{3}{8})$	2.9967E-7	$(1,\frac{3}{8})$	4.09293E-7
$(\frac{1}{8},\frac{1}{2})$	1.80912E-8	$(\frac{1}{4},\frac{1}{2})$	1.05384E-8	$(\frac{3}{8},\frac{1}{2})$	7.37387E-9
$(\frac{1}{2},\frac{1}{2})$	3.31E-8	$(\frac{5}{8},\frac{1}{2})$	1.00733E-7	$(\frac{3}{4},\frac{1}{2})$	6.12193E-8
$(\frac{7}{8},\frac{1}{2})$	9.93533E-8	$(1,\frac{1}{2})$	3.34101E-7	$(\frac{1}{8},\frac{5}{8})$	5.94095E-8
$(\frac{1}{4},\frac{5}{8})$	9.84102E-8	$(\frac{3}{8},\frac{5}{8})$	1.54047E-7	$(\frac{1}{2},\frac{5}{8})$	1.08379E-7
$(\frac{5}{8},\frac{5}{8})$	1.60087E-7	$(\frac{3}{4},\frac{5}{8})$	2.38162E-7	$(\frac{7}{8},\frac{5}{8})$	3.53861E-7
$(1,\frac{5}{8})$	3.36553E-7	$(\frac{1}{8},\frac{3}{4})$	2.41853E-8	$(\frac{1}{4},\frac{3}{4})$	3.32353E-8
$(\frac{3}{8},\frac{3}{4})$	8.08178E-9	$(\frac{1}{2},\frac{3}{4})$	1.01562E-7	$(\frac{5}{8},\frac{3}{4})$	1.92346E-7
$(\frac{3}{4},\frac{3}{4})$	1.76453E-7	$(\frac{7}{8},\frac{3}{4})$	2.70557E-7	$(1,\frac{3}{4})$	1.09973E-6
$(\frac{1}{8},\frac{7}{8})$	6.26277E-8	$(\frac{1}{4},\frac{7}{8})$	9.61575E-8	$(\frac{3}{8},\frac{7}{8})$	1.49274E-7
$(\frac{1}{2},\frac{7}{8})$	8.1272E-9	$(\frac{5}{8},\frac{7}{8})$	1.37606E-7	$(\frac{3}{4},\frac{7}{8})$	7.28689E-8
$(\frac{7}{8},\frac{7}{8})$	5.28157E-8	$(1,\frac{7}{8})$	4.2853E-8	$(\frac{1}{8},1)$	2.46473E-10
$(\frac{1}{4},1)$	2.28465E-9	$(\frac{3}{8},1)$	6.28643E-9	$(\frac{1}{2},1)$	4.04543E-9
$(\frac{5}{8},1)$	1.25729E-8	$(\frac{3}{4},1)$	2.48547E-8	$(\frac{7}{8},1)$	2.45054E-8
$(1,1)$	0	$(1,1)$	6.73463E-8		

However, in the reproducing kernel space, $\psi_i(t,x)$ can be represented easily by using the reproducing kernel function. (see Lemma 5.4.2).

(2) In the Fourier series expansion, we need compute inner product, which is defined by integral. However, in our method, the computation of inner product can be replaced by the function value at one point.

(3) In Theorem 5.4.2, we prove that the convergence in space ${}^oW_2^{(3,3)}(\mathcal{D})$ is stronger than the convergence in space $C^2(\mathcal{D})$. Hence, the approximate solution and its derivatives are convergent to exact solution and its derivatives, respectively.

5.5 Solving a Nonlinear Second Order System

5.5.1 Introduction

We consider the following nonlinear system of second order boundary value problems in the reproducing kernel space

$$\begin{cases} u_1'' + a_1u_1' + a_2u_1 + a_3u_2'' + a_4u_2' + a_5u_2 + N_1(u_1, u_2) = f_1(x), \ \ 0 \le x \le 1, \\ u_1'' + b_1u_1' + b_2u_1 + b_3u_2'' + b_4u_2' + b_5u_2 + N_2(u_1, u_2) = f_2(x), \ \ 0 \le x \le 1, \\ u_1(0) = u_1(1) = 0, \quad u_2(0) = u_2(1) = 0, \end{cases} \tag{5.5.1}$$

where N_i, f_i $(i = 1, 2)$, $a_j(x)$, $b_j(x)$ $(j = 1, 2, 3, 4, 5)$ are sufficiently regular given functions. Ordinary differential systems are important tools in solving real-world problems. A wide variety of natural phenomena are modelled by second order ordinary differential systems. Ordinary differential systems have been applied to many problems, in physics, engineering, biology and so on. For example, the so-called Emden–Fowler equations arises in the study of gas dynamics, fluid mechanics, relativistic mechanics, nuclear physics and also in the study of chemically reacting systems. However, many classical numerical methods used with second order initial value problems can not be applied to second order boundary value problems. We all know that the finite difference method can be used to solve linear second order boundary value problems, but it may be difficult to solve nonlinear second order boundary value problems using this method. For nonlinear second order boundary value problems, there are few valid methods to obtain numerical solutions. In [158]–[162] the authors discussed the existence of solutions to second order systems, including the approximation of solutions via finite difference equations. Valanarasu and Ramannujam suggested a method to solve a system of singularly linear second order ordinary differential equations [163].

5.5.2 Several Reproducing Kernel Spaces and Lemmas

1) The reproducing kernel space ${}^oW_2^3[0, 1])$ is defined as

$${}^oW_2^3[0, 1] = \Big\{u(x)|\ u(x) \in W_2^3[0, 1]\ (1.3.3),\ u(0) = 0, u(1) = 0\Big\}.$$

   Its reproducing kernel is $R_y(x)$ given by (5.1.6).

2) The reproducing kernel space $W_2^1[0, 1]$ is defined as (1.3.3), its reproducing kernel is $r_y(x)$ by (1.4.1).

Here, we will give the representation of analytical solution to Eq. (5.5.1) in the reproducing kernel space under the assumption that the solution to Eq. (5.5.1) is unique.

Define linear operators $\mathbb{A}_{ij} : {}^oW_2^3[0,1] \longrightarrow W_2^1[0,1]$, $i,j = 1,2$ by

$$\mathbb{A}_{11}u_1 = u_1'' + a_1(x)u_1' + a_2(x)u_1, \quad \mathbb{A}_{12}u_2 = a_3(x)u_2'' + a_4(x)u_2' + a_5(x)u_2,$$
$$\mathbb{A}_{21}u_1 = u_1'' + b_1(x)u_1' + b_2(x)u_1, \quad \mathbb{A}_{22}u_2 = b_3(x)u_2'' + b_4(x)u_2' + b_5(x)u_2.$$

It is obvious to $\mathbb{A}_{ij}$ is bounded. Define the spaces

$$\begin{aligned}\mathbf{W}_3[0,1] &= \Big\{\mathbf{u} = (u_1,u_2)^\top |\ u_i \in {}^oW_2^3[0,1],\ i = 1,2\Big\},\\ \mathbf{W}_1[0,1] &= \Big\{\mathbf{u} = (u_1,u_2)^\top |\ u_i \in W_2^1[0,1],\ i = 1,2\Big\}.\end{aligned}$$

Respectively, the inner product by

$$\begin{aligned}\langle \mathbf{u},\mathbf{v}\rangle_{\mathbf{W}_3} &= \sum_{i=1}^{2}\langle u_i, v_i\rangle_{{}^oW_2^3},\\ \langle \mathbf{u},\mathbf{v}\rangle_{\mathbf{W}_1} &= \sum_{i=1}^{2}\langle u_i, v_i\rangle_{W_2^1}.\end{aligned}$$

Clearly, they are both Hilbert space.

Lemma 5.5.1. *A linear operator* $\mathbb{A} : \mathbf{W}_3[0,1] \longrightarrow \mathbf{W}_1[0,1]$*, where*

$$\mathbb{A} = \begin{pmatrix} \mathbb{A}_{11} & \mathbb{A}_{12} \\ \mathbb{A}_{21} & \mathbb{A}_{22} \end{pmatrix}$$

is bounded.

Then Eq. (5.5.1) can be converted into the following form

$$\begin{cases} \mathbb{A}\mathbf{u} = \mathbf{F}(x) - \mathbf{N}(u_1,u_2), \ \ 0 \le x \le 1,\\ \mathbf{u}(0) = \mathbf{u}(1) = \mathbf{0}, \end{cases} \tag{5.5.2}$$

where $\mathbf{u} = (u_1,u_2)^\top$, $\mathbf{F} = (f_1,f_2)^\top$, $\mathbf{N} = (N_1,N_2)^\top$.

5.5.3 The Analytical and Approximate Solutions of Eq. (5.5.2)

Put

$$\Phi_{ij}(x) = r_{x_i}(x)\mathbf{e}_j = \begin{cases}(r_{x_i}(x),0)^\top, & j = 1,\\ (0, r_{x_i}(x))^\top, & j = 2,\end{cases}$$
$$\mathbf{e}_1 = (1,0)^\top, \quad \mathbf{e}_2 = (0,1)^\top, \ \ x_i \in [0,1],$$
$$\Psi_{ij}(x) = \mathbb{A}^*\Phi_{ij}(x),$$

where $\mathbb{A}^*$ is the is the conjugated operator of $\mathbb{A}$.

Theorem 5.5.1. *If* $\{x_i\}_{i=1}^{\infty}$ *is dense on* $[0,1]$*, then* $\{\Psi_{ij}(x)_{j=1,2}\}_{i=1}^{\infty}$ *is the complete system of* $\mathbf{W}_3[0,1]$.

The proof is omitted.

The normal orthogonal basis $\{\overline{\Psi}_{ij}(x)_{j=1,2}\}_{i=1}^{\infty}$ of $\mathbf{W}_3[0,1]$ can be derived from Gram–Schmidt orthogonalization process of $\{\Psi_{ij}(x)_{j=1,2}\}_{i=1}^{\infty}$,

$$\overline{\Psi}_{ij}(x) = \sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}\Psi_{lk}(x), \quad i = 1,2,\ldots, \quad j = 1,2.$$

Theorem 5.5.2. *If* $\{x_i\}_{i=1}^{\infty}$ *is dense on* $[0,1]$ *and the solution of Eq.* (5.5.2) *is unique, then the solution of Eq.* (5.5.2) *satisfies the form*

$$\mathbf{u}(x) = \sum_{i=1}^{\infty}\sum_{j=1}^{2}\sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}F_k(x_l, u_1(x_l), u_2(x_l))\overline{\Psi}_{ij}(x), \tag{5.5.3}$$

where

$$\mathbf{F}(x, u_1(x), u_2(x)) = \mathbf{F}(x) - \mathbf{N}(u_1(x), u_2(x)) = (F_1, F_2)^{\top}.$$

The proof is omitted.

Remark 5.5.1. (i) If Eq. (5.5.2) is linear, that is, $\mathbf{N}(u_1(x), u_2(x)) = \mathbf{0}$. Then the analytical solution to Eq. (1.2) can be obtained directly from (5.5.3).

(ii) If Eq. (5.5.2) is nonlinear. In this case, the analytical solution to Eq. (5.5.2) can be obtained using following method.

5.5.3-1 The Implementation Method

Write

$$\mathbf{F}(x, \mathbf{u}(x)) = \mathbf{F}(x, u_1(x), u_2(x))$$

simply. (5.5.3) can be denoted by

$$\mathbf{u}(x) = \sum_{i=1}^{\infty}\sum_{j=1}^{2}C_{ij}\overline{\Psi}_{ij}(x), \tag{5.5.4}$$

where

$$C_{ij} = \sum_{l=1}^{i}\sum_{k=1}^{j}\beta_{lk}^{ij}F_k(x_l, \mathbf{u}(x_l)).$$

Let $x_1 = 0$, it follows that $\mathbf{u}(x_1)$ is known from the boundary conditions of Eq. (5.5.2). So $\mathbf{F}(x_1, \mathbf{u}(x_1))$ is known. Considering the numerical computation, we put $\mathbf{u}_0(x_1) = \mathbf{u}(x_1)$ and define the n-term approximation to $\mathbf{u}(x)$ by

$$\mathbf{u}_n(x) = \sum_{i=1}^{n}\sum_{j=1}^{2}B_{ij}\overline{\Psi}_{ij}(x), \tag{5.5.5}$$

where

$$\begin{aligned}
B_{1j} &= \sum_{k=1}^{j} \beta_{1k}^{1,j} F_k(x_1, \mathbf{u}_0(x_1)), \quad j = 1, 2, \\
\mathbf{u}_1(x) &= \sum_{j=1}^{2} B_{1j} \overline{\Psi}_{1j}(x), \\
B_{2j} &= \sum_{l=1}^{2} \sum_{k=1}^{j} \beta_{lk}^{2j} F_k(x_l, \mathbf{u}_{l-1}(x_l)), \quad j = 1, 2, \\
\mathbf{u}_2(x) &= \sum_{i=1}^{2} \sum_{j=1}^{2} B_{ij} \overline{\Psi}_{ij}(x), \\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
\mathbf{u}_{n-1}(x) &= \sum_{i=1}^{n-1} \sum_{j=1}^{2} B_{ij} \overline{\Psi}_{ij}(x), \\
B_{nj} &= \sum_{l=1}^{n} \sum_{k=1}^{j} \beta_{lk}^{nj} F_k(x_l, \mathbf{u}_{l-1}(x_l)), \quad j = 1, 2.
\end{aligned} \tag{5.5.6}$$

Theorem 5.5.3. $\mathbf{u}_n(x)$ *is convergent in* $\mathbf{W}_3[0, 1]$.

The proof is omitted.

Lemma 5.5.2. *If* $\lim\limits_{n\to\infty} \mathbf{u}_n(x) = \overline{\mathbf{u}}(x)$, $\|\mathbf{u}_n\|$ *is bounded,* $x_n \underset{n\to\infty}{\longrightarrow} y$ *and* $f(x, \mathbf{u}(x))$ *is continuous, then* $f(x_n, \mathbf{u}_{n-1}(x_n)) \underset{n\to\infty}{\longrightarrow} f(y, \overline{\mathbf{u}}(y))$

The proof is omitted.

Theorem 5.5.4. *Let* $\lim\limits_{n\to\infty} \mathbf{u}_n(x) = \overline{\mathbf{u}}(x)$, *then* $\overline{\mathbf{u}}(x)$ *is the solution of Eq.* (5.5.2).

Proof. Taking limits in (5.5.5), we get

$$\overline{\mathbf{u}}(x) = \sum_{i=1}^{\infty} \sum_{j=1}^{2} B_{ij} \overline{\Psi}_{ij}(x). \tag{5.5.7}$$

Note here that

$$(\mathbb{A}\overline{\mathbf{u}})(x) = \sum_{i=1}^{\infty} \sum_{j=1}^{2} B_{ij} \mathbb{A}\overline{\Psi}_{ij}(x)$$

and

$$(\mathbb{A}\overline{\mathbf{u}})_k(x_l) = \sum_{i=1}^{\infty} \sum_{j=1}^{2} B_{ij} \langle \mathbb{A}\overline{\Psi}_{ij}(x), \Phi_{lk}(x) \rangle_{\mathbf{W}_1}$$

$$= \sum_{i=1}^{\infty}\sum_{j=1}^{2} B_{ij}\langle\overline{\Psi}_{ij}(x), \mathbb{A}^*\Phi_{lk}(x)\rangle_{\mathbf{W}_3}$$
$$= \sum_{i=1}^{\infty}\sum_{j=1}^{2} B_{ij}\langle\overline{\Psi}_{ij}(x), \Psi_{lk}(x)\rangle_{\mathbf{W}_3}, \quad k = 1, 2.$$

Therefore,

$$\sum_{l'=1}^{l}\sum_{k'=1}^{k}\beta_{l'k'}^{lk}(\mathbb{A}\overline{\mathbf{u}})_{k'}(x_{l'}) = \sum_{i=1}^{\infty}\sum_{j=1}^{2} B_{ij}\left\langle\overline{\Psi}_{ij}(x), \sum_{l'=1}^{l}\sum_{k'=1}^{k}\beta_{l'k'}^{lk}\Psi_{l'k'}(x)\right\rangle_{\mathbf{W}_3}$$
$$= \sum_{i=1}^{\infty}\sum_{j=1}^{2} B_{ij}\left\langle\overline{\Psi}_{ij}(x), \overline{\Psi}_{lk}(x)\right\rangle_{\mathbf{W}_3} = B_{lk}, \quad k = 1, 2. \quad (5.5.8)$$

If $l = 1$, then

$$(\mathbb{A}\overline{\mathbf{u}})_k(x_1) = F_k(x_1, \mathbf{u}_0(x_1)), \quad k = 1, 2,$$

that is,

$$\mathbb{A}\overline{\mathbf{u}}(x_1) = \mathbf{F}(x_1, \mathbf{u}_0(x_1)).$$

If $l = 2$, then

$$(\mathbb{A}\overline{\mathbf{u}})_k(x_2) = F_k(x_2, \mathbf{u}_1(x_2)), \quad k = 1, 2,$$

that is,

$$\mathbb{A}\overline{\mathbf{u}}(x_2) = \mathbf{F}(x_2, u_1(x_2)).$$

In the same way,

$$\mathbb{A}\overline{\mathbf{u}}(x_n) = \mathbf{F}(x_n, \mathbf{u}_{n-1}(x_n)).$$

Since $\{x_i\}_{i=1}^{\infty}$ is dense on [0,1], for $\forall\, y \in [0, 1]$, there exists a subsequence $\{x_{n_j}\}_{j=1}^{\infty}$ such that $x_{n_j} \longrightarrow y$ $(j \longrightarrow \infty)$. We have known that

$$\mathbb{A}\overline{\mathbf{u}}(x_{n_j}) = \mathbf{F}\big(x_{n_j}, \mathbf{u}_{n_j-1}(x_{n_j})\big).$$

Let $j \longrightarrow \infty$, by Lemma 5.5.2 and the continuity of $\mathbf{F}$, we have

$$(\mathbb{A}\overline{\mathbf{u}})(y) = \mathbf{F}(y, \overline{\mathbf{u}}(y)). \quad (5.5.9)$$

From (5.5.9), it follows that $\overline{\mathbf{u}}(x)$ satisfies Eq. (5.5.2).

Since $\overline{\Psi}_{ij}(x) \in \mathbf{W}_3$, clearly, $\overline{\mathbf{u}}(x)$ satisfies the boundary conditions of Eq. (5.5.2).

That is, $\overline{\mathbf{u}}(x)$ is the solution of Eq. (5.5.2). The application of the uniqueness of solution to Eq. (5.5.2) then yields that

$$\mathbf{u}(x) = \sum_{i=1}^{\infty}\sum_{j=1}^{2} B_{ij}\overline{\Psi}_{ij}(x). \quad (5.5.10)$$

The proof is complete. □

5.5.4 Numerical Experiments

Some numerical examples are studied to demonstrate the accuracy of the present method. The results obtained by the method are compared with the analytical solution of each example and are found to be in good agreement with each other.

Example 5.5.1. Consider the equation

$$\begin{cases} u''(x) + u'(x) + xu(x) + v'(x) + 2xv(x) = f_1(x), \\ v''(x) + v(x) + 2u'(x) + x^2u(x) = f_2(x), \\ u(0) = u(1) = 0, \quad v(0) = v(1) = 0, \end{cases}$$

where $0 \le x \le 1$,

$$f_1(x) = -2(1+x)\cos(x) + \pi\cos(\pi x) + 2x\sin(\pi x) + (4x - 2x^2 - 4)\sin(x),$$

and

$$f_2(x) = -4(x-1)\cos(x) + 2(2 - x^2 + x^3)\sin(x) - (\pi^2 - 1)\sin(\pi x).$$

The true solutions

$$u(x) = 2\sin(x)(1-x) \text{ and } v(x) = \sin(\pi x),$$

respectively. Using our method, we choose 11 points and 21 points on $[0,1]$ and obtain approximate solutions $u_{11}(x)$, $v_{11}(x)$, $u_{21}(x), v_{21}(x)$ on $[0,1]$. The numerical results are given in the following Table 5.10–5.13.

Example 5.5.2. Consider the equation

$$\begin{cases} u''(x) + xu(x) + 2xv(x) + xu^2(x) = f_1(x), \\ v'(x) + v(x) + x^2u(x) + \sin(x)v^2(x) = f_2(x), \\ u(0) = u(1) = 0, \quad v(0) = v(1) = 0, \end{cases}$$

where $0 \le x \le 1$,

$$\begin{aligned} f_1(x) &= -2x\sin(x) - 2 + x^2 - 2x^4 + x^5, \\ f_2(x) &= (1-x)x^3 + (1-\pi^2)\sin(\pi x) + \sin(x)\sin(\pi x). \end{aligned}$$

The true solutions

$$u(x) = x - x^2 \text{ and } v(x) = \sin(\pi x),$$

respectively. Using our method, we choose 21 points on $[0,1]$ and obtain approximate solutions $u_{21}(x)$, $v_{21}(x)$ on $[0,1]$. The numerical results are given in the following able 5.14, 5.15.

Table 5.10: The numerical results of example 1($u_{11}(x)$)

x	True solution $u(x)$	Approximate solution u_{11}	absolute error
0.08	0.147043	0.143698	0.0033
0.24	0.361308	0.353524	0.0077
0.40	0.467302	0.457538	0.0097
0.56	0.467444	0.457875	0.0095
0.72	0.369255	0.361953	0.0073
0.88	0.184977	0.181565	0.0034
0.96	0.065535	0.064371	0.0011

Table 5.11: The numerical results of example 1($v_{11}(x)$)

x	True solution $v(x)$	Approximate solution v_{11}	absolute error
0.08	0.24869	0.240926	0.0077
0.24	0.684547	0.664188	0.0203
0.40	0.951057	0.923921	0.0271
0.56	0.982287	0.955155	0.0271
0.72	0.770513	0.749933	0.0205
0.88	0.368125	0.35865	0.0094
0.96	0.125333	0.122171	0.0031

Table 5.12: The numerical results of example 1($u_{21}(x)$)

x	True solution $u(x)$	Approximate solution u_{21}	absolute error
0.08	0.147043	0.146188	0.0008
0.24	0.361308	0.359319	0.0019
0.40	0.467302	0.464806	0.0024
0.56	0.467444	0.465001	0.0024
0.72	0.369255	0.36739	0.0018
0.88	0.184977	0.184106	0.0008
0.96	0.065535	0.065245	0.0002

Table 5.13: The numerical results of example 1($v_{21}(x)$)

x	True solution $v(x)$	Approximate solution v_{21}	absolute error
0.08	0.24869	0.246706	0.0019
0.24	0.684547	0.679363	0.0051
0.40	0.951057	0.944133	0.0071
0.56	0.982287	0.975374	0.0069
0.72	0.770513	0.765269	0.0052
0.88	0.368125	0.365713	0.0024
0.96	0.125333	0.124531	0.0008

Table 5.14: The numerical results of example 2($u_{21}(x)$)

x	True solution $u(x)$	Approximate solution u_{21}	absolute error
0.08	0.0736	0.073047	0.0005
0.24	0.1824	0.180903	0.0014
0.40	0.2400	0.237879	0.0021
0.56	0.2464	0.244116	0.0022
0.72	0.2016	0.199711	0.0018
0.88	0.1056	0.104645	0.0009
0.96	0.0384	0.038065	0.0003

Table 5.15: The numerical results of example 2($v_{21}(x)$)

x	True solution $v(x)$	Approximate solution v_{21}	absolute error
0.08	0.24869	0.246634	0.0020
0.24	0.684547	0.678860	0.0056
0.40	0.951057	0.943094	0.0079
0.56	0.982287	0.974003	0.0082
0.72	0.770513	0.763992	0.0065
0.88	0.368125	0.365010	0.0031
0.96	0.125333	0.124274	0.0010

5.6 To Solve a Class of Nonlinear Differential Equations

5.6.1 Introduction

Nonlinear problems are puzzles in computational mathematics at all times. In recent years, a large of mathematical researchers make a great deal efforts to solve nonlinear differential equations. Now there are some methods for quasilinear differential equations [164]–[168]. However, there are few valid methods for solving true nonlinear differential equations.

Considering the following nonlinear differential equations in the reproducing kernel space

$$\begin{cases} H(u''(x)) + a(x)u'(x) + b(x)u(x) = f(x), \ \ 0 \le x \le 1, \\ u(0) = 1, \quad u(1) = 0 \end{cases} \tag{5.6.1}$$

where $H(x)$ is a continuous function, $a(x)$, $b(x)$, $f(x)$ are continuous functions on $[0, 1]$ and $a(0) = 0$, $f(0) - b(0) \neq 0$.

Note here that Eq. (5.6.1) is a true nonlinear differential equation. That is the highest order derivative is nonlinear.

We may equivalently rewriting Eq. (5.6.1) as

$$\begin{cases} v(x) + a(x)u'(x) + b(x)u(x) = f(x), \ \ 0 \le x \le 1, \\ v(x) = H(u''(x)), \\ u(0) = 1, \quad u(1) = 0. \end{cases} \tag{5.6.2}$$

5.6.2 Linear Operator on Reproducing Kernel Spaces

Given the reproducing kernel space $W_2^1[0,1]$ (1.3.2), its reproducing kernel $Q_y(x)$ is form (1.4.1).

Let $\mathcal{D} = [0,1] \times [0,1]$, the reproducing kernel space

$$^oW_2^{(3,1)}(\mathcal{D}) = \Big\{ u(x,y) |\ u(x,y) \in W_2^{(3,1)}(\mathcal{D}),\ u(1,y) = 0 \Big\},$$

its reproducing kernel is $K_{(\xi,\eta)}(x,y) = R_\xi(x)Q_\eta(y)$, where

$$\begin{aligned} R_\xi(x) &= \frac{1}{264960}\Big(2208\xi^3(\xi^2 - 5\xi x + 10x^2) \\ &+ 8(\xi-1)\Big(- 120(156 + \xi(2+\xi)(18 + (\xi-6)\xi)) \\ &- 120(-120 + \xi(2+\xi)(18 + (\xi-6)\xi))x \\ &- 30\big(- 120 + \xi(-240 + \xi(6 + (\xi-4)\xi))\big)x^2 \\ &- 10\big(- 120 + \xi(-240 + \xi(6 + (\xi-4)\xi))\big)x^3 \\ &+ 5\big(- 120 + \xi(2+\xi)(18 + (\xi-6)\xi)\big)x^4 \\ &- (156 + \xi(2+\xi)(18 + (\xi-6)\xi))x^5\Big) - 11040\xi^2x^2(\xi + x + |\xi - x|) \\ &+ 1380\xi x(\xi + x + |\xi - x|)^3 - 69(\xi + x + |\xi - x|)^5\Big), \end{aligned}$$

$Q_y(x)$ is form (1.4.1).

To solve Eq. (5.6.2), one introduces a linear operator $\mathbb{L} : {}^oW_2^{(3,1)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$. For every $w(x,y) \in {}^oW_2^{(3,1)}(\mathcal{D})$,

$$(\mathbb{L}w(s,t))(x) = w(0,x) + c_0 a(x)\frac{\partial w(x,0)}{\partial x} + c_0 b(x) w(x,0), \tag{5.6.3}$$

where $c_0 = (f(0) - b(0))^{-1}$.

Lemma 5.6.1. *$\mathbb{L} : {}^oW_2^{(3,1)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$ is a bounded linear operator.*

Theorem 5.6.1. *If $(u(x), v(x))$ is the solution to Eq. (5.6.2), then $u(x)v(y)$ is the solution of following operator equation*

$$\begin{cases} (\mathbb{L}w(s,t))(x) = f(x), \\ w(s,t) = c_0 w(s,0)w(0,t), \\ w(0,t) = H\left(c_0 \dfrac{\partial^2 w(\circ,0)}{\partial \circ^2}\Big|_{\circ=t} \right). \end{cases} \tag{5.6.4}$$

Whereas, if $w(s,t)$ is the solution to Eq. (5.6.4)*, then $(c_0 w(x,0), w(0,x))$ is the solution of Eq.* (5.6.2)*.*

Proof. Let $(u(x), v(x))$ is the solution to Eq. (5.6.2), then $w(s,t) = u(s)v(t)$ is the solution of Eq. (5.6.4). In fact, (note that $c_0 v(0) = 1$)

$$\begin{aligned}(\mathbb{L}u(s)v(t))(x) &= u(0)v(x) + c_0 a(x)u'(x)v(0) + c_0 b(x)u(x)v(0)\\ &= v(x) + a(x)u'(x) + b(x)u(x) = f(x)\end{aligned}$$

and

$$\begin{aligned}w(s,t) &= u(s)v(t) = c_0(u(s)v(0))(u(0)v(t)) = c_0 w(s,0)w(0,t),\\ w(0,t) &= u(0)v(t) = v(t) = H(u''(t)) = H(c_0 u''(t)v(0))\\ &= H\left(c_0 \frac{\partial^2 w(\circ,0)}{\partial \circ^2}\bigg|_{\circ=t}\right).\end{aligned}$$

Whereas, if $w(s,t)$ is the solution to Eq. (5.6.4), thus

$$w(0,x) + a(x)\frac{\partial(c_0 w(x,0))}{\partial x} + b(x)(c_0 w(x,0)) = (\mathbb{L}w(s,t))(x) = f(x).$$

Clearly, $c_0 w(1,0) = 0$ (from the definition of space ${}^oW_2^{(3,1)}(\mathcal{D})$), $c_0 w(0,0) = 1$ and $w(0,x) = H(c_0 \frac{\partial^2 w(x,0)}{\partial x^2})$. □

5.6.3 Direct Sum of ${}^oW_2^{(3,1)}(\mathcal{D})$

Using the adjoint operator $\mathbb{L}^*$ of $\mathbb{L}$ given in (5.6.3), one will choose and fix a countable dense subset $\mathrm{T} = \{x_1, x_2, \ldots\}$ on $[0,1]$ and define $\psi_i(s,t)$ by

$$\psi_i(x,y) \stackrel{def}{=} (L^* Q_{x_i})(x,y), \quad i = 1,2,\ldots.$$

By Gram–Schmidt process, we obtain an orthogonal set $\{\widetilde{\psi}_i(x,y)\}_{i=1}^{\infty}$ of ${}^oW_2^{(3,1)}(\mathcal{D})$, such that

$$\widetilde{\psi}_i(x,y) = \sum_{k=1}^{i} \beta_{ik}\psi_k(x,y),$$

where β_{ik} are orthogonal coefficients.

Let

$$\mathrm{S} = \left\{ w(x,y) |\, w(x,y) = \sum_{i=1}^{m} c_i \widetilde{\psi}_k(x,y)\ (c_1, c_2, \ldots) \in \ell^2 \right\}$$

and $\mathrm{S}^{\perp}$ be the orthogonal complement of S in ${}^oW_2^{(3,1)}(\mathcal{D})$. We choose a countable dense subset

$$\mathrm{B} = \Big\{ (x_1, y_1), (x_2, y_2), \ldots \Big\}$$

of $\mathcal{D}$. It is easy to show that

$$\sigma_j(x,y) \stackrel{def}{=} R_{x_j}(x)Q_{y_j}(y), \;\; j=1,2,\ldots$$

constitutes a basis of the space ${}^oW_2^{(3,1)}(\mathcal{D})$. Again we orthogonalize $\{\widetilde{\psi}_1,\widetilde{\psi}_2,\ldots,\sigma_1,\sigma_2,\ldots\}$ to obtain

$$\widetilde{\sigma}_j(x,y)=\frac{\sigma_j-\sum\limits_{k=1}^{\infty}\langle\sigma_j,\widetilde{\psi}_k\rangle_{{}^oW_2^{(3,1)}}\widetilde{\psi}_k-\sum\limits_{l=1}^{j-1}\langle\sigma_j,\widetilde{\sigma}_l\rangle_{{}^oW_2^{(3,1)}}\widetilde{\sigma}_l}{\big\|\sigma_j-\sum\limits_{k=1}^{\infty}\langle\sigma_j,\widetilde{\psi}_k\rangle_{{}^oW_2^{(3,1)}}\widetilde{\psi}_k-\sum\limits_{l=1}^{j-1}\langle\sigma_j,\widetilde{\sigma}_l\rangle_{{}^oW_2^{(3,1)}}\widetilde{\sigma}_l\big\|_{{}^oW_2^{(3,1)}}},$$

that is

$$\widetilde{\sigma}_j(x,y)\stackrel{def}{=}\sum_{k=1}^{\infty}\beta_{ik}\psi_k(x,y)+\sum_{l=1}^{j}\beta^*_{jl}\sigma_l(x,y),\;\; j=1,2,\ldots. \tag{5.6.5}$$

Hence, we have ${}^oW_2^{(3,1)}(\mathcal{D})=\mathrm{S}\oplus\mathrm{S}^{\perp}$, and $\{\widetilde{\psi}_1,\widetilde{\psi}_2,\ldots,\widetilde{\rho}_1,\widetilde{\rho}_2,\ldots\}$ constitutes an orthogonal basis for ${}^oW_2^{(3,1)}(\mathcal{D})$.

5.6.4 Solution of $(\mathbb{L}w)(x)=f(x)$

Theorem 5.6.1 shows that finding a solution of Eq. (5.6.1) is equivalent to finding the solution of Eq. (5.6.4).

Theorem 5.6.2.

$$w(x,y)=\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)\widetilde{\psi}_i(x,y)+\sum_{j=1}^{\infty}\lambda_j\widetilde{\sigma}_j(x,y) \tag{5.6.6}$$

is a solution of $(\mathbb{L}w)(x)=f(x)$ *for every* $(\lambda_1,\lambda_2,\ldots)\in\ell^2$.

Proof. Applying to $\mathbb{L}$ of both sides (5.6.6), we have

$$(\mathbb{L}w)(x)=\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)(\mathbb{L}\widetilde{\psi}_i)(x)+\sum_{j=1}^{\infty}\lambda_j(\mathbb{L}\widetilde{\sigma}_j)(x).$$

For every $x_l\in\mathrm{T}$, we have

$$(\mathbb{L}\widetilde{\sigma}_j)(x_l)=\langle\mathbb{L}\widetilde{\sigma}_j,Q_{x_l}\rangle_{W_2^1}=\langle\widetilde{\sigma}_j,\mathbb{L}^*Q_{x_l}\rangle_{{}^oW_2^{(3,1)}}=\langle\widetilde{\sigma}_j,\psi_l\rangle_{{}^oW_2^{(3,1)}}=0.$$

Hence

$$(\mathbb{L}w)(x_l)\;\;=\;\;\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)(\mathbb{L}\widetilde{\psi}_i)(x_l)$$

$$= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)\langle \mathbb{L}\widetilde{\psi}_i,\, Q_{x_l}\rangle_{W_2^1}$$
$$= \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)\langle \widetilde{\psi}_i,\, \psi_l\rangle_{{}_oW_2^{(3,1)}}.$$

Multiplying both sides of the above equality by β_{nl} and summing with respect to l, $1 \le l \le n$, we have

$$\sum_{l=1}^{n}\beta_{nl}(\mathbb{L}w)(x_l) = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)\langle\widetilde{\psi}_i, \widetilde{\psi}_n\rangle_{{}_oW_2^{(3,1)}} = \sum_{k=1}^{n}\beta_{nk}f(x_k).$$

We claim that $(\mathbb{L}w)(x_m) = f(x_m)$ holds for all m. For $n = 1$, it is easy to show that $(\mathbb{L}w)(x_1) = f(x_1)$. For induction, we assume that $(\mathbb{L}w)(x_n) = f(x_n)$, $n \le m$. Since

$$\sum_{l=1}^{m+1}\beta_{(m+1)l}(Lw)(x_l) = \sum_{k=1}^{m+1}\beta_{(m+1)k}f(x_k)$$

and

$$\sum_{l=1}^{m}\beta_{(m+1)l}(Lw)(x_l) + \beta_{(m+1)(m+1)}(Lw)(x_{m+1}) = \sum_{k=1}^{m+1}\beta_{(m+1)k}f(x_k),$$

we have $(Lw)(x_{m+1}) = f(x_{m+1})$. Hence $(\mathbb{L}w)(x_m) = f(x_m)$ holds for every $x_m \in \mathrm{T}$ and T is dense on $[0,1]$. And therefore we conclude $(\mathbb{L}w)(x) = f(x)$ holds for all $x \in [0,1]$. Therefore, our assertion is proved. □

By Theorem 5.6.1, we have know that $c_0 w(x)$ is the solution of Eq. (5.6.2). Furthermore, the following theorem holds.

Theorem 5.6.3. *If $w(x,y)$ of* (5.6.6) *is the solution for Eq.* (5.6.4)*, then*

$$u(x) \overset{def}{=} c_0\left[\sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)\widetilde{\psi}_k(x,0) + \sum_{j=1}^{\infty}\lambda_j\widetilde{\sigma}_j(x,0)\right] \tag{5.6.7}$$

is the solution of Eq. (5.6.2)

From Theorem 5.6.2 and Theorem 5.6.3, to solve Eq. (5.6.1), we only need to determine $\lambda_1, \lambda_2, \ldots$ such that

$$\begin{cases} w(x,y) = \sum_{i=1}^{\infty}\sum_{k=1}^{i}\beta_{ik}f(x_k)\widetilde{\psi}_k(x,y) + \sum_{j=1}^{\infty}\lambda_j\widetilde{\sigma}_j(x,y), \\ w(x,y) = c_0 w(x,0)w(0,y), \\ w(0,x) = H\left(c_0\dfrac{\partial^2 w(x,0)}{\partial x^2}\right). \end{cases}$$

Then (5.6.7) is the series representation of solution to Eq. (5.6.1).

An approximate solution $u_{nm}(x)$ is obtained by truncating the series (5.6.7),

$$u_{nm}(x) = c_0 \left[\sum_{i=1}^{n} \sum_{k=1}^{i} \beta_{ik} f(x_k) \widetilde{\psi}_k(x,0) + \sum_{j=1}^{m} \lambda_j \widetilde{\sigma}_j(x,0) \right] \tag{5.6.8}$$

for each n, $m \in \mathbb{N}$. In order to obtain the function $u_{nm}(x)$, we have to determine the values of $\lambda_1, \lambda_2, \ldots, \lambda_m$. So, one obtains $\lambda_1, \lambda_2, \ldots, \lambda_m$ when the following function G reaches the minimum.

$$\begin{aligned} G(\lambda_1, \lambda_2, \ldots, \lambda_m) &\overset{def}{=} \left\| w_{nm}(x,x) - c_0 w_{nm}(x,0) w_{nm}(0,x) \right\|_{W_2^1} \\ &+ \left\| w_{nm}(0,x) - H\left(c_0 \frac{\partial^2 w_{nm}(x,0)}{\partial x^2} \right) \right\|_{W_2^1}, \end{aligned} \tag{5.6.9}$$

where $w_{nm}(x,y)$ is the partial sum (5.6.6). Running Mathematica 5.0 for a concrete example, it can be easily confirmed that our result is effective.

Table 5.16: The error of function $u(x)$

x	$u_{nm}(x)$	$u(x)$	absolute error	x	$u_{nm}(x)$	$u(x)$	absolute error
0	1	1	1.22869E-7	$\frac{1}{10}$	0.987741	0.987688	0.0000531468
$\frac{2}{10}$	0.950981	0.951057	0.0000753896	$\frac{3}{10}$	0.890889	0.891007	0.000117249
$\frac{4}{10}$	0.808844	0.809017	0.000173416	$\frac{5}{10}$	0.706872	0.707107	0.000234296
$\frac{6}{10}$	0.587457	0.587785	0.000328057	$\frac{7}{10}$	0.453656	0.45399	0.000334482
$\frac{8}{10}$	0.308568	0.309017	0.000448943	$\frac{9}{10}$	0.155709	0.156434	0.000725822
1	3.05928E-14	0	3.05928E-14				

5.6.5 Example

Example 5.6.1. Consider the nonlinear equation

$$\begin{cases} \sin[u''(x)] + xu'(x) + u(x) = \cos\dfrac{\pi x}{2} - \dfrac{1}{2}\pi x \sin\dfrac{\pi x}{2} - \sin\left(\dfrac{1}{4}\pi^2 \cos\dfrac{\pi x}{2}\right), \\ u(0) = 1, \quad u(1) = 0. \end{cases}$$

Its true solution $u(x) = \cos\frac{\pi x}{2}$. By using Mathematica 5.0 and the method of Section 5.6.4, where $n = 10$, $m = 100$ in (5.6.8), we calculate the approximate solution $u_{nm}(x)$. The numerical results are given in Table 5.16.

Chapter 6

The Exact Solution of Nonlinear Operator Equation $\mathbb{A}u\mathbb{B}u + \mathbb{C}u = f$

The operator theory is the important content of functional analysis. The research on operator equation will be of more extensive significance.

This chapter provides numerical algorithm for two types of operator equations: 1. nonlinear operator equation $\mathbb{A}u\mathbb{B}u + \mathbb{C}u = f$; 2. ill-posed operator equation.

6.1 Introduction

Solving operator equation is a difficult but meaningful problem, including how to solve linear operator equation and nonlinear operator equation.

With regard to linear operator equation $\mathbb{A}u = f$, by supposing the equation has solutions, Cui discussed the structure of its solution space. In this paper, without any assumption, we discuss the structure of the solution space of system of linear operator equations $\mathbb{A}_i u = f_i$ where $i = 1, 2, \ldots, n$. and n is a positive integer. And we get an easy-to-compute criterion about the existence of solutions. If the system of equations has solutions, we obtain the analytic representation of its minimum norm solution u_0. At the same time, we represent its solution space as $u_0 + \bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i)$, where $\mathrm{N}(\mathbb{A}_i)$ is the null space of operator $\mathbb{A}_i$, that is, $\mathrm{N}(\mathbb{A}_i) = \{u|\ \mathbb{A}_i u = 0\}$. Thus we have a complete understanding about linear operator equation.

As to a general nonlinear operator equation, no analytic algorithm on how to get its exact solution can be obtained until now. Only some papers [169], [170] give the exact solution of some special nonlinear equation; some papers [171]–[175] discuss the existence and uniqueness of solutions of some kind of operator equation. In [23], the authors give the minimal norm solution of linear operator equation.

6.1.1 Preliminary Knowledge

The reproducing kernel space $W_2^1[a,b]$ is defined by (1.3.2) and its reproducing kernel is

$$R_\xi(x) = \begin{cases} x+1-a, & x \le \xi, \\ \xi+1-a, & x > \xi. \end{cases} \tag{6.1.1}$$

Let $\mathcal{D} = [a,b] \times [a,b]$. The reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$ is defined by (1.5.1) and its reproducing kernel is $R_\xi(x)R_\eta(y)$, where $R_\xi(x)$ by (6.1.1).

In this section, we will discuss how to solve the following quadratic nonlinear operator equation

$$\mathbb{A}u\mathbb{B}u + \mathbb{C}u = f,\ f, u \in W_2^1[a,b] \tag{6.1.2}$$

in reproducing kernel space $W_2^1[a,b]$, where $\mathbb{A}$, $\mathbb{B}$ and $\mathbb{C}$ are bounded linear operators of $W_2^1[a,b] \longrightarrow W_2^1[a,b]$ and we obtain the representation of its exact solution.

In symbol $(\mathbb{L}_x f(x))(z)$,the subscript x by operator $\mathbb{L}_x$ indicates that the operator $\mathbb{L}$ applies to functions of x. And z means that it is the function of z after the operator $\mathbb{L}$ being applied to the function $f(x)$. Sometimes we use symbols $\langle\cdot,\cdot\rangle_{x,W_2^1}$ and $\langle\cdot,\cdot\rangle_{\mathbf{p},W_2^{(1,1)}}$, where the subscript x and $\mathbf{p}$ by the inner product indicate that the inner product acts on functions of x in space W_2^1 and functions of $\mathbf{p}$ in space $W_2^{(1,1)}$. Correspondingly, we have the forms $\|\cdot\|_{x,W_2^1}$ and $\|\cdot\|_{\mathbf{p},W_2^{(1,1)}}$.

6.1.2 Operator $\mathbb{K}$

Let $\mathbb{A},\mathbb{B},\mathbb{C}$ be bounded linear operators of $W_2^1[a,b] \longrightarrow W_2^1[a,b]$. In order to obtain the solution of Eq. (6.1.2), we first study the property of operator $\mathbb{K} : W_2^{(1,1)}(\mathcal{D}) \longrightarrow W_2^1[a,b]$, which is defined by

$$\mathbb{K}u(t,\tau)(x) \equiv (\mathbb{L}u)(x) + (\mathbb{H}u)(x),\ \ u \in W_2^{(1,1)}(\mathcal{D}), \tag{6.1.3}$$

$$(\mathbb{L}u)(x) \stackrel{\text{def}}{=} \mathbb{A}_t\big[\mathbb{B}_\tau[u(t,\tau)](x)\big](x) \tag{6.1.4}$$

For any $y_0 \in [a,b]$ is fixed,

$$(\mathbb{H}u)(x) \stackrel{\text{def}}{=} [\mathbb{C}_t u(t,y_0)](x). \tag{6.1.5}$$

Let us first discuss how to solve

$$\mathbb{K}u = f,\ \ u \in W_2^{(1,1)},\ \ f(x) \in W_2^1. \tag{6.1.6}$$

Denote by $\mathrm{N}(\mathbb{K})$ the null space of operator $\mathbb{K}$ and decompose $W_2^{(1,1)}(\mathcal{D}) = \mathrm{N}(\mathbb{K}) \oplus \Psi$, where $\Psi = (\mathrm{N}(\mathbb{K}))^\perp$. Suppose that equation $\mathbb{K}u = f$ has solutions. Then it's easy to prove that $\mathbb{K}u = f$ has a unique solution $u_0(x,y)$ on Hilbert space Ψ and all solutions of $\mathbb{K}u = f$ can be denoted by $u = u_0 + v$, $v \in \mathrm{N}(\mathbb{K})$.

Let $\{x_i\}_{i=1}^{\infty}$ be a dense subset of interval $[a, b]$. Put

$$\psi_i(x, y) = \big[\mathbb{L}_{(t,\tau)} R_x(t) R_y(\tau)\big](x_i) + \big[\mathbb{C}_t R_x(t) R_y(y_0)\big](x_i) \in W_2^{(1,1)}(\mathcal{D}).$$

Theorem 6.1.1. *Function system $\{\psi_i(x, y)\}_{i=1}^{\infty}$ is complete in Ψ.*

Proof. (1) Taking any $v(x, y) \in \mathrm{N}(\mathbb{K})$ and using Lemma 1.2 again, we get that

$$\begin{aligned}
\langle v, \psi_i\rangle_{W_2^{(1,1)}} &= \langle v(x,y), (A_\xi R_x(\xi))(x_i)(B_\eta R_y(\eta))(x_i) \\
&+ (C_\xi R_x(\xi))(x_i) R_{y_0}(y)\rangle_{W_2^{(1,1)}} \\
&= \langle v(x,y), (A_\xi R_x(\xi))(x_i)(B_\eta R_y(\eta))(x_i)\rangle_{W_2^{(1,1)}} \\
&+ \langle v(x,y), (C_\xi R_x(\xi))(x_i) R_{y_0}(y)\rangle_{W_2^{(1,1)}} \\
&= \Big[A_\xi B_\eta \langle v(x,y), R_x(\xi) R_y(\eta)\rangle_{W_2^{(1,1)}}\Big](x_i) \\
&+ \Big[C_\xi \langle v(x,y), R_x(\xi) R_{y_0}(y)\rangle_{W_2^{(1,1)}}\Big](x_i) \\
&= \big[A_\xi B_\eta v(\xi,\eta)\big](x_i) + \big[C_\xi v(\xi, y_0)\big](x_i) \\
&= \big[\mathbb{L}_{\xi,\eta} v(\xi,\eta)\big](x_i) + \big[\mathbb{H}_{\xi,\eta} v(\xi,\eta)\big](x_i) \\
&= [\mathbb{K} v(\xi,\eta)](x_i) = 0. \qquad (6.1.7)
\end{aligned}$$

So $\psi_i(x, y) \perp \mathrm{N}(\mathbb{K})$, that is, $\psi_i(x, y) \in \Psi$.

(2) Take any $u(x, y) \in \Psi$ such that $\langle u, \psi_i\rangle_{W_2^{(1,1)}} = 0$. Then from (1) it follows that $\mathbb{K}u(x_i) = 0$. Since $\{x_i\}$ is dense in $[a, b]$, $\mathbb{K}u = 0$. Noting that $\mathbb{K}u = 0$ has a unique solution on Ψ, we obtain $u(x, y) \equiv 0$.

□

By Gram–Schmidt process, we obtain an orthogonal basis $\{\overline{\psi}_i(x, y)\}_{i=1}^{\infty}$ of $W_2^{(1,1)}(\mathcal{D})$,

$$\overline{\psi}_i(x, y) = \sum_{l=1}^{i} \sum_{k=1}^{i} \beta_{ik} \psi_k(x, y), \qquad (6.1.8)$$

where β_{ik} are orthogonal coefficients.

Theorem 6.1.2. *Let $u_0(x, y)$ is the solution of $\mathbb{K}u = f$ in Ψ, then*

$$u_0(x, y) = \sum_{i=1}^{\infty} \widetilde{f}_i \overline{\psi}_i(x, y),$$

where

$$\widetilde{f}_i = \sum_{i=1}^{i} \beta_{ik} f(x_k), \quad i = 1, 2, \ldots. \qquad (6.1.9)$$

Proof. From (6.1.7) it follows that

$$\big\langle u_0(x,y), \psi_i(x,y)\big\rangle_{W_2^{(1,1)}} = \mathbb{K}u_0(x_i) = f(x_i).$$

So

$$\big\langle u_0(x,y), \overline{\psi}_i\big\rangle_{W_2^{(1,1)}} = \sum_{k=1}^{i} \beta_{ik}\langle u_0, \psi_k\rangle_{W_2^{(1,1)}} = \sum_{i=1}^{i} \beta_{ik} f(x_k) = \widetilde{f}_i.$$

Therefore,

$$u_0(x,y) = \sum_{i=1}^{\infty} \big\langle u_0(x,y), \overline{\psi}_i(x,y)\big\rangle_{W_2^{(1,1)}} \overline{\psi}_i(x,y) = \sum_{i=1}^{\infty} \widetilde{f}_i \overline{\psi}_i(x,y). \qquad \square$$

6.1.3 About Eq. (6.1.10) and Eq. (6.1.6)

Rewrite Eq. (6.1.2) as follows:

$$\begin{cases} \mathbb{A}u\mathbb{B}u + \mathbb{C}u = f, \\ u(y_0) = 1, \end{cases} \qquad u,\ f \in W_2^1, \tag{6.1.10}$$

where $\mathbb{A}$, $\mathbb{B}$, $\mathbb{C}$ are bounded linear operators of $W_2^1[a,b] \longrightarrow W_2^1[a,b]$ and y_0 is a fixed point of interval $[a,b]$.

The following theorem shows that solving Eq. (6.1.10) can be turned into solving Eq. (6.1.6).

Theorem 6.1.3. *Suppose nonzero function $u(x) \in W_2^1[a,b]$. In order that $u(x)$ is one solution of Eq.* (6.1.10)*, it's necessary and sufficient that $u(y_0) = 1$ and $u(x)u(y)$ is one solution of Eq.* (6.1.6)*.*

Proof.

$$[\mathbb{K}u(t)u(\tau)](x) = (\mathbb{A}u)(x)(\mathbb{B}u)(x) + u(y_0)(\mathbb{C}u)(x). \tag{6.1.11}$$

Necessity. If $u(x)$ is one solution of (6.1.10), then $u(y_0) = 1$

$$(\mathbb{A}u)(x)(\mathbb{B}u)(x) + (\mathbb{C}u)(x) = f(x). \tag{6.1.12}$$

Hence,

$$[\mathbb{K}u(t)u(\tau)](x) = f(x).$$

It means that $u(x)u(y)$ is one solution of Eq. (6.1.6).

Sufficiency. By (6.1.11), $u(y_0) = 1$ and Eq. (6.1.6), we obtain that $u(x)$ is one solution of Eq. (6.1.10). $\square$

6.1.4 Solving Eq. (6.1.10)

Lemma 6.1.1. *Let $u(x,y) \in W_2^{(1,1)}(\mathcal{D})$ and y_0 be a fixed point in interval $[a,b]$. Then the following conditions are equivalent.*

(a) *There exist a function $v(x) \in W_2^1$, such that $v(y_0) = 1$ and $u(x,y) = v(x)v(y)$.*

(b) $0 \neq u(x,y) = u(x,y_0)u(y,y_0)$.

What's more, if condition (b) *is valid, then function $v(x)$ in condition* (a) *equals to $u(x,y_0)$.*

Proof. (1) (a) implies (b). If $u(x,y) = 0$, then from $0 = u(x,x) = v^2(x)$, it follows that $v(x) = 0$. Thus $v(y_0) = 0$. This is a contradiction with $v(y_0) = 1$. Further, from $v(y_0) = 1$ and $u(x,y) = v(x)v(y)$, condition (b) follows.

(2) To show (b) implies (a). Write $u(x,y_0) = v(x)$. Then $v(x) \in W_2^1$ and condition (b) changed into

$$0 \neq u(x,y) = v(x)v(y). \tag{6.1.13}$$

By formula (6.1.13), we have $v(x) \neq 0$. Replacing y by y_0 in formula (6.1.13), we get $v(x) = v(x)v(y_0)$. Therefore $v(y_0) = 1$. Besides, the last assertion in this theorem follows from the fact that $v(x) = v(x)v(y_0) = u(x,y_0)$.

□

Theorem 6.1.4. *Suppose that Eq.* (6.1.6) *has solutions and denote by* U *the solution set of Eq.* (6.1.6). *Then the solution set of Eq.* (6.1.10) *is*

$$\mathrm{S} = \Big\{ u(x,y_0) |\ 0 \neq u(x,y) = u(x,y_0)u(y,y_0),\ u \in \mathrm{U} \Big\}. \tag{6.1.14}$$

Proof. Denote by T the solution set of Eq. (6.1.10). Then it's sufficient to prove that T = S.

Prove T ⊂ S. Suppose $v(x) \in$ T, namely, $v(x) \in W_2^1$ is one solution of Eq. (6.1.10). From Theorem 6.1.3, it follows $v(y_0) = 1$ and $u(x,y) = v(x)v(y)$ is one solution of Eq. (6.1.6), namely, $u(x,y) \in$ U. Further, by Lemma 6.1.1, we have $0 \neq u(x,y) = u(x,y_0)u(y,y_0)$. Therefore, $v(x) = v(x)v(y_0) = u(x,y_0) \in$ S.

Prove S ⊂ T. Suppose that $w(x) \in$ S. From the definition of set S, it follows that there exist $u(x,y) \in$ U, such that $w(x) = u(x,y_0) \in W_2^1$ and $0 \neq u(x,y) = u(x,y_0)u(y,y_0)$. By Lemma 6.1.1, we get $w(y_0) = 1$ and $w(x)w(y) = u(x,y) \in$ U, namely, $w(x)w(y)$ is one solution of Eq. (6.1.6). Then it follows from Theorem 6.1.3 that $w(x)$ is one solution of Eq. (6.1.10), namely, $w(x) \in$ T.

Thus, T = S. □

Theorem 6.1.5. *If Eq.* (6.1.10) *has solutions and if Eq.* (6.1.6) *has a unique solution on space* $W_2^{(1,1)}(\mathcal{D})$*, then Eq.* (6.1.10) *has a unique solution on space* $W_2^1[a,b]$*. And its solution* $u(x)$ *can be denoted by*

$$u(x) = \sum_{i=1}^{\infty} \widetilde{f}_i \overline{\psi}_i(x, y_0), \tag{6.1.15}$$

where $\overline{\psi}_i(x,y)$ *and* $\widetilde{f}_i$ *are defined by* (6.1.8), (6.1.9) *respectively.*

Proof. Suppose that $u(x)$ is a solution of Eq. (6.1.10). By Theorem 6.1.3, $u(y_0) = 1$ and $u(x)u(y)$ is one solution of Eq. (6.1.6). By assumption,

$$u(x)u(y) = u_0(x,y) = \sum_{i=1}^{\infty} \left[\sum_{k=1}^{i} \beta_{ik} f(x_k) \right] \overline{\psi}_i(x,y).$$

From equality $u(x) = u(x)u(y_0) = u_0(x, y_0)$ the equality (6.1.15) follows.

Let us prove the uniqueness. If $u(x)$, $v(x)$ are both solutions of Eq. (6.1.10) and $u(x) \neq v(x)$. Then $u(y_0) = v(y_0) = 1$ and $u(x)u(y)$, $v(x)v(y)$ are both solutions of Eq. (6.1.6). By assumption, $u(x)u(y) = v(x)v(y)$. Let $y = y_0$, we get $u(x) = v(x)$. This is a contradiction with $u(x) \neq v(x)$. □

6.1.5 Numerical Experiments

Find the approximate solution $u_n(x)$ of the following two equations in space $W_2^1[0, 2\pi]$. Here

$$u_n(x) = \sum_{i=1}^{n} \widetilde{f}_i \overline{\psi}_i(x, y_0), \tag{6.1.16}$$

which is obtained by computing the front n terms of formula (6.1.15).

Example 6.1.1. Solve integral equation

$$\int_0^{2\pi} K_1(x,y)u(y)dy \int_0^{2\pi} K_2(x,y)u(y)dy + \int_0^{2\pi} K_3(x,y)u(y)dy = f(x), \tag{6.1.17}$$

where the true solution of Eq. (6.1.17) is

$$u(y) = \frac{\cos(y) - ch(2\pi - y)sech(2\pi)}{1 - sech(2\pi)}, \quad y_0 = 2\pi,$$

$$K_1(x,y) = \cos(3x+y), \quad K_2(x,y) = \cos(5x+y), \quad K_3(x,y) = \cos(4x+y)$$

and

$$ch(x) = \frac{e^x + e^{-x}}{2}, \quad sech(x) = \frac{1}{ch(x)}.$$

Applying formula (6.1.16) and taking $n = 200$, the numerical results are given in Table 6.1.

Table 6.1: The error of Example 6.1.1

iterate n	maximum error max_n	error norm r_n
1	6.4E-01	1.1416
2	5.8E-01	1.1357
4	5.4E-01	1.0610
6	8.55E-02	1.88E-01
8	2.97E-08	6.75E-08
11	9.38E-10	2.37E-09

Table 6.2: The error of Example 6.1.2

iterate n	maximum error max_n	error norm r_n
1	2.30E-04	2.55E-04
12	2.03E-04	2.23E-04
24	2.01E-04	2.04E-04
36	1.75E-04	1.82E-04
48	1.28E-04	1.36E-04
60	9.43E-05	1.01E-04

Example 6.1.2. Solve equation

$$\mathbb{A}u\mathbb{B}u + \mathbb{C}u = f, \tag{6.1.18}$$

where

$$\begin{aligned}
\mathbb{A}u &= 71\int_0^{2\pi}\cos(4x+y)u(y)dy + u(x),\\
\mathbb{B}u &= 100\int_0^{2\pi}\cos(6x+y)u(y)dy + u(x),\\
\mathbb{C}u &= \int_0^{2\pi}\cos(4x+y)u(y)dy,\\
u(y) &= \frac{\cos(y) - ch(2\pi - y)sech(2\pi)}{1 - sech(2\pi)} \text{ is the true solution of is Eq. (6.1.18),}
\end{aligned}$$

$y_0 = 2\pi$ and $ch(x) = \frac{e^x+e^{-x}}{2}$, $sech(x) = \frac{1}{ch(x)}$.

Applies formula (6.1.16) and taking $n = 200$. The numerical results are given in Table 6.2.

6.2 All Solutions of System of Ill-Posed Operator Equations of the First Kind

6.2.1 Introduction

Let $\mathcal{H}$, $\mathcal{H}_1$ be separable Hilbert spaces, and let $A : \mathcal{H} \longrightarrow \mathcal{H}_1$ be a bounded linear operator. Consider operator equations of the first kind

$$\mathbb{A}u = f, \quad u \in \mathcal{H}, \quad f \in \mathcal{H}_1. \tag{6.2.1}$$

Usually, operator equations of the first kind is ill-posed, so the problem of how to solve Eq. (6.2.1) becomes very important. Here, ill-posed means that at least one of the following three conditions can't be satisfied:

1. Eq. (6.2.1) has solutions for every $f \in \mathcal{H}_1$;

2. Eq. (6.2.1) has a unique solution for every $f \in \mathcal{H}_1$;

3. Inverse operator $\mathbb{A}^{-1}$ is continuous.

Usually, in order to solve Eq. (6.2.1), it needs to suppose that $\mathbb{A}^{-1}$ is single-valued. In this paper, only supposing ill-posed Eq. (6.2.1) has solutions, we discuss the structure of its solution space and obtained a complete orthonormal system of $\mathrm{N}(\mathbb{A})$.

For generality, we directly discuss the following system of linear operator equations (I) in Hilbert spaces,

$$\text{(I)} \begin{cases} \mathbb{A}_1 u = f_1 \\ \mathbb{A}_2 u = f_2 \\ \vdots \\ \mathbb{A}_n u = f_n \end{cases} \qquad u \in \mathcal{H}, \quad f_i \in \mathcal{H}_1, \quad i = 1, 2, \ldots, n,$$

where $\mathcal{H}$, $\mathcal{H}_1$ are separable Hilbert spaces, for every $i = 1, 2, \ldots, n$, $\mathbb{A}_i : \mathcal{H} \longrightarrow \mathcal{H}_1$ is a bounded linear operator.

6.2.2 Lemmas

Throughout this section, we suppose that $\mathbb{A}_i : \mathcal{H} \longrightarrow \mathcal{H}_1$ is a bound linear linear operator, $\mathrm{N}(\mathbb{A}_i)$ is the null space of operator $\mathbb{A}_i$, namely, $\mathrm{N}(\mathbb{A}_i) = \{u \in \mathcal{H} |\ \mathbb{A}_i u = 0\}$. For $\mathrm{M} \subset \mathcal{H}_1$, we denote *span*M by [M] and the closure of M by $\overline{\mathrm{M}}$. Suppose $\{r_i\}_{i=1}^{\infty} \subset \mathcal{H}$, and we use $\{\overline{r}_i\}_{i=1}^{\infty}$ to denote the normal orthonormal system that is obtained from $\{r_i\}_{i=1}^{\infty}$ by Gram–Schmidt process of orthonormalization.

We can easily obtain the following lemma.

Lemma 6.2.1. *Suppose* $\{0\} \neq \{r_i\}_{i=1}^{\infty}, \{v_i\}_{i=1}^{\infty} \subset \mathcal{H}$. *Define* $\{\overline{r}_i\}_{i=1}^{\infty}$ *and* $\{\widetilde{v}_i\}_{i=1}^{\infty}$ *as follows:*

$$\overline{r}_1 = \frac{r_1}{\|r_1\|_{\mathcal{H}}}, \tag{6.2.2}$$

$$\overline{r}_{k+1} = \frac{r_{k+1} - \sum_{i=1}^{k} \langle r_{k+1}, \overline{r}_i \rangle_{\mathcal{H}} \overline{r}_i}{\left\| r_{k+1} - \sum_{i=1}^{k} (r_{k+1}, \overline{r}_i) \overline{r}_i \right\|_{\mathcal{H}}}, \quad k = 1, 2, \ldots, \tag{6.2.3}$$

$$\widetilde{v}_1 = \frac{v_1 - \sum_{i=1}^{\infty} \langle v_1, \overline{r}_i \rangle_{\mathcal{H}} \overline{r}_i}{\left\| v_1 - \sum_{i=1}^{\infty} \langle v_1, \overline{r}_i \rangle_{\mathcal{H}} \overline{r}_i \right\|_{\mathcal{H}}}, \tag{6.2.4}$$

$$\widetilde{v}_{k+1} = \frac{v_{k+1} - \sum_{i=1}^{k} \langle v_{k+1}, \widetilde{v}_i \rangle_{\mathcal{H}} \widetilde{v}_i - \sum_{i=1}^{\infty} \langle v_{k+1}, \overline{r}_i \rangle_{\mathcal{H}} \overline{r}_i}{\left\| v_{k+1} - \sum_{i=1}^{k} \langle v_{k+1}, \widetilde{v}_i \rangle_{\mathcal{H}} \widetilde{v}_i - \sum_{i=1}^{\infty} \langle v_{k+1}, \overline{r}_i \rangle_{\mathcal{H}} \overline{r}_i \right\|_{\mathcal{H}}}, \quad k = 1, 2, \ldots. \tag{6.2.5}$$

To formula (6.2.2)*, we follow the following rules: if for some positive integer* k*,*

$$r_{k+1} - \sum_{i=1}^{k} (r_{k+1}, \overline{r}_i) \overline{r}_i = 0,$$

then we delete r_{k+1} *from the sequence* $\overline{r}_1, \ldots, \overline{r}_k, r_{k+1}, r_{k+2}, \ldots, v_1, v_2, \ldots$ *and rewrite* $r_{k+2}, r_{k+3}, \ldots, v_1, v_2, \ldots$ *as* $r_{k+1}, r_{k+2}, \ldots, v_1, v_2, \ldots$*, after that, we compute* $\overline{r}_{k+1}$ *by* (6.2.3) *from the new sequence* $\overline{r}_1, \ldots, \overline{r}_k, r_{k+1}, r_{k+2}, \ldots, v_1, v_2, \ldots$*. We will repeat the same process until*

$$r_{k+1} - \sum_{i=1}^{k} (r_{k+1}, \overline{r}_i) \overline{r}_i \neq 0.$$

To formula (6.2.4), (6.2.5)*, we follow the same rules. Then* $\{\overline{r}_i\}_{i=1}^{\infty} \cup \{\widetilde{v}_i\}_{i=1}^{\infty}$ *is an orthonormal system of* $\mathcal{H}$ *which is obtained from the sequence* $r_1, r_2, \ldots, r_n, \ldots, v_1, v_2, \ldots, v_n, \ldots$ *by Gram–Schmidt process.*

With respect to null space $\mathrm{N}(\mathbb{A}_i)$ and conjugate operator $\mathbb{A}_i^*$ of operator $\mathbb{A}$, we have the following properties.

Proposition 6.2.1. $\mathrm{N}(\mathbb{A}_i)$ *is a closed linear subspace of* $\mathcal{H}$.

Proposition 6.2.2. *If* $\{\varphi_i\}_{i=1}^{\infty}$ *satisfied* $\overline{[\{\varphi_i\}_{i=1}^{\infty}]} = \mathcal{H}_1$*, then*

$$\overline{\left[\bigcup_{i=1}^{n} \{\mathbb{A}_i^* \varphi_j\}_{j=1}^{\infty}\right]} = \overline{\mathbb{A}_1^* \mathcal{H}_1 + \mathbb{A}_2^* \mathcal{H}_1 + \cdots + \mathbb{A}_n^* \mathcal{H}_1}$$

.

Proposition 6.2.3. $\overline{\mathbb{A}_1^*\mathcal{H}_1 + \mathbb{A}_2^*\mathcal{H}_1 + \cdots + \mathbb{A}_n^*\mathcal{H}_1}^{\perp} = \bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i)$.

Proof.

$$\begin{aligned}
& x \in \overline{\mathbb{A}_1^*\mathcal{H}_1 + \mathbb{A}_2^*\mathcal{H}_1 + \cdots + \mathbb{A}_n^*\mathcal{H}_1}^{\perp} \\
\Longleftrightarrow\ & (x, \mathbb{A}_1^* z_1 + \mathbb{A}_2^* z_2 + \cdots + \mathbb{A}_n^* z_n) = 0, \ \ \forall\, z_i \in \mathcal{H}_1, \ \ i = 1, 2, \ldots, n \\
\Longleftrightarrow\ & (x, \mathbb{A}_i^* z_i) = 0, \ \ \forall\, z_i \in \mathcal{H}_1, \ \ i = 1, 2, \ldots, n \\
\Longleftrightarrow\ & x \in \mathrm{N}(\mathbb{A}_i), \ \ i = 1, 2, \ldots, n \\
\Longleftrightarrow\ & x \in \bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i).
\end{aligned}$$

The proof is complete. □

In this section, we suppose

$$\overline{[\{\varphi_i\}_{i=1}^{\infty}]} = \mathcal{H}_1. \tag{6.2.6}$$

Arrange the order of set

$$\left\{ \mathbb{A}_m^* \varphi_i |\ m = 1, 2, \ldots, n; \ \ i = 1, 2, \ldots \right\}$$

and set

$$\left\{ \langle f_m, \varphi_i \rangle_{\mathcal{H}_1} |\ m = 1, 2, \ldots, n; \ i = 1, 2, \ldots \right\}$$

as follows: first, fix i and arrange $\{\mathbb{A}_m^* \varphi_i\}_{m=1}^{n}$, $\{\langle f_m, \varphi_i \rangle_{\mathcal{H}_1}\}_{m=1}^{n}$ as

$$\mathbb{A}_1^* \varphi_i, \mathbb{A}_2^* \varphi_i, \ldots, \mathbb{A}_n^* \varphi_i \ \text{ and } \ \langle f_1, \varphi_i \rangle_{\mathcal{H}_1}, \langle f_2, \varphi_i \rangle_{\mathcal{H}_1}, \ldots, \langle f_n, \varphi_i \rangle_{\mathcal{H}_1}$$

separately, then for the next i, we do the same work, then for the next next $i, \ldots$, generally, we can write

$$r_{n(i-1)+m} = \mathbb{A}_m^* \varphi_i, \ l_{n(i-1)+m} = \langle f_m, \varphi_i \rangle_{\mathcal{H}_1} \tag{6.2.7}$$

for every $m = 1, 2, \ldots, n$ and $i = 1, 2, \ldots$. Denote by $\{\overline{r}_i\}_{i=1}^{\infty}$ the result that is obtained from $\{r_i\}_{i=1}^{\infty} \subset \mathcal{H}$ by Gram–Schmidt process, namely

$$\overline{r}_i = \sum_{k=1}^{i} \beta_{ik} r_k, \ \ \beta_{ii} > 0, \ \ i = 1, 2, \ldots. \tag{6.2.8}$$

In order to get the main results, we give another lemma.

Lemma 6.2.2. *If u is one solution of system of equations (I), then u satisfied*

$$\langle u, \overline{r}_i \rangle_{\mathcal{H}} = \sum_{k=1}^{i} \beta_{ik} l_k.$$

for every $i = 1, 2, \ldots$, where $\overline{r}_i$, β_{ik}, l_k are defined in (6.2.7), (6.2.8).

Proof. From hypothesis (6.2.6), it follows that $\{\varphi_i\}_{i=1}^{\infty}$ is complete in $\mathcal{H}_1$. Hence equation $\mathbb{A}_1 u = f_1$ is valid if and only if $\langle \mathbb{A}_1 u, \varphi_i\rangle_{\mathcal{H}_1} = \langle f_1, \varphi_i\rangle_{\mathcal{H}_1}$ is valid for every $i = 1, 2, \ldots$, namely, $\langle u, \mathbb{A}_1^* \varphi_i\rangle_{\mathcal{H}} = \langle f_1, \varphi_i\rangle_{\mathcal{H}_1}$ for every $i = 1, 2, \ldots$.

Similarly, for every $1 \le l \le n$, equation $A_l u = f_l$ is valid if and only if $\langle u, A_l^* \varphi_i\rangle_{\mathcal{H}} = \langle f_l, \varphi_i\rangle_{\mathcal{H}_1}$ for every $i = 1, 2, \ldots$. Therefore, u is one solution of equations (I) if and only if u satisfied $\langle u, A_l^* \varphi_i\rangle_{\mathcal{H}} = \langle f_l, \varphi_i\rangle_{\mathcal{H}_1}$ for every $i = 1, 2, \ldots$ and $l = 1, 2, \ldots, n$. By notation (6.2.7), the last system of equations is $\langle u, r_i\rangle_{\mathcal{H}} = l_i$, where $i = 1, 2, \ldots$. Then the conclusion follows from the fact that $\langle u, r_i\rangle_{\mathcal{H}} = l_i$ being valid for every $i = 1, 2, \ldots$ implies

$$\langle u, \overline{r}_i\rangle_{\mathcal{H}} = \sum_{k=1}^{i} \beta_{ik} \langle u, r_k\rangle_{\mathcal{H}} = \sum_{k=1}^{i} \beta_{ik} l_k$$

for every $i = 1, 2, \ldots$, where $\overline{r}_i, \beta_{ik}$ and l_k are defined in (6.2.8), (6.2.7). □

Lemma 6.2.3. *If equations* (I) *has solutions, then*

$$u_0 = \sum_{i=1}^{\infty} \left[\sum_{k=1}^{i} \beta_{ik} l_k \right] \overline{r}_i$$

is one solution of it, where $\overline{r}_i$, β_{ik} *and* l_k *are defined in* (6.2.8), (6.2.7) *separately.*

Proof. Let u is one solution of equations (I) and write $\overline{\mathbb{A}_1^* \mathcal{H}_1 + \mathbb{A}_2^* \mathcal{H}_1 + \cdots + \mathbb{A}_n^* \mathcal{H}_1} = \mathcal{H}_2 \subset \mathcal{H}$. Let

$$u = u_0' + u_1,$$

where $u_0' \in \mathcal{H}_2$, $u_1 \in \mathcal{H}_2^{\perp}$ and $\mathcal{H}_2^{\perp}$ is the orthogonal complement space which satisfied $\mathcal{H} = \mathcal{H}_2 \oplus \mathcal{H}_2^{\perp}$. Property 6.2.2 shows $\overline{[\{r_i\}_{i=1}^{\infty}]} = \mathcal{H}_2$. Therefore, $\{\overline{r}_i\}_{i=1}^{\infty}$ is complete in $\mathcal{H}_2$. Hence, by Lemma 2.2,

$$\begin{aligned} u_0' &= \sum_{i=1}^{\infty} \langle u_0', \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i = \sum_{i=1}^{\infty} \langle u_0', \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i + \sum_{i=1}^{\infty} \langle u_1, \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i \\ &= \sum_{i=1}^{\infty} \langle u, \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i = \sum_{i=1}^{\infty} \left[\sum_{k=1}^{i} \beta_{ik} l_k \right] \overline{r}_i = u_0. \end{aligned} \tag{6.2.9}$$

Consequently, $u_0 = u_0' \in \mathcal{H}_2$. On the other hand, from Property 6.2.3, $u_1 \in \mathcal{H}_2^{\perp} = \bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i)$. Hence for every $i = 1, 2, \ldots, n$, $\mathbb{A}_i(u_1) = 0$. It follows that for every $i = 1, 2, \ldots, n$,

$$f = \mathbb{A}_i(u) = \mathbb{A}_i(u_0) + \mathbb{A}_i(u_1) = \mathbb{A}_i(u_0).$$

So, u_0 is one solution of equations (I). □

Main Results

Theorem 6.2.1. (*The representation of solutions*) *If system of equations* (I) *has solutions, then*

$$u_0 = \sum_{i=1}^{\infty} \left[\sum_{k=1}^{i} \beta_{ik} l_k \right] \overline{r}_i$$

is its minimal norm solution. And the solution set of equations (I) *is*

$$u_0 + \overline{\mathbb{A}_1^* \mathcal{H}_1 + \mathbb{A}_2^* \mathcal{H}_1 + \cdots + \mathbb{A}_n^* \mathcal{H}_1}^{\perp} = u_0 + \bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i).$$

Proof. Property (6.2.3) shows the last equality in this theorem is valid. Next we will prove the other assertions. By Lemma (6.2.3), u_0 is one solution of equations (I). Denote by A the solution set of equations (I). Write

$$u_0 + \overline{\mathbb{A}_1^* \mathcal{H}_1 + \mathbb{A}_2^* \mathcal{H}_1 + \cdots + \mathbb{A}_n^* \mathcal{H}_1}^{\perp} = \mathrm{B}$$

and

$$\overline{\mathbb{A}_1^* \mathcal{H}_1 + \mathbb{A}_2^* \mathcal{H}_1 + \cdots + \mathbb{A}_n^* \mathcal{H}_1} = \mathcal{H}_2 \subset \mathcal{H}.$$

Then it's sufficient to prove that A = B.

Prove A $\subset$ B. Suppose $u \in$ A, namely, u is one solution of equations (I). Then $u = u_0' + u_1$, where $u_0' \in \mathcal{H}_2$, $u_1 \in \mathcal{H}_2^{\perp}$ and $H_2^{\perp}$ satisfied $\mathcal{H} = \mathcal{H}_2 \oplus \mathcal{H}_2^{\perp}$. Since $\{\overline{r}_i\}_{i=1}^{\infty}$ is complete in $\mathcal{H}_2$(This has been proved in Lemma (6.2.3)), using (6.2.9), we have $u_0' = u_0$. It follows that $u = u_0 + u_1$,where $u_1 \in \mathcal{H}_2^{\perp}$. Therefore $A \subset B$.

Prove $B \subset$ A. If $u \in$ B, then $u = u_0 + u_1$, $u_1 \in \mathcal{H}_2^{\perp}$. From property (6.2.3), $u_1 \in \bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i)$, namely, for every $i = 1, 2, \ldots, n$, $\mathbb{A}_i(u_1) = 0$, hence,

$$\mathbb{A}_i(u) = \mathbb{A}_i(u_0) + \mathbb{A}_i(u_1) = f + 0 = f, \quad i = 1, 2, \ldots, n.$$

Therefore $u \in$ A. Thus B $\subset$ A.

So, A = B. Besides, if u is a solution of equations (I), then $u = u_0 + u_1$, where $u_0 \in \mathcal{H}_2$ and $u_1 \in \mathcal{H}_2^{\perp}$. It follows that

$$\|u\|_{\mathcal{H}}^2 = \|u_0\|_{\mathcal{H}}^2 + \|u_1\|_{\mathcal{H}}^2 \geq \|u_0\|_{\mathcal{H}}^2.$$

Therefore, u_0 has the smallest norm. □

Theorem 6.2.2. (*One complete orhtonormal system of the null space* $\bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i)$). *Let* $\{v_i\}_{i=1}^{\infty}$ *be complete in* $\mathcal{H}$. *By Lemma* (6.2.1), *we get* $\{\overline{r}_i\}_{i=1}^{\infty}$ *and* $\{\widetilde{v}_i\}_{i=1}^{\infty}$ *from the sequence* $r_1, r_2, \ldots, r_n, \ldots, v_1, v_2, \ldots, v_n, \ldots$ *by Gram–Schmidt process, where* $\{r_i\}_{i=1}^{\infty}$ *is defined in* (6.2.7). *Then* $\{\widetilde{v}_i\}_{i=1}^{\infty}$ *is a complete orthonormal set of the space* $\bigcap_{i=1}^{n} \mathrm{N}(\mathbb{A}_i) = \overline{[\{r_i\}_{i=1}^{\infty}]}^{\perp}$.

Proof. From notation (6.2.7), properties (6.2.2), (6.2.3), it follows that the last equality in this theorem is valid. Next we will prove the other assertions. Write $M = \{\widetilde{v}_i\}_{i=1}^{\infty}$, $H_2 = \overline{[\{r_i\}_{i=1}^{\infty}]}$. Then $\{\overline{r}_i\}_{i=1}^{\infty}$ is complete in H_2. From Lemma 2.1, $\{\overline{r}_i\}_{i=1}^{\infty} \cup \{\widetilde{v}_i\}_{i=1}^{\infty}$ is an orhtonormal function set of H. So, set M is an orthonormal function set of $H_2^{\perp}$. Then we only need to show that M is complete in $H_2^{\perp}$. Suppose that $w \in \mathcal{H}_2^{\perp}$ satisfied $\langle w, \widetilde{v}_i\rangle_{\mathcal{H}} = 0$ for every $i = 1, 2, \ldots$. Then, by using $\langle w, \overline{r}_i\rangle_{\mathcal{H}} = 0$, $i = 1, 2, dots$ and the definition (6.2.4), (6.2.5) of $\widetilde{v}_i$, we have $\langle w, v_k\rangle_{\mathcal{H}} = \langle w, \widetilde{v}_k\rangle_{\mathcal{H}} = 0$ for every $k = 1, 2, \ldots$. Therefore, from the completeness of $\{v_i\}_{i=1}^{\infty}$, we have $w = 0$. □

Remark 6.2.1. Write

$$\widetilde{v}_1^n = \frac{v_1 - \sum_{i=1}^{n} \langle v_1, \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i}{\left\| v_1 - \sum_{i=1}^{n} \langle v_1, \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i \right\|\rangle_{\mathcal{H}}}, \tag{6.2.10}$$

$$\widetilde{v}_{k+1}^n = \frac{v_{k+1} - \sum_{i=1}^{k} \langle v_{k+1}, \widetilde{v}_1^n\rangle_{\mathcal{H}} - \sum_{i=1}^{n} \langle v_{k+1}, \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i}{\left\| v_{k+1} - \sum_{i=1}^{k} \langle v_{k+1}, \widetilde{v}_1^n\rangle_{\mathcal{H}} - \sum_{i=1}^{n} \langle v_{k+1}, \overline{r}_i\rangle_{\mathcal{H}} \overline{r}_i \right\|\rangle_{\mathcal{H}}}, \quad k = 1, 2, \ldots. \tag{6.2.11}$$

Then it is not difficult to prove: for every $i = 1, 2, \ldots$, we have $\widetilde{v}_i^n \overset{\|\bullet\|_{\mathcal{H}}}{\longrightarrow} \widetilde{v}_i$, as $n \longrightarrow \infty$.

6.2.3 Solving $\mathbb{A}u = f$ in Reproducing Kernel Sapce

Let $W(\mathcal{D})$ be a reproducing kernel space which is composed by some continuous functions defined on domain $\mathcal{D} \subset \mathbb{R}^n$, where n is a positive integer. Two concrete examples are space $W_2^1[a, b]$ (1.3.2) and space $W_2^{(1,1)}(\mathcal{D})$ (1.5.1).

Now we discuss how to solve the equation

$$\mathbb{A}u(\mathbf{p}) = f(\mathbf{p}), \tag{6.2.12}$$

where $u, f \in W(\mathcal{D})$, $\mathbb{A}$ is a bounded linear operator of $W(\mathcal{D}) \to W(\mathcal{D})$.

Let $R_{\mathbf{p}'}(\mathbf{p})$ is the reproducing kernel of $W(\mathcal{D})$, where $\mathbf{p}, \mathbf{p}' \in \mathcal{D}$. We need a property of $R_{\mathbf{p}'}(\mathbf{p})$: Write

$$\varphi_i(\mathbf{p}) = R_{\mathbf{p}_i}(\mathbf{p})$$

for every $i = 1, 2, \ldots$, where $\{\mathbf{p}_i\}_{i=1}^{\infty}$ is dense in domain $\mathcal{D}$. Then $\{\varphi_i(\mathbf{p})\}_{i=1}^{\infty}$ is a complete orthonormal system of $W(\mathcal{D})$.

From here to the end of this section, we suppose that $\{\mathbf{p}_i\}_{i=1}^{\infty}$ is dense in domain $\mathcal{D}$.

In equations (I), we set $n = 1$, $\mathcal{H} = \mathcal{H}_1 = W(\mathcal{D})$, $f_1(\mathbf{p}) = f(\mathbf{p})$. Then equations (I) changed into Eq. (6.2.12). Rewrite (6.2.7), (6.2.8) as (6.2.13), (6.2.14),

$$\varphi_i(\mathbf{p}) = R_{\mathbf{p}_i}(\mathbf{p}), \;\; r_i = \mathbb{A}^*\varphi_i, \;\; l_i = \langle f, \varphi_i\rangle_W = f(\mathbf{p}_i), \;\; i = 1, 2, \ldots . \tag{6.2.13}$$

Here the last equality of formula (6.2.13) is obtained from the property of reproducing kernel.

$$\overline{r}_i(\mathbf{p}) = \sum_{k=1}^{i} \beta_{ik} r_k(\mathbf{p}), \;\; \beta_{ii} > 0, \;\; i = 1, 2, \ldots . \tag{6.2.14}$$

Then we get Theorems 6.2.3, 6.2.4, Remark 6.2.2 from Theorems 6.2.1, 6.2.2 and Remark 6.2.1.

Theorem 6.2.3. (*Structure of the solution space*). *Suppose that Eq.* (6.2.12) *has solutions, then*

$$u_0(\mathbf{p}) = \sum_{i=1}^{\infty} \left[\sum_{k=1}^{i} \beta_{ik} f(\mathbf{p}_k)\right] \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}) \tag{6.2.15}$$

is its minimal norm solution and the solution set of Eq. (6.2.12) *is*

$$u_0 + \overline{\mathbb{A}^*[W(\mathcal{D})]} = u_0 + \mathrm{N}(\mathbb{A}),$$

where $\{\overline{\mathbb{A}^*\varphi_i}\}_{i=1}^{\infty}$ *is the result that is obtained from* $\{\mathbb{A}^*\varphi_i\}_{i=1}^{\infty}$ *by Gram–Schmidt process,* $\mathrm{N}(\mathbb{A}) = \{w \in W(\mathcal{D}) | \mathbb{A}w = 0\}$.

Since $\{\mathbf{p}_i\}_{i=1}^{\infty}$ is dense in domain $\mathcal{D}$, $\{\varphi_i(\mathbf{p})\}_{i=1}^{\infty} = \{R_{\mathbf{p}_i}(\mathbf{p})\}_{i=1}^{\infty}$ is complete in $W(\mathcal{D})$. Replacing $\{v_i(\mathbf{p})\}_{i=1}^{\infty}$ by $\{\varphi_i(\mathbf{p})\}_{i=1}^{\infty}$ in Theorem 6.2.2, we'll get

Theorem 6.2.4. (*One complete orthonormal system of* $\mathrm{N}(\mathbb{A})$). $\{\widetilde{v}_i\}_{i=1}^{\infty}$ *defined as follows is a complete orthonormal system of* $\mathrm{N}(\mathbb{A})$.

$$\widetilde{v}_1 = \frac{\varphi_1 - \sum\limits_{i=1}^{\infty} \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_1) \cdot \overline{\mathbb{A}^*\varphi_i}}{\left\|\varphi_1 - \sum\limits_{i=1}^{\infty} \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_1) \cdot \overline{\mathbb{A}^*\varphi_i}\right\|_W}, \tag{6.2.16}$$

$$\widetilde{v}_{k+1} = \frac{\varphi_{k+1} - \sum\limits_{i=1}^{k} \widetilde{v}_i(\mathbf{p}_{k+1}) \cdot \widetilde{v}_i - \sum\limits_{i=1}^{\infty} \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_{k+1}) \cdot \overline{\mathbb{A}^*\varphi_i}}{\left\|\varphi_{k+1} - \sum\limits_{i=1}^{k} \widetilde{v}_i(\mathbf{p}_{k+1}) \cdot \widetilde{v}_i - \sum\limits_{i=1}^{\infty} \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_{k+1}) \cdot \overline{\mathbb{A}^*\varphi_i}\right\|_W}, \;\; k = 1, 2, \ldots . \tag{6.2.17}$$

where φ_i *is defined by* (6.2.13), $\{\overline{\mathbb{A}^*\varphi_i}\}_{i=1}^{\infty}$ *is the result that is obtained from* $\{\mathbb{A}^*\varphi_i\}_{i=1}^{\infty}$ *by Gram–Schmidt process and*

$$\mathbb{A}^*\varphi_i(\mathbf{p}) = \left\langle \mathbb{A}^*\varphi_i(\mathbf{p}'), R_{\mathbf{p}}(\mathbf{p}')\right\rangle_{\mathbf{p}', W}$$

$$\begin{aligned} &= \big\langle \varphi_i(\mathbf{p}''), \mathbb{A}_{\mathbf{p}'} R_{\mathbf{p}}(\mathbf{p}')(\mathbf{p}'') \big\rangle_{\mathbf{p}'',W} \\ &= \big[\mathbb{A}_{\mathbf{p}'} R_{\mathbf{p}}(\mathbf{p}')\big](\mathbf{p}_i), \end{aligned}$$

where $\mathbf{p}, \mathbf{p}', \mathbf{p}'' \in \mathcal{D}$.

Remark 6.2.2. Write

$$\widetilde{v}_1^n = \frac{\varphi_1 - \sum_{i=1}^{n} \overline{\mathbb{A}^*\varphi_i} \cdot \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_1)}{\big\|\varphi_1 - \sum_{i=1}^{n} \overline{\mathbb{A}^*\varphi_i} \cdot \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_1)\big\|_W}, \tag{6.2.18}$$

$$\widetilde{v}_{k+1}^n = \frac{\varphi_{k+1} - \sum_{i=1}^{k} \widetilde{v}_i^n(\mathbf{p}_{k+1}) \cdot \widetilde{v}_i^n - \sum_{i=1}^{n} \overline{\mathbb{A}^*\varphi_i} \cdot \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_{k+1})}{\big\|\varphi_{k+1} - \sum_{i=1}^{k} \widetilde{v}_i^n(\mathbf{p}_{k+1}) \cdot \widetilde{v}_i^n - \sum_{i=1}^{n} \overline{\mathbb{A}^*\varphi_i} \cdot \overline{\mathbb{A}^*\varphi_i}(\mathbf{p}_{k+1})\big\|_W}, \quad k = 1, 2, \ldots. \tag{6.2.19}$$

Then it is not difficult to prove: for every $i = 1, 2, \ldots$, we have $\widetilde{v}_i^n \xrightarrow{\|\bullet\|_W} \widetilde{v}_i$, as $n \longrightarrow \infty$.

6.2.4 Numerical Experiments

We will give all solutions of the following Eq. (6.2.20) in space $W_2^1[0, 2\pi]$. In fact, if u_0 is one solution of Eq. (6.2.20), then the set

$$\left\{ u_0 + \sum_{i=1}^{\infty} \alpha_i \widetilde{v}_i \middle| \text{ real sequence } \{\alpha_i\}_{i=1}^{\infty} \text{ satisfied } \sum_{i=1}^{\infty} \alpha_i^2 < \infty \right\}$$

give all solutions of Eq. (6.2.20), where $\{\widetilde{v}_i\}_{i=1}^{\infty}$ is defined by (6.2.16), (6.2.17). In fact, (6.2.16) and (6.2.17) can't be finished off by computer, but we can approximate (6.2.16), (6.2.16) by (6.2.18), (6.2.19).

We need some symbols.

Write

$$\max_n = \max_{t\in[a,b]} |u_{0n}(t) - u(t)|,$$

where u is the exact solution of Eq. (6.2.20), and u_{0n} is the approximate solution of Eq. (6.2.20), which is obtained by computing the first n terms of formula (6.2.15). The first $4m + 1$ terms of $\{\mathbf{p}_i\}_{i=1}^{\infty}$ is taken as

$$\left\{a + \frac{i(b-a)}{m}\right\}_{i=0}^{m}, \quad \left\{a + \frac{(2i-1)(b-a)}{2m}\right\}_{i=1}^{m},$$

$$\left\{a + \frac{(4i-3)(b-a)}{4m}, a + \frac{(4i-1)(b-a)}{4m}\right\}_{i=1}^{m}$$

successively.

Table 6.3:

$n = 2$
where n means the sequence $\overline{r}_1, \ldots, \overline{r}_n$.
$\|u_{0n}(x) - u(x)\|_{W_2^1} = 7.25987\text{E-}15$
$max_n = 8.0324\text{E-}16$

Table 6.4:

$n = 2, K = 90$

where n, K means the sequence $\overline{r}_1, \ldots, \overline{r}_n, \widetilde{v}_1^n, \widetilde{v}_2^n, \ldots, \widetilde{v}_K^n$.

$b_k \triangleq \max\limits_{x\in[0,2\pi]} |A\widetilde{v}_k^n(x)|, \quad c_k \triangleq \|A\widetilde{v}_k^n(x)\|_{W_2^1}$

k	$\frac{b_k}{10^{-15}}$	$\frac{c_k}{10^{-15}}$	k	$\frac{b_k}{10^{-14}}$	$\frac{c_k}{10^{-14}}$	k	$\frac{b_k}{10^{-14}}$	$\frac{c_k}{10^{-14}}$
1	4.44	8.59	31	4.30	8.93	61	2.15	3.03
2	5.11	6.88	32	5.64	11.0	62	1.78	3.26
3	5.80	7.44	33	6.25	10.9	63	2.69	6.63
4	5.18	11.6	34	7.63	14.1	64	2.96	6.01
5	4.70	7.59	35	7.65	13.9	65	3.72	6.67
6	5.01	6.59	36	7.09	12.5	66	4.54	8.99
7	4.41	10.5	37	5.99	9.72	67	6.57	13.4
8	5.40	9.76	38	3.39	6.31	68	2.11	14.6
9	5.76	7.90	39	3.04	4.62	69	8.11	14.4
10	6.60	12.4	40	2.01	4.83	70	7.40	12.4
11	4.53	10.0	41	1.18	2.86	71	7.98	13.2
12	3.47	6.90	42	1.15	2.12	72	9.68	20.5
13	5.57	6.73	43	1.34	3.09	73	9.68	19.5
14	5.75	8.42	44	1.93	3.37	74	9.44	13.4
15	4.49	12.0	45	2.83	5.13	75	7.81	14.2
16	5.92	7.95	46	2.77	5.31	76	7.86	18.8
17	5.94	5.26	47	3.08	5.47	77	7.30	11.8
18	6.02	6.93	48	3.77	7.25	78	6.11	9.17
19	5.38	11.6	49	4.14	6.30	79	4.12	7.09
20	7.58	15.4	50	4.19	6.83	80	2.09	4.90
21	4.36	7.73	51	4.60	8.27	81	4.17	7.58
22	9.78	19.0	52	4.86	8.77	82	2.39	6.93
23	10.2	16.0	53	4.79	7.33	83	1.72	2.85
24	10.3	29.1	54	4.14	7.32	84	1.75	4.95
25	13.1	18.2	55	3.19	6.67	85	2.26	5.72
26	13.9	32.4	56	3.41	6.37	86	2.13	4.80
27	20.3	34.2	57	3.11	5.35	87	2.33	5.01
28	23.4	56.6	58	2.88	3.99	88	2.99	6.25
29	28.3	52.1	59	2.25	3.78	89	3.57	6.28
30	37.7	74.7	60	0.98	2.81	90	3.14	6.46

Solve equation

$$\int_0^{2\pi} \cos(3x + y)u(y)dy = \pi \cos(3x). \tag{6.2.20}$$

(a) To prove u_0 defined by (6.2.15) is one solution of Eq. (6.2.20). In fact, $u(y) = \cos(y)$ is one solution of Eq. (6.2.20).

Take $m = 40$. And the numerical results are given in Table 6.3.

(b) In order to show that when n, K are large enough, $\{\widetilde{v}_i\}_{i=1}^K$ will be an approximate complete system of $\mathrm{N}(\mathbb{A})$, we do numerical experiment with $w(y) = y + 2\sin(y)$, which is an element of $\mathrm{N}(\mathbb{A})$. We will verify: when n, K are large enough,

$$\begin{cases} A\widetilde{v}_i^n \approx 0, \quad i = 1, 2, \ldots, K, \\ w \approx \sum_{i=1}^{K} (w, \widetilde{v}_i^n)\widetilde{v}_i^n. \end{cases}$$

And the numerical results are listed in Table 6.4 and Table 6.5 separately.

Table 6.5:

$n = 2, K = 90$ where n, K means the sequence $\overline{r}_1, \ldots, \overline{r}_n, \widetilde{v}_1^n, \widetilde{v}_2^n, \ldots, \widetilde{v}_K^n$.			
$W_K^{(n)} \triangleq \sum_{i=1}^{K} (w, \widetilde{v}_i^n)\widetilde{v}_i^n$			
$\|w(x) - W_K^{(n)}(x)\|_{W_2^1} = 0.098057$			
x	$w(x)$	$W_K^{(n)}(x)$	$w(x) - W_K^{(n)}(x)$
0	0	3.41061E-13	-3.41061E-13
0.5	1.45885	1.45925	-6.68839E-4
1	2.68294	2.68367	-7.30621E-4
1.5	3.49499	3.49421	7.75098E-4
2	3.81859	3.81385	4.74355E-3
2.5	3.69694	3.69322	3.72081E-3
3	3.28224	3.27754	4.70195E-3
3.5	2.79843	2.79162	6.81791E-3
4	2.4864	2.48469	1.43312E-3
4.5	2.54494	2.54224	2.70109E-3
5	3.08215	3.08102	1.13172E-3
5.5	4.08892	4.08894	-2.46183E-5
6	5.44117	5.44166	-4.86433E-4
6.2	6.03382	6.03389	-7.30908E-5

So, every element of set $\left\{ u_{0n} + \sum_{i=1}^{K} \alpha_i \widetilde{v}_i^n | \text{ sequence } \{\alpha_i\}_{i=1}^K \text{is real} \right\}$ is an approximate solution of Eq.(20).

Chapter 7

Solving the Inverse Problems

We have mentioned in Chapter 5 that differential equation is an important mathematical model to reflect the natural science. In general, the known conditions of a differential equation are composed of the following factors: 1. coefficient; 2. right-side item; 3. geometric region; 4. boundary condition; 5. initial condition. If one or several of the above five conditions become the object for solution, this kind of problem will be called inverse problem of differential equation.

If there is no other restrictive condition, the solution of the inverse problem will not be unique. Therefore, the inverse problem is actually a common ill-posed problem, and there will always be additional conditions or subsidiary conditions accompanying the inverse problem. The inverse problem exists extensively. For instance, if the internal parameter of a system attracts the interests of people when they are making research on the system, relevant inverse problem will be raised.

The research of inverse problem includes two parts: 1. theory research; 2. numerical algorithm research. This chapter proposes a unique numerical algorism for two the types of coefficient inverse problems.

7.1 Solving the Coefficient Inverse Problem

7.1.1 Introduction

In this section, we consider the coefficient inverse problem of differential equation with initial-value, boundary-value and additional condition

$$\begin{cases} \dfrac{\partial}{\partial x}\left[k(x)\dfrac{\partial u(x,t)}{\partial x}\right] - \dfrac{\partial^2 u(x,t)}{\partial t^2} = f(x,t), \\ u(x,0) = h(x), \ \dfrac{\partial}{\partial t}u(x,0) = 0, \\ u(0,t) = 0, \ \dfrac{\partial}{\partial x}u(1,t) = 0, \\ u(1,t) = g(t) \ \text{(additional condition)}, \end{cases} \quad 0 \le x \le 1, \ 0 \le t \le 1, \qquad (7.1.1)$$

where $f(x,t)$, $h(x)$, $g(t)$ are sufficiently regular given functions, $u(x,t), k(x)$ are unknowns.

If the conditions in Eq. (7.1.1) is inhomogeneous, for example

$$\begin{cases} \dfrac{\partial}{\partial x}\left[k(x)\dfrac{\partial u(x,t)}{\partial x}\right] - \dfrac{\partial^2 u(x,t)}{\partial t^2} = f(x,t), \\ u(x,0) = h(x), \quad \dfrac{\partial}{\partial t}u(x,0) = b(x), \\ u(0,t) = c(t), \dfrac{\partial}{\partial x}u(1,t) = d(t), \\ u(1,t) = g(t) \ \text{(additional condition)}, \end{cases} \quad 0 \le x \le 1, \ \ 0 \le t \le 1, \tag{7.1.2}$$

where $f(x,t)$, $h(x)$, $b(x)$, $c(t)$, $d(t)$, $g(t)$ are sufficiently regular given functions, $u(x,t), k(x)$ are unknown.

We can give the following homogenization, let

$$\widetilde{u}(x,t) = u(x,t) + \alpha(x,t)b(x) + \beta(x,t)c(t) + \gamma(x,t)d(t),$$

where

$$\begin{aligned} \alpha(x,t) &= \begin{cases} -x(1-x)^2 e^{-\frac{t}{x(1-x)^2}}, & 0 < x < 1, \\ 0, & x = 0 \ \text{ or } \ 1. \end{cases} \\ \beta(x,t) &= \begin{cases} (1-x)^2 t^2 - e^{-\frac{x(1-x)^2}{t}}, & 0 < t < 1, \\ 0, & t = 0. \end{cases} \\ \gamma(x,t) &= \begin{cases} x(1-x)^2 t^2 - x e^{-\frac{x(1-x)^2}{t}}, & 0 < t < 1, \\ 0, & t = 0. \end{cases} \end{aligned}$$

Put

$$H(x,t) = \alpha(x,t)b(x) + \beta(x,t)c(t) + \gamma(x,t)d(t).$$

The Eq. (7.1.2) can transformed into the following form:

$$\begin{cases} \dfrac{\partial}{\partial x}\left[k(x)\dfrac{\partial(\widetilde{u}(x,t) - H(x,t))}{\partial x}\right] - \dfrac{\partial^2 \widetilde{u}(x,t)}{\partial t^2} = \widetilde{f}(x,t), \\ \qquad\qquad 0 \le x \le 1, \ \ 0 \le t \le 1, \\ \widetilde{u}(x,0) = \widetilde{h}(x), \quad \dfrac{\partial}{\partial t}\widetilde{u}(x,0) = 0, \\ \widetilde{u}(0,t) = 0, \quad \dfrac{\partial}{\partial x}\widetilde{u}(1,t) = 0, \\ \widetilde{u}(1,t) = \widetilde{g}(t) \ \text{(additional condition)}. \end{cases} \tag{7.1.3}$$

We can solve Eq. (7.1.3) in the same way as the solution procedure of Eq. (7.1.1).

The problem of determining the variable coefficients of partial differential equations from information on solutions to the corresponding boundary problems is called

a coefficient inverse problem. Such problems arise in many applied areas, such as remote sensing, geophysics, operation research, etc. In the sense of Hadamard, these problems are nonlinear and ill-posed. Due to practical requires, studies of inverse problems have been further developed. Therefore, many algorithms have been proposed, such as GPST method [176], [177], regularized nonlinear least squares and iterative methods [178], [179] and convexification algorithm [180], [181], etc. However, such methods are extremely time-consuming, and some assumptions and restrictions about known conditions are added, such as, well initial value, well background value. Here, a new method is presented to solve the coefficient inverse problems. The method is established in reproducing kernel space which is a special Hilbert space. Using some special technics in the reproducing kernel space, the problem of solving the coefficient inverse problems can be converted into the simple problem of solving linear equations. After the coefficient $k(x)$ is obtained, the solution $u(x,t)$ of the partial differential equation can be obtained directly by means of the simple four arithmetic operations of several given functions.

7.1.2 The Reproducing Kernel Spaces

The reproducing kernel space $W_2^1[0,1]$ is defined as (1.3.2), its reproducing kernel is $Q_y(x)$ by (1.4.1).

About this inverse problem Eq. (7.1.1), we need that the second order derivatives of $u(x,t)$ with respect to x and t are continuous. So we can consider the solution $u(x,t)$ to Eq. (7.1.1) in the following reproducing kernel space ${}^oW_2^{(3,3)}(\mathcal{D})$.

Let $\mathcal{D}=[0,1]\times[0,1]$,

$$ {}^oW_2^{(3,3)}(\mathcal{D})=\Big\{u(x,t)|\ u\in W_2^{(3,3)}(\mathcal{D}) \text{ by (1.5.1)},\ u(0,t)=\partial_x u(1,t)=\partial_t u(x,0)=0\Big\}. $$

Similar to Section 1.5.2, we can obtain its reproducing kernel $K_{(\xi,\eta)}(x,t) = R_\xi(x)G_\eta(t)$, where

$$\begin{aligned} R_\xi(x) &= \frac{1}{26880}\,(224\xi^3(10x^2-5x\xi+\xi^2) \\ &+ 4x\Big(56x^4-120(\xi-2)\xi(16-(\xi-2)\xi)+5x^3(\xi-2)\xi(16-(\xi-2)\xi) \\ &+ 60x(\xi-2)\xi(12+(\xi-2)\xi)+20x^2(\xi-2)\xi(12+(\xi-2)\xi)\Big) \\ &- 1120x^2\xi^2(x+\xi+|x-\xi|)+140x\xi(x+\xi+|x-\xi|)^3 \\ &- 7(x+\xi+|x-\xi|)^5), \qquad (7.1.4) \\ G_\eta(t) &= \begin{cases} \frac{1}{120}\left(120+t^5-5t^4\eta+10t^2(3+t)\eta^2\right), & t\le\eta, \\ \frac{1}{120}\left(120-5t\eta^4+\eta^5+10t^2\eta^2(3+\eta)\right), & t>\eta. \end{cases} \end{aligned}$$

7.1.3 Transformation of Eq. (7.1.1)

Integrating both sides of Eq. (7.1.1) from x to 1 ($0 \le x \le 1$), Eq. (7.1.1) is equivalent to

$$-k(x)\frac{\partial}{\partial x}u(x,t) - \int_x^1 u_t''(z,t)dz = \int_x^1 f(z,t)dz.$$

Linear operator $\mathbb{L} : {}^oW_2^{(3,3)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$ is defined by

$$(\mathbb{L}u)(x) \stackrel{def}{=} \int_x^1 u_t''(z,0)dz. \tag{7.1.5}$$

Theorem 7.1.1. *The operator $\mathbb{L} : {}^oW_2^{(3,3)}(\mathcal{D}) \longrightarrow W_2^1[0,1]$ defined by (7.1.5) is a bounded operator.*

$$\begin{aligned}
\left\|(\mathbb{L}u)(x)\right\|_{W_2^1}^2 &= \int_0^1 \left[\left(\int_x^1 u_t''(z,0)dz\right)^2 + (u_t''(x,0))^2\right] dx \\
&\le \int_0^1\int_x^1 (u_t''(z,0))^2\, dz\, dx + \int_0^1 (u_t''(x,0))^2\, dx \\
&\le \int_0^1\int_0^1 (u_t''(z,0))^2\, dz\, dx + \int_0^1 (u_t''(x,0))^2\, dx \\
&= \int_0^1 (u_t''(z,0))^2 dz + \int_0^1 (u_t''(x,0))^2 dx \\
&= 2\int_0^1 (u_t''(x,0))^2 dx.
\end{aligned}$$

Note that

$$\begin{aligned}
u(x,t) &= \left\langle u(\xi,\eta), R_x(\xi)G_t(\eta)\right\rangle_{\xi,\eta,{}^oW_2^{(3,3)}}, \\
u_t''(x,t) &= \left\langle u(\xi,\eta), R_x(\xi)\frac{\partial^2}{\partial t^2}G_t(\eta)\right\rangle_{\xi,\eta,{}^oW_2^{(3,3)}}, \\
u_t''(x,0) &= \left\langle u(\xi,\eta), R_x(\xi)\left.\frac{\partial^2}{\partial t^2}G_t(\eta)\right|_{t=0}\right\rangle_{\xi,\eta,{}^oW_2^{(3,3)}}, \\
|u_t''(x,0)| &\le \|u\|_{{}^oW_2^{(3,3)}} \left\|R_x(\xi)\left.\frac{\partial^2}{\partial t^2}G_t(\eta)\right|_{t=0}\right\|_{\xi,\eta,{}^oW_2^{(3,3)}}
\end{aligned}$$

and $\left\|R_x(\xi)\frac{\partial^2}{\partial t^2}G_t(\eta)|_{t=0}\right\|_{\xi,\eta,{}^oW_2^{(3,3)}}$ is a continuous function about x on the interval $[0,1]$. Therefore, we have

$$|u_t''(x,0)| \le \|u\|_{{}^oW_2^{(3,3)}} M,$$

that is,

$$\left\|(\mathbb{L}u)(x)\right\|_{W_2^1}^2 \le 2M\|u\|_{{}^oW_2^{(3,3)}}.$$

In the following, we will discuss Eq. (7.1.1) in the reproducing kernel space $W(\mathcal{D})$, by converting Eq. (7.1.1) into the operator equation,

$$\begin{cases}(\mathbb{L}u)(x) = F(x)\\ u(x,0) = h(x), \quad u(1,t) = g(t) \ \text{ (additional condition)}\end{cases} \tag{7.1.6}$$

where

$$F(x) \overset{def}{=} -k(x)h'(x) - \int_x^1 f(z,0)dz.$$

It is obvious that the solution $(k(x), u(x,t))$ to (7.1.1) is the solution to (7.1.6).

Lemma 7.1.1. *Using the adjoint operator $\mathbb{L}^*$ of $\mathbb{L}$, then $\mathbb{L}^*$ is also a bounded operator form $W_2^1[0,1]$ to ${}^oW_2^{(3,3)}(\mathcal{D})$. Then it follows that*

$$(\mathbb{L}^*Q_s(\cdot\,))(x,t) = \frac{d^2}{d\diamond^2}G_t(\diamond)\Big|_{\diamond=0}\int_s^1 R_x(\star)\,d\star, \tag{7.1.7}$$

$\cdot$, $\star$ and $\diamond$ denote the variables corresponding to functions respectively.

Proof. By virtue of Theorem 2.1

$$\begin{aligned}(\mathbb{L}^*Q_s(\cdot\,))(x,t) &= \left\langle(\mathbb{L}^*Q_s(\cdot\,))(\star,\diamond), Q_x(\star)G_t(\diamond)\right\rangle_{{}^oW_2^{(3,3)}}\\ &= \left\langle Q_s(x), \mathbb{L}(R_x(\star)G_t(\diamond))(x)\right\rangle_{W_2^1}\\ &= \mathbb{L}(R_x(\star)G_t(\diamond))(s)\\ &= \frac{d^2}{d\diamond^2}G_t(\diamond)\Big|_{\diamond=0}\int_s^1 R_x(\star)\,d\star\,.\end{aligned}$$

The proof is complete. □

7.1.4 Decomposition into Direct Sum of ${}^oW_2^{(3,3)}(\mathcal{D})$

Fix a countable dense subset $\mathrm{T} = \{s_1, s_2, \ldots\}$ of interval $[0,1]$, and put $\varphi_i(x) \overset{def}{=} Q_{s_i}(x)$. And let

$$\psi_i(x,t) \overset{def}{=} (\mathbb{L}^*\varphi_i)(x,t), \ \ i = 1,2,\ldots.$$

By virtue of (7.1.7), we have

$$\psi_i(x,t) \quad = \quad \frac{d^2}{d\eta^2} G_t(\eta)\Big|_{\eta=0} \int_{s_i}^{1} R_x(\xi)d\xi, \ \ i = 1,2,\ldots,$$

$$\psi t(t) \quad \overset{def}{=} \quad \frac{d^2}{d\eta^2} G_t(\eta)\Big|_{\eta=0}, \ \ \psi x_i(x) \overset{def}{=} \int_{s_i}^{1} R_x(\xi)d\xi,$$

then

$$\psi_i(x,t) = \psi t(t)\psi x_i(x).$$

And

$$m \overset{def}{=} \|\psi t\|^2_{{}^oW_2^{(3,3)}} = \langle \psi t(t), \psi t(t)\rangle_{{}^oW_2^{(3,3)}} \approx 0.505315.$$

By virtue of Lemma 2.1, we have

$$\big\langle \psi_i(x,t), \psi_j(x,t)\big\rangle_{{}^oW_2^{(3,3)}} = m\big\langle \psi x_i(x), \psi x_j(x)\big\rangle_{{}^oW_2^{(3,3)}}.$$

Using Gram–Schmidt process, we orthonormalize the sequence $\{\psi_i(x,t)\}_{i=1}^{\infty}$, to obtain an orthogonal system $\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty}$, and

$$\overline{\psi}_i(x,t) = \sum_{k=1}^{i} \beta_{ik}\psi_k(x,t), \ \ i = 1,2,\ldots,$$

where β_{ik} is the coefficient of orthonormalization. Then S is a subspace of ${}^oW_2^{(3,3)}(\mathcal{D})$.

$$\mathrm{S} = \left\{ u(x,t) |\ u(x,t) = \sum_{i=1}^{\infty} c_i\overline{\psi}_i(x,t),\ \{c_i\} \in \ell^2 \right\}, \tag{7.1.8}$$

and $\mathrm{S}^{\perp}$ denote the orthogonal complement space of S in ${}^oW_2^{(3,3)}(\mathcal{D})$.

Lemma 7.1.2. *$v_0(x,t) \in \mathrm{S}^{\perp} \subseteq {}^oW_2^{(3,3)}(\mathcal{D})$ if and only if $(\mathbb{L}v_0)(x) = 0$*

Proof. Note that it holds that

$$\langle v_0, \psi_i\rangle_{{}^oW_2^{(3,3)}} = \langle v_0, \mathbb{L}^*\varphi_i\rangle_{{}^oW_2^{(3,3)}} = \langle \mathbb{L}v_0, \varphi_i\rangle_{W_2^1} = (\mathbb{L}v_0)(s_i)$$

for each $s_i \in \mathrm{T}$. Therefore, $v_0(x,t) \in \mathrm{S}^{\perp}$ if and only if $(\mathbb{L}v_0)(x) = 0$. □

Lemma 7.1.3. *$v_0(x,t) \in \mathrm{S}^{\perp} \subseteq {}^{o}W_2^{(3,3)}(\mathcal{D})$ if and only if*

$$\left.\frac{\partial^2 v_0(x,t)}{\partial t^2}\right|_{t=0} = 0.$$

The proof is omitted.

The function space

$$\mathrm{S}^{\perp} = \left\{ u(x,t) |\ u(x,t) \in {}^{o}W_2^{(3,3)}(\mathcal{D}),\ \frac{\partial^2}{\partial t^2} u(x,0) = 0 \right\}$$

is a reproducing kernel space. It possesses the reproducing kernel function

$$\rho_{(\xi,\eta)}(x,t) = R_{\xi}(x) G_{\eta}^{\perp}(t),$$

where $R_{\xi}(x)$ by (7.1.4),

$$G_{\eta}^{\perp}(t) = \begin{cases} \dfrac{1}{120}\,(120 + 10s^2t^3 - 5st^4 + t^5), & t \le s, \\ \dfrac{1}{120}\,(120 + s^5 - 5s^4t + 10s^3t^2), & t > s. \end{cases}$$

Take a set of points $\mathrm{B} = \{p_1(\xi_1,\eta_1), p_2(\xi_2,\eta_2), \ldots\}$ as a dense set of region $\mathcal{D} = [0,1] \times [0,1]$, and put

$$\rho_j(x,t) = R_{\xi_j}(x) G_{\eta_j}^{\perp}(t), \quad j = 1, 2, \ldots .$$

Then $\{\rho_j(x,t)\}_{j=1}^{\infty}$ is the basis of $\mathrm{S}^{\perp}$. The normal orthogonal basis $\{\overline{\rho}_j(x,t)\}_{j=1}^{\infty}$ can derived from the Gram–Schmidt orthogonalization process of $\{\rho_j(x,t)\}_{j=1}^{\infty}$

$$\overline{\rho}_j(x,t) = \sum_{m=1}^{j} \beta_{jm}^{*} \rho_m(x,t), \quad j = 1, 2, \ldots .$$

Note that

$${}^{o}W_2^{(3,3)}(\mathcal{D}) = \mathrm{S} \oplus \mathrm{S}^{\perp}$$

and $\{\overline{\psi}_i(x,t)\}_{i=1}^{\infty} \cup \{\overline{\rho}_j(x,t)\}_{j=1}^{\infty}$ is an orthonormal basis of $W(\mathcal{D})$.

Corollary 7.1.1. *Let $\widetilde{\psi x}_i(x) = \sum_{n=1}^{i} \beta_{in} \psi x_n(x)$, $i = 1, 2, \ldots$, then $\{\widetilde{\psi x}_i(x)\}_{i=1}^{\infty}$ is an orthogonal function system, and*

$$\overline{\psi_i}(x,t) = \psi t(t) \widetilde{\psi x}_i(x), \quad \|\widetilde{\psi x}_i(x)\|^2_{{}_{o}W_2^{(3,3)}} = \frac{1}{m}, \quad j = 1, 2, \ldots . \tag{7.1.9}$$

7.1.5 The Method of Solving Eq. (7.1.6)

Theorem 7.1.2. *Assume $(k(x), u(x,t))$ be the solution of Eq. (7.1.1), then*

$$\begin{aligned} u(x,t) &= \psi t(t)\sum_{i=1}^{\infty}\sum_{n=1}^{i}\beta_{in}\left[-k(s_n)h'(s_n) - \int_{s_n}^{1} f(z,0)dz\right]\widetilde{\psi x}_i(x) \\ &+ \sum_{i=1}^{\infty}\langle u, \overline{\rho}_i\rangle_{{}^{o}W_2^{(3,3)}}\overline{\rho}_i(x,t), \end{aligned} \tag{7.1.10}$$

where $h(x)$, $f(x,t)$ are given functions in (7.1.1).

Proof. Due to $u(x,t) \in {}^{o}W_2^{(3,3)}(\mathcal{D})$, and $\{\overline{\psi}_i\}_{i=1}^{\infty} \cup \{\overline{\rho}_i\}_{i=1}^{\infty}$ is an normal orthogonal basis, it holds that

$$\begin{aligned} u(x,t) &= \sum_{i=1}^{\infty}\big\langle u(x,t), \overline{\psi}_i(x,t)\big\rangle_{{}^{o}W_2^{(3,3)}}\overline{\psi}_i(x,t) \\ &+ \sum_{i=1}^{\infty}\big\langle u(x,t), \overline{\rho}_i(x,t)\big\rangle_{{}^{o}W_2^{(3,3)}}\overline{\rho}_i(x,t). \end{aligned}$$

On the other hand, since $(k(x), u(x,t))$ is also the solution of Eq. (7.1.6), it holds that

$$\begin{aligned} \big\langle u(x,t), \overline{\psi}_i(x,t)\big\rangle_{{}^{o}W_2^{(3,3)}} &= \sum_{n=1}^{i}\beta_{in}\big\langle u(x,t), \psi_i(x,t)\big\rangle_{{}^{o}W_2^{(3,3)}} \\ &= \sum_{n=1}^{i}\beta_{in}\big\langle u(x,t), (\mathbb{L}^*\varphi)_i(x,t)\big\rangle_{{}^{o}W_2^{(3,3)}} \\ &= \sum_{n=1}^{i}\beta_{in}\big\langle \mathbb{L}u, \varphi_i\big\rangle_{W_2^1} = \sum_{n=1}^{i}\beta_{in}F(s_n) \\ &= \sum_{n=1}^{i}\beta_{in}\left[-k(s_n)h'(s_n) - \int_{s_n}^{1} f(z,0)dz\right]. \end{aligned}$$

The proof is complete. □

By (7.1.9), (7.1.10) holds. Let

$$\alpha_i = \big\langle u(x,t), \overline{\rho}_i(x,t)\big\rangle_{{}^{o}W_2^{(3,3)}},$$

then

$$u(x,t) = \psi t(t)\sum_{i=1}^{\infty}\sum_{n=1}^{i}\beta_{in}\left[-k(s_n)h'(s_n) - \int_{s_n}^{1} f(z,0)dz\right]\widetilde{\psi x}_i(x)$$

$$+ \quad \sum_{i=1}^{\infty} \alpha_i \overline{\rho}_i(x,t). \tag{7.1.11}$$

Using the additional condition of Eq. (7.1.6), we have

$$\begin{aligned} g(t) = u(1,t) &= \psi t(t) \sum_{i=1}^{\infty} \sum_{n=1}^{i} \beta_{in} \left[-k(s_n) h'(s_n) - \int_{s_n}^{1} f(z,0) dz \right] \widetilde{\psi x}_i(1) \\ &+ \sum_{i=1}^{\infty} \alpha_i \overline{\rho}_i(1,t) \end{aligned}$$

and

$$\begin{aligned} g'(t) &= \psi t'(t) \sum_{i=1}^{\infty} \sum_{n=1}^{i} \beta_{in} \left[-k(s_n) h'(s_n) - \int_{s_n}^{1} f(z,0) dz \right] \widetilde{\psi x}_i(1) \\ &+ \sum_{i=1}^{\infty} \alpha_i \frac{\partial}{\partial t} \overline{\rho}_i(1,t). \end{aligned}$$

Note that

$$\left. \frac{\partial^2}{\partial t^2} \overline{\rho}_i(x,t) \right|_{t=0} = 0, \quad \left. \frac{\partial^2}{\partial t^2} \psi t(t) \right|_{t=0} \neq 0,$$

so

$$\sum_{i=1}^{\infty} \sum_{n=1}^{i} \beta_{in} \left[-k(s_n) h'(s_n) - \int_{s_n}^{1} f(z,0) dz \right] \widetilde{\psi x}_i(1) = \lim_{t \to 0} \frac{g'(t)}{\psi t'(t)} \stackrel{def}{=} A$$

and

$$\sum_{i=1}^{\infty} \alpha_i \overline{\rho}_i(1,t) = g(t) - A \psi t(t).$$

Taking $t_l \in [0,1]$, $l = 1, 2, \ldots$ we get the infinite linear system about α_i:

$$\sum_{i=1}^{\infty} \alpha_i \overline{\rho}_i(1,t_l) = g(t_l) - A \psi t(t_l), \;\; l = 1, 2, \ldots . \tag{7.1.12}$$

Replacing (7.1.11) with the solution α_i of (7.1.12), we have

$$u(x,t) = \psi t(t) \sum_{i=1}^{\infty} \sum_{n=1}^{i} \beta_{in} \left[-k(s_n) h'(s_n) - \int_{s_n}^{1} f(z,0) dz \right] \widetilde{\psi x}_i(x)$$

$$+ \quad \sum_{i=1}^{\infty} \alpha_i \overline{\rho}_i(x,t). \tag{7.1.13}$$

Again using the boundary condition of (7.1.12), we can obtain

$$\begin{aligned} h(x) = u(x,0) &= \psi t(0) \sum_{i=1}^{\infty} \sum_{n=1}^{i} \beta_{in} \left[-k(s_n)h'(s_n) - \int_{s_n}^{1} f(z,0)dz \right] \widetilde{\psi x}_i(x) \\ &+ \sum_{i=1}^{\infty} \alpha_i \overline{\rho}_i(x,0). \end{aligned} \tag{7.1.14}$$

By the virtue of Corollary 7.1.1, making inner product on both sides of (7.1.14) with $\overline{\psi x}_j(x)$, it holds that

$$\begin{aligned} \langle h(x), \widetilde{\psi x}_j(x) \rangle_{W_2} &= \frac{\psi t(0)}{m} \sum_{n=1}^{j} \beta_{jn} \left[-k(s_n)h'(s_n) - \int_{s_n}^{1} f(z,0)dz \right] \\ &+ \sum_{i=1}^{\infty} \alpha_i \left\langle \overline{\rho}_i(x,0), \overline{\psi x}_j(x) \right\rangle_{W_2}, \quad j = 1, 2, \ldots . \end{aligned}$$

Note that this is the infinite linear system about $k(s_i)$ and its coefficient matrix is a infinite lower triangular matrix. So it is easy to solve the unique solution $k(s_i)$. Replacing (7.1.13) by it, we obtain the solution $u(x,t)$ directly.

7.1.6 Numerical Experiments

Example 7.1.1. A coefficient inverse problem of differential equation:

$$\begin{cases} \dfrac{\partial}{\partial x}\left[k(x)\dfrac{\partial u(x,t)}{\partial x}\right] - \dfrac{\partial^2 u(x,t)}{\partial t^2} = f(x,t), \\ u(x,0) = 5\sin\dfrac{\pi x}{2}, \quad \dfrac{\partial}{\partial t}u(x,0) = 0, \\ u(0,t) = 0, \quad \dfrac{\partial}{\partial x}u(1,t) = 0, \\ u(1,t) = 5 + t^2 \ \text{(additional condition)} \end{cases} \qquad 0 \le x \le 1, \ \ 0 \le t \le 1$$

where

$$f(x,t) = e^{5+2x}\pi(5+t^2)\cos\frac{\pi x}{2} - \frac{1}{4}\left(8 + e^{5+2x}\pi^2(5+t^2)\right)\sin\frac{\pi x}{2}.$$

The true coefficient function is $k(x) = e^{2x+5}$ and the true solution is $u(x,t) = \sin\frac{\pi}{2}x(t^2+2)$. By using Mathematica 4.2 and the method of Section 5, we calculate the approximate coefficient of the inverse problem $\widetilde{k}(x)$ and the approximate solution (7.1.5) $\widetilde{u}(x,t)$. The numerical results are given in Table 7.1, and left side of Figure 7.1, respectively.

Table 7.1: The error of coefficient $k(x)$ of Example 7.1.1

x	$k(x)$	$\widetilde{k}(x)$	relative error	x	$k(x)$	$\widetilde{k}(x)$	relative error
$\frac{1}{16}$	168.174	167.708	0.002278158	$\frac{2}{16}$	190.566	190.125	0.002322
$\frac{3}{16}$	215.94	215.534	0.00188237	$\frac{4}{16}$	244.692	244.329	0.00148641
$\frac{5}{16}$	277.272	276.955	0.00114515	$\frac{6}{16}$	314.191	313.92	0.000860991
$\frac{7}{16}$	356.025	355.8	0.00063093	$\frac{8}{16}$	403.429	403.248	0.000449094
$\frac{9}{16}$	457.145	457.004	0.000308648	$\frac{10}{16}$	518.013	517.908	0.000202875
$\frac{11}{16}$	586.985	586.912	0.000125643	$\frac{12}{16}$	665.142	665.094	0.0000715061
$\frac{13}{16}$	753.704	753.677	0.0000356879	$\frac{14}{16}$	854.059	854.047	0.0000140617
$\frac{15}{16}$	967.775	967.772	3.1378236E-6				

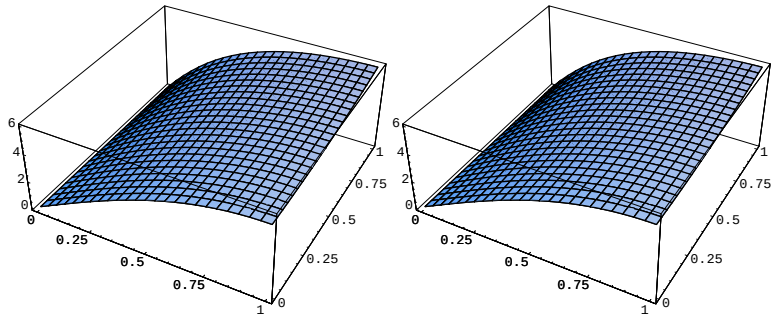


Figure 7.1: Left side: The above image is $\widetilde{u}(x,t)$ and the below image is $u(x,t)$ in Example 7.1.1. Right side: The above image is $\widetilde{u}(x,t)$ and the below image is $u(x,t)$ in Example 7.1 2.

In order to demonstrate the stability of our algorithm, we give a small perturbation ε to the right side function $f(x,t)$ in Example 7.1.2. Results obtained by the algorithm indicate the algorithm is stable.

Example 7.1.2. We shall give a perturbation $\varepsilon = 10^{-4}$ to the right side function $f(x,t)$ in Example 1, that is $\widetilde{f}(x,t) = f(x,t) + \varepsilon$. The numerical results are given in Table 7.2, and right side of Figure 7.1, respectively.

7.2 A Determination of an Unknown Parameter in Parabolic Equations

7.2.1 Introduction

Many physical phenomena can be described in terms of parabolic partial differential equations with source control parameters. These type of equations appear for example, in the study of heat condition processes, chemical diffusion, thermoelasticity and control theory [183]–[188]. In general, these problems are ill-posed. Therefore,

Table 7.2: The error of coefficient $k(x)$ of Example 7.1.2

x	$k(x)$	$\widetilde{k}(x)$	relative error	x	$k(x)$	$\widetilde{k}(x)$	relative error
$\frac{1}{16}$	168.174	167.708	0.002278165	$\frac{2}{16}$	190.566	190.125	0.00232206
$\frac{3}{16}$	215.94	215.534	0.00188242	$\frac{4}{16}$	244.692	244.329	0.00148645
$\frac{5}{16}$	277.272	276.955	0.00114518	$\frac{6}{16}$	314.191	313.92	0.000861022
$\frac{7}{16}$	356.025	355.8	0.000630956	$\frac{8}{16}$	403.429	403.248	0.000449117
$\frac{9}{16}$	457.145	457.004	0.000308667	$\frac{10}{16}$	518.013	517.908	0.000202891
$\frac{11}{16}$	586.985	586.912	0.000125657	$\frac{12}{16}$	665.142	665.094	0.0000715186
$\frac{13}{16}$	753.704	753.677	0.0000356988	$\frac{14}{16}$	854.059	854.047	0.0000140713
$\frac{15}{16}$	967.775	967.772	3.14621E-6				

a variety of numerical techniques based on regularization, finite difference, finite element and finite volume methods are given to approximate solutions of the equations [189], [190].

In this section, we present a new algorithm for the following inverse source problem for a parabolic equation

$$\begin{cases} w_t = w_{xx} + p(t)w(t,x) + f(t,x), \\ w(0,x) = \varphi(x), \\ w(t,0) = w(t,1) = 0, \\ w(t,x^*) = E(t), \ \ 0 \le x \le 1, \ \ 0 \le t \le T. \end{cases} \tag{7.2.1}$$

where $f(t,x)$, $\varphi(x)$ and $E(t)$ are sufficiently regular given functions, and x^* is a fixed prescribed interior point in $(0,1)$, while the functions $w(t,x)$ and $p(t)$ are unknown. If $w(t,x)$ is a temperature, then $Eq.(1.1)$ can be regarded as a control problem finding the control function $p(t)$ such that the internal constraint is satisfied. The existence and uniqueness of the equations have been proved [191].

Employing a pair of transformations

$$r(t) = \exp\left(-\int_0^t p(s)ds\right), \tag{7.2.2}$$

$$u(t,x) = r(t)w(t,x) - \varphi(x), \tag{7.2.3}$$

(7.2.1) will be

$$\begin{cases} u_t = u_{xx} + r(t)f(t,x) + \varphi''(x), \\ u(0,x) = 0, \\ u(t,0) = u(t,1) = 0, \\ u(t,x^*) = E(t)r(t) - \varphi(x^*), \ \ 0 \le x \le 1, \ \ 0 \le t \le T. \end{cases} \tag{7.2.4}$$

7.2.2 The Exact Solution of Eq. (7.2.4)

Let $\mathcal{D} = [0,T] \times [0,1]$, $T > 0$, the reproducing kernel space $W_2^{(1,1)}(\mathcal{D})$ is defined by (1.5.1). $Q_{(\eta,\xi)}(t,x)$ is its reproducing kernel given by (1.5.7). The reproducing kernel space

$$W_2^{(2,3)}(\mathcal{D}) = \Big\{ u(t,x) |\ u \in {}^{o}W_2^{(2,3)}(\mathcal{D}) \text{ by (1.5.1)},\ u(0,x) = u(t,0) = u(t,1) = 0 \Big\}.$$

The reproducing kernel function is

$$K_{(\eta,\xi)}(t,x) = G_\eta(t) R_\xi(x),$$

where $G_\eta(t)$, $R_\xi(x)$ by (5.1.5), (5.1.6), respectively.

If $r(t)$ is known, then we can give the exact solution of Eq. (7.2.4). Let a operator $\mathbb{L} : {}^{o}W_2^{(2,3)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ by

$$\mathbb{L}u = u_t - u_{xx},$$

it is clear that $\mathbb{L} : {}^{o}W_2^{(2,3)}(\mathcal{D}) \longrightarrow W_2^{(1,1)}(\mathcal{D})$ is bounded linear operator. Put $\mathbf{q} = (t,x), \mathbf{q}_i = (t_i, x_i)$,

$$\varphi_i(\mathbf{q}) = Q_{\mathbf{q}_i}(\mathbf{q}) \ \text{ and } \ \psi_i(\mathbf{q}) = \mathbb{L}^* \varphi_i(\mathbf{q}),$$

$\mathbb{L}^*$ is the conjugate operator of $\mathbb{L}$.

Theorem 7.2.1. *For If $\{\mathbf{q}_i\}_{i=1}^{\infty}$ is dense on $\mathcal{D}$, then $\{\psi_i(\mathbf{q})\}_{i=1}^{\infty}$ is a complete system of ${}^{o}W_2^{(2,3)}(\mathcal{D})$.*

The proof is omitted.

$\{\overline{\psi}_i(\mathbf{q})\}_{i=1}^{\infty}$ derives from Gram–Schmidt orthonormalization of $\{\psi_i(\mathbf{q})\}_{i=1}^{\infty}$,

$$\overline{\psi}_i(\mathbf{q}) = \sum_{k=1}^{i} \beta_{ik} \psi_k(\mathbf{q}) \ \ (\beta_{ii} > 0, \ \ i = 1, 2, \ldots).$$

Theorem 7.2.2. *If $\{\mathbf{q}_i\}_{i=1}^{\infty}$ is dense on $\mathcal{D}$ and the solution of Eq.* (7.2.4) *is unique, then the solution of Eq.* (7.2.4) *satisfies the form*

$$u(\mathbf{q}) = \sum_{i=1}^{\infty} \sum_{k=1}^{i} \beta_{ik} F(\mathbf{q}_k, r(t_k)) \overline{\psi}_i(\mathbf{q}), \tag{7.2.5}$$

where

$$F(\mathbf{q}, r(t)) = r(t) f(t,x) + \varphi''(x), \ \ \mathbf{q} = (t,x).$$

Proof. By Theorem 7.2.1, it is clear that $\{\overline{\psi}_i(\mathbf{q})\}_{i=1}^{\infty}$ is the normal orthogonal basis of ${}^oW_2^{(2,3)}(\mathcal{D})$. Then

$$\begin{aligned} u(\mathbf{q}) &= \sum_{i=1}^{\infty} \big\langle u(\mathbf{q}), \overline{\psi}_i(\mathbf{q}) \big\rangle_{{}_oW_2^{(2,3)}} \overline{\psi}_i(\mathbf{q}) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} \big\langle u(\mathbf{q}), \mathbb{L}^*\varphi_k(\mathbf{q}) \big\rangle_{{}_oW_2^{(2,3)}} \overline{\psi}_i(\mathbf{q}) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} \big\langle F(\mathbf{q}, r(t)), \varphi_k(\mathbf{q}) \big\rangle_{W_2^{(1,1)}} \overline{\psi}_i(\mathbf{q}) \\ &= \sum_{i=1}^{\infty}\sum_{k=1}^{i} \beta_{ik} F(\mathbf{q}_k, r(t_k)) \overline{\psi}_i(\mathbf{q}), \end{aligned}$$

which proves the theorem. □

7.2.3 An Iteration Procedure

Next, a new algorithm of obtaining the solution (7.2.5) is presented. (7.2.5) can be denoted by

$$u(\mathbf{q}) = \sum_{i=1}^{\infty} A_i \overline{\psi}_i(\mathbf{q}),$$

where

$$A_i = \sum_{k=1}^{i} \beta_{ik} F(\mathbf{q}_k, r(t_k)).$$

Let $\mathbf{q}_1 = (0, 0)$, it follows that $r(0) = 1$, then $F(\mathbf{q}_1, r(t_1))$ is known. Considering the numerical computation, we put $r_0(t_1) = r(t_1)$. The iteration algorithm is performed through the following two equations

$$u_n(\mathbf{q}) = \sum_{i=1}^{n}\sum_{k=1}^{i} \beta_{ik} F(\mathbf{q}_k, r_{k-1}(t_k)) \overline{\psi}_i(\mathbf{q}), \tag{7.2.6}$$

$$r_n(t) = \frac{u_n(t, x^*) + \varphi(x^*)}{E(t)}. \tag{7.2.7}$$

Let

$$B_i = \sum_{k=1}^{i} \beta_{ik} F(\mathbf{q}_k, r_{k-1}(t_k)), \tag{7.2.8}$$

then

$$u_n(\mathbf{q}) = \sum_{i=1}^{n} B_i \overline{\psi}_i(\mathbf{q}). \tag{7.2.9}$$

Due to (7.2.7), the convergence of $u_n(\mathbf{q})$ can lead to that of $r_n(t)$. So we only need to show the convergence of $u_n(\mathbf{q})$. Now, two lemmas are needed.

Lemma 7.2.1. *If $r(t) \in W_2^{(1,1)}(\mathcal{D})$, then there exists a $M > 0$, such that $|r(t)| \leq M\|r\|_{W_2^{(1,1)}}$.*

It is easy to obtain from the definition of reproducing kernel and Schwarz inequality.

Lemma 7.2.2. *If $r_n(t) \xrightarrow{\|\cdot\|} \overline{r}(t)$ $(n \longrightarrow \infty)$, $\mathbf{q}_n = (t_n, x_n) \longrightarrow \mathbf{q} = (t, x)$ $(n \longrightarrow \infty)$, $\|r_n\|$ is bounded and $F(\mathbf{q}, r(t))$ is continuous in $\mathbf{q}$, then $F(\mathbf{q}_n, r_{n-1}(t_n)) \longrightarrow F(\mathbf{q}, \overline{r}(t))$ as $n \longrightarrow \infty$.*

Proof. From the given condition $r_n(t) \longrightarrow \overline{r}(t)$ $(n \longrightarrow \infty)$, and Lemma 7.2.1, it follows that, for any $\mathbf{q} \in \mathcal{D}$, $r_n(t)$ converges uniformly to $\overline{r}(t)$. So we can easily prove the conclusion of Lemma 7.2.2. □

Theorem 7.2.3. *Suppose that $\|u_n\|$ is bounded in* (7.2.6)*, if $\{\mathbf{q}_i\}_{i=1}^{\infty}$ is dense in $\mathcal{D}$, then the n-term approximate solution $u_n(\mathbf{q})$ derived from the above method converges to the exact solution $u(\mathbf{q})$ of Eq.* (7.2.4) *and $u(\mathbf{q}) = \sum\limits_{i=1}^{\infty} B_i \overline{\psi}_i(\mathbf{q})$, where B_i is given by* (7.2.8)*.*

Proof. (1) First, we will prove the convergence of $u_n(\mathbf{q})$. By (7.2.9), we infer that

$$u_{n+1}(\mathbf{q}) = u_n(\mathbf{q}) + B_{n+1}\overline{\psi}_{n+1}(\mathbf{q}).$$

From the orthonormality of $\{\overline{\psi}_i(\mathbf{q})\}_{i=1}^{\infty}$ and (7.2.6), it follows that

$$\|u_{n+1}\|^2_{{}_oW_2^{(2,3)}} = \|u_n\|^2_{{}_oW_2^{(2,3)}} + B_{n+1}^2. \tag{7.2.10}$$

From (7.2.10), it holds that $\|u_{n+1}\|_{{}_oW_2^{(2,3)}} \geq \|u_n\|_{{}_oW_2^{(2,3)}}$. Due to the condition that $\|u_n\|_{{}_oW_2^{(2,3)}}$ is bounded, $\|u_n\|_{{}_oW_2^{(2,3)}}$ is convergent and there exists a constant c such that

$$\sum_{i=1}^{\infty} B_i^2 = c.$$

This implies that $B_i \in \ell^2$, $i = 1, 2, \ldots$. If $m > n$, then

$$\|u_m - u_n\|^2_{{}_oW_2^{(2,3)}} = \big\|u_m - u_{m-1} + u_{m-1} - u_{m-2} + \cdots + u_{n+1} - u_n\big\|^2_{{}_oW_2^{(2,3)}}.$$

From $(u_m - u_{m-1}) \perp (u_{m-1} - u_{m-2}) \perp \cdots \perp (u_{n+1} - u_n)$, it follows that

$$\|u_m - u_n\|^2_{{}_oW_2^{(2,3)}} = \|u_m - u_{m-1}\|^2_{{}_oW_2^{(2,3)}} + \cdots + \|u_{n+1} - u_n\|^2_{{}_oW_2^{(2,3)}}.$$

Furthermore, $\|u_m - u_{m-1}\|^2_{{}_oW_2^{(2,3)}} = B_m^2$. Consequently,

$$\|u_m - u_n\|^2_{{}_oW_2^{(2,3)}} = \sum_{i=n+1}^{m} B_i^2 \longrightarrow 0 \ \ (n \longrightarrow \infty).$$

For(7.2.6), let $n \longrightarrow \infty$, then

$$\overline{u}(\mathbf{q}) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(\mathbf{q}).$$

(2) Secondly, we will prove $\overline{u}(\mathbf{q})$ is the solution of (7.2.4). Due to the orthogonality of $\{\overline{\psi}_i(\mathbf{q})\}_{i=1}^{\infty}$,

$$\big\langle \overline{u}(\mathbf{q}), \overline{\psi}_i(\mathbf{q}) \big\rangle_{{}_oW_2^{(2,3)}} = B_i.$$

So

$$\begin{aligned}
\sum_{k=1}^{i} \beta_{ik} (\mathbb{L}\overline{u})(\mathbf{q}_k) &= \sum_{k=1}^{i} \beta_{ik} \big\langle \overline{u}(\mathbf{q}), (\mathbb{L}^* \varphi_k)(\mathbf{q}) \big\rangle_{{}_oW_2^{(2,3)}} \\
&= \left\langle \overline{u}(\mathbf{q}), \sum_{k=1}^{i} \beta_{ik} \psi_k(\mathbf{q}) \right\rangle_{{}_oW_2^{(2,3)}} \\
&= \big\langle \overline{u}(\mathbf{q}), \overline{\psi}_i(\mathbf{q}) \big\rangle_{{}_oW_2^{(2,3)}} = B_i.
\end{aligned}$$

If $i = 1$, then

$$(\mathbb{L}\overline{u})\mathbf{q}_1 = F(\mathbf{q}_1, r_0(t_1)).$$

If $i = 2$, then

$$\beta_{21}(\mathbb{L}\overline{u}(\mathbf{q}_1) + \beta_{22}(\mathbb{L}\overline{u}(\mathbf{q}_2) = \beta_{21} F(\mathbf{q}_1, r_0(t_1)) + \beta_{21} F(\mathbf{q}_2, r_1(t_2)).$$

It is clear that

$$(\mathbb{L}\overline{u}(\mathbf{q}_2) = F(\mathbf{q}_2, r_1(t_2)).$$

In the same way,

$$(\mathbb{L}\overline{u}(\mathbf{q}_j) = F(\mathbf{q}_j, r_{j-1}(t_j)), \ \ j = 1, 2, \ldots . \tag{7.2.11}$$

Since $\{\mathbf{q}_i\}_{i=1}^{\infty}$ is dense in $\mathcal{D}$, for $\forall\, \mathbf{s} = (t, x) \in \mathcal{D}$, there exists a subsequence $\{\mathbf{q}_{nj}\}_{j=1}^{\infty}$ such that

$$\mathbf{q}_{nj} \longrightarrow \mathbf{s} = (t, x) \ \ (j \longrightarrow \infty).$$

For (7.2.11), let $j \longrightarrow \infty$, by the convergence of u_n and Lemma 7.2.2, we have

$$(\mathbb{L}\overline{u})(\mathbf{s}) = F(\mathbf{s}, r(t)). \tag{7.2.12}$$

That is, $\overline{u}(\mathbf{q})$ is the solution of $Eq.(1.4)$ and

$$u(\mathbf{q}) = \sum_{i=1}^{\infty} B_i \overline{\psi}_i(\mathbf{q}), \tag{7.2.13}$$

then the proof is complete.

□

From the proof of Theorem (7.2.6) and Eq. (7.2.12), we can see that for arbitrary initial value, the iteration sequence is convergent and the limiting function is the solution of Eq. (7.2.6). Furthermore, this immediately leads to the following conclusion.

Corollary 7.2.1. *The n-term approximate solution $u_n(\mathbf{q})$ in* (7.2.9) *can converge to exact solution in the large.*

Theorem 7.2.4. *Assume $u(\mathbf{q})$ is the solution of Eq.* (7.2.4) *and $\gamma_n(\mathbf{q})$ is the approximate error of $u_n(\mathbf{q})$, where $u_n(\mathbf{q})$ is given by* (7.2.6)*, then the error $\gamma_n(\mathbf{q})$ is monotone decreasing in the sense of $\|\cdot\|_{{}_oW_2^{(2,3)}}$.*

Proof. From (7.2.9),(7.2.13), it follows that

$$\|\gamma_n\|^2_{{}_oW_2^{(2,3)}} = \left\| \sum_{i=n+1}^{\infty} B_i \overline{\psi}_i(x) \right\|^2_{{}_oW_2^{(2,3)}} = \sum_{i=n+1}^{\infty} B_i^2,$$

which shows that the error γ_n is monotone decreasing in the sense of $\|\cdot\|_{{}_oW_2^{(2,3)}}$. □

7.2.4 Numerical Experiments

To illustrate the description above and to test iteration algorithm developed in this article for solving the one-dimensional diffusion with a control function, we include a numerical example for which the exact solution is known.

Example 7.2.1. Consider Eq. (7.2.1) with

$$\varphi(x) = \sin(\pi x), \quad f(t,x) = e^t(\pi^2 - t^2)\sin(\pi x), \quad E(t) = e^t \sin(\pi x^*),$$

for which the true solution is

$$w(t,x) = e^t \sin(\pi x),$$

and

$$p(t) = 1 + t^2.$$

In the process of computation, all the symbolic and numerical computations performed by using Mathematica 5.0. Taking $T = 1.0$, $x^* = 0.4$ and choosing 81 and 144 points on $\mathcal{D} = [0, T] \times [0, 1]$, respectively, then we obtain the approximate solution u_{81} and u_{144}on $\mathcal{D}$.The numerical results are given in Table 7.3–7.4 and 7.5–7.6. Results obtained by the method have been compared with the exact solution of each example and are found to be in good agreement with each other.

Table 7.3: $n = 81$, The error of $u(t, x)$

(t,x)	True solution $u(t, x)$	Approximate solution u_{81}	Absolute error	Relative error
(0.06,0.06)	0.198968	0.198968	5.57763E-07	2.80328E-06
(0.12,0.12)	0.415059	0.415057	2.43845E-06	5.87495E-06
(0.18,0.18)	0.641501	0.641495	6.4493E-06	1.00535E-05
(0.24,0.24)	0.87023	0.870219	1.07722E-05	1.23786E-05
(0.30,0.30)	1.09206	1.09205	1.32012E-05	1.20883E-05
(0.36.0.36)	1.29692	1.29091	8.76693E-06	6.75983E-06
(0.42,0.42)	1.47415	1.47415	2.37183E-06	1.60895E-06
(0.48,0.48)	1.61289	1.61292	3.44639E-05	2.13678E-05
(0.54,0.54)	1.70248	1.70254	6.76741E-05	3.97504E-05
(0.60,0.60)	1.73294	1.73308	1.41218E-04	8.14904E-05
(0.66,0.66)	1.69547	1.69566	1.90033E-04	1.12083E-04
(0.72.0.72)	1.58297	1.58326	2.93295E-04	1.85281E-04
(0.78,0.78)	1.39052	1.39085	3.23009E-04	2.32293E-04
(0.84,0.84)	1.11592	1.11627	3.49175E-04	3.12904E-04
(0.90,0.90)	0.760059	0.760378	3.19067E-04	4.19792E-04

Table 7.4: $n = 81$, The error of $p(t)$

t	True solution $p(t)$	Approximate solution p_{81}	Absolute error	Relative error
0.06	1.0036	1.00321	3.86715E-04	3.85328E-04
0.12	1.0144	1.01284	1.56012E-03	1.53798E-03
0.18	1.0324	1.03006	2.34023E-03	2.26679E-03
0.24	1.0576	1.05373	3.87024E-03	3.65946E-03
0.30	1.09	1.08429	5.7076E-03	5.23633E-03
0.36	1.1296	1.12177	7.82698E-03	6.92898E-03
0.42	1.1764	1.16501	1.13855E-02	9.67827E-03
0.48	1.2304	1.21605	1.43484E-02	1.16616E-02
0.54	1.2916	1.27135	2.02475E-02	1.56763E-02
0.60	1.36	1.33567	2.4331E-02	1.78905E-02
0.66	1.4356	1.40241	3.3189E-02	2.31186E-02
0.72	1.5184	1.47951	3.88941E-02	2.56152E-02
0.78	1.6084	1.55938	4.90215E-02	3.04784E-02
0.84	1.7056	1.64771	5.78872E-02	3.39395E-02
0.90	1.81	1.73684	7.31555E-02	4.04174E-02

Table 7.5: $n = 144$, The error of $u(t, x)$

(t, x)	True solution $u(t, x)$	Approximate solution u_{144}	Absolute error	Relative error
(0.06,0.06)	0.198968	0.198968	2.88968E-07	1.45233E-06
(0.12,0.12)	0.415059	0.415058	1.0469E-06	2.52229E-06
(0.24,0.24)	0.87023	0.870226	4.34356E-06	4.99128E-06
(0.30,0.30)	1.09206	1.09205	4.64225E-06	4.25091E-06
(0.36.0.36)	1.29692	1.29691	1.85604E-06	1.43112E-06
(0.42,0.42)	1.47415	1.47415	3.63194E-06	2.46376E-06
(0.48,0.48)	1.61289	1.6129	1.30126E-05	8.0679E-06
(0.54,0.54)	1.70248	1.70251	3.14264E-05	1.84592E-05
(0.60,0.60)	1.73294	1.73299	5.45239E-05	3.14633E-05
(0.66,0.66)	1.69547	1.69554	7.32428E-05	4.31991E-05
(0.72.0.72)	1.58297	1.58305	8.44003E-05	5.33178E-05
(0.78,0.78)	1.39052	1.39061	9.10671E-05	6.54913E-05
(0.84,0.84)	1.11592	1.11597	5.27523E-05	4.72726E-05
(0.90,0.90)	0.760059	0.760046	1.339E-05	1.76171E-05

Table 7.6: $n = 144$, The error of $p(t)$

t	True solution $p(t)$	Approximate solution p_{144}	Absolute error	Relative error
0.06	1.0036	1.00329	3.07732E-04	3.06629E-04
0.12	1.0144	1.01372	6.83165E-04	6.73467E-04
0.18	1.0324	1.03107	1.32526E-03	1.28367E-03
0.24	1.0576	1.05506	2.53737E-03	2.39918E-03
0.30	1.09	1.0868	3.19584E-03	2.93197E-03
0.36	1.1296	1.12532	4.27506E-03	3.78458E-03
0.42	1.1764	1.17032	6.08278E-03	5.17067E-03
0.48	1.2304	1.22143	8.96667E-03	7.28761E-03
0.54	1.2916	1.28076	1.0843E-02	8.39503E-03
0.60	1.36	1.34643	1.35727E-02	9.97992E-03
0.66	1.4356	1.41814	1.74614E-02	1.21631E-02
0.72	1.5184	1.49537	2.3028E-02	1.51659E-02
0.78	1.6084	1.58134	2.70618E-02	1.68253E-02
0.84	1.7056	1.6743	3.13025E-02	1.83528E-02
0.90	1.81	1.76897	4.10324E-02	2.26699E-02

Bibliography

[1] Zaremba S. (1908), Sur le calcul numerique des founctions demandess dans le problems de dirichlet et le problems hydrodynamique. *Bulletin International de I Academie des Sciences de Cracovie*, 125–195.

[2] Bergman S. (1922) Uber die entwicklung der harmonischen funcktionen der Ebene unddes raumes nach orthogonal funktionen. *Math. Ann.* 86: 238–271.

[3] Bergman S. (1930) Über Kurvenintegale von Funktionen zweier komplexen Veranderlichen, die Differentialgleichungen befriedigen. *Math.Z.* 32: 386–406.

[4] Bergman S. (1936) Über ein verfahren zur konstruktion der naherungslÖsungslÖsungen der gleichung $\Delta u + \tau^2 u = 0$. *Prikl. Mat. Meh.*, 97–107.

[5] Bergman S. (1937) Zur theorie der funktion, die eine linear partielle differentialgleichung berriedigen. *Mat. Sb.* 44: 1169–1198.

[6] Bergman S. (1937) Zur theorie der funktion,die eine linear partielle differentialgleichung. berriedigen. *Soviet. Math. Dokl.* 15: 227–230.

[7] Bergman S. (1937) Sur un lien entre lal théorie des equations aux derives partielles elliptiques et celle des functions d'une variable complexe. *C.R. Acad. Sci., Paris* 205: 1198–1200.

[8] Bergman S. (1940) The approximation of functions satisfying a linear partial differential equation. *Duke Math.J.* 6: 537–561.

[9] Mercer J. (1909) Function of positive and negative type and their connection with the theory of integral equation. *Philos. Trans.Roy. Soe.London Ser. A* 209: 415–446.

die Differentialgleichungen befriedigen,*Math.Z.*,32(1930)386-406.

[10] Aronszajn N. (1950) Theory of reproducing kernels. *Trans. Amer. Math. Soc.* 68: 337–404.

[11] Brylinski R. (1998) Geometric quantization of real minimal nilpotent orbits. *Differential Geometry and its Applications* 9(1-2): 5–58.

[12] Wilson R. (2001) Composition series for analytic continuations of holomrphic discrete series representations of SUp,q. *Differential Geometry and its Applications* 15(3): 221–252.

[13] Alpay D., Bolotnikov V., Kaptanğlu H.T. (2002) The Schur algorithm and reproducing kernel Hilbert spaces in the ball. *Linear Algebra and its Applications* 342(1–3): 163–186.

[14] Larkin F.M. (1992) Optimal approximation in Hilbert space with reproducing kernel function. *Math.Comp.* 22(4): 911–921.

[15] Veenakaul M.M.Ch. (1974) Optimal rules with polynomial precesion for Hilbert space possessing reproducing kernel function. *Numer Math* 22: 207–218.

[16] Alpay D., Dym H. (1996), On a new class of realization formulas and their application. *Linear Algebra and its Applications* 241-243: 3–84.

[17] Hult H. (2003) Approximating some Volterra type stochastic integrals with applications to parameter estimation. *Stochastic Processes and their Applications* 105(1): 1–32.

[18] Chong Gu (1998) Penalized likelihood estimation: convergence under incorrect model. *Statistics and Probability Letters* 36(4): 359–364.

[19] Park B.U., Kim W.C., Jones M.C. (1997) On identity reproducing nonparametric regression estimators. *Statistics and Probability Letters* 32(3): 279–290.

[20] Strauss D.J., Steidl G. (2002) Hybrid wavelet-support vector classification of waveforms. *Journal of Computational and Applied Mathematics* 148(2): 375–400.

[21] Cui M., Deng M. (1986) On the best operator of interpolation. *Math. Numerica Sinica (Chinese)* 8(2): 209–216.

[22] Lin Y., Cui M., Zheng Yi (2005),Representation of the exact solution for infinite system of linear equation. *Applied Mathematics and Computation* 168: 636–650.

[23] Li Ch., Cui M. (2002) How to solve the equation $AuBu + Cu = f$. *Applied Mathematics and Computation* 133: 643–653.

[24] Li Ch., Cui M. (2003) The exact solution for a class nonlinear operator equations in the reproducing kernel space. *Applied Mathematics and Computation* 143: 393–399.

[25] Geng H., Cui M. (2007) Solving singular nonlinear second-order periodic boundary value problems in the reproducing kernel space. *Applied Mathematics and Computation* 192: 389–398.

[26] Cui M., Geng F. (2007) Solving singular two-point boundary value problem in reproducing kernel space. *Journal of Computational and Applied Mathematics* 205: 6–15.

[27] Yao H., Cui M. (2007) A new algorithm for a class of singular boundary value problems. *Applied Mathematics and Computation* 186: 1183–1191.

[28] Yang L., Cui M., New algorithm for a class of nonlinear intergro-differential equations in the reproducing kernel space. *Applied Mathematics and Computation* 174: 942–960.

[29] Li Y., Geng F., Cui M. (2007) The analytical solution of singular linear periodic boundary value problem. *Applied Mathematical Science* 1(2): 77–87.

[30] Cui M., Lin Y. (2007) A new method of solving the coefficient inverse problem of differential equation. *Science in China Series A–Mathematics* 50(4): 561–572.

[31] Smirnov V.E. *High Mathematics Tutorial*. G.E.T.B., 1960.

[32] Cui M., Wu B. (2004) Numerical Analysis in Reproducing Kernel Space. *Science Press (China), Beijing*, 14–27 (in Chinese).

[33] Mohanty R.K., Jha N. (2005) A Class of variable mesh spline in compression methods for singularly perturbed two point singular boundary value problems. *Appl. Math. Comput.* 168: 704–716.

[34] Mohanty R.K., Arora U. (2006) A family of non-uniform mesh tension spline methods for singularly perturbed two point singular boundary value problems with significant first derivatives. *Appl. Math. Comput.* 172: 531–544.

[35] Ling L., Trummer M.R. (2006) Adaptive multiquadric collocation for boundary layer problems. *Journal of Comutational and Applied Mathematics.* 188: 265–282.

[36] Wang X., Jiang Y. (2005) A general methods for solving singular perturbed impulsive differential equations with two-poing boundary conditions. *Appl. Math. Comput.* 171: 775–806.

[37] Zhang X., Liu L. (2006) Positive solution of superlinear semipositone singular Dirichlet boundary value problems. *J. Math. Anal.* 316: 525–537.

[38] Yang H. (2006) On a singular perturbation problem with two second–order turning points. *Journal of Computational and Applied Mathematics.* 190: 287–303.

[39] Wong R., Yang H. (2002) On an internal boundary layer problem. *J. Comp. Math.* 144: 301–323.

[40] Wong R., Yang H. (2002) On a boundary-layer problem. *Studies in Appl. Math.* 108: 369–398.

[41] Wong R., Yang H. (2003) On the Ackerberg-O'Malley resonance. *Studies in Appl. Math.* 110.

[42] Cui M., Deng Zh. (1986) Solutions to the definite solution problem of differential equations in space $W_2^l[0,1]$. *Advances in Mathematics* 17(3): 327–328.

[43] Noor M.A. (1994) Variational inequalities in physical oceanography. In: M. Rahman (Ed.), *Ocean Waves Engineering, Computational Mechanics Publications*, England: Southampton, 201–226.

[44] Noor M.A. (1997) Some recent advances in variational inequalities part I, basic concepts. *New Zealand Journal of Mathematics* 26: 53–80.

[45] Lewy H., Stampacchia G. (1969) On the regularity of the solutions of the variational inequalities. *Communication in Pure and Applied Mathematics* 22: 153–188.

[46] Tuck E.O., Schwartz L.W. (1990) A Numerical and asymptotic study of some third-order ordinary differential equations relevant to draining and coating flows. *Siam Review* 32: 453–469.

[47] Al–Said E.A. (2001) Numerical solutions of third–order boundary value problems. *Int. J. Comput. Math.* 78: 111–122.

[48] Noor M.A., Khalifa A.K. (1994) A numerical approach for odd-order obstacle problems. *Int. J. Comput. Math.* 54: 109–116.

[49] Noor M.A., Al-Said E.E. (2004) Quartic splines solutions of third-order obstacle problems Muhammad. *AMC* 153: 307–316.

[50] Al-Said E.A., Noor M.A., Rassias Th.M. (1998) Numerical solutions of third-order obstacle problems. *Int. J. Comput. Math.* 69: 75–84.

[51] Noor M.A., Al-Said E.A. (2002) Finite difference method for system of third-order boundary-value problems. *J. Optimiz. Theory Appl.* 122: 627–637.

[52] Siraj-ul-Islam, Khan M.A., Tirmizi I.A., Twizell E.H. (2005) Non polynomial spline approach to the solution of a system of third-order boundary-value problems. *Appl. Math. Comput.* 168: 152–163.

[53] Al-Said E.A., Noor M.A. (2003) Cubic splines method for a system of third-order boundary-value problems. *Appl. Math. Comput.* 142: 195–204.

[54] Al-Said E.A., Noor M.A., Khalifa A.K. (1996) Finite-difference scheme for variational inequalities. *J. Optim. Theory. Appl.* 89: 453–459.

[55] Gao F., Chi Ch. (2006) Solving third-order obstacle problems with quartic B-splines. *Appl. Math. Comput.*180: 270–274.

[56] Roos H.G., Stynes M., Tobiska L. *Numerical methods for singularly perturbed differential equations*. Springer–Verlag, 1996.

[57] Howes F.A. (1976) Singular perturbations and differential inequalities. *Memoirs of the American Mathematical Society, Providence*, Rhode Island, 168.

[58] Ascher V., Weiss R. (1984) Collocation for singular-perturbation problems, III: Nonlinear problems without turning points. *SIAM Journal on Scientific and Statistical Computing* 5: 811–829.

[59] Kadalbajoo M., Aggarwal V.K. (2005) Fitted mesh B-spline collocation method for solving self-adjoint singularly perturbed boundary value problems. *Applied Mathematics and Computation* 161: 973–987.

[60] Roos H.G. (1986) A second-order monotone upwind scheme. *Computing* 36: 57–67.

[61] Ilicasu F.O., Schultz D.H. (2004) High-Order finite-difference techniques for linear singular perturbation boundary value problems. *Computers and Mathematics with Applications* 47: 391–417.

[62] Stynes M., O'Riordan E. (1986) A uniformly accurate finite-element method for a singular-perturbation problem in conservative form, *SIAM Journal on Numerical Analysis* 23: 369–375.

[63] Vigo-Aguiar J., Natesan S. (2004) A parallel boundary value techniques for singualrly perturbed two-point boundary value problems. *The Journal of Supercomputing* 27: 195–206.

[64] Reddy Y.N., Chakravarthy P.P. (2004) An initial-value approach for singularly perturbed two-point boundary value problems. *Applied Mathematics and Computation* 155: 95–110.

[65] Valanarasu T., Ramanujam N. (2003) Asymptotic initial-value method for singularly-perturbed boundary-value problems for second-order ordinary differential equations. *Journal of Optimization Theory and Applications* 116: 167–182.

[66] Bawa R.K. (2005) Spline based computational technique for linear singularly perturbed boundary value problems. *Applied Mathematics and Computation* 167: 225–236.

[67] Aziz T., Khan A. (2002) A spline method for second-order singularly perturbed boundary-value problems. *Journal of Computational and Applied Mathematics* 147: 445–452.

[68] Howes F.A. (1982) Differential inequalities of higher order and the asymptotic solution of nonlinear boundary value problems. *SIAM J.math.Anal.* 13(1): 61–80.

[69] Howes F.A. (1983) The asympototic solution of a class of third-order boundary problem arising in the theory of thin film flow. *SIAM J. Appl. Math.* 43(5): 993–1004.

[70] Weili Z. (1990) Singular perturbations of boundary value problems for a class of third order nonlinear ordinary differential equations. *J. Differential Equations* 88(2): 265–278.

[71] Roberts S.M. (1988) Further examples of the boundary value technique in singular perturbation problems. *j. Math. Anal. Appl.* 133: 411–436.

[72] Valarmathi S., Ramanujam N. (2002) A computational method for solving boundary value problems for third-order singualrly perturbed ordinary differential equations. *Applied Mathematics and Computation* 129: 345–373.

[73] Nayfeh A.H. *Introduction to perturbation method*. Wiley: New York, 1981.

[74] Natesan S., Ramanujam N. (1998) A computational method for solving singularly perturbed turing point problems exhibiting twin boundary layer. *Appl. Math. Comput.* 93: 259–275.

[75] Abrahamsson L.R. (1977) A priori estimates for solutions of singular perturbation with a turning point. *Stud. Appl. Math.* 56: 51–69.

[76] Ibiejugba M.A., Aderibigbe F.M., Adegboye O.S. (1990) Parameter selection for a delay equation. *J. Nigerian Math. Soc.* 9: 77–93.

[77] Kordonis I.-G.E. Philos Ch.G. (1999) The behavior of solutions of linear integro-differential equations with unbounded delay. *Computers and Mathematics with Applications* 38: 45–50.

[78] Koto T. (2002) Stability of Runge-Kutta methods for delay integor-differential equations. *Journal of Computational and Applied Mathenatics* 145: 483–492.

[79] Liang J., Xiao T. (2004) Solvability of the Cauchy problem for infinite delay equations. *Nonlinear Analysis* 58: 271–297.

[80] Enright W.H., Hu M. (1997) Continuous Runge-Kutta methods for neutral Volterra integor-differential eqnations euth delay. *Applied Numerical Mathematics* 24: 175–190.

[81] Kantorovich L.B. (1962) *Approximation method of higher analysis.* Moskva: GEF-MR, 37–38.

[82] Kontorovich L.B., Akilov G.P. (1982), Functional analysis (Chinese Translation). *Higher Education Publishing House*, Beijing, 99–103.

[83] Dewilde P., Olshevsky V., Sayed A.H. (2002), Special issue on structured and infinite systems of linear equations. *Linear Algebra and its Applications* 1(4): 343–344.

[84] Zayed A.I. *Advances in Shannon's Sampling Theory.* FL: CRC Press, Boca Raton, 1993.

[85] Li T.Y., Sauer T., Yorke J.A. (1991) Numerical solution of a class of deficient polynomial system. *SIAM Numerical Math.* 51: 481–500.

[86] Li T.Y. (1983) On Chow mallet-Paret and Yorke homotopy for solving systems of polynomials. *Bulletin of the Institute of Mathematics Acad. Sin* 11: 433–437.

[87] Alpay D., Bolotnikov V., Kaptanoğlu H.T. (2002) The Schur algorithm and reproducing kernel Hilbert spaces in the ball. *Linear Algebra Appl.* 342: 163–186.

[88] Ball J.A., Vinnikov V. (2003) Formal reproducing kernel Hilbert spaces: the commutative and noncommutative settings. *Reproducing Kernel Spaces and Applications* 1(43): 77–134.

[89] Kailath T. (1965) Some applications of reproducing kernel Hilbert spaces. *Proc. Third Annual Allerton Conf. on Circuit and System Theory* 182: 320–328.

[90] Kon M.A., Raphael L.A. (2006) Approximating functions in reproducing kernel Hilbert spaces via statistical learning theory. *Mod. Methods Math.* 182: 271–286.

[91] Groestch C.W., *Inverse Problems In The Mathematical Sciences.* Braunschweig: Vieweg, 1993.

[92] Morimoto J., Fudamoto M., Tashiro S., Arai M., Miyakawa T., Bube R.H.(1988) *Jpn. J. Appl. Phys.* 27: 2256.

[93] Winterhalter J., Maier D., Grabowski D., Honerkamp J., Müller S., Schmidt C. (1999) *J. Chem. Phys.* 110: 4035.

[94] Hansen J., Maier D., Honerkamp J., Richtering W., Horn M.F., Senff H. (1999) *J. Coll. Interf. Sci.* 215: 72.

[95] Hadamard J., Lectures on the Cauchy problems in partial differential equation. New Haven: Yale University Press, 1923.

[96] Tikhonov A.N., Arsenin, *Solutions of ill–posed problems.* John Wiley and Sons: New York, 1977.

[97] Roths T., Marth M., Weese J., Honerkamp J.(2001) Ageneralization regularization method for nonlinear ill-posed problems enhanced for nonlinear regularization terms. *Computer Physics Communication* 139: 279–296.

[98] Yildiz B., Yetiskin H., Sever A. (2003) A stability estimate on the regularized solution of the backward heat equation. *Applied mathematics and computation* 135(2-3): 561–567.

[99] Xiong X. (2004) Central difference schemes in time and error estimate on a non-standard inverse heat conduction problem. *Applied Mathematics and Computation* 157(1): 77–99.

[100] Kirsch A. *An introduction to the mathematical theory of inverse problems.* New York: Springer–Verleg New York Inc., 1996.

[101] Maleknejad K., Hadizadeh M. (1999) A new computational method for Volterra–Fredholm integral equations. *J. Comput. Math. Appl.* 37: (1999)1–8(1994)339.

[102] Kauthen P.J. (1989) Continous time collocation methods for Volterra–Fredholm integral equations. *Numer. Math.* 56: 409–424.

[103] Brunner H. (1990) On the numerical solution of nonlinear Volterra–Fredholm integral equation by collocation methods. *SIAM J. Numer. Anal.* 27(4): 987.

[104] Guoqiang H. (1995) Asymptotic error expansion for the Nystrom method for a Volterra–Fredholm integral equations. *J. Comput. Appl. Math.* 59: 49–59.

[105] Adomian G. *Solving frontier problems of physics: the decomposition method.* Dordrecht: Kluwer, 1994.

[106] Cherruault Y., Saccomandi G., Some B. (1992) New results for convergent of Adomian's method applied to integral equation. *Math. Comput. Modelling* 16(2): 85.

[107] Kanwal R.P., Liu K.C. (1989) A Taylor expansion approach for solving integral equations. *Int. J. Math. Educ. Sci. Technol.* 20(3): 411.

[108] Sezer M. (1994) Taylor polynomial solution of Volterra integral equations. *Int. J. Math. Educ. Sci. Technol.* 25(5): 625.

[109] Yalcinbas S. (2002) Taylor polynomial solutions of nonlinear Volterra–Fredholm integral equations. *Applied Mathematics and Computation* 127: 195–206.

[110] Hacia L. (1996) On approximate solution for integral equatons of mixed type. *ZAMM.* 76: 415–416.

[111] Diekman O. (1978) Thresholds and traveling waves for the geographical spread of infection. *J. Math. Biol.* 6: 109–130.

[112] Thieme H.R. (1977) A model for the spatio spread of an epidemic. *J. Math. Biol.* 4: 337–351.

[113] Guoqiang H., Liqing Z. (1994) Asymptotic expansion for the trapazoidal Nystrom method of linear Volterra–Fredholm integral equations. *J. Comput. Appl. Math.* 51: 339–348.

[114] Cui M., Deng Zh. *Numerical Functional Method In Reproducing Kernel space.* The Publication of Harbin Institute of Technology, 1988.

[115] Li Sh., Xu D., Zhao H. (2000) Stability region of nonlinear integrodifferential equations. *Applied Mathematics Letters* 13: 77–82.

[116] Liu L., Wu C., Guo F. (2004) Existence theorems of global solutions of initial value problems for nonlinear integrodifferential equations of mixed type in Banach spaces and applications. *Computers and Mathematics with Applications* 47: 13–22.

[117] Pachpatle D.G. (1994) On certain boundary value problems for nonlinear integrodifferential equations. *Acta Math. Sci. (Eng.ed.)* 412: 226–234.

[118] Wang Y. (2001) Asymptotic behavior of the numerical solutions for a system of nonlinear integrodifferential reaction-diffusion equations. *Applied Numerical Mathematics* 39: 205–223.

[119] Cui M.G., Yang L. (2005) The exact solution of a kind of nonlinear operator equations. *Applied mathematics and Computation* 167: 607–615.

[120] Debtnath L. *Nonlinear Partial Differential Equations for Scientist and Engineers*. Boston: Birkhauser, 1997.

[121] Logan J.D. *An introduction to nonlinear partial differential equations*. New York: Wiley–Interscience, 1994.

[122] Adomian G. (1995) The diffusion-Brusselator equation, *Comput. Math. Appl.* 29: 1–3.

[123] Burger J.M. (1948) A mathematical model illustrating the theory of turbulence. *Adv. Appl. Mech.* 1: 171–199.

[124] Wang J., Warnecke G. (2003) Existence and uniqueness of solutions for a nonuniformly parabilic equation. *Journal of Differential Equations* 189: 1–16.

[125] Gyöngy I. (1998) Existence and uniqueness results for semilinear stochastic partial differential equations, *Stochastic Process and their Applications* 73: 271–299.

[126] Fletcher C.A.J. (1983) Generating exact solutions of the two-dimensional Burgers' equation. *Int. J. Numer. Meth. Fluids* 3: 213–216.

[127] Benton E., Platzman G.W. (1972) A table of solutions of the one-dimensional Burgers' equation. *Quart. Appl. Math.* 30: 195–212.

[128] Kutluay S., Bahadir A.R. (1999) Numerical solution of one-dimensinal Burgers' equation: explicit and exact-explicit finite-difference methods. *J. Comput. Appl.* 103: 251–261.

[129] Caldwell J., Wanless P., Cook E. (1981) A finite element approach to Burgers'equation. *Appl. Math. Model.* 5: 239–253.

[130] Dağ İ., Irk D., Saka B. (2005) A numerical solution od the Burgers equation using cubic *B*-splines. *Appl. Math. Comp.* 163: 199–211.

[131] Bahadir A.R., Sağlam M. (2005) A mixed finite difference and boundary element approach to one-dimensional Burgers' equation, *Applied Mathematics and Computation* 160: 663–673.

[132] Ramadan M.A., El-Danaf T.S. (2005) Numerical treatment for the modified Burgers equation. *Mathematics and Computers in Simulation* 70: 90–98.

[133] Miskinis P. (2001) New exact solutions of one-dimensional inhomogeneous Burgers equation. *Report on Mathematical Physics* 48: 175–181.

[134] Sophocleous C. (2004) Transformation properties of a variable-coefficient Burgers equation. *Chao. Solitions and Fractals* 20: 1047–1057.

[135] Abia L.M., López-Marcos J.C. (1995) Runge–Kutta Methods for age-structured population Models. *Appl. Numer. Math.* 17: 1–17.

[136] Iannelli M., Kim M. (1997) Splitting method for the numerical approximation of some models of age–structured population dynamics and epidemiology. *Appl. Math. Comp.* 87: 69–93.

[137] Abia L.M., López-Marcos J.C. (1999) On the numerical integration of non-local terms for age-structured population model. *Math. Biosciences* 157: 147–167.

[138] Huang J. (1999) The existence and uniqueness of global solutions for Verhulst's partial differential equation on the population mathematical model. *Yunnan Normal University.* 19(6): 28–39.

[139] Cui M., Yan Y. (1995) The representation of the solution of a kind of operator equation $Au = f$. *Numerical Mathematics A journal of Chinese University* 1: 82–86.

[140] Kim M.Y., Park E.J. (1995) An unwind scheme for a nonlinear model in age-structured population dynamics. *Computers Math. Applic.* 30(8): 5–17.

[141] Ayaz F. (2003) On the two-dimensional differential transform method. *Applied Mathematics and Computation* 143: 361–374.

[142] Chen Y., Li B., Zhang H. (2003) Exact solutions for a new class of nonlinear evolution equations with nonlinear term of any order. *Chaos, Solitions and Fractals* 17: 675–682.

[143] Khalifa M.E., Elgamal M. (2005) A numerical solution to Klein–Gordon equation with Dirichlet boundary condition. *Applied Mathematics and Computation* 160: 451–475.

[144] Jörgens K. (1961) Das aufangswertproblem in Grossen Füureine klasse nichtliner wallenglerchungen. *Math. Z.* 295–308.

[145] Kevrekidis P.G., Konotop V.V. (2003) Compactons in discrete nonlinear Klein–Gordon models. *Math. Comput. Simulat.* 62: 79–89.

[146] Temam R. Infinite-dimensional dynamical systems in mechanics and physics. *Applied Math. Sci.*, v. 68, Springer–Verlag, 1988.

[147] Lions J.L. *Quelques methods der solution des problemes aux limites non lineaires.* Paris: Dunod/Gauthier-Villars, 1969.

[148] Temam R. Infinite-dimensional dynamical systems in mechanics and physics (2nd ed). *Applied Math. Sci.*, v. 68, Springer–Verlag, 1997.

[149] Fusaoka H. (1989) Higher-order numerical solution of a $2+1d$ nonlinear Klein–Gordon equation. *J. Phys. Soc., Japan* 58(10): 3509–3513.

[150] Fusaoka H. (1990) Numerical solution of Klein–Gordon equation with quadratic and cubic nonlinearity. *J. Phys. Soc., Japan* 59(2): 455–463.

[151] Kaya D., El–Sayed S.M. (2004) A numerical solution of the Klein–Gordon equation and convergence of the decomposition method. *Applied Mathematics and Computation* 156(2): 341–353.

[152] Lee I.J. (1995) Numerical solution for nonlinear Klein–Gordon equation by collocation method with respect to spectral method. *J. Korean Math. Soc.* 32(3): 541–551.

[153] Strauss W., Vazquez L. (1978) Numerical solution of a nonlinear Klein–Gordon equation. *J. Comput. Phys.* 28: 271–278.

[154] Nakagiri S., Ha J. (2000) Optimal control problems for the damped Klein–Gordon equations. *Nonlinear Anal.* 47: 89–100.

[155] Park J.Y., Jeong J.U. (2007) Optimal control of damped Klein–Gordon equations whit state constraints. *J. Math. Anal. Appl.* 334: 11–27.

[156] Ha J., Nakagiri S.I. (2004) Identification problems for the damped Klein–Gordon equations. *J. Math. Anal. Appl.* 289: 77–89.

[157] Wang Q., Cheng D. (2005) Numerical solution of damped nonlinear Klein–Gordon equations using variational method and finite element approach.*J. Applied Mathematics and Computation*162: 381–401.

[158] Thompson H.B., Tisdell C. (2000) Systems of difference equations associated with boundary value problems for second order systems of ordinary differential equations. *Journal of Mathematical Analysis and Application* 248: 333–347.

[159] Cheng X., Zhong Ch. (2005) Existence of positive solutions for a second order ordinary differential system. *J. Math. Anal. Appl.* 312: 14–23.

[160] Thompson H.B., Tisdell (2002) C. Boundary value problems for systems of diffference equations associated with systems of second-order ordinary differential equations. *Appl. Math. Lett.* 15(6): 761–766.

[161] Thompson H.B., Tisdell C. (2003) The nonexistence of spurious solutions to discrete, two-point boundary value problems. *Appl. Math. Lett.* 16(1): 79–84.

[162] Mawhin J., Tisdell C. (2003) A note on the uniqueness of solutions to nonlinear,discrete,vector boundary value problems. *Nonlinear Analysis and Application: to V.Lakshmikantham on His 80th Birthday* 1(2): 789–798.

[163] Valanarasu T., Ramanujam N. (2004) An asymptotic initial value method for boundary value problems for a system of singularly perturbed second–order ordinary differential equations. *Applied mathematics and Computation* 147: 227–240.

[164] Guo B., Wang Y. (1998) An almost monotone approximation for a nonlinear two-point boundary value problem. *Advances in Computational Mathematics* 8: 65–96.

[165] Baluška I., Osborm J.E. (1983) Generalized finite element method: their performence and their ratlation to mixed methods. *SIAM J. Numer. Anal.* 20: 510–536.

[166] Boglaev I.P., Miller J.J.H. (1988) Petror–Galerkin and Green's function methods for nonlinear boundary value problem with application to semiconductor devices. *Proc. Roy. Irish Acad. Sect. A* 88: 153–168.

[167] Okrasinski W., Vila S. (1999) Approximations of solutions to some second order nonlinear differential equations. *Nonlinear Analysis* 35: 1061–1072.

[168] Ixaru L.Gr., De Meyer H., Berghe G.V., Daele M.V. (1997) EXPFIT4-a FORTRAN program for the numerical solution of systems of nonlinear second-order initial-value problems. *Computer Physics Communications* 100: 71–80.

[169] Lin C., Zhang X. (2000) The new exact solution of nonlinear wave equation. *Journal of Northwest Normal University (Nartual Science)* 36(4): 42–45.

[170] Yang K., Luo H. (1995) Exact solutions of a class nonlinear equations. *Journal of Lan Zhou University* 31(1): 35–37.

[171] Zhang X., Liu L. (2001) Exisence theorems of iterate solutions for nonlinear operator equations and applications. *Acta Mathematica Scientia*, 3, 398–404.

[172] Liu Z., Xu Y. (2001) Iterate solution of nonlinear equations with π-strongly accretive operators. *Archiv der Mathematik* 77(6): 508–516.

[173] Liu Z. (2000) On the solvability of degenarate quasilinear parabolic equations of second order. *Acta Mathematica Sinica* 16(2): 313–324.

[174] He Y., Li K. (1998) Talor expansion algorithm for nonlinear operator equation. *Acta Mathematica Sinica* 2: 317–326.

[175] Gabriella M. (1997) Almost global existence for solutions of nonlinear fourth order hyperbolic equation. *Nonlinear Differential Equations And Applications* 4(2): 249–270.

[176] Chen Y.M. (1985) Generalized Pulse-Spectrum Technique. *Geophysics* 50: 1664–1675.

[177] Han B., Zhang M.L., Liu J.Q. (1997) A widely convergent Genevalized Pulse-Spectrum Technique for the coefficient converse problem of Differential equations. *Applied Mathematics and Computation* 81: 97–112.

[178] Bakushinsky A.B., Goncharsky A.V. *Ill–posed problem: theory and applications*. Dordrecht: Kluwer Academic, 1994.

[179] Tikhonov A.N., Leonov A.S., Yagola A.G., *Nonlinear ill-posed problems*. London: Chapman and Hall, 1998.

[180] Klibanov M.V., Timonov A. (2001) A new slant on the inverse problems of electromagnetic frequency sounding: "convexification" of a multi-extremal objective function via the Carleman weight functions. *Inverse Problem* 17: 1865–1887.

[181] Klibanov M.V., Timonov A. (2003) A globally convergent convexification algorithm for the inverse problem of electromagnetic frequency sounding in one dimension. *Numer. Methods Programming* 4: 52–81.

[182] Wen S., Cui M. (1997) The best approximating interpolation operator in the space W. *Chinasa J. of Numerical Math.* 3: 177–184.

[183] Day W.A. (1982) Extension of a property of the heat equation to linear thermoelasticity and other theories. *Quartly Appl. Math.* 40: 319–330.

[184] Macbain J.A. (1987) Inversion theory for parametrized diffusion problem. *SIAMI. Appl. Math.* 18: 1386–1391.

[185] Cannon J.R., Lin Y. (1990) An inverse problem of finding a parometer in a semilinear heat equation. *J. Math. Anal. Appl.* 145(2): 470–484.

[186] Cannon J.R., Lin Y., Xu S. (1994) Numerical procedures for the determination of an unknown coefficient in semi-linear parabolic differential equations. *Inverse Probl.* 10: 227–243.

[187] Cannon J.R., Yin H.M. (1990) Numerical solutions of some parabolic inverse problems. *Numer. Meth. Partial Differential Equations* 2: 177–191.

[188] Cannon J.R., Yin H.M. (1989) On a class of non-classical parabolic problems. *J. Differential Equations* 79(2): 266–288.

[189] Dehghan M. (2003) Numerical solution of one dimensional parabolic inverse problem. *Applied Mathematics and Computation* 136: 333–344.

[190] Fatullayev A., Can E. (2000) Numerical procedures for determining unknown source parameter in parabolic equations. *Mathematics and Computers in Simulation* 54: 159–167.

[191] Cannon J.R., Lin Y., Wang S. (1992) Determination of source parameter in parabolic equations. *Meccanica* 27: 85–94.

[192] Cui M., Geng F. (2007) A computational method for solving third-order singularly perturbed boundary value problems. *Appl. Math. Comput.*, doi:10.1016/j.amc.2007.09.023

[193] Lin Y., Cui M. *Representation of exact solution for a class of variable delay integro-differential equations*. China: Journal of Harbin Institute of Technology, 2008.

[194] Lin Y., Chung P., Cui M. (2007) A solution of an infinite system of quadratic equations in reproducing kernel space. *Complex Analysis and Operator Theory*, online 19 May.

[195] Du H., Cui M. (2006) Representation of the exact solutions and a stability analysis on the Fredholm integral equations of the first kind in reproducing kernel space. *Applied Mathematics and Computation* 182: 1608–1614.

[196] Cui M., Du H. (2006) Representation of exact solution for the nonlinear Volterra–Fredholm integral equations. *Applied Mathematics and Computation* 182: 1795–1802.

[197] Du H., Cui M. (2007) A method of solving a class of nonlinear integral equation in the reproducing kernel space. *Research Journal of Applied Sciences* 2(6): 659–665.

[198] Cui M., Geng F. (2007) A computational method for solving one-dimensional variable-coefficient Burgers equation. *AMC* 188: 1389–1401.

[199] Cui M., Chen Zh. (2007) The exact solution of nonlinear age-structured population model. *Nonlinear Analysis: RealWorld Applications* 8: 1096–1112.

[200] Lin Y., Cui M., Yang L. (2006) Representation of the exact solution for a kind of nonlinear partial differential equation. *Applied Mathematics Letters* 19: 808–813.

[201] Lin Y., Cui M. *A new method to solve the damped nonlinear Klein–Gordon equation*. Science in China Series A, 2008.

[202] Geng F., Cui M. (2007) Solving a nonlinear system of second order boundary value problems. *J. Math. Anal. Appl.* 327: 1167–1181.

[203] Cui M., Chen Zh. (2006) The exact solution of nonlinear operator equation $AuBu + Cu = f$. *J. Math. Anal. Appl.* 317: 113–126.

[204] Cui M., Lin Y., Yang L. (2007) A new method for solving the coefficient inverse problem. *Science in China Series A* 50(4): 561–572.

[205] Fang Y.S., Cui M. *New algorithm for the determination of an unknown parameter in parabolic equations*. The pure and applied mathematics, 2008.

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